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Qi et al.

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(54) **SYSTEMS AND METHODS FOR
SUPPRESSING SOUND LEAKAGE**

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patent is extended or adjusted under 35
U.S.C. 154(b) by 195 days.

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claimer.

(21) Appl. No.: **18/361,844**

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(Continued)

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G10K 9/13 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H04R 25/505** (2013.01); **G10K 9/13**
(2013.01); **G10K 9/22** (2013.01); **G10K 11/175**
(2013.01);

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(58) **Field of Classification Search**
CPC H04R 25/505; H04R 1/2811; H04R 9/066;
H04R 1/2876; H04R 17/00;
(Continued)

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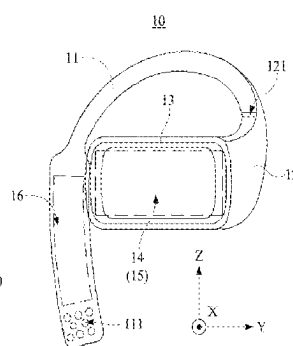
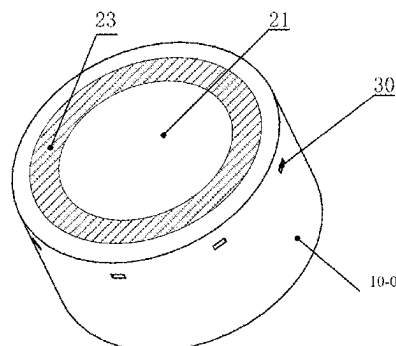
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(57) **ABSTRACT**

A speaker comprises a housing, a transducer residing inside the housing, and at least one sound guiding hole located on the housing. The transducer generates vibrations. The vibrations produce a sound wave inside the housing and cause a leaked sound wave spreading outside the housing from a portion of the housing. The at least one sound guiding hole guides the sound wave inside the housing through the at least one sound guiding hole to an outside of the housing. The guided sound wave interferes with the leaked sound wave in a target region. The interference at a specific frequency relates to a distance between the at least one sound guiding hole and the portion of the housing.

19 Claims, 38 Drawing Sheets



Related U.S. Application Data

which is a continuation of application No. 17/804,611, filed on May 31, 2022, now Pat. No. 11,659,341, which is a continuation of application No. 17/170,874, filed on Feb. 8, 2021, now Pat. No. 11,363,392, which is a continuation-in-part of application No. 17/074,762, filed on Oct. 20, 2020, now Pat. No. 11,197,106, which is a continuation-in-part of application No. 16/813,915, filed on Mar. 10, 2020, now Pat. No. 10,848,878, which is a continuation of application No. 16/419,049, filed on May 22, 2019, now Pat. No. 10,616,696, which is a continuation of application No. 16/180,020, filed on Nov. 5, 2018, now Pat. No. 10,334,372, which is a continuation of application No. 15/650,909, filed on Jul. 16, 2017, now Pat. No. 10,149,071, which is a continuation of application No. 15/109,831, filed as application No. PCT/CN2014/094065 on Dec. 17, 2014, now Pat. No. 9,729,978, said application No. 17/170,874 is a continuation-in-part of application No. 16/833,839, filed on Mar. 30, 2020, now Pat. No. 11,399,245, which is a continuation of application No. 15/752,452, filed as application No. PCT/CN2015/086907 on Aug. 13, 2015, now Pat. No. 10,609,496, application No. 18/361,844, filed on Jul. 29, 2023 is a continuation-in-part of application No. 17/457,258, filed on Dec. 2, 2021, now Pat. No. 11,838,705, which is a continuation-in-part of application No. PCT/CN2021/109154, filed on Jul. 29, 2021.

(51) **Int. Cl.**

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G10K 11/175 (2006.01)
G10K 11/178 (2006.01)
G10K 11/26 (2006.01)
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H04R 9/06 (2006.01)
H04R 17/00 (2006.01)

(52) **U.S. Cl.**

CPC **G10K 11/178** (2013.01); **G10K 11/26** (2013.01); **H04R 1/2811** (2013.01); **H04R 9/066** (2013.01); **G10K 2210/3216** (2013.01); **H04R 1/2876** (2013.01); **H04R 17/00** (2013.01); **H04R 2460/13** (2013.01)

(58) **Field of Classification Search**

CPC H04R 2460/13; H04R 1/2888; G10K 9/13; G10K 9/22; G10K 11/175; G10K 11/178; G10K 11/26; G10K 2210/3216; G10K 11/17827; G10K 11/17881; G10K 15/04
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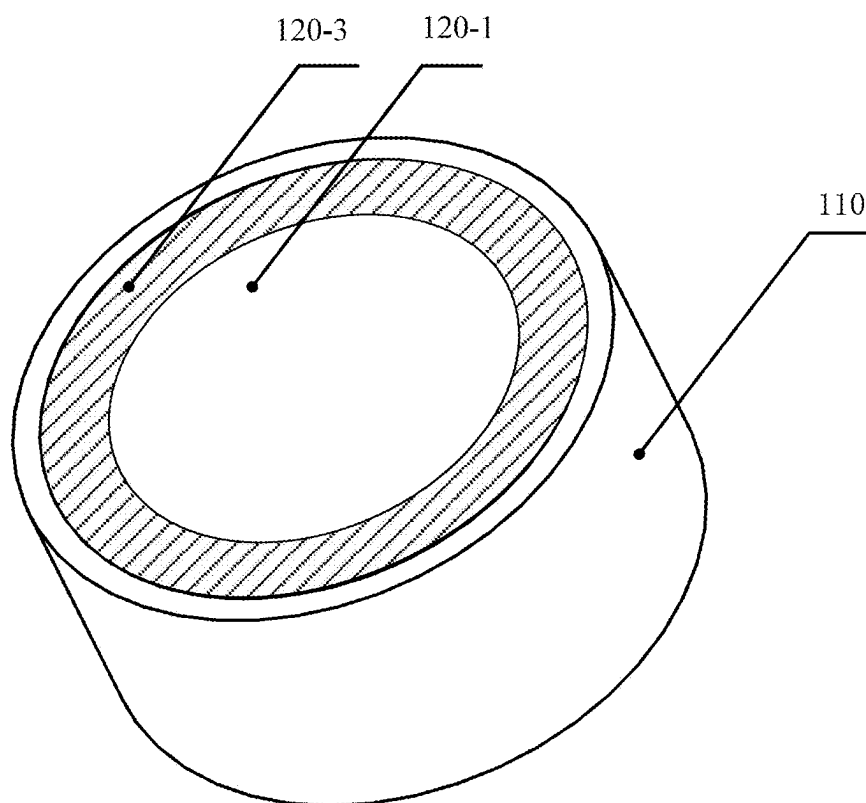


FIG. 1A

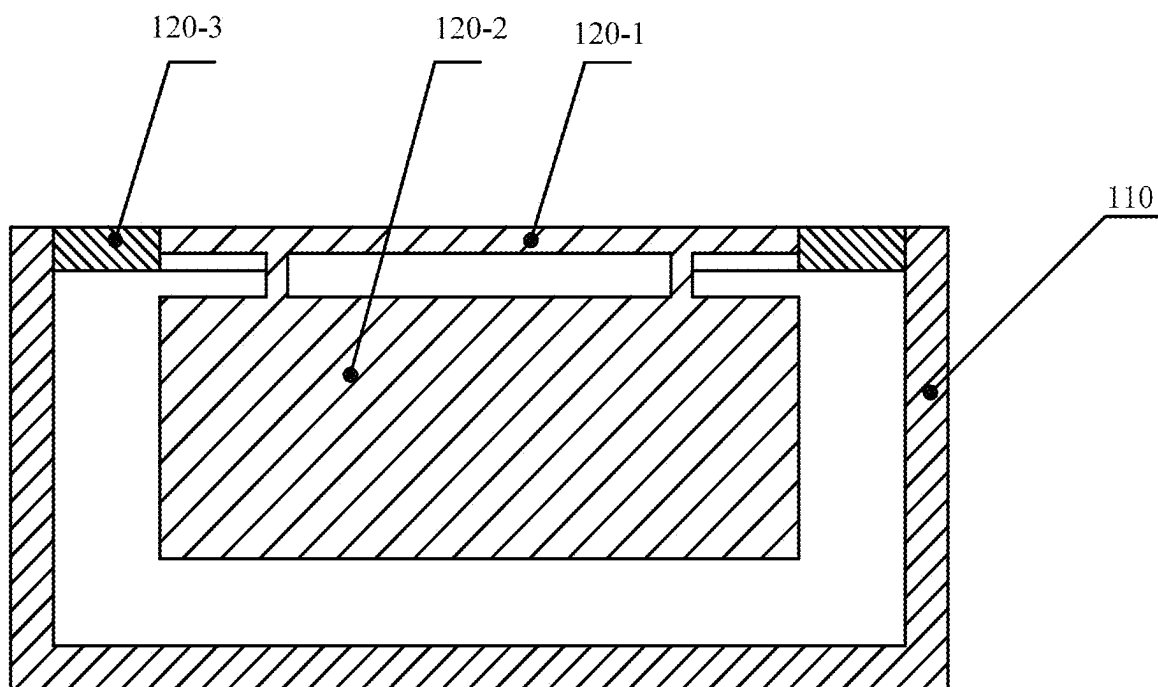


FIG. 1B

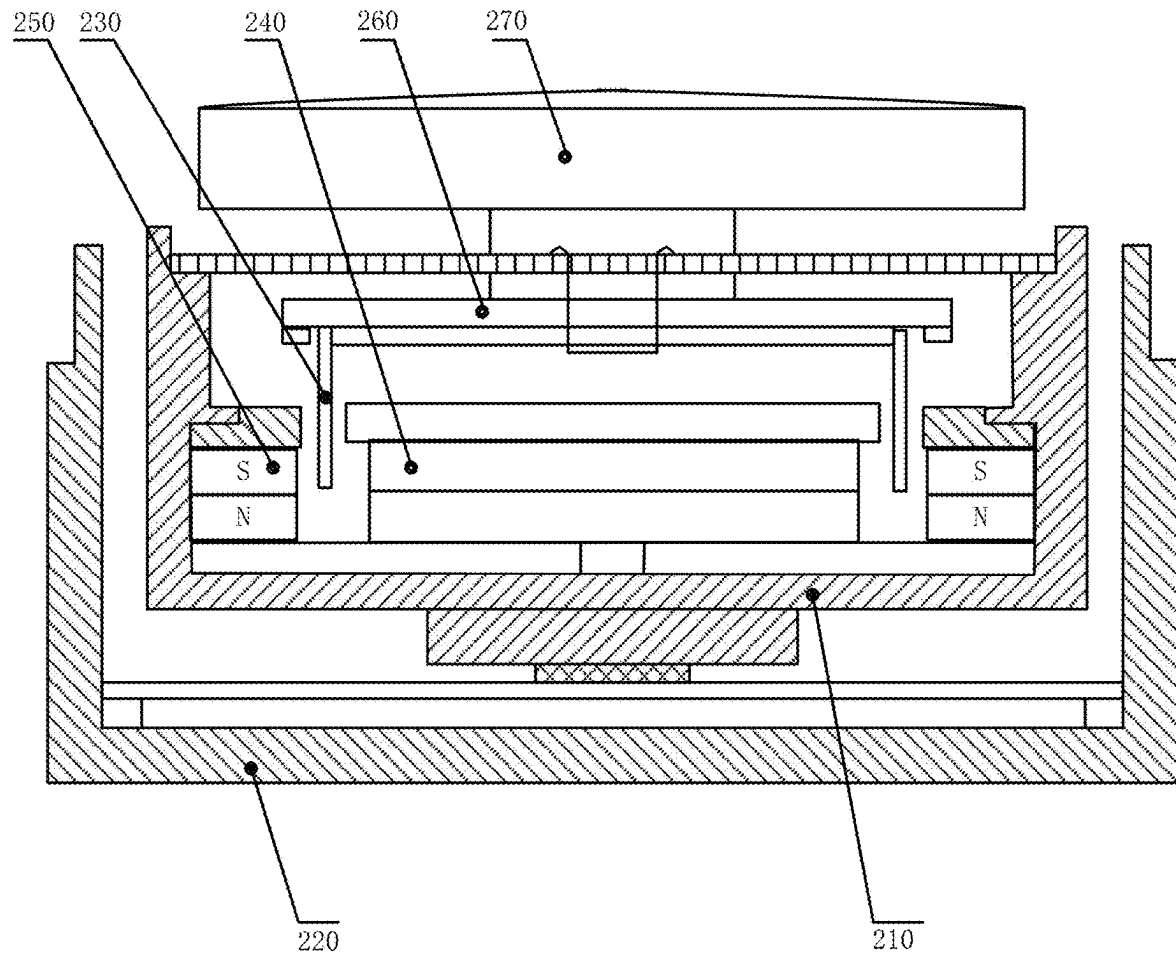


FIG. 2

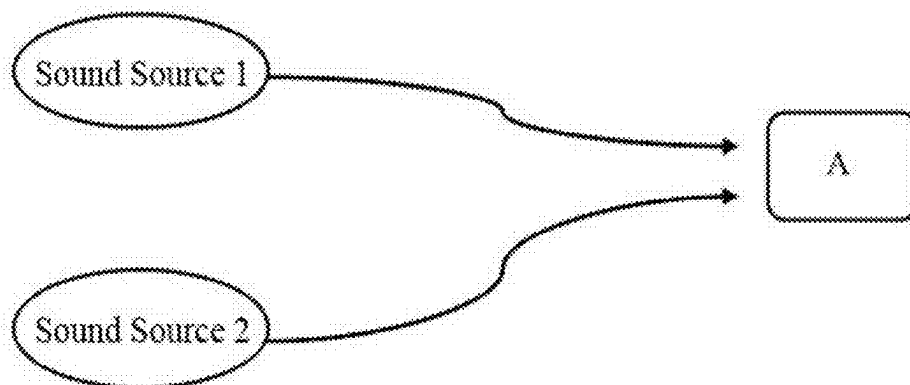


FIG. 3

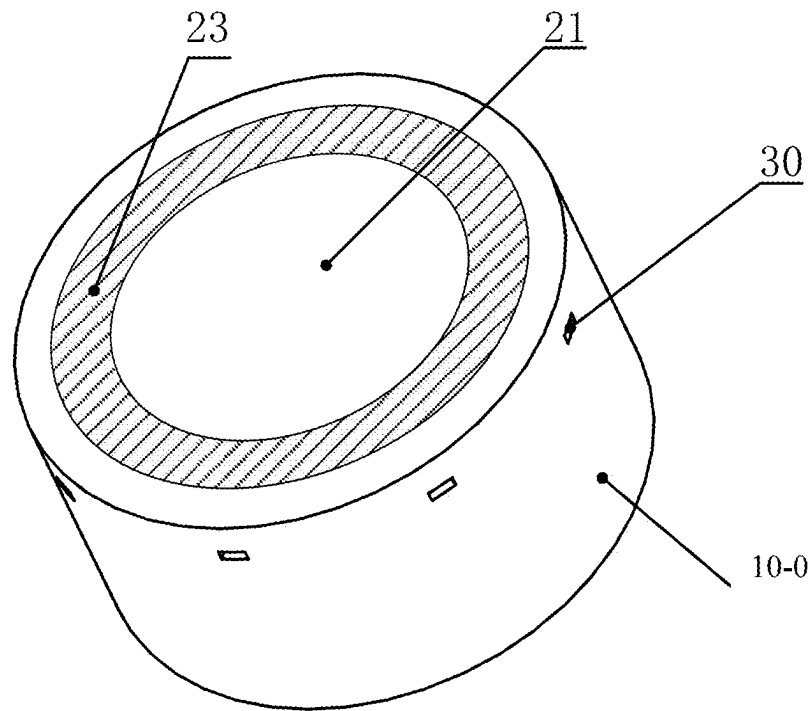


FIG. 4A

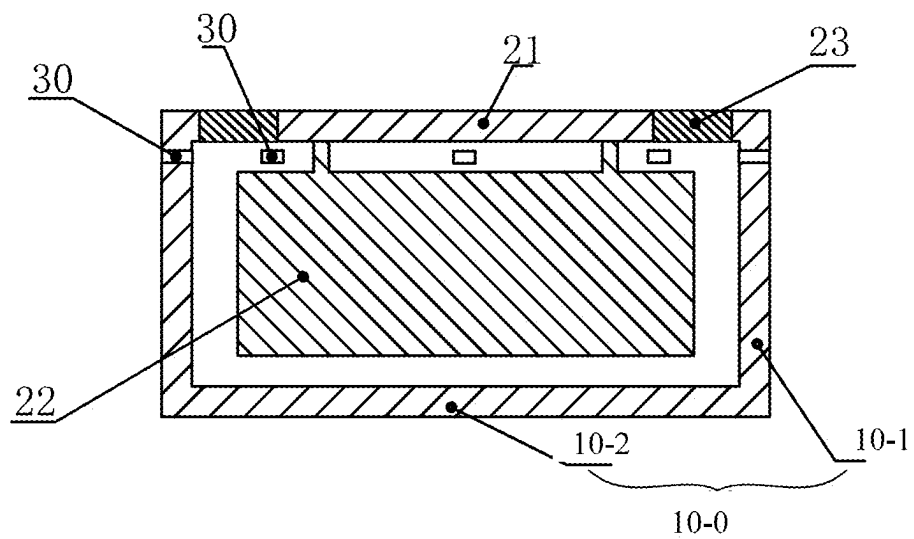


FIG. 4B

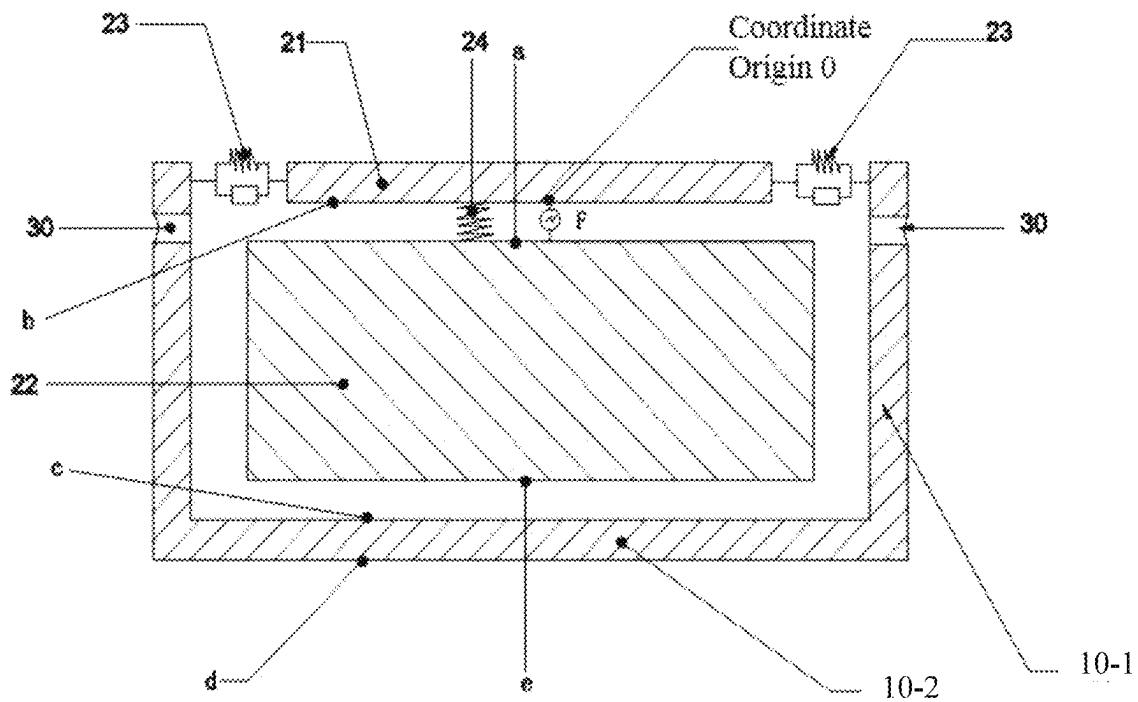


FIG. 4C

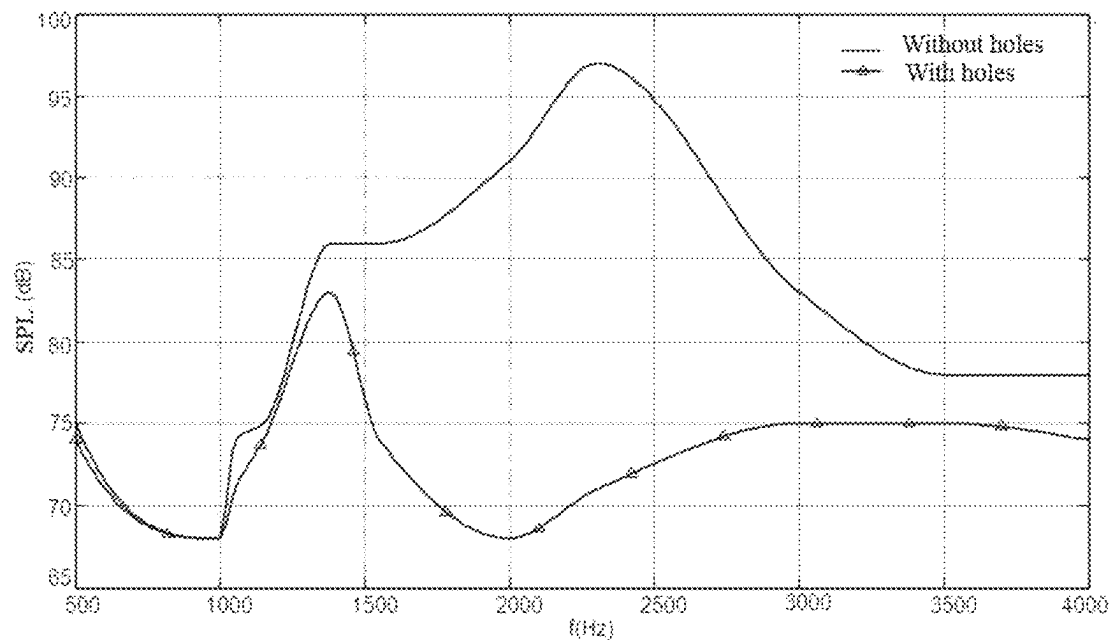


FIG. 4D

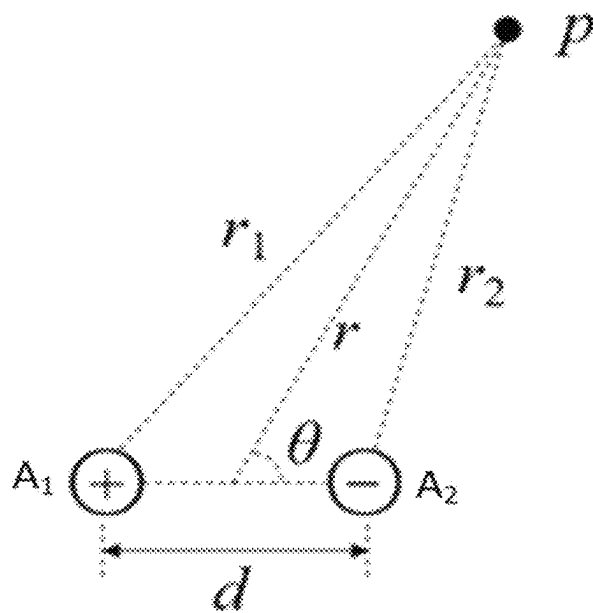


FIG. 4E

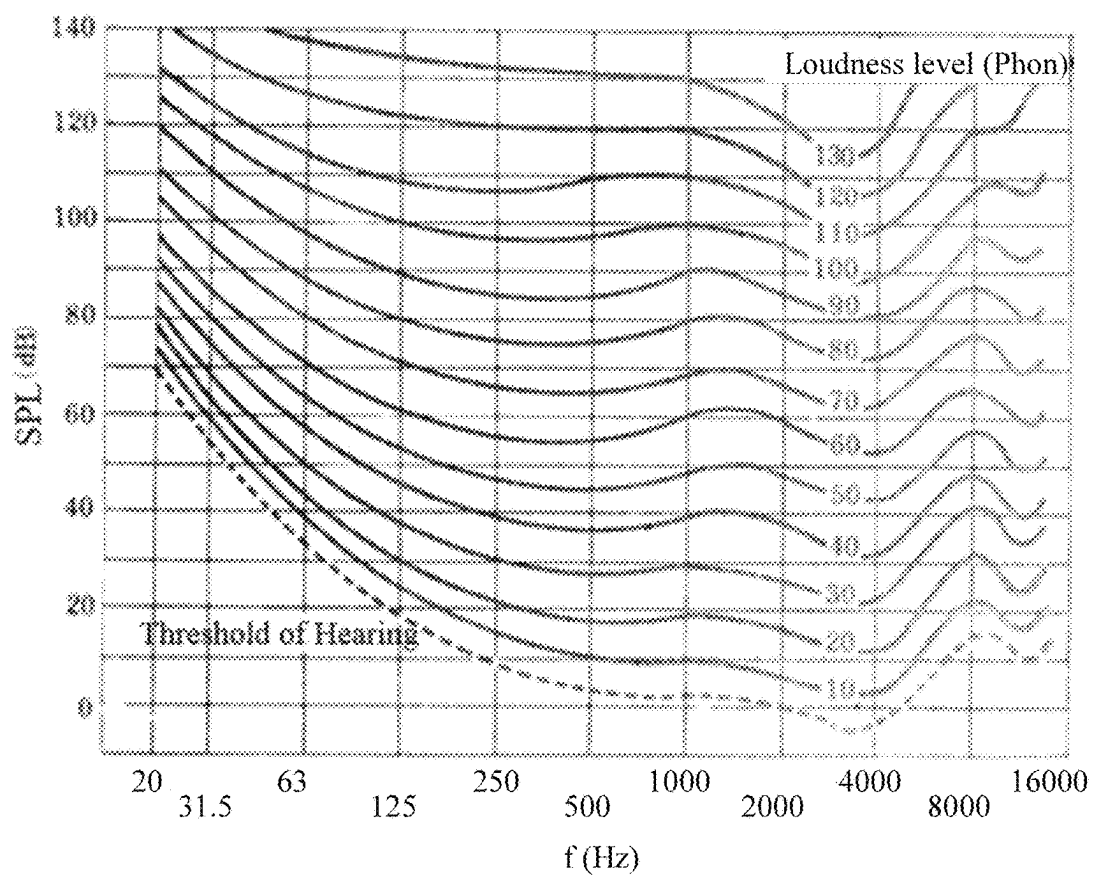


FIG. 5

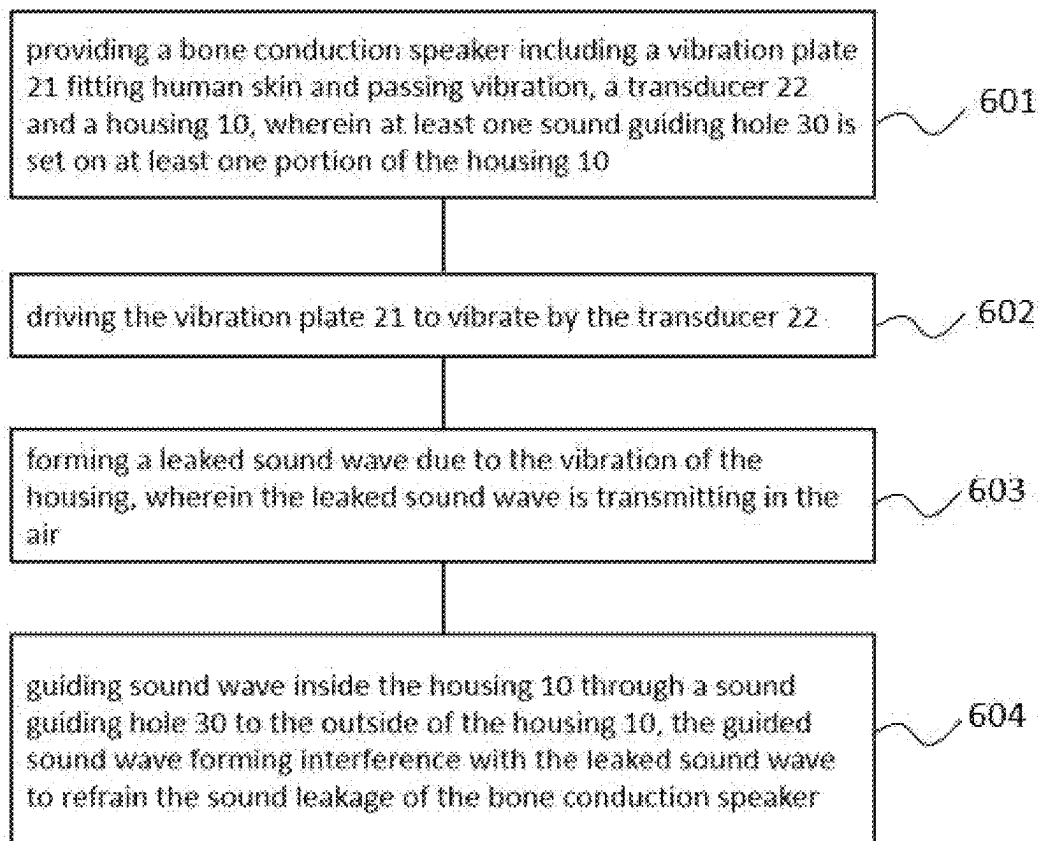


FIG. 6

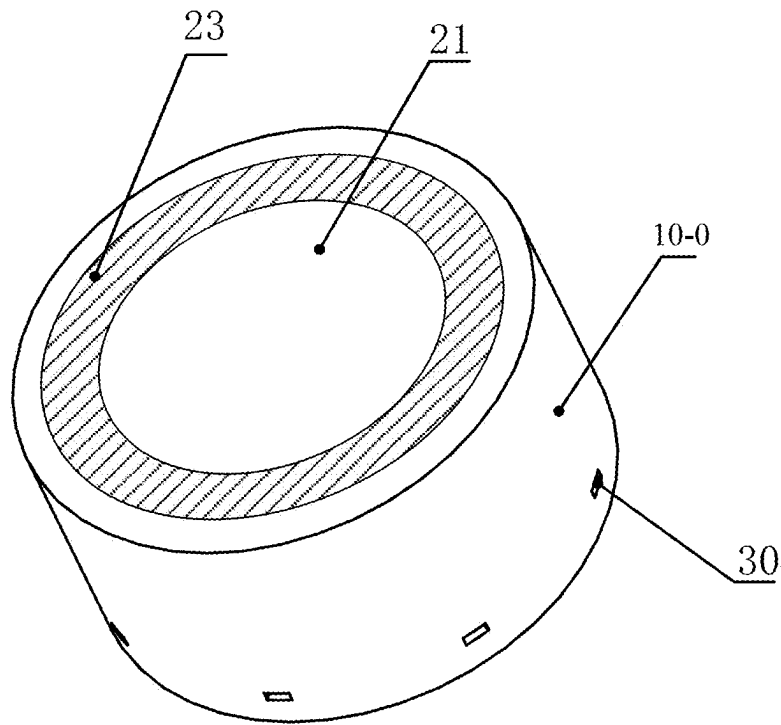


FIG. 7A

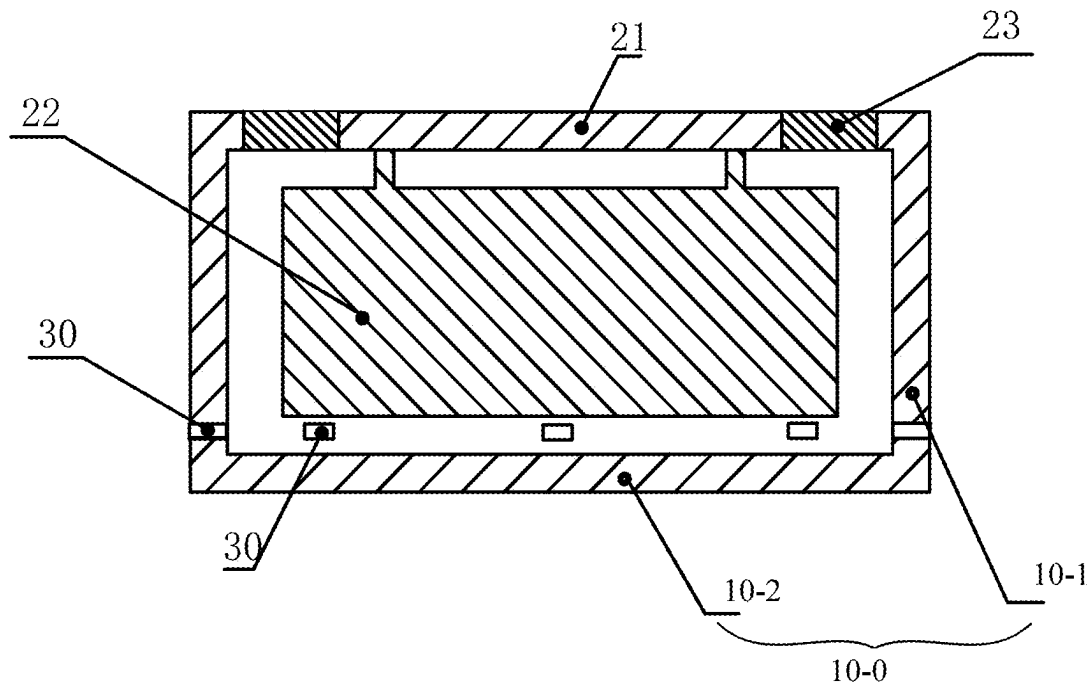


FIG. 7B

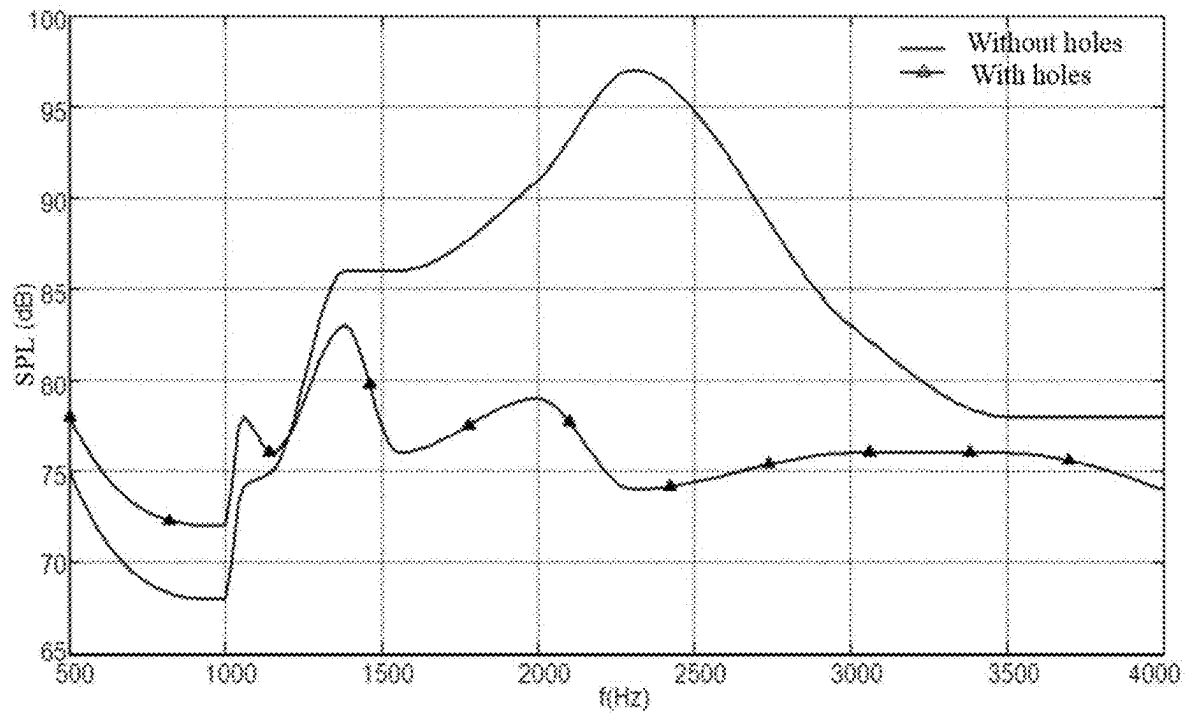


FIG. 7C

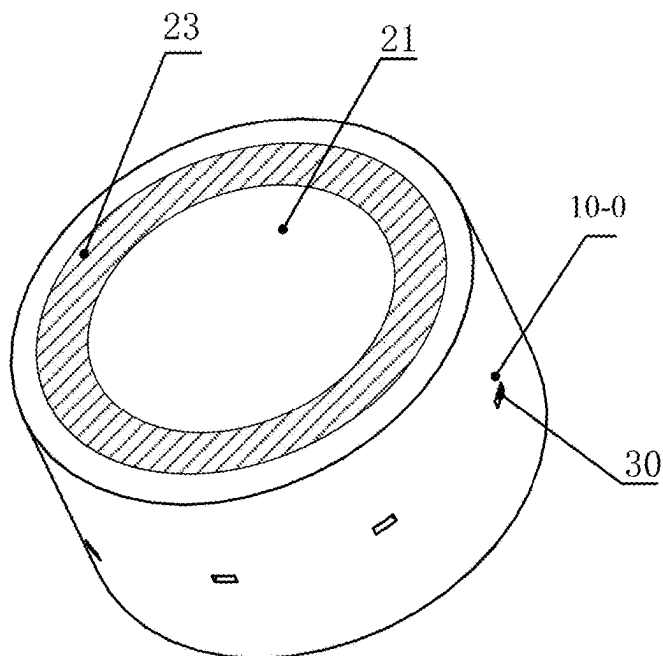


FIG. 8A

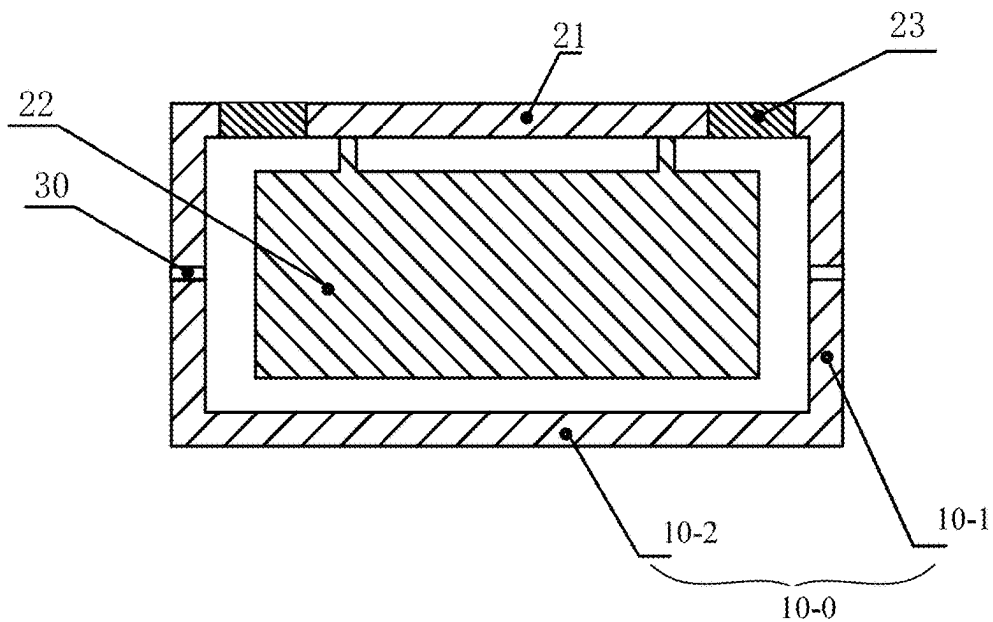


FIG. 8B

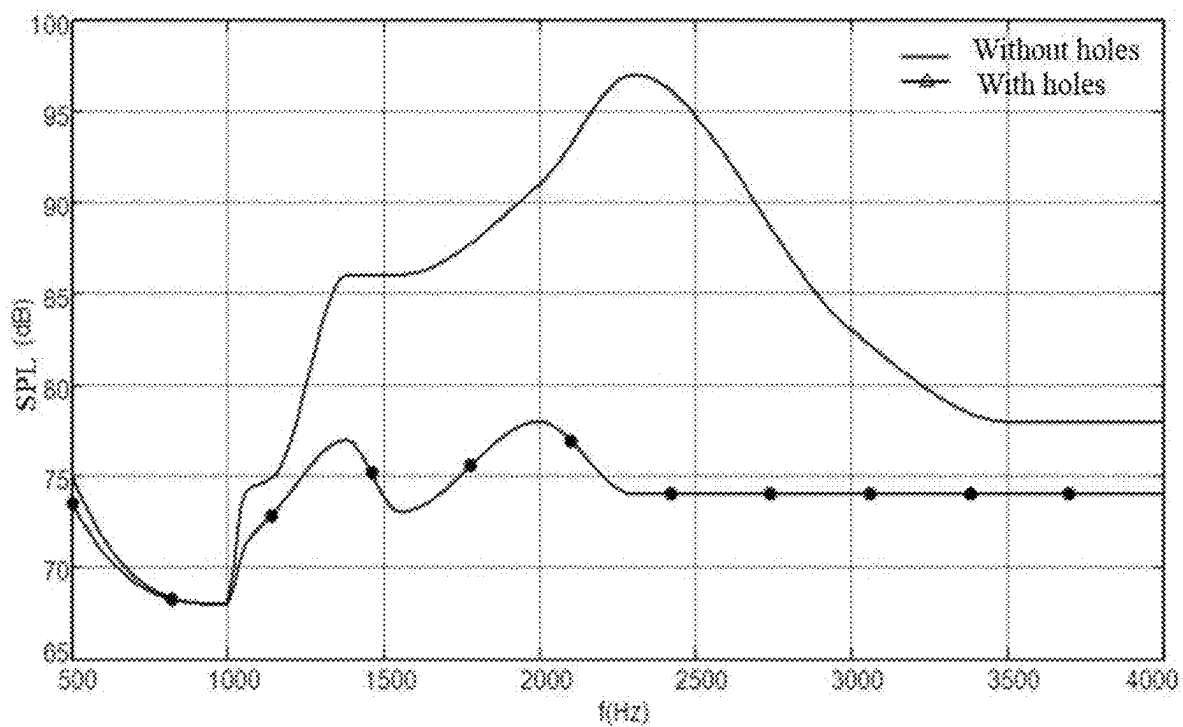


FIG. 8C

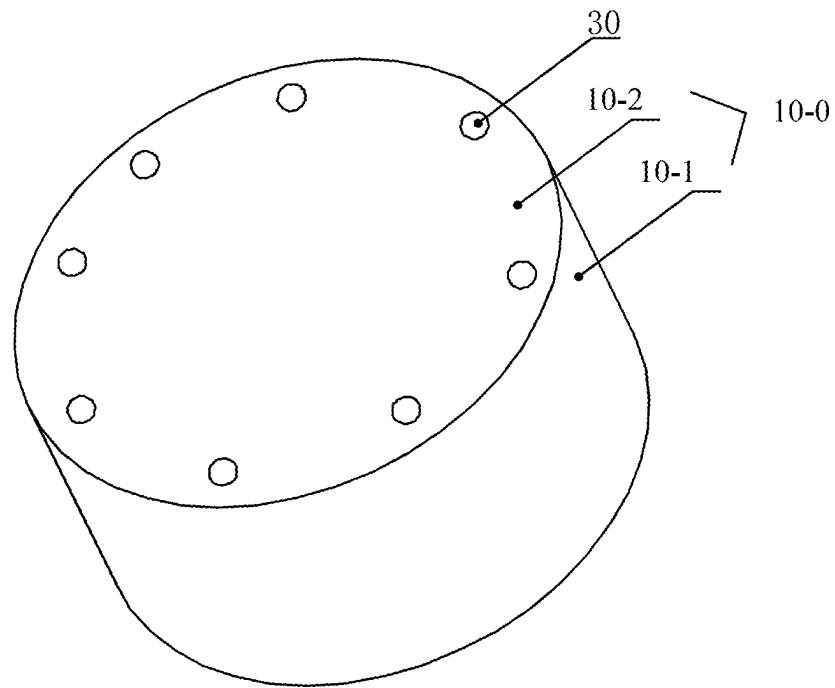


FIG. 9A

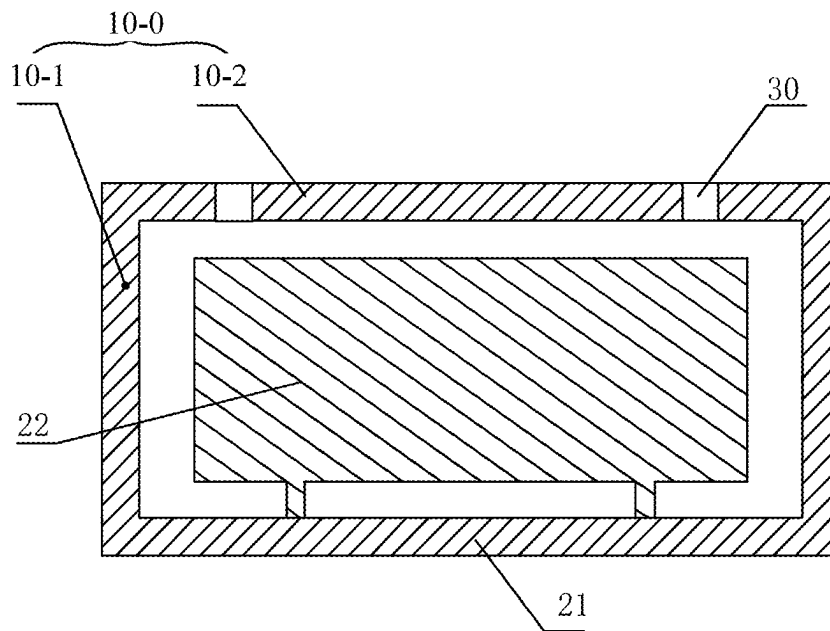


FIG. 9B

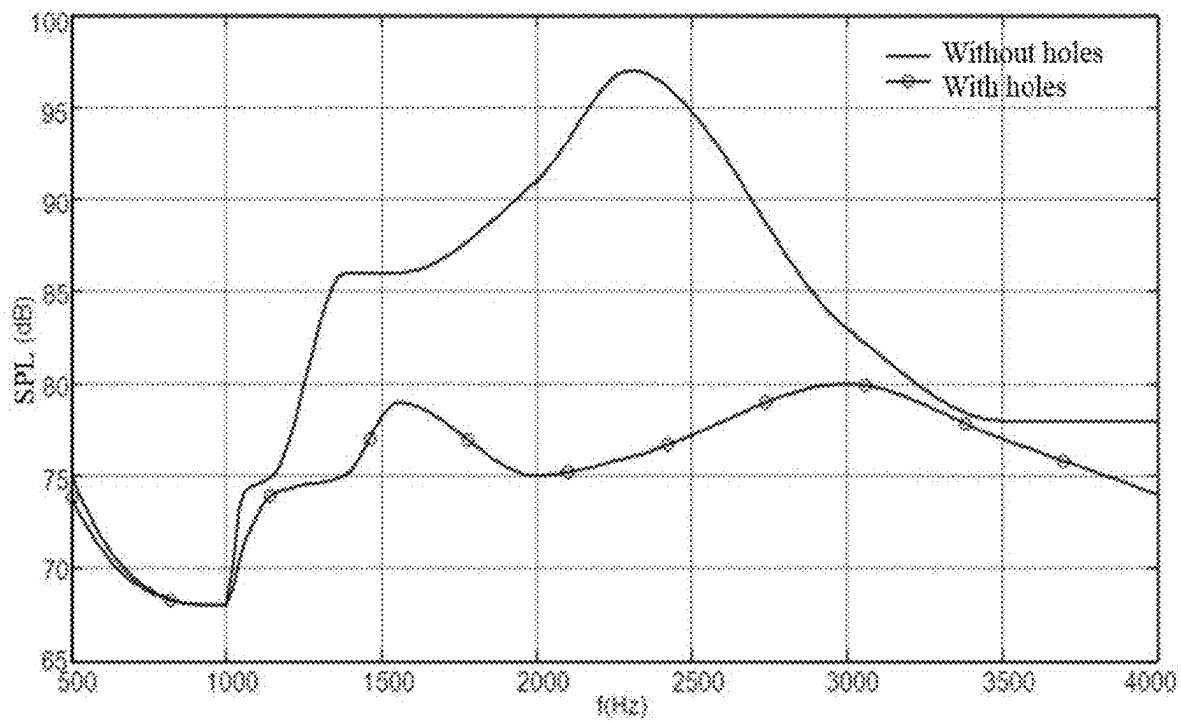


FIG. 9C

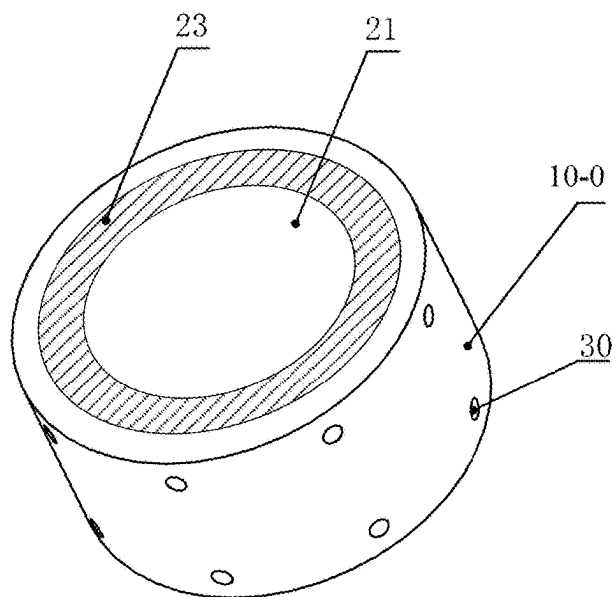
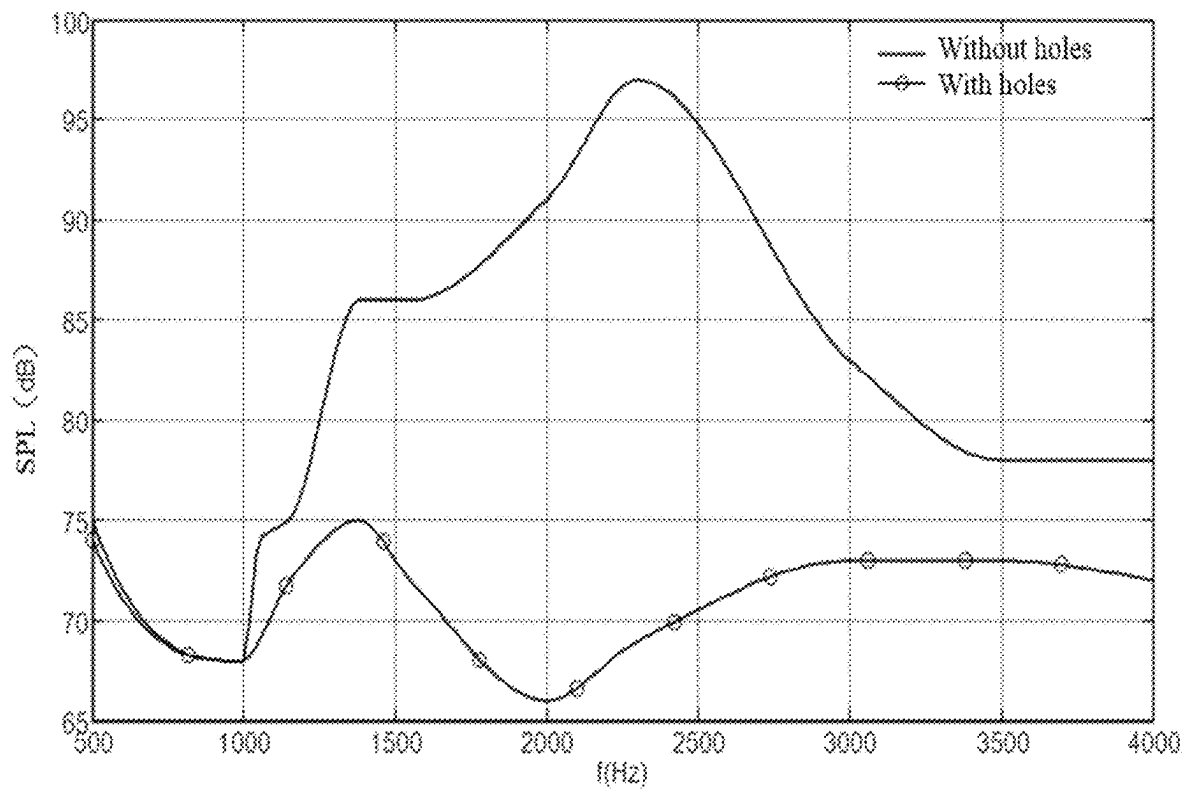
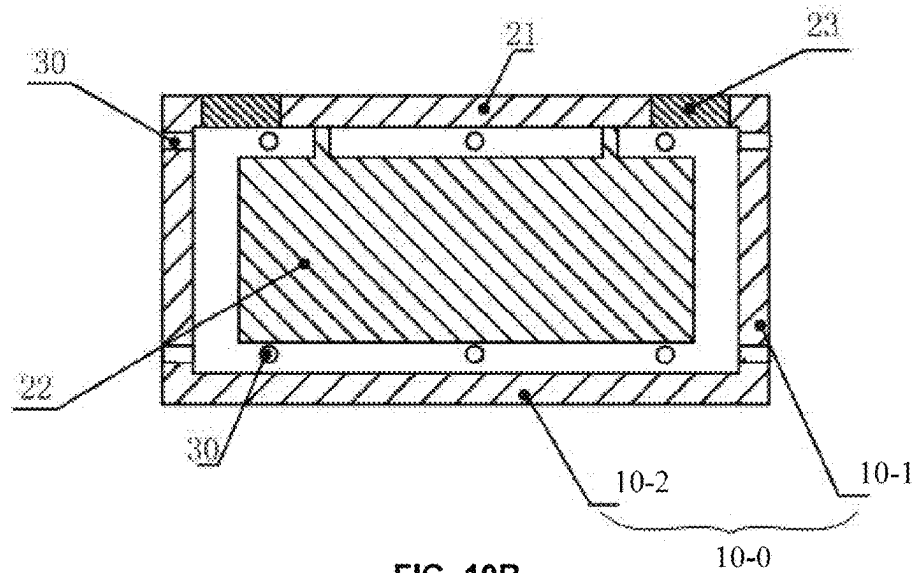


FIG. 10A



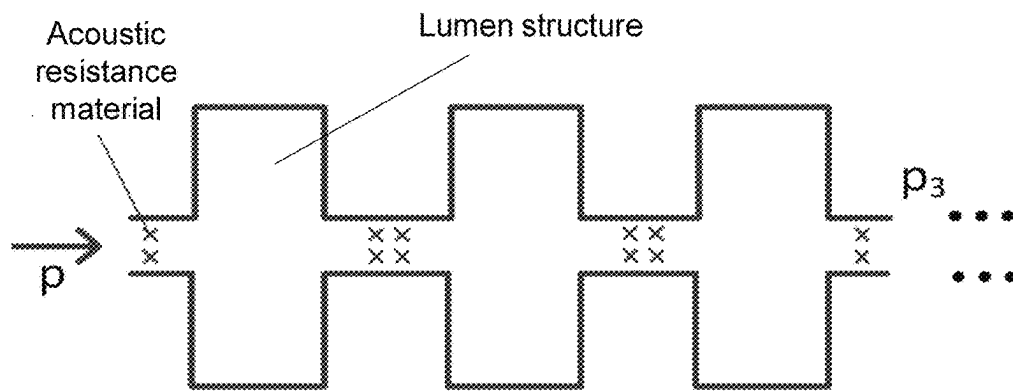


FIG. 10D

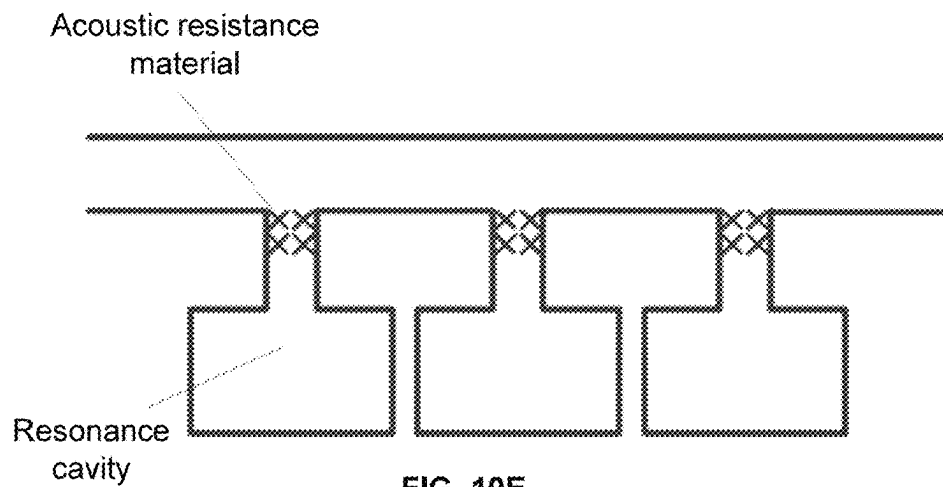


FIG. 10E

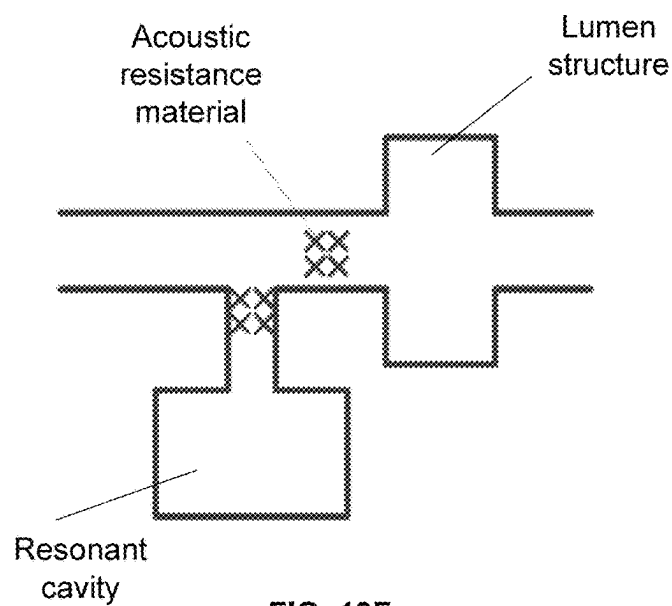


FIG. 10F

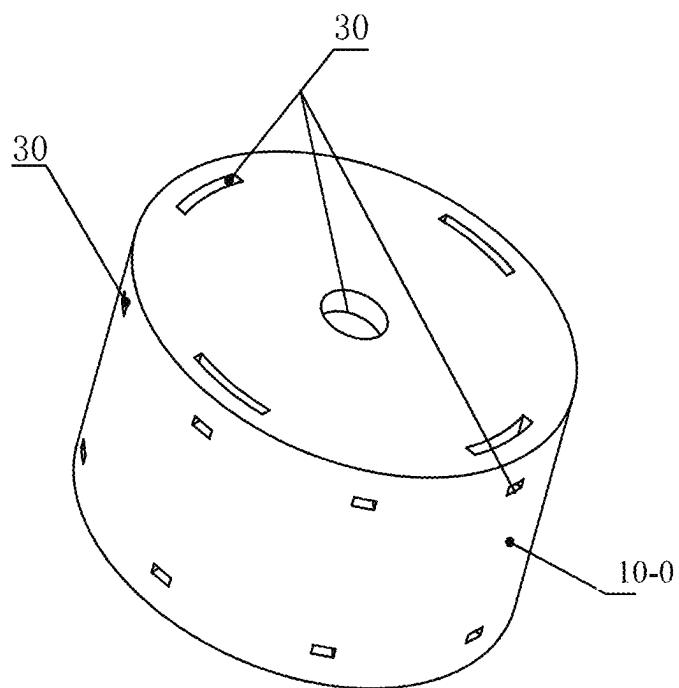


FIG. 11A

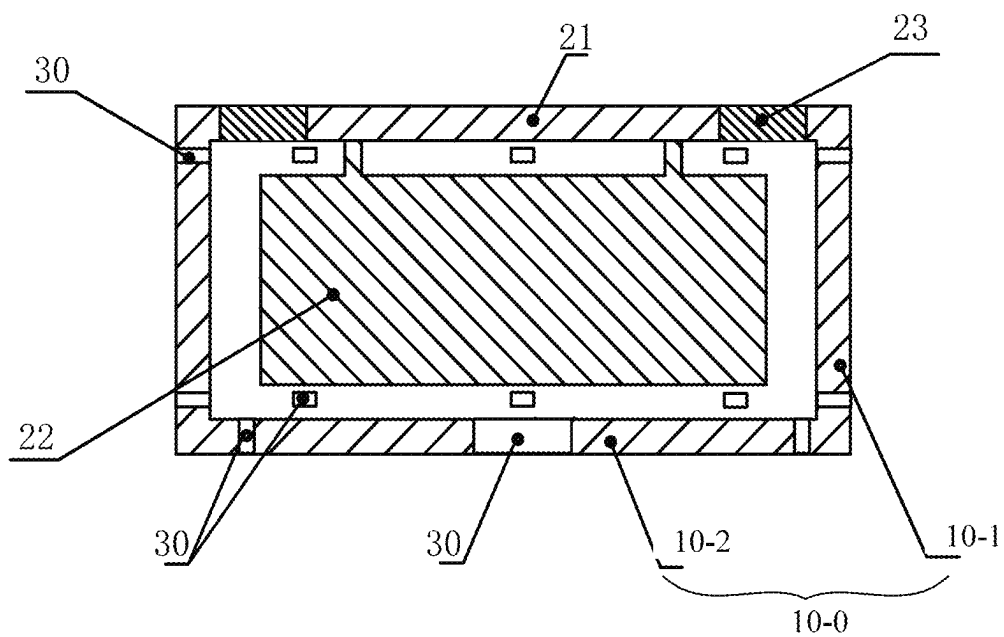


FIG. 11B

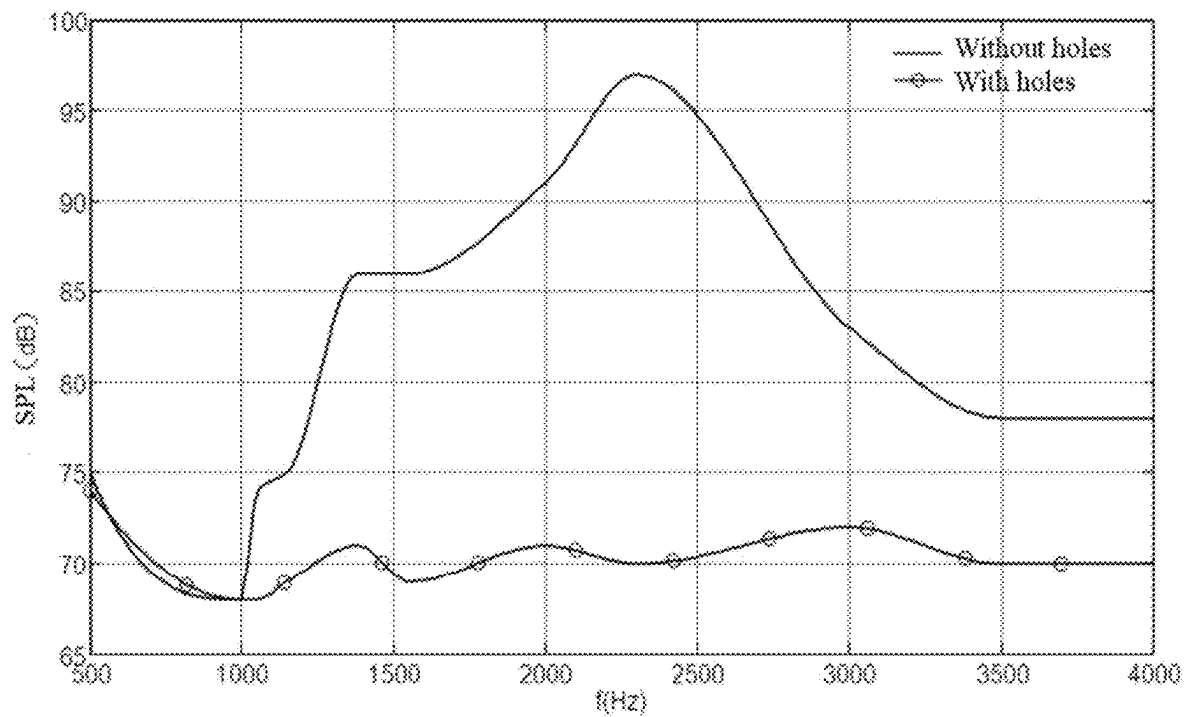


FIG. 11C

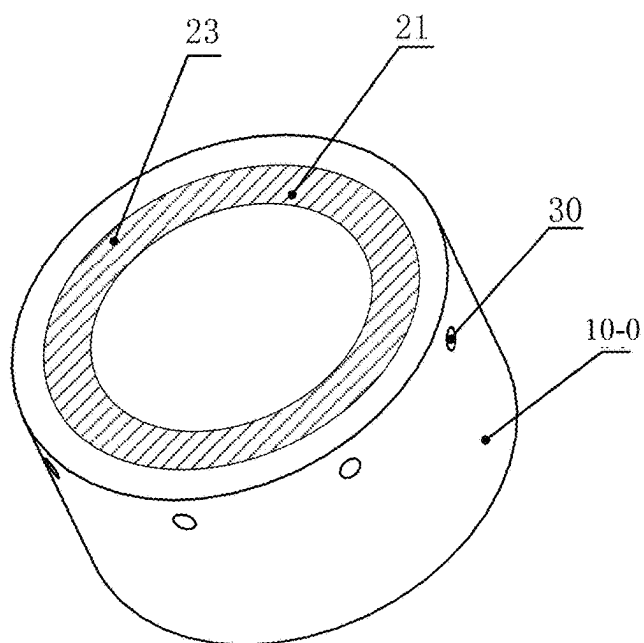
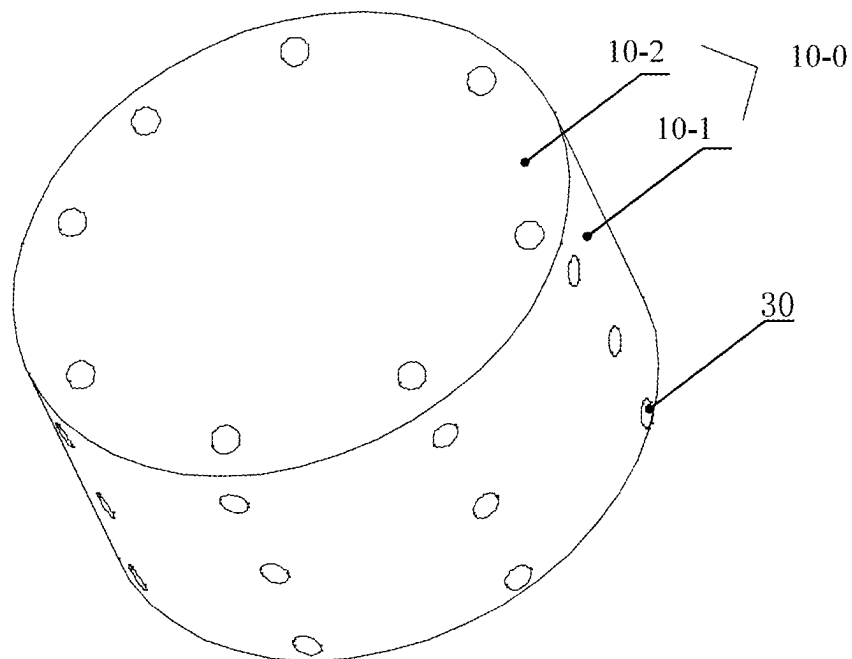
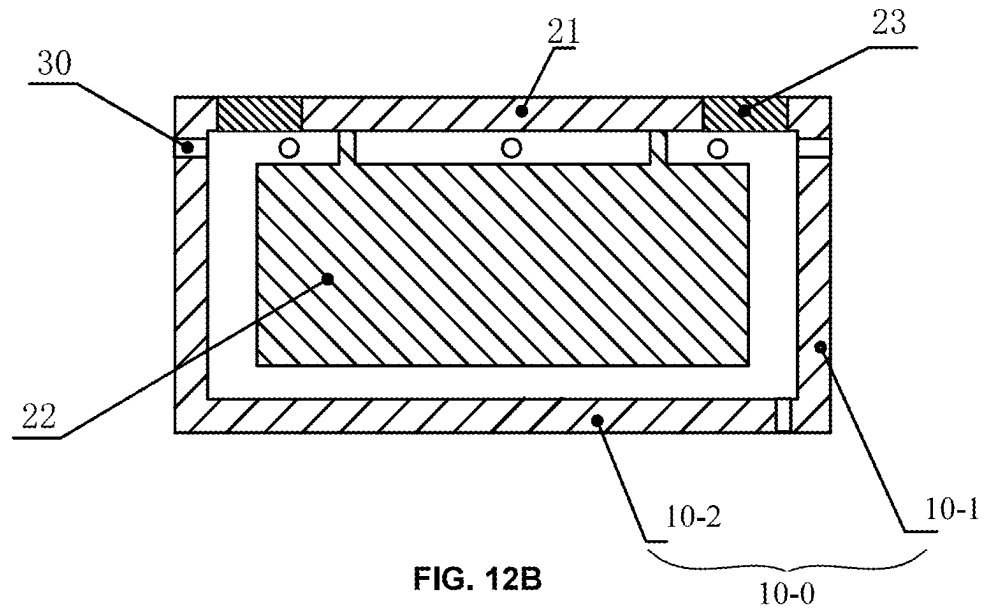


FIG. 12A



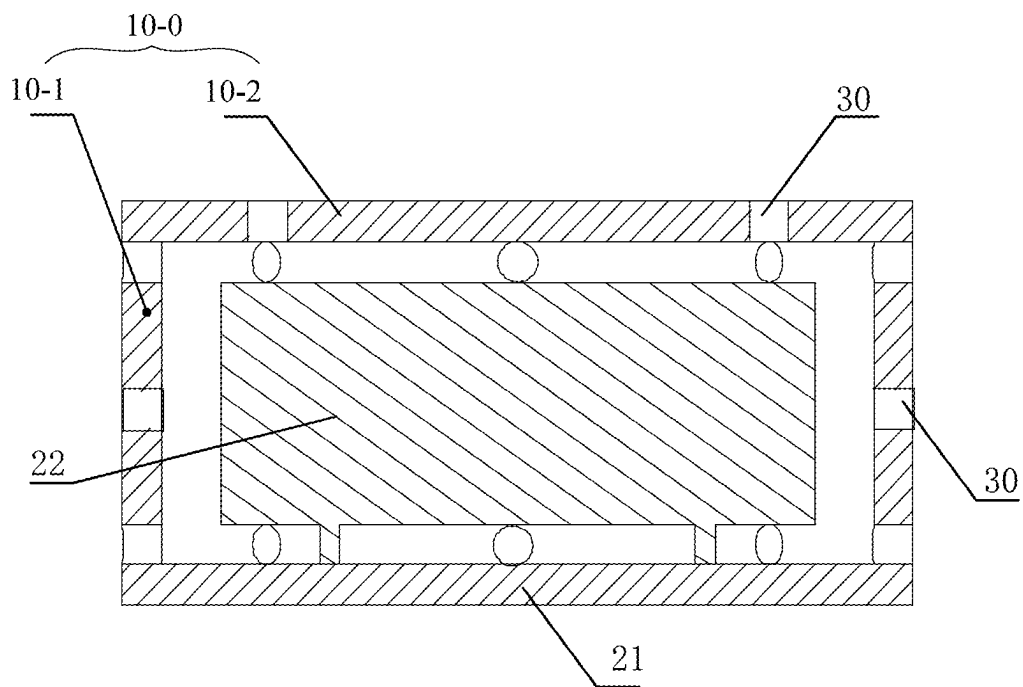


FIG. 13B

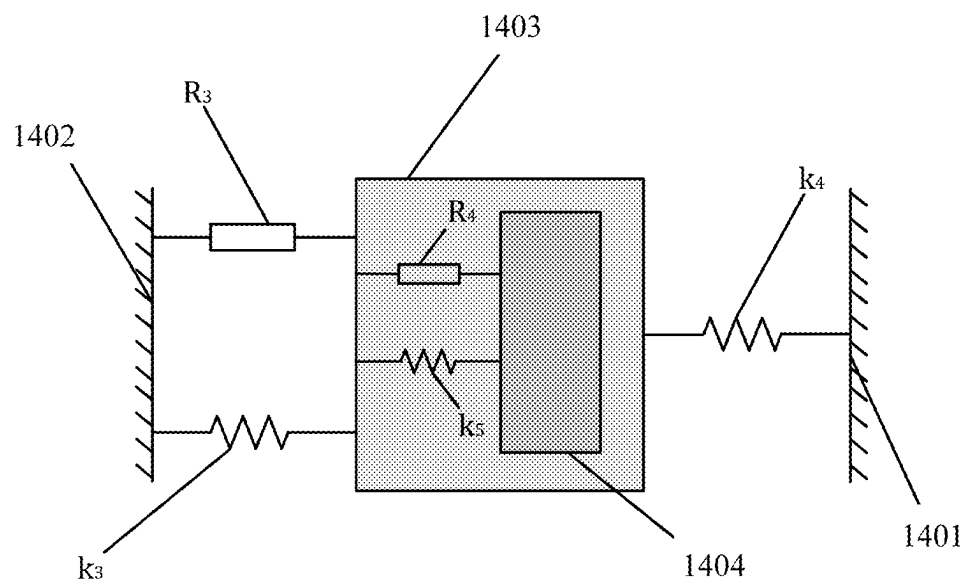


FIG. 14

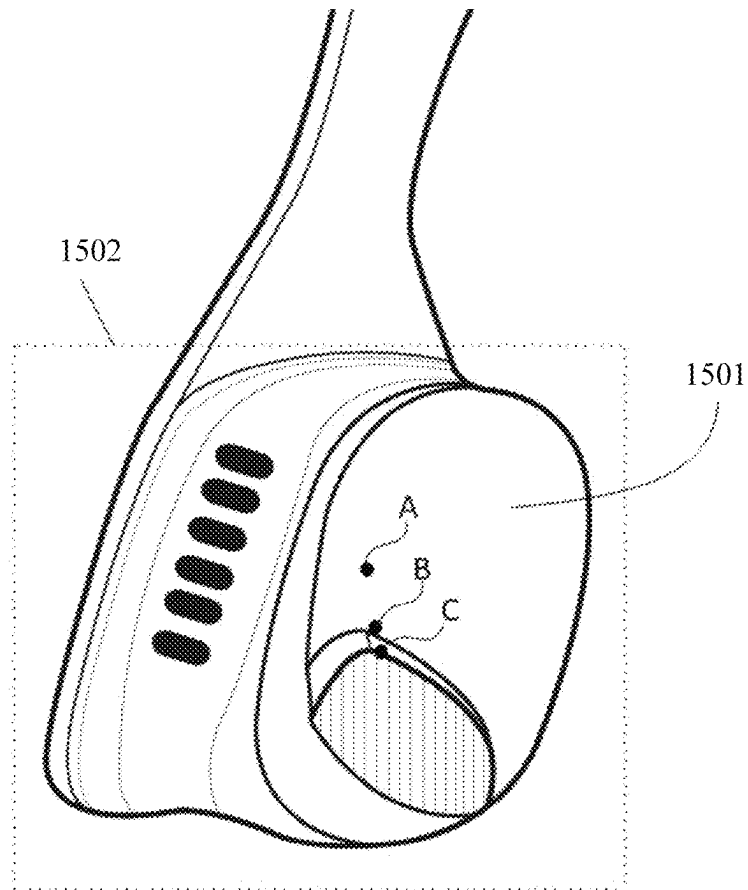


FIG. 15A

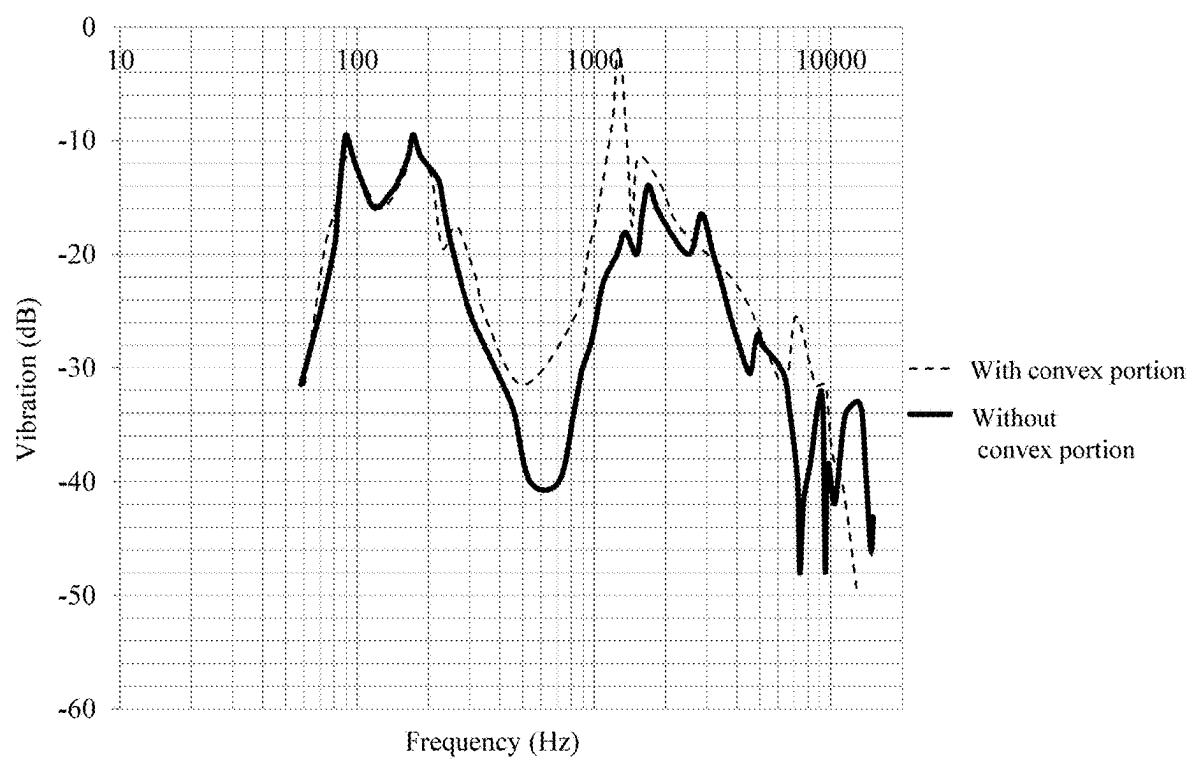


FIG. 15B

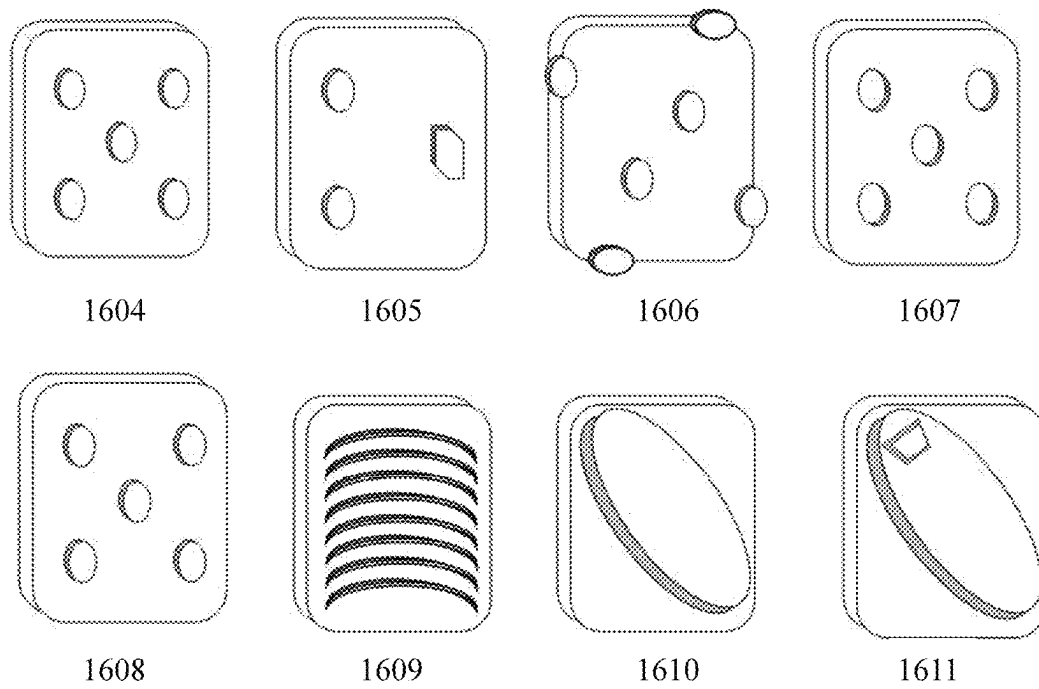


FIG. 16

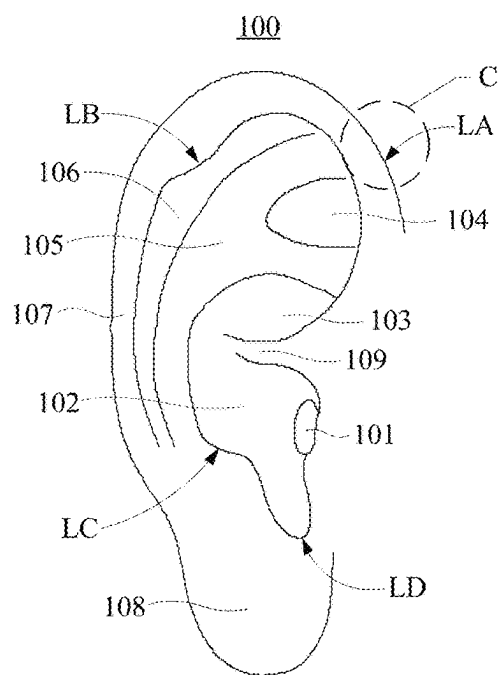


FIG. 17

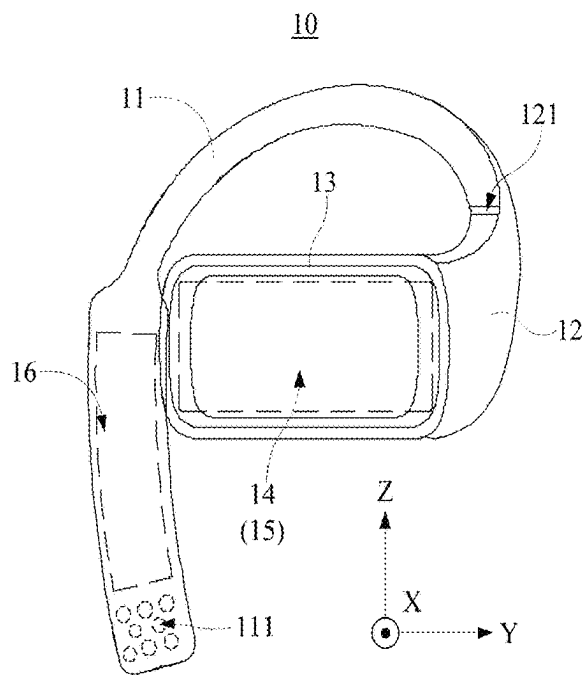


FIG. 18

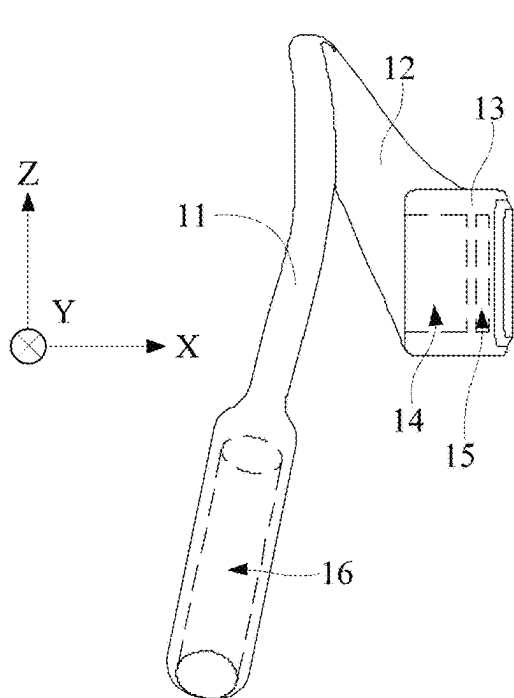


FIG. 19

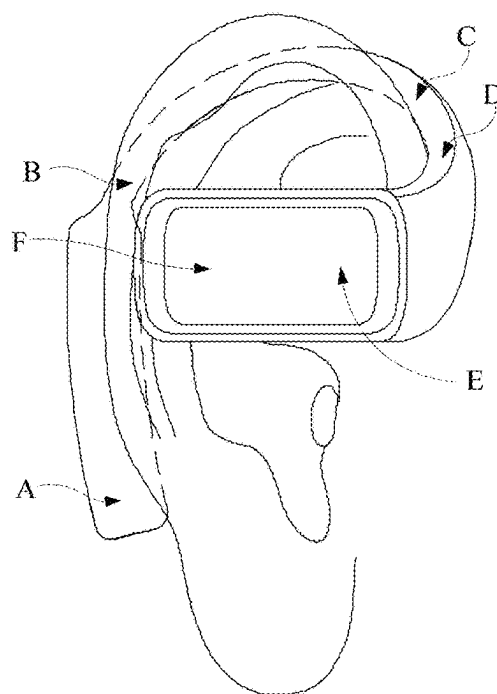


FIG. 20

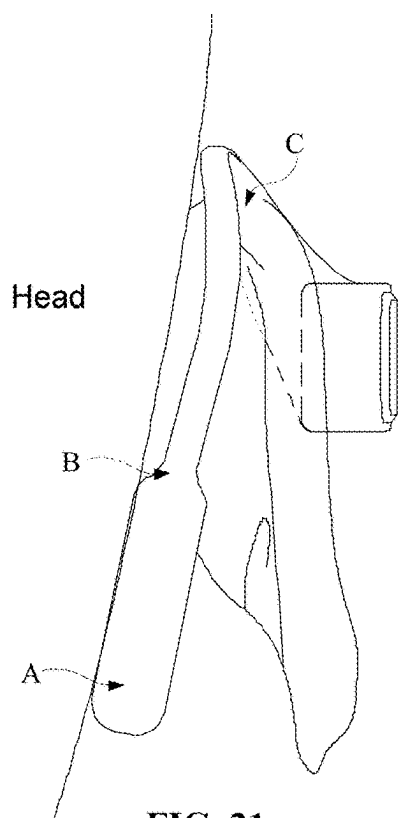


FIG. 21

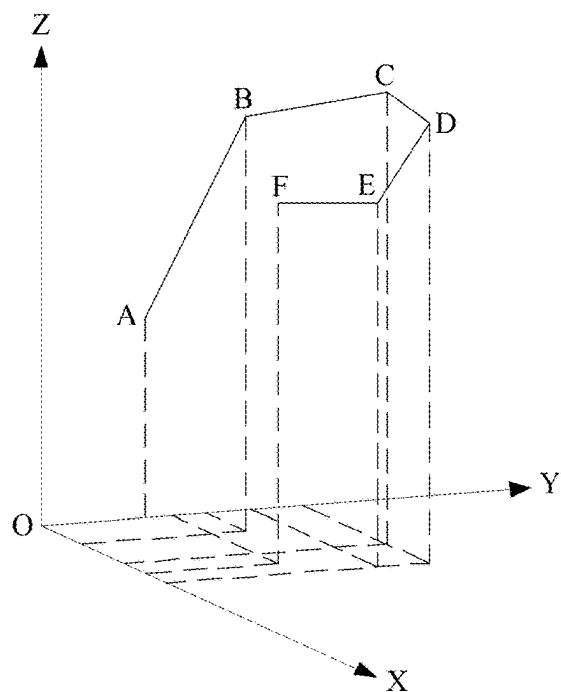


FIG. 22

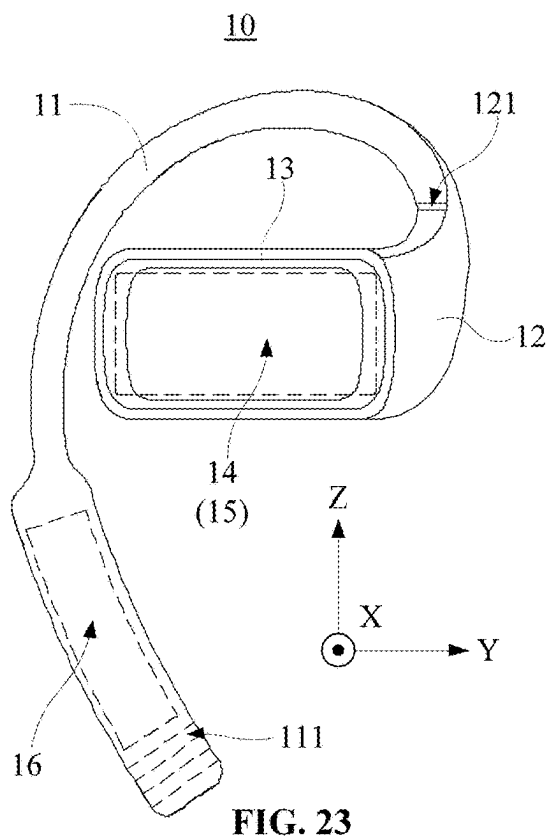


FIG. 23

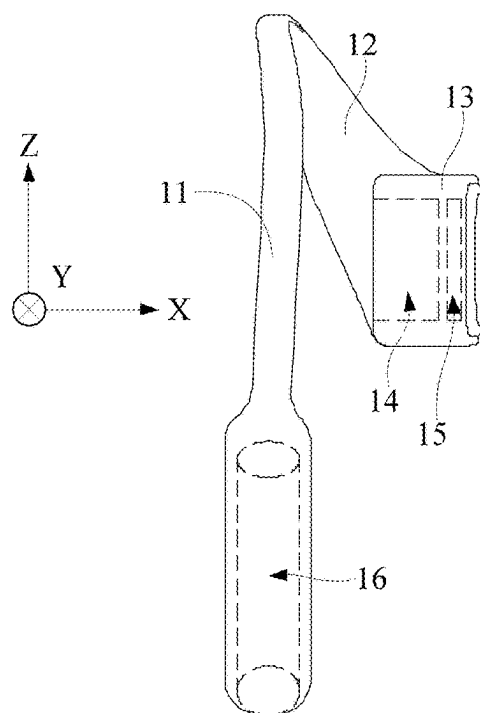


FIG. 24

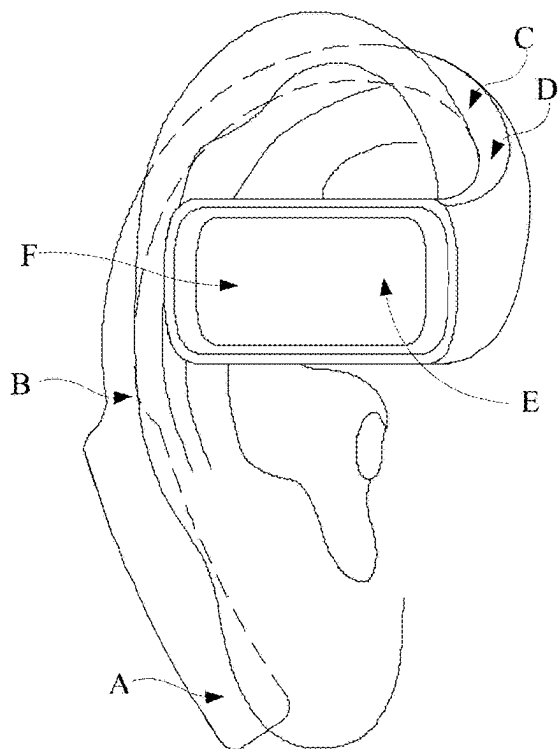


FIG. 25

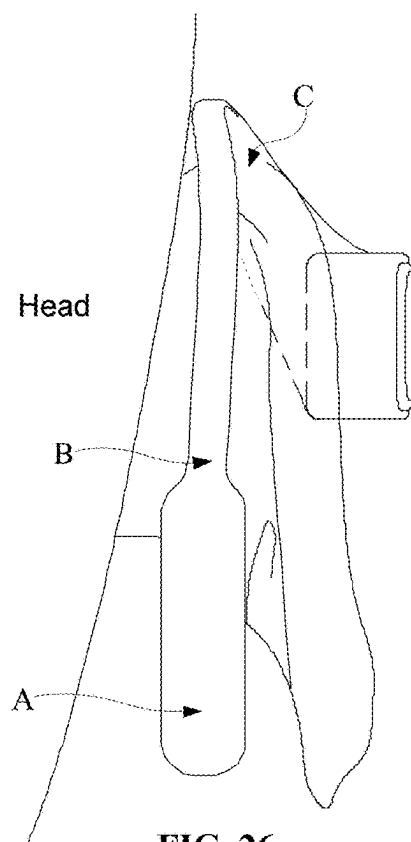


FIG. 26

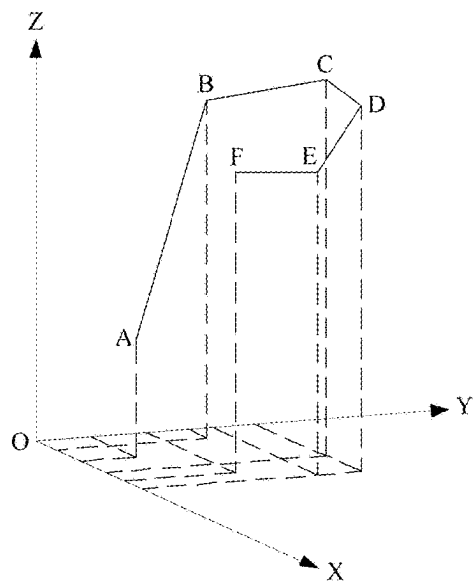


FIG. 27

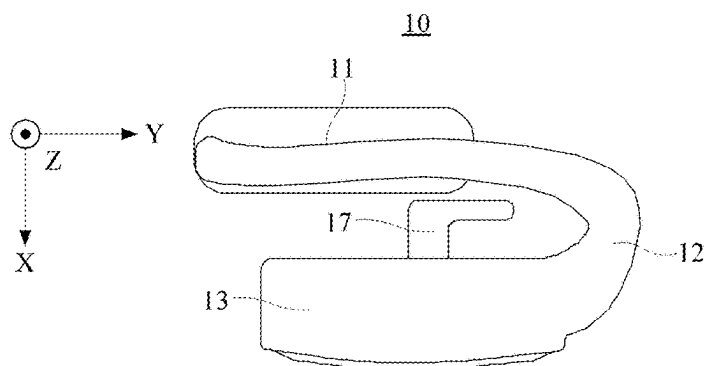


FIG. 28

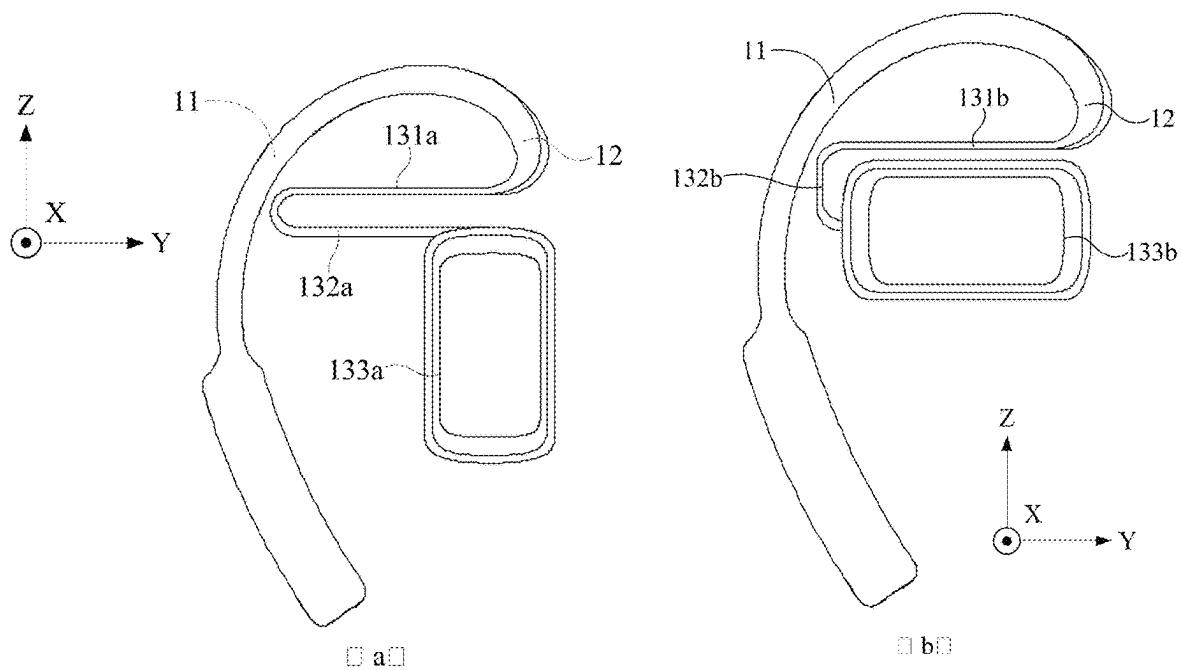


FIG. 29

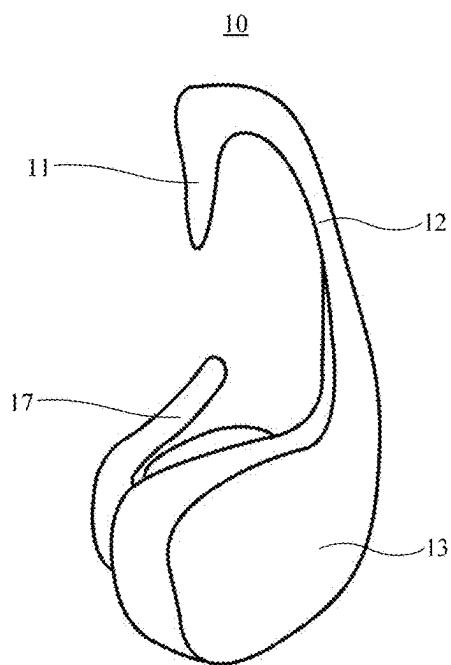


FIG. 30

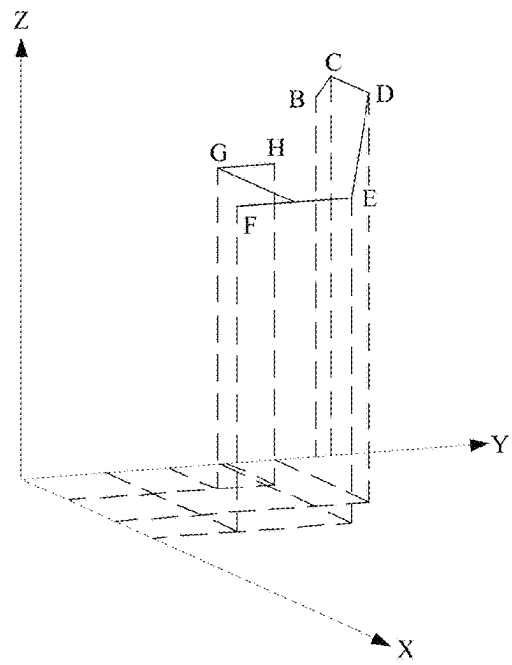


FIG. 31

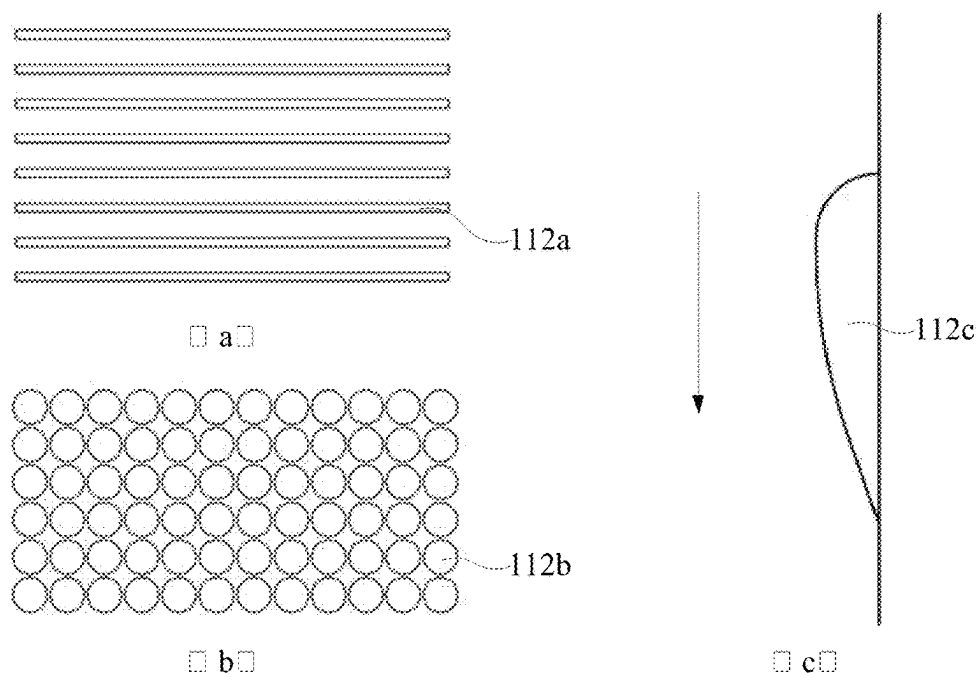


FIG. 32

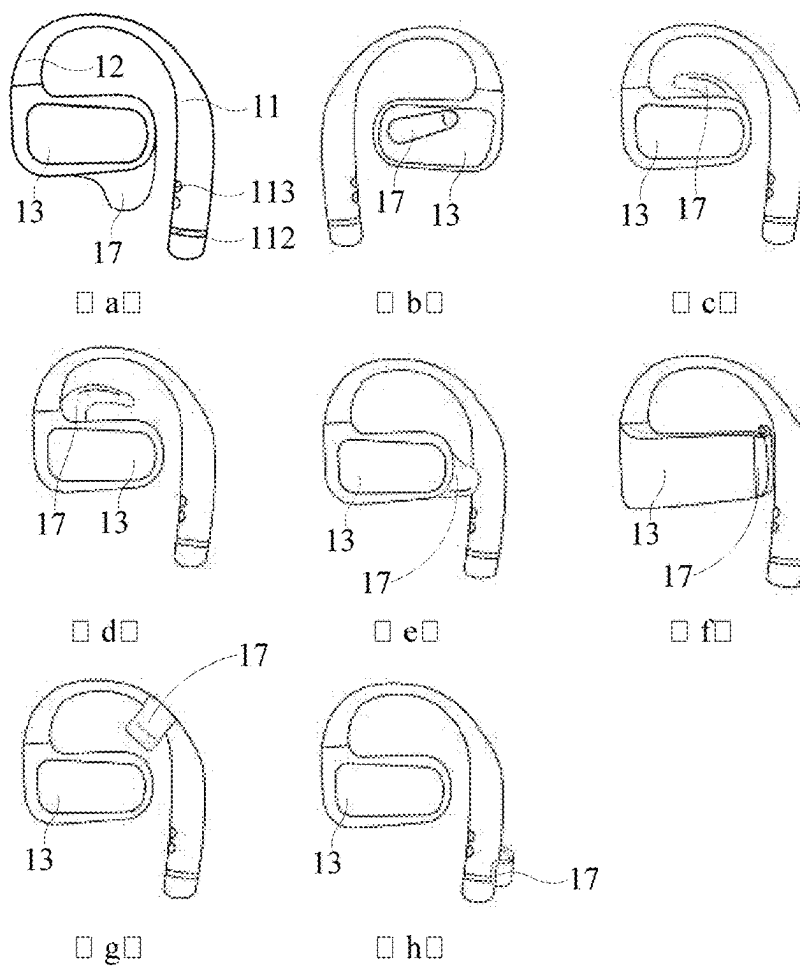


FIG. 33

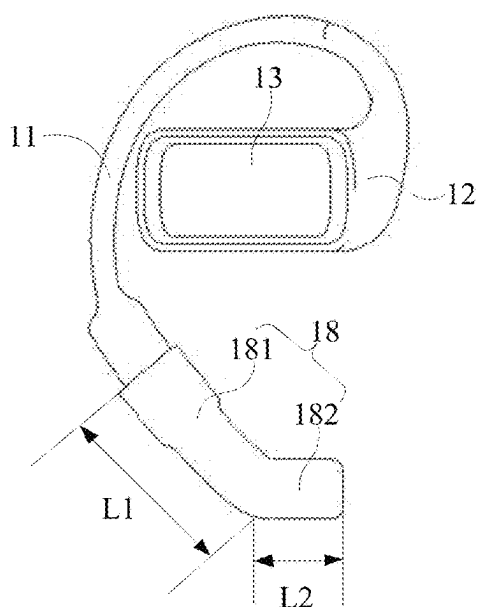


FIG. 34

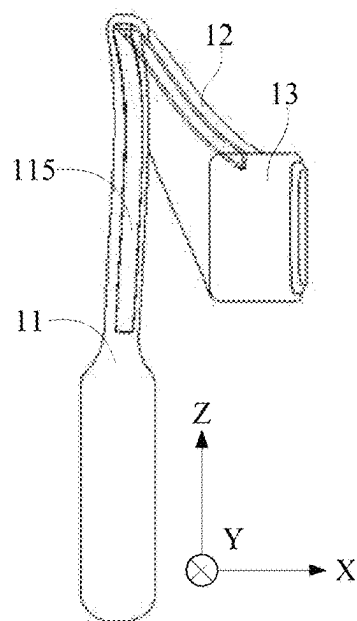


FIG. 35

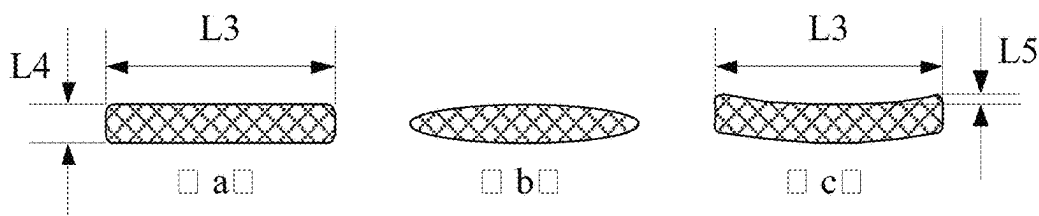


FIG. 36

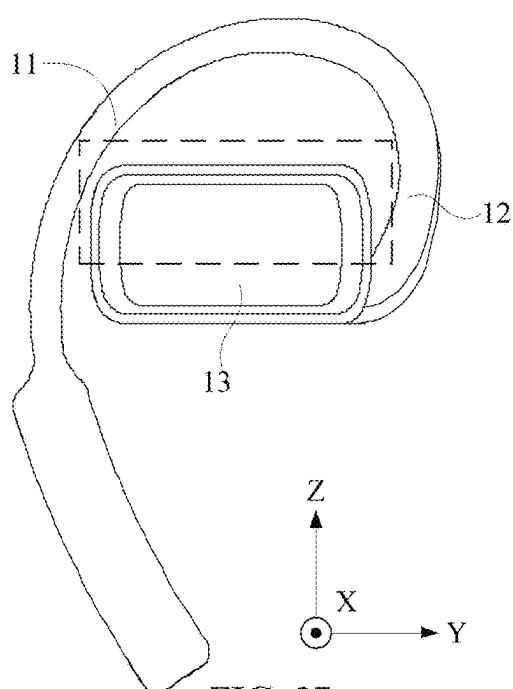


FIG. 37

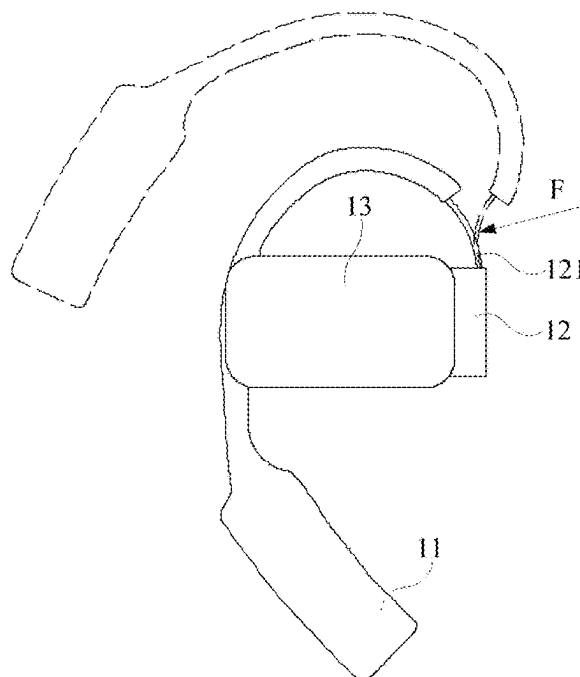


FIG. 38

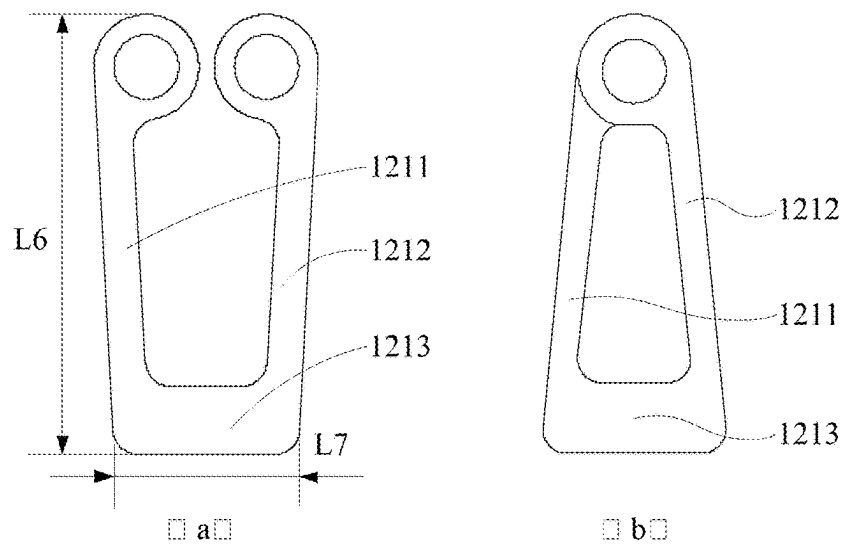


FIG. 39

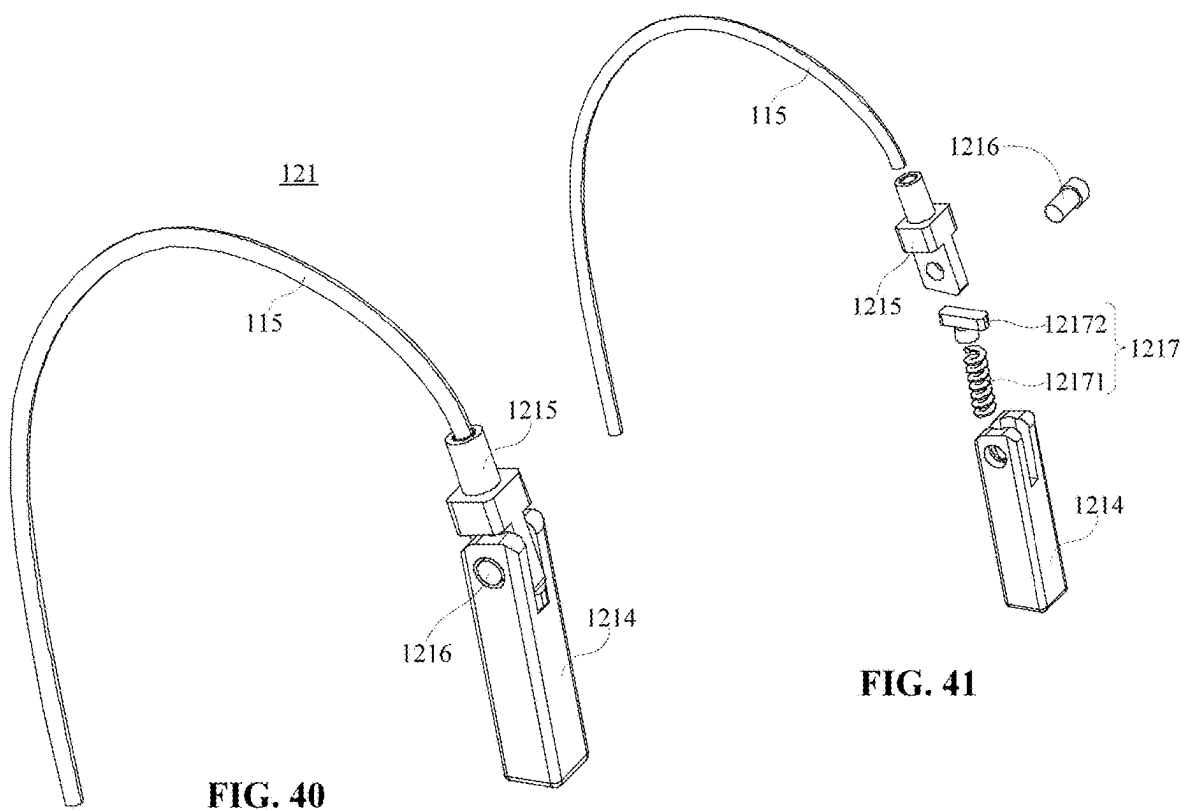


FIG. 41

FIG. 40

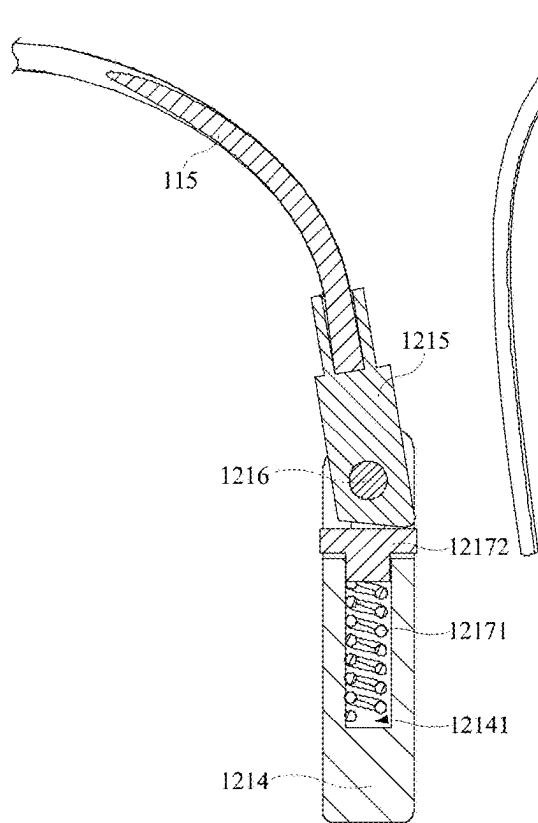


FIG. 42

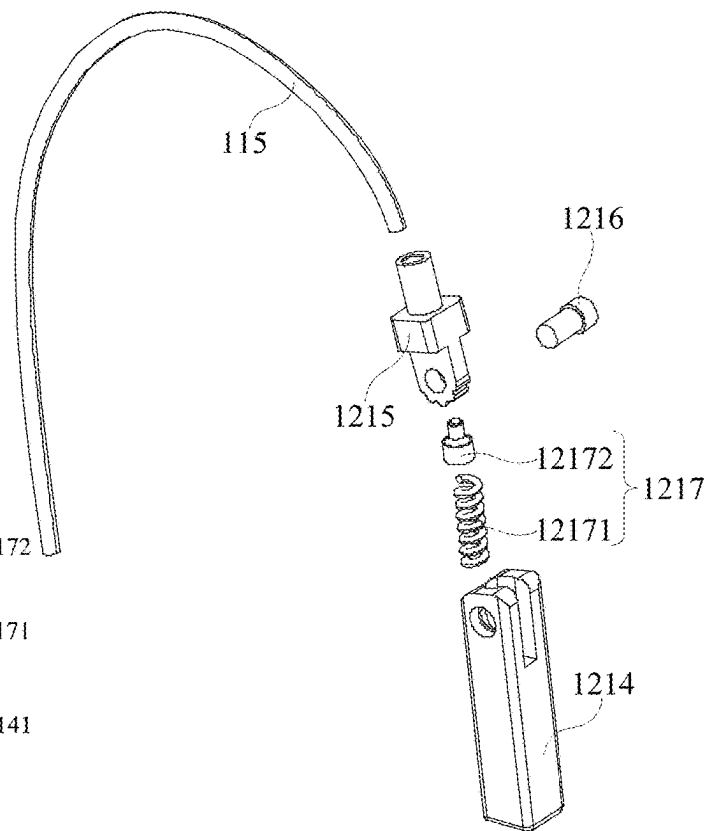


FIG. 43

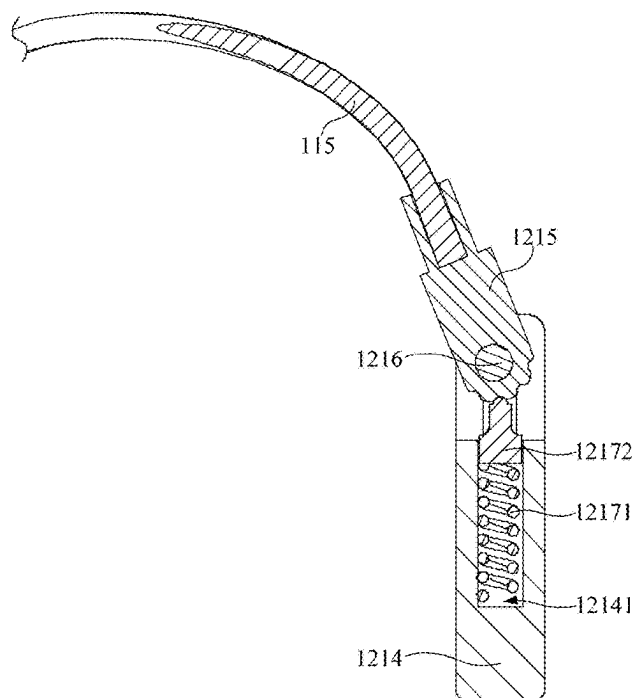


FIG. 44

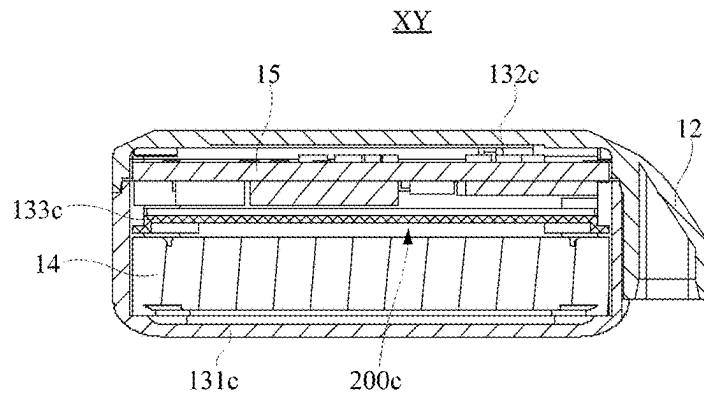


FIG. 45

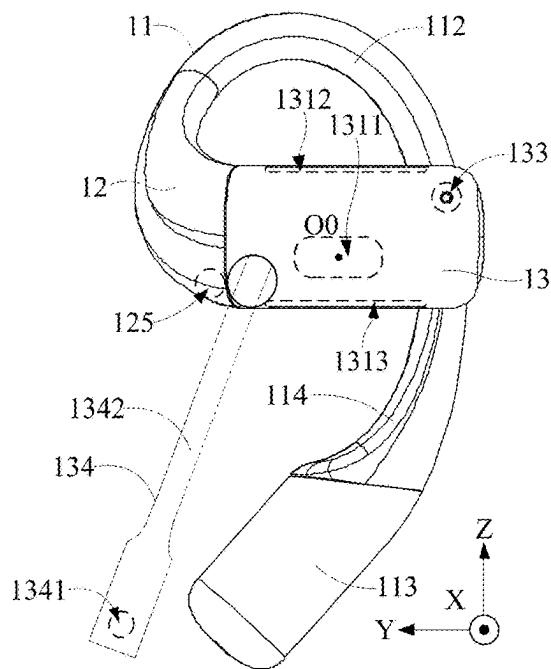


FIG. 46

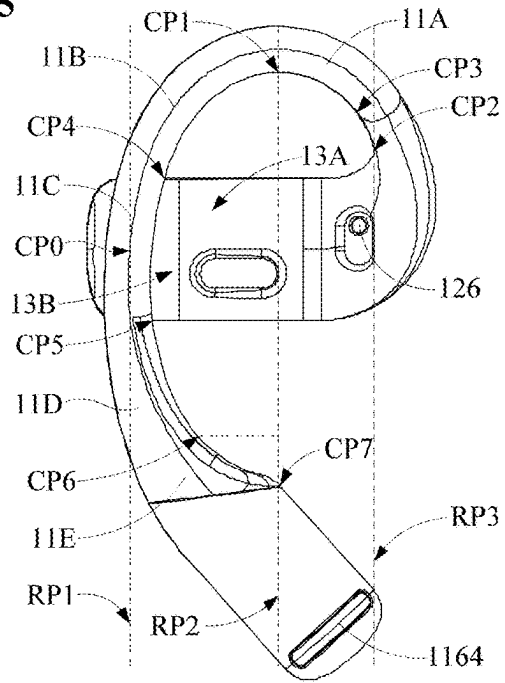


FIG. 47

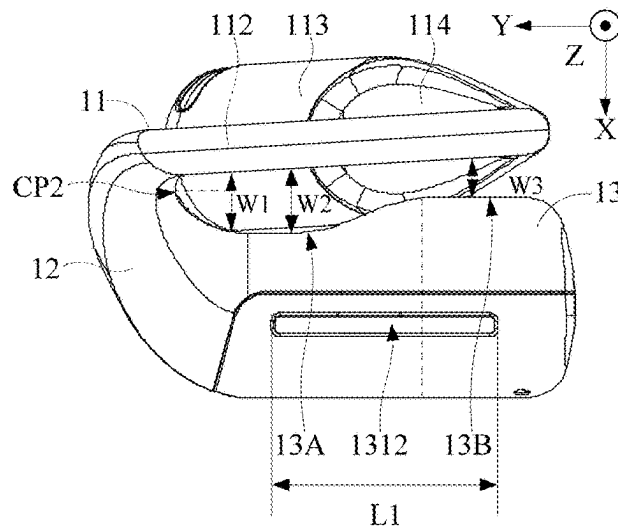


FIG. 48

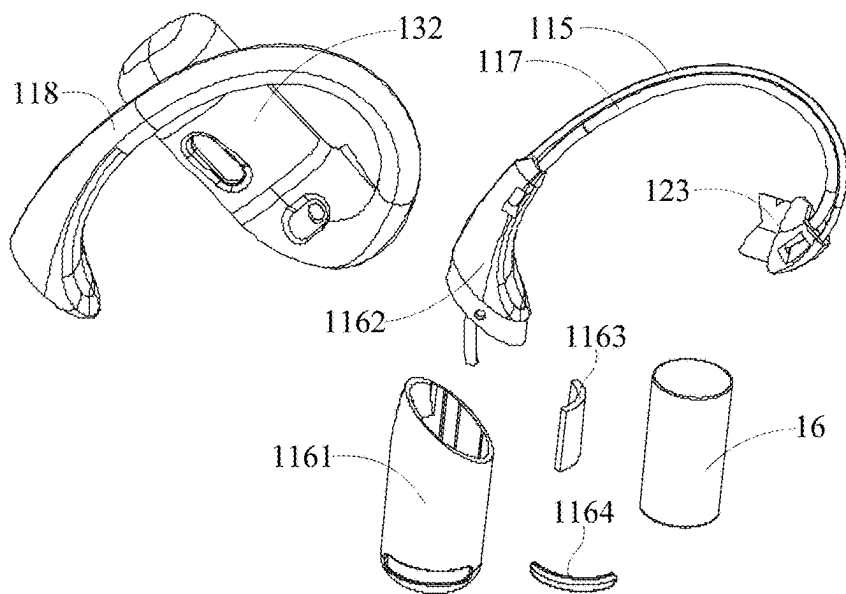


FIG. 49

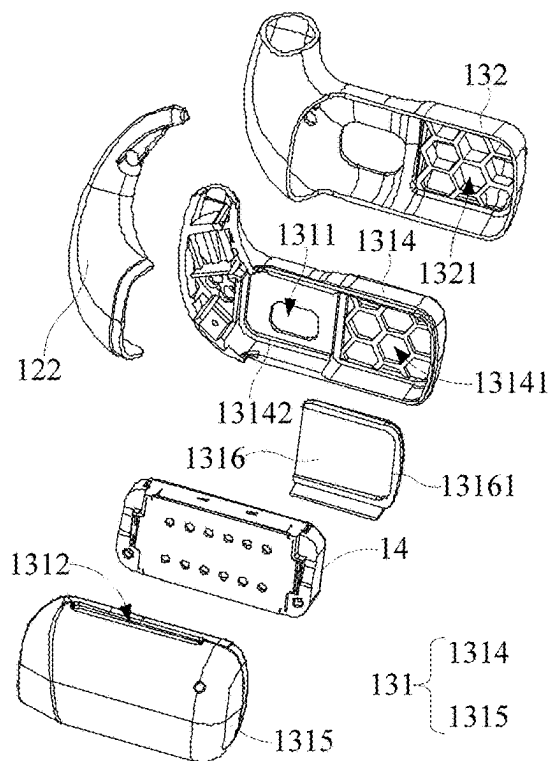


FIG. 50

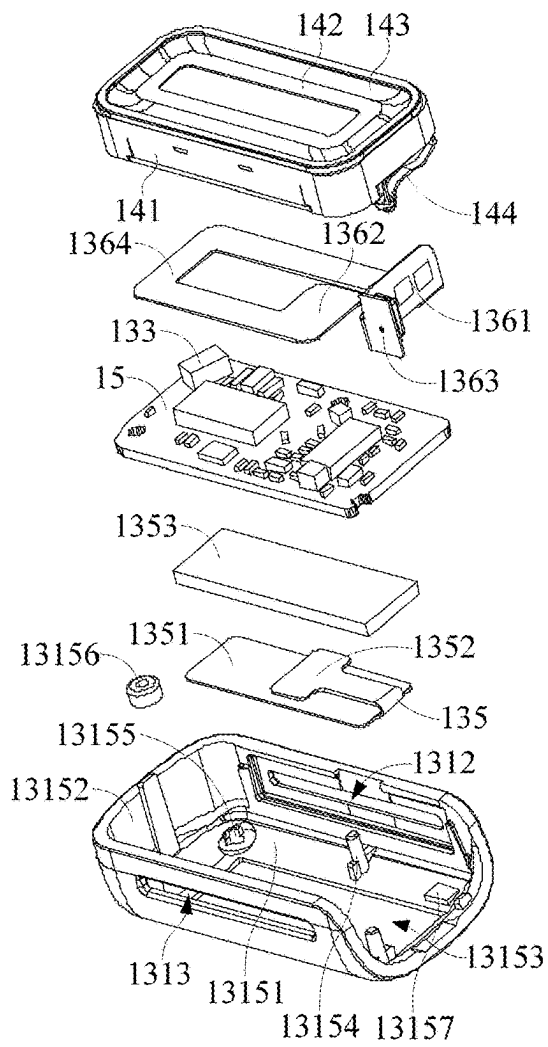


FIG. 51

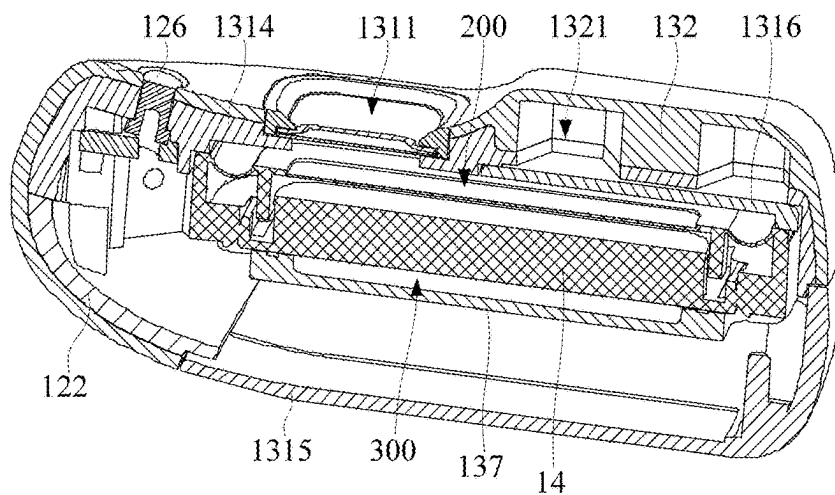


FIG. 52

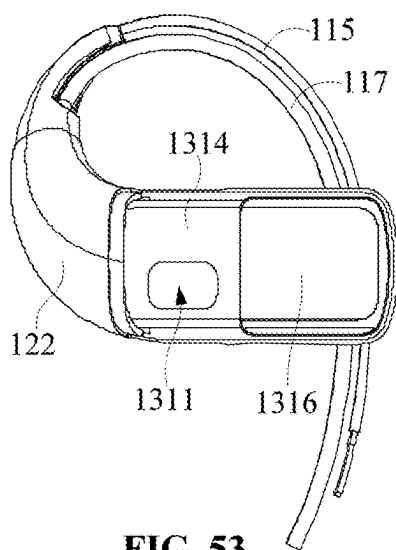


FIG. 53

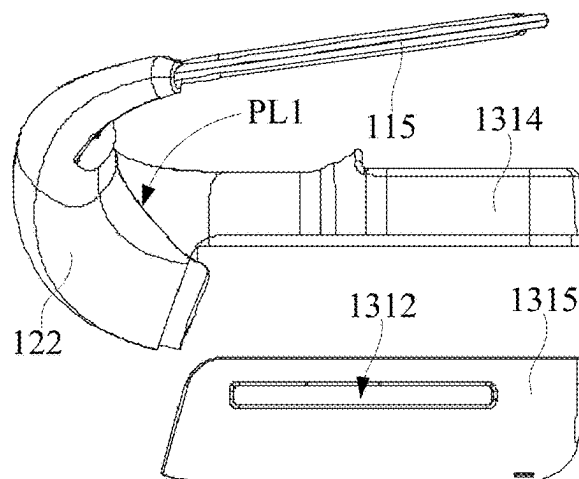


FIG. 54

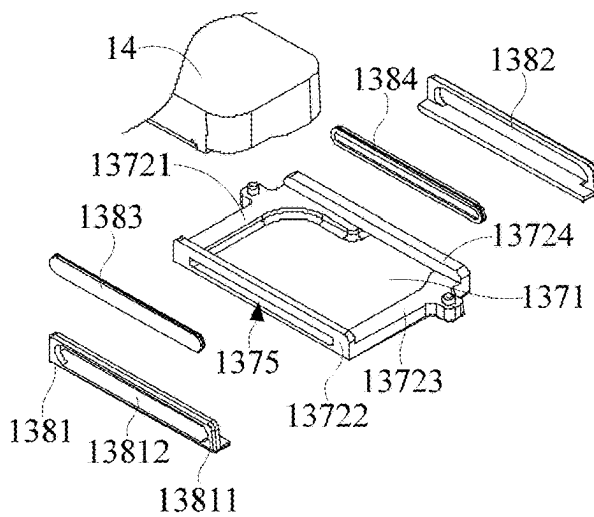


FIG. 55

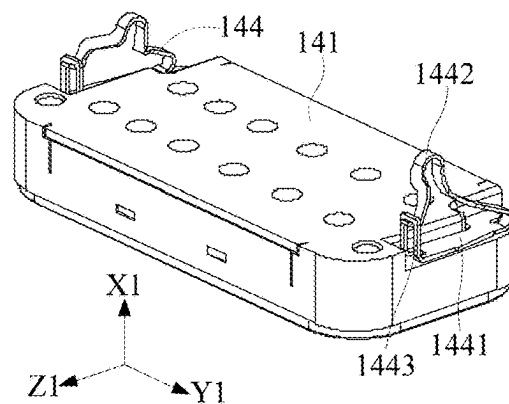


FIG. 56

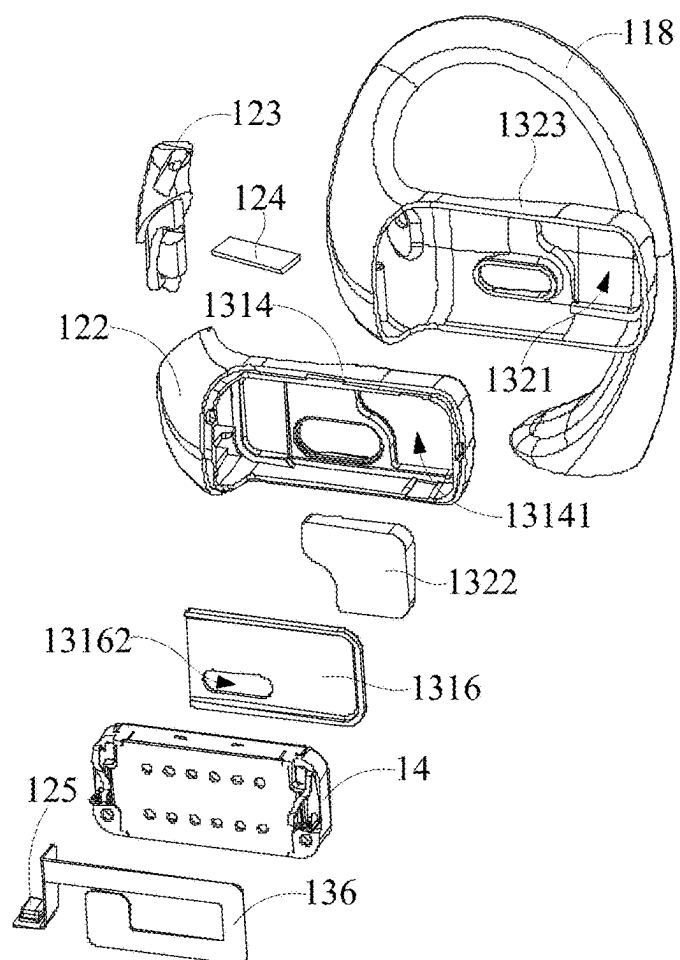


FIG. 57

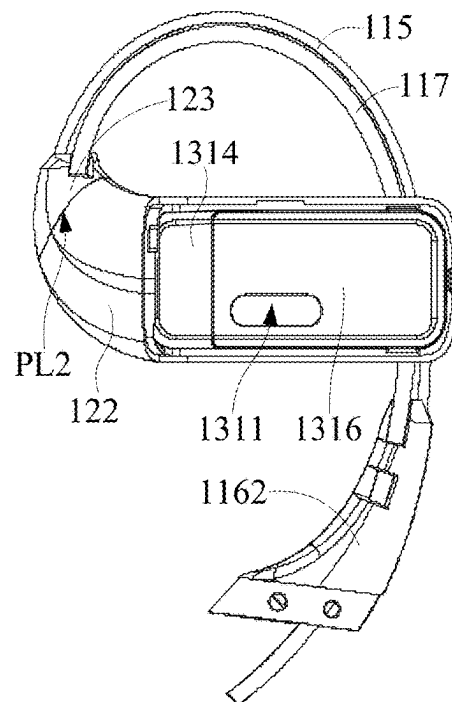


FIG. 58

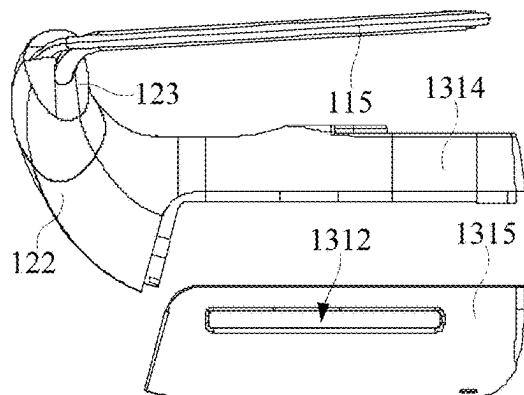


FIG. 59

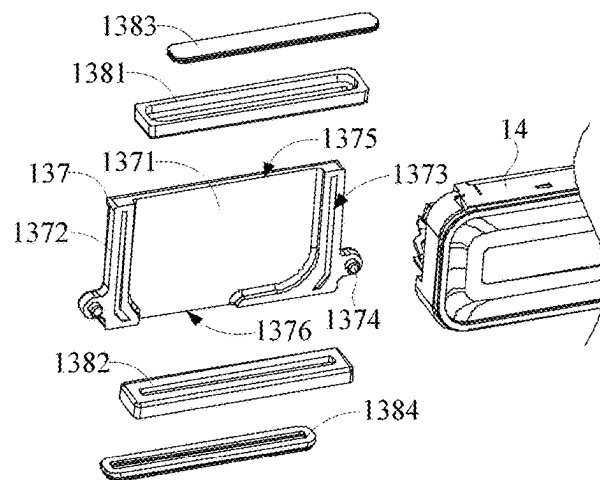


FIG. 60

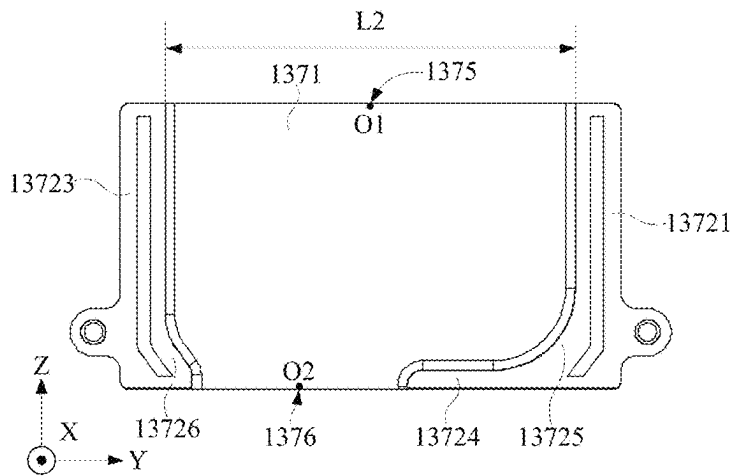


FIG. 61

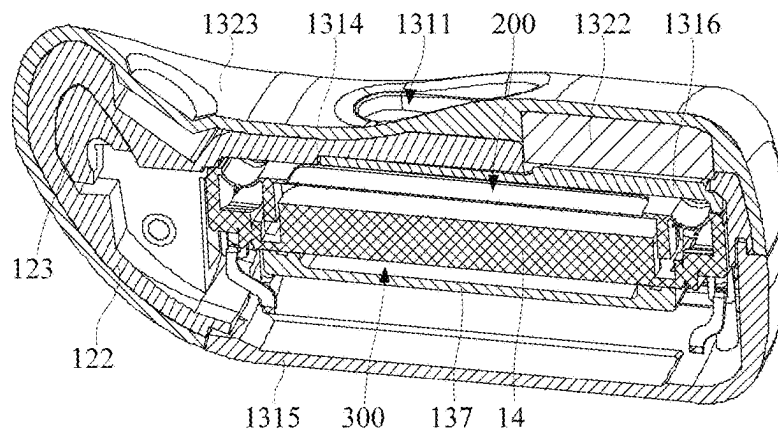


FIG. 62

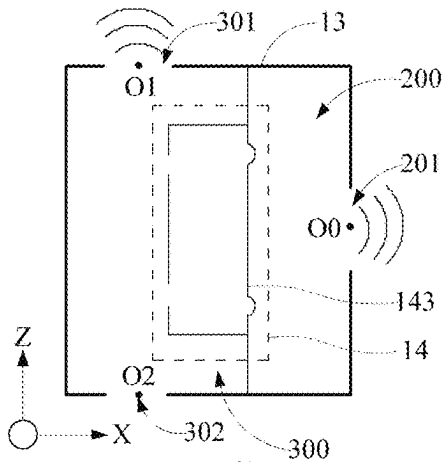


FIG. 63

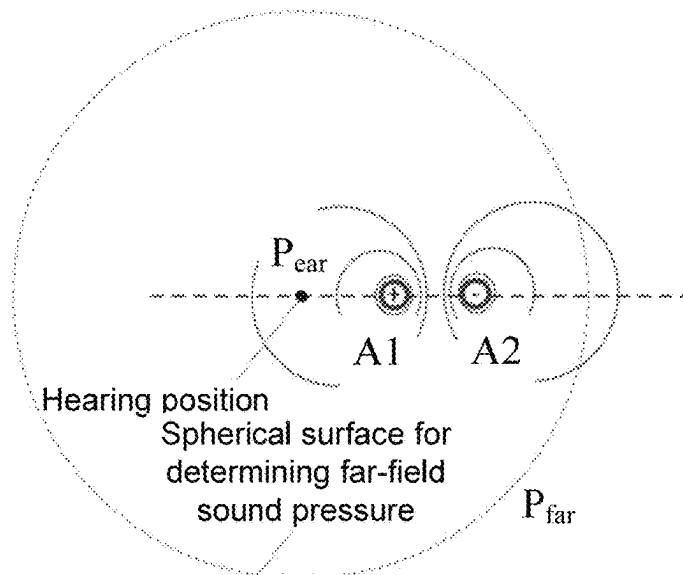


FIG. 64

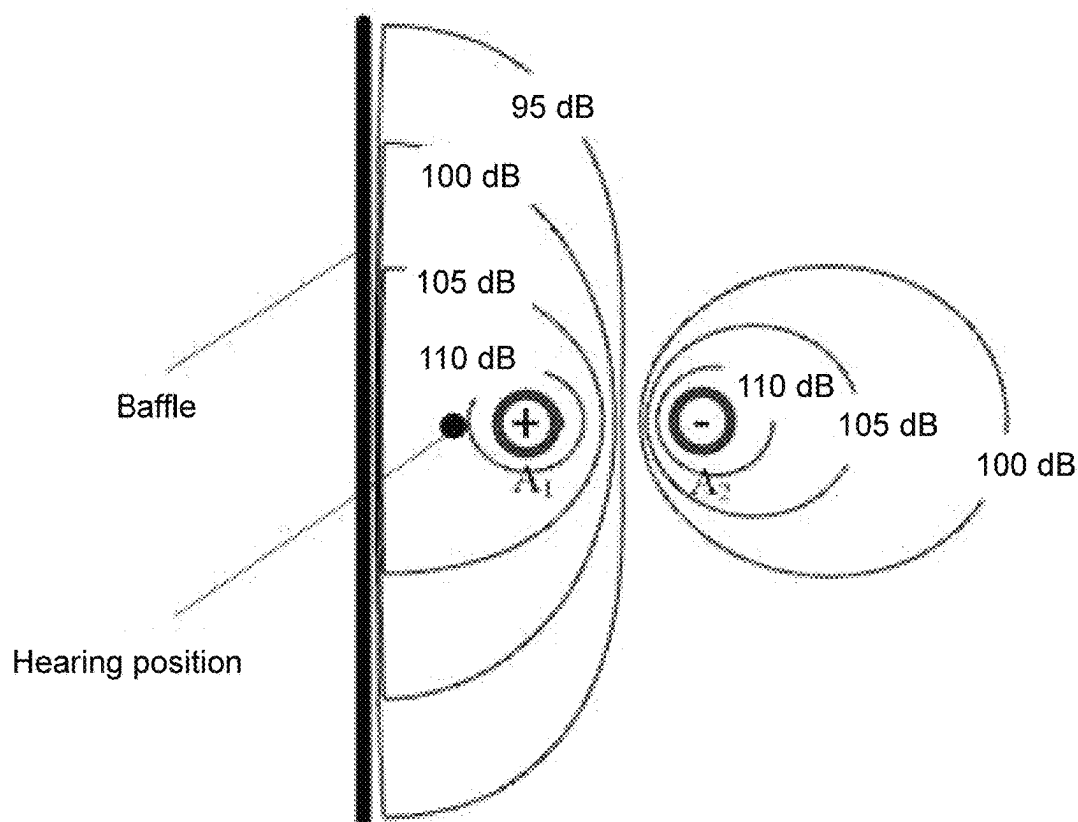


FIG. 65

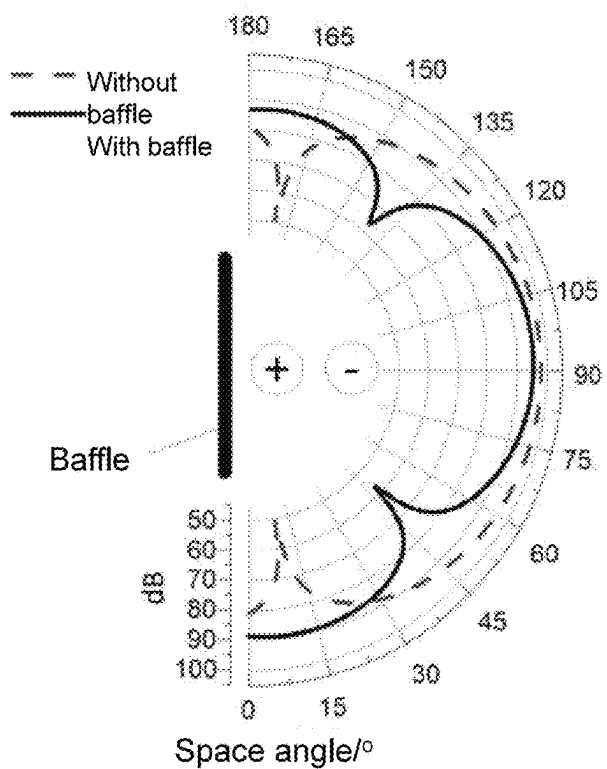


FIG. 66

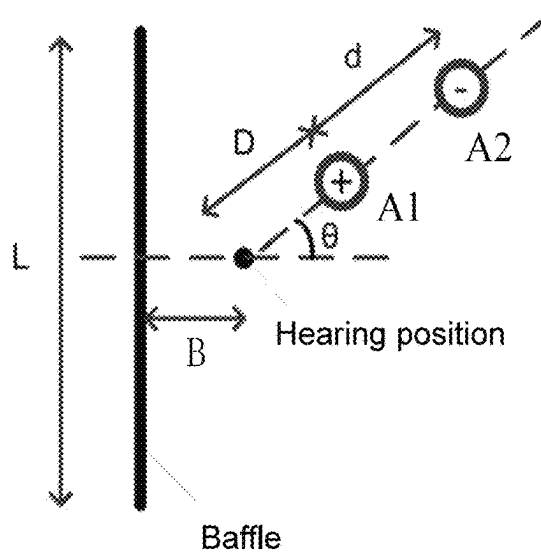


FIG. 67

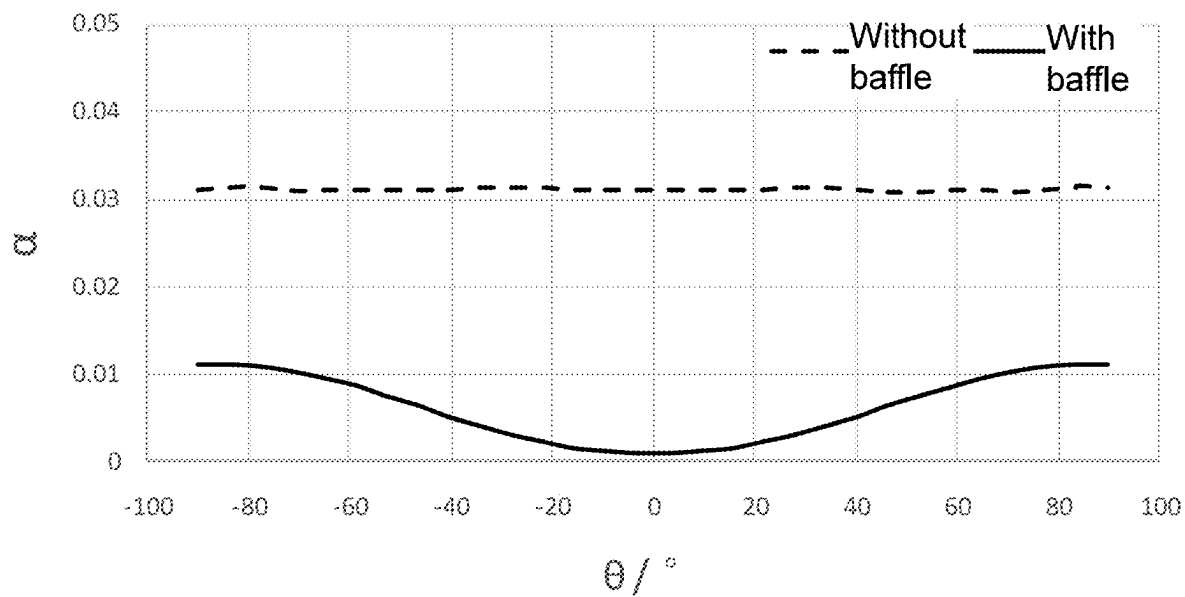


FIG. 68

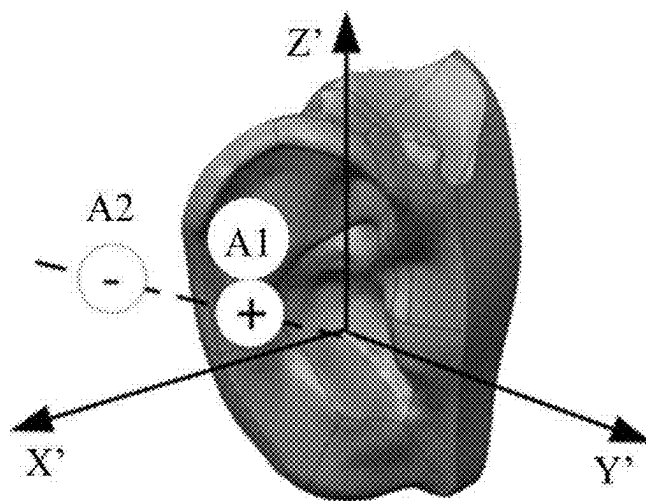


FIG. 69

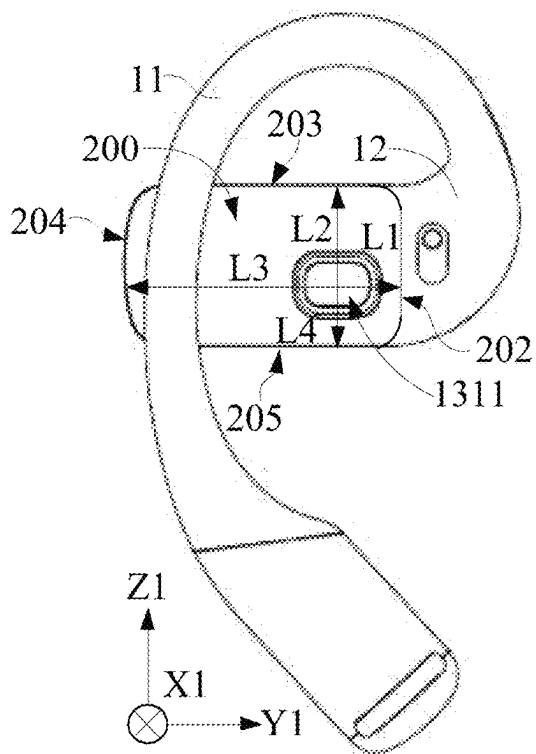


FIG. 70

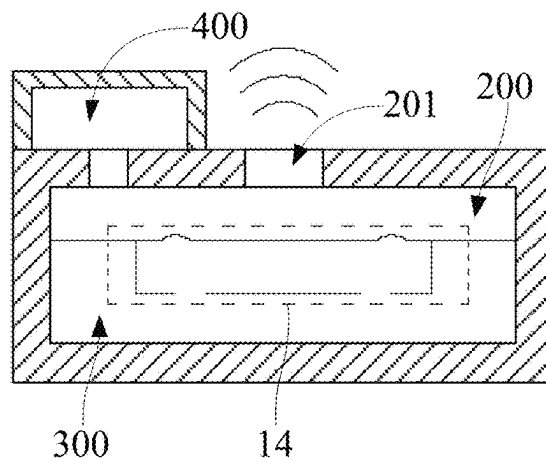


FIG. 71

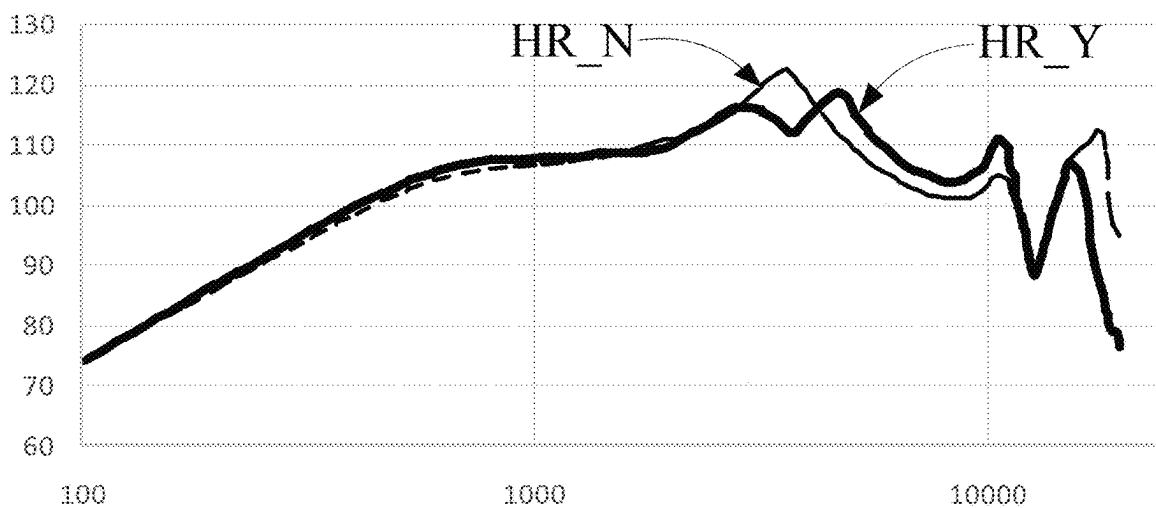


FIG. 72

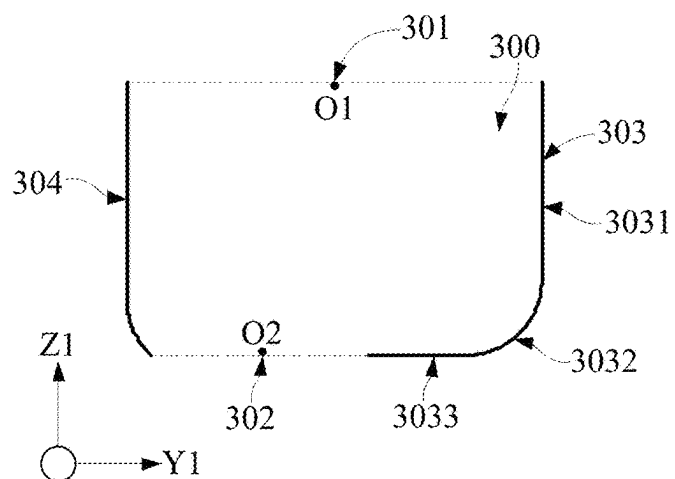


FIG. 73

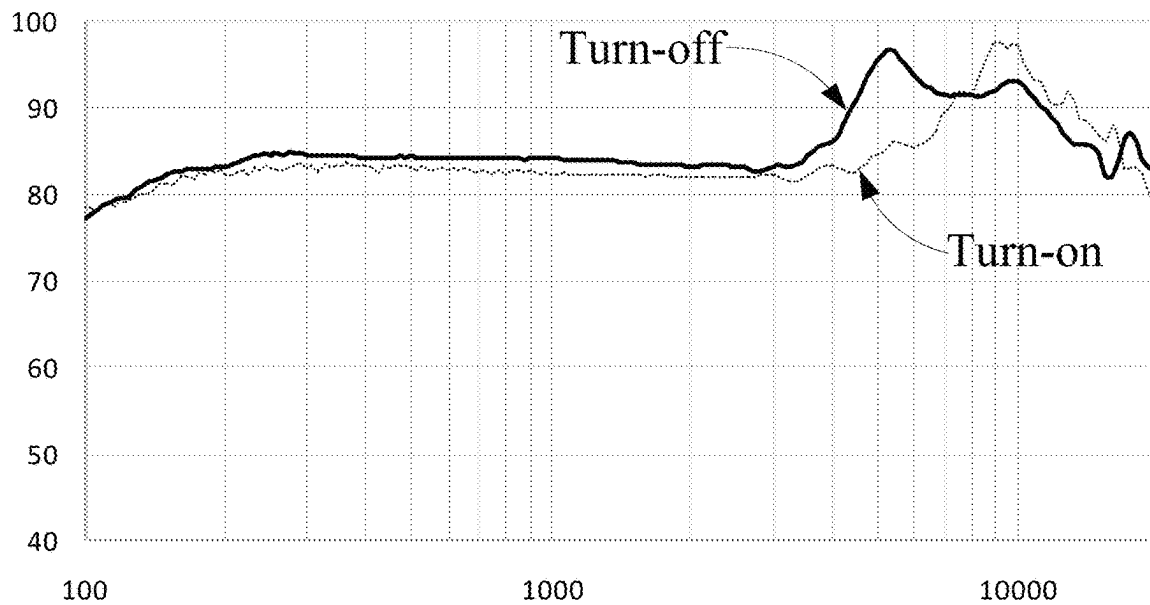


FIG. 74

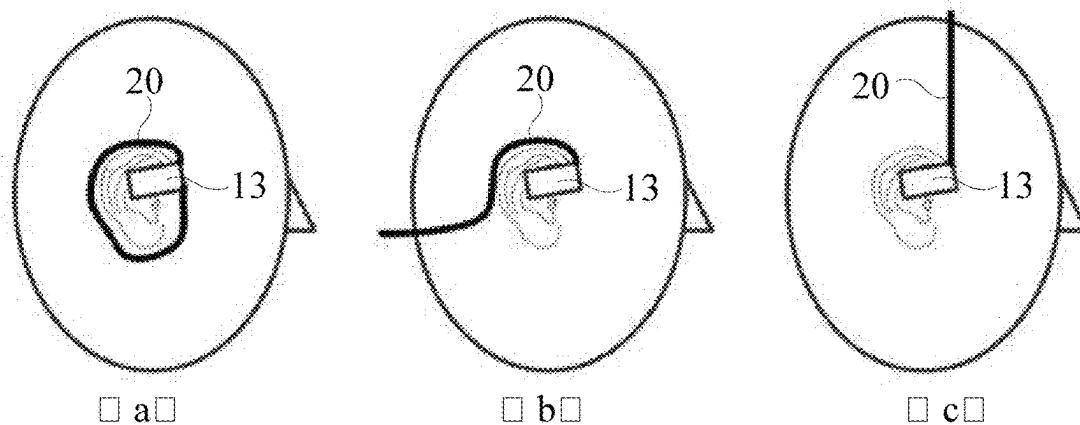


FIG. 75

1

SYSTEMS AND METHODS FOR SUPPRESSING SOUND LEAKAGE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation-in-part of U.S. patent application Ser. No. 18/308,760, filed on Apr. 28, 2023, which is a continuation of U.S. patent application Ser. No. 17/804,611 (now U.S. Pat. No. 11,659,341), filed on May 31, 2022, which is a continuation of U.S. patent application Ser. No. 17/170,874 (now U.S. Pat. No. 11,363,392), filed on Feb. 8, 2021, which is a continuation-in-part application of U.S. patent application Ser. No. 17/074,762 (now U.S. Pat. No. 11,197,106), filed on Oct. 20, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/813,915 (now U.S. Pat. No. 10,848,878), filed on Mar. 10, 2020, which is a continuation of U.S. patent application Ser. No. 16/419,049 (issued as U.S. Pat. No. 10,616,696), filed on May 22, 2019, which is a continuation of U.S. patent application Ser. No. 16/180,020 (issued as U.S. Pat. No. 10,334,372), filed on Nov. 5, 2018, which is a continuation of U.S. patent application Ser. No. 15/650,909 (issued as U.S. Pat. No. 10,149,071), filed on Jul. 16, 2017, which is a continuation of U.S. patent application Ser. No. 15/109,831 (issued as U.S. Pat. No. 9,729,978), filed on Jul. 6, 2016, which is a U.S. National Stage entry under 35 U.S.C. § 371 of International Application No. PCT/CN2014/094065, filed on Dec. 17, 2014, designating the United States of America, which claims priority to Chinese Patent Application No. 201410005804.0, filed on Jan. 6, 2014; U.S. patent application Ser. No. 17/170,874 is also a continuation-in-part application of U.S. patent application Ser. No. 16/833,839 (now U.S. Pat. No. 11,399,245), filed on Mar. 30, 2020, which is a continuation of U.S. application Ser. No. 15/752,452 (issued as U.S. Pat. No. 10,609,496), filed on Feb. 13, 2018, which is a national stage entry under 35 U.S.C. § 371 of International Application No. PCT/CN2015/086907, filed on Aug. 13, 2015, the entire contents of each of the above-referenced applications is hereby incorporated by reference; the present application is also a continuation-in-part of U.S. patent application Ser. No. 17/457,258, filed on Dec. 2, 2021, which is a continuation-in-part of International Patent Application No. PCT/CN2021/109154, filed on Jul. 29, 2021, which claims priority of Chinese Patent Application No. 202010743396.4, filed on Jul. 29, 2020, and Chinese Patent Application No. 202011328519.4, filed on Nov. 24, 2020, the entire contents of each of which are hereby incorporated by reference.

FIELD OF THE INVENTION

This application relates to a bone conduction device, and more specifically, relates to methods and systems for reducing sound leakage by a bone conduction device.

BACKGROUND

A bone conduction speaker, which may be also called a vibration speaker, may push human tissues and bones to stimulate the auditory nerve in cochlea and enable people to hear sound. The bone conduction speaker is also called a bone conduction headphone.

An exemplary structure of a bone conduction speaker based on the principle of the bone conduction speaker is shown in FIGS. 1A and 1B. The bone conduction speaker may include an open housing **110**, a panel **120-1**, a trans-

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ducer **120-2**, and a linking component **120-3**. The transducer **120-2** may transduce electrical signals to mechanical vibrations. The panel **120-1** may be connected to the transducer **120-2** and vibrate synchronically with the transducer **120-2**.

The panel **120-1** may stretch out from the opening of the housing **110** and contact with human skin to pass vibrations to auditory nerves through human tissues and bones, which in turn enables people to hear sound. The linking component **120-3** may reside between the transducer **120-2** and the housing **110**, configured to fix the vibrating transducer **120-2** inside the housing **110**. To minimize its effect on the vibrations generated by the transducer **120-2**, the linking component **120-3** may be made of an elastic material.

However, the mechanical vibrations generated by the transducer **120-2** may not only cause the panel **120-1** to vibrate, but may also cause the housing **110** to vibrate through the linking component **120-3**. Accordingly, the mechanical vibrations generated by the bone conduction speaker may push human tissues through the bone board **121**, and at the same time a portion of the vibrating board **121** and the housing **110** that are not in contact with human tissues may nevertheless push air. Air sound may thus be generated by the air pushed by the portion of the vibrating board **121** and the housing **110**. The air sound may be called “sound leakage.” In some cases, sound leakage is harmless. However, sound leakage should be avoided as much as possible if people intend to protect privacy when using the bone conduction speaker or try not to disturb others when listening to music.

Attempting to solve the problem of sound leakage, Korean patent KR10-2009-0082999 discloses a bone conduction speaker of a dual magnetic structure and double-frame. As shown in FIG. 2, the speaker disclosed in the patent includes: a first frame **210** with an open upper portion and a second frame **220** that surrounds the outside of the first frame **210**. The second frame **220** is separately placed from the outside of the first frame **210**. The first frame **210** includes a movable coil **230** with electric signals, an inner magnetic component **240**, an outer magnetic component **250**, a magnet field formed between the inner magnetic component **240**, and the outer magnetic component **250**. The inner magnetic component **240** and the outer magnetic component **250** may vibrate by the attraction and repulsion force of the coil **230** placed in the magnet field. A vibration board **260** connected to the moving coil **230** may receive the vibration of the moving coil **230**. A vibration unit **270** connected to the vibration board **260** may pass the vibration to a user by contacting with the skin. As described in the patent, the second frame **220** surrounds the first frame **210**, in order to use the second frame **220** to prevent the vibration of the first frame **210** from dissipating the vibration to outsides, and thus may reduce sound leakage to some extent.

However, in this design, since the second frame **220** is fixed to the first frame **210**, vibrations of the second frame **220** are inevitable. As a result, sealing by the second frame **220** is unsatisfactory. Furthermore, the second frame **220** increases the whole volume and weight of the speaker, which in turn increases the cost, complicates the assembly process, and reduces the speaker’s reliability and consistency.

SUMMARY

The embodiments of the present application disclose methods and system of reducing sound leakage of a bone conduction speaker.

In one aspect, the embodiments of the present application disclose a method of reducing sound leakage of a bone conduction speaker, including:

- providing a bone conduction speaker including a panel fitting human skin and passing vibrations, a transducer, and a housing, wherein at least one sound guiding hole is located in at least one portion of the housing;
- the transducer drives the panel to vibrate;
- the housing vibrates, along with the vibrations of the transducer, and pushes air, forming a leaked sound wave transmitted in the air;
- the air inside the housing is pushed out of the housing through the at least one sound guiding hole, interferes with the leaked sound wave, and reduces an amplitude of the leaked sound wave.

In some embodiments, one or more sound guiding holes may locate in an upper portion, a central portion, and/or a lower portion of a sidewall and/or the bottom of the housing.

In some embodiments, a damping layer may be applied in the at least one sound guiding hole in order to adjust the phase and amplitude of the guided sound wave through the at least one sound guiding hole.

In some embodiments, sound guiding holes may be configured to generate guided sound waves having a same phase that reduce the leaked sound wave having a same wavelength; sound guiding holes may be configured to generate guided sound waves having different phases that reduce the leaked sound waves having different wavelengths.

In some embodiments, different portions of a same sound guiding hole may be configured to generate guided sound waves having a same phase that reduce the leaked sound wave having same wavelength. In some embodiments, different portions of a same sound guiding hole may be configured to generate guided sound waves having different phases that reduce leaked sound waves having different wavelengths.

In another aspect, the embodiments of the present application disclose a bone conduction speaker, including a housing, a panel and a transducer, wherein:

- the transducer is configured to generate vibrations and is located inside the housing;
- the panel is configured to be in contact with skin and pass vibrations;

At least one sound guiding hole may locate in at least one portion on the housing, and preferably, the at least one sound guiding hole may be configured to guide a sound wave inside the housing, resulted from vibrations of the air inside the housing, to the outside of the housing, the guided sound wave interfering with the leaked sound wave and reducing the amplitude thereof.

In some embodiments, the at least one sound guiding hole may locate in the sidewall and/or bottom of the housing.

In some embodiments, preferably, the at least one sound guiding sound hole may locate in the upper portion and/or lower portion of the sidewall of the housing.

In some embodiments, preferably, the sidewall of the housing is cylindrical and there are at least two sound guiding holes located in the sidewall of the housing, which are arranged evenly or unevenly in one or more circles. Alternatively, the housing may have a different shape.

In some embodiments, preferably, the sound guiding holes have different heights along the axial direction of the cylindrical sidewall.

In some embodiments, preferably, there are at least two sound guiding holes located in the bottom of the housing. In some embodiments, the sound guiding holes are distributed evenly or unevenly in one or more circles around the center

of the bottom. Alternatively or additionally, one sound guiding hole is located at the center of the bottom of the housing.

In some embodiments, preferably, the sound guiding hole is a perforative hole. In some embodiments, there may be a damping layer at the opening of the sound guiding hole.

In some embodiments, preferably, the guided sound waves through different sound guiding holes and/or different portions of a same sound guiding hole have different phases or a same phase.

In some embodiments, preferably, the damping layer is a tuning paper, a tuning cotton, a nonwoven fabric, a silk, a cotton, a sponge, or a rubber.

In some embodiments, preferably, the shape of a sound guiding hole is circle, ellipse, quadrangle, rectangle, or linear. In some embodiments, the sound guiding holes may have a same shape or different shapes.

In some embodiments, preferably, the transducer includes a magnetic component and a voice coil. Alternatively, the transducer includes piezoelectric ceramic.

The design disclosed in this application utilizes the principles of sound interference, by placing sound guiding holes in the housing, to guide sound wave(s) inside the housing to the outside of the housing, the guided sound wave(s) interfering with the leaked sound wave, which is formed when the housing's vibrations push the air outside the housing. The guided sound wave(s) reduces the amplitude of the leaked sound wave and thus reduces the sound leakage. The design not only reduces sound leakage, but is also easy to implement, doesn't increase the volume or weight of the bone conduction speaker, and barely increase the cost of the product.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A and 1B are schematic structures illustrating a bone conduction speaker of prior art;

FIG. 2 is a schematic structure illustrating another bone conduction speaker of prior art;

FIG. 3 illustrates the principle of sound interference according to some embodiments of the present disclosure;

FIGS. 4A and 4B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 4C is a schematic structure of the bone conduction speaker according to some embodiments of the present disclosure;

FIG. 4D is a diagram illustrating reduced sound leakage of the bone conduction speaker according to some embodiments of the present disclosure;

FIG. 4E is a schematic diagram illustrating exemplary two-point sound sources according to some embodiments of the present disclosure;

FIG. 5 is a diagram illustrating the equal-loudness contour curves according to some embodiments of the present disclosure;

FIG. 6 is a flow chart of an exemplary method of reducing sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 7A and 7B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 7C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

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FIGS. 8A and 8B are schematic structure of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 8C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 9A and 9B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 9C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 10A and 10B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 10C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIG. 10D is a schematic diagram illustrating an acoustic route according to some embodiments of the present disclosure;

FIG. 10E is a schematic diagram illustrating another acoustic route according to some embodiments of the present disclosure;

FIG. 10F is a schematic diagram illustrating a further acoustic route according to some embodiments of the present disclosure;

FIGS. 11A and 11B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 11C is a diagram illustrating reduced sound leakage of a bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 12A and 12B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIGS. 13A and 13B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure;

FIG. 14 illustrates an equivalent model of a vibration generation and transferring system of a bone conduction speaker according to some embodiments of the present disclosure;

FIG. 15A illustrates a structure of a contact surface of a vibration unit of a bone conduction speaker according to some embodiments of the present disclosure;

FIG. 15B illustrates a vibration response curve of a bone conduction speaker according to some embodiments of the present disclosure;

FIG. 16 illustrates a structure of a contact surface of a vibration unit of a bone conduction speaker according to some embodiments of the present disclosure;

FIG. 17 is a schematic diagram illustrating a front view of a structure of a contour of a user's ear according to some embodiments of the present disclosure;

FIG. 18 is a schematic diagram illustrating a front view of a structure of an exemplary earphone according to some embodiments of the present disclosure;

FIG. 19 is a schematic diagram illustrating a left side view of the structure of the earphone in FIG. 18;

FIG. 20 is a schematic diagram illustrating a front side view of the earphone in FIG. 18 in a wearing state;

FIG. 21 is a schematic diagram illustrating a rear side view of the earphone in FIG. 18 in a wearing state;

FIG. 22 is a schematic diagram illustrating a mechanical model of the earphone in FIG. 18 in a wearing state;

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FIG. 23 is a schematic diagram illustrating a front view of a structure of an earphone according to some embodiments of the present disclosure;

FIG. 24 is a schematic diagram illustrating a left side view of the earphone in FIG. 23;

FIG. 25 is a schematic diagram illustrating a front side view of the earphone in FIG. 23 in a wearing state;

FIG. 26 is a schematic diagram illustrating a rear side view of the earphone in FIG. 23 in a wearing state;

FIG. 27 is a schematic diagram illustrating a mechanical model of the earphone in FIG. 23 in a wearing state;

FIG. 28 is a schematic diagram of a top view of a structure of an earphone according to some embodiments of the present disclosure;

FIG. 29 is a schematic diagram illustrating a front view of a structure of an earphone according to some embodiments of the present disclosure;

FIG. 30 is a schematic diagram illustrating a structure of an earphone according to some embodiments of the present disclosure;

FIG. 31 is a schematic diagram illustrating a mechanical model of the earphone in FIG. 30 in a wearing state;

FIG. 32 is a schematic diagram illustrating surface structures of a skin contact area of a battery part according to some embodiments of the present disclosure;

FIG. 33 is a schematic diagram illustrating different structures of the extending component in FIG. 28;

FIG. 34 is a schematic diagram illustrating a structure of an earphone according to some embodiments of the present disclosure;

FIG. 35 is a schematic diagram illustrating a perspective view of a structure of the hook-shaped component in FIG. 24;

FIG. 36 is a schematic diagram illustrating a cross-sectional structure of the elastic metal wire in FIG. 35 on a reference plane perpendicular to an extending direction of the hook-shaped component;

FIG. 37 is a schematic diagram illustrating a front view of a structure of an earphone according to some embodiments of the present disclosure;

FIG. 38 is a schematic diagram illustrating a structure of a rotating shaft assembly according to an embodiment of the present disclosure;

FIG. 39 is a schematic diagram illustrating a structure of the rotating shaft assembly in FIG. 38 before and after assembly;

FIG. 40 is a schematic diagram illustrating a structure of a rotating shaft assembly according to another embodiment of the present disclosure;

FIG. 41 is a schematic diagram illustrating a disassembled structure of the rotating shaft assembly in FIG. 40 according to an embodiment of the present disclosure;

FIG. 42 is a schematic diagram illustrating a cross-sectional structure of the rotating shaft assembly in FIG. 41;

FIG. 43 is a schematic diagram illustrating a disassembled structure of the rotating shaft assembly in FIG. 40 according to another embodiment of the present disclosure;

FIG. 44 is a schematic diagram illustrating a cross-sectional structure of the rotating shaft assembly in FIG. 43;

FIG. 45 is a schematic diagram illustrating a cross-sectional structure of an earphone in an XY plane according to some embodiments of the present disclosure;

FIG. 46 is a schematic diagram illustrating a structure of an earphone on a side facing away from an ear according to some embodiments of the present disclosure;

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FIG. 47 is a schematic diagram illustrating a structure of an earphone on a side facing an ear according to some embodiments of the present disclosure;

FIG. 48 is a schematic diagram illustrating a structure of an earphone viewed from a side on a top of a user's head according to some embodiments of the present disclosure;

FIG. 49 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure;

FIG. 50 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure;

FIG. 51 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure;

FIG. 52 is a schematic diagram illustrating a cross-sectional structure of an earphone according to some embodiments of the present disclosure;

FIG. 53 is a schematic diagram illustrating a structure of an earphone at a side facing away from an ear according to some embodiments of the present disclosure;

FIG. 54 is a schematic diagram illustrating a structure of an earphone viewed from a side on a top of a user's head according to some embodiments of the present disclosure;

FIG. 55 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure;

FIG. 56 is a schematic diagram illustrating a structure of a core facing a side of a main board according to some embodiments of the present disclosure;

FIG. 57 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure;

FIG. 58 is a schematic diagram illustrating a structure of an earphone on a side facing away from an ear according to some embodiments of the present disclosure;

FIG. 59 is a schematic diagram illustrating a structure of an earphone viewed from a side on a top of a user's head according to some embodiments of the present disclosure;

FIG. 60 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure;

FIG. 61 is a schematic diagram illustrating a structure of a baffle facing a side of a core according to some embodiments of the present disclosure;

FIG. 62 is a schematic diagram illustrating a cross-sectional structure of an earphone according to some embodiments of the present disclosure;

FIG. 63 is a schematic diagram illustrating a cross-sectional structure of an earphone according to some embodiments of the present disclosure;

FIG. 64 is a schematic diagram illustrating a sound field distribution of an acoustic dipole according to some embodiments of the present disclosure;

FIG. 65 is a schematic diagram illustrating a sound field distribution of an acoustic dipole with a baffle according to some embodiments of the present disclosure;

FIG. 66 is a schematic diagram illustrating a sound pressure in far-field of when an acoustic dipole with and without a baffle according to some embodiments of the present disclosure;

FIG. 67 is a schematic diagram illustrating a theoretical model of an acoustic dipole with a baffle according to some embodiments of the present disclosure;

FIG. 68 is a schematic diagram illustrating a relationship between a parameter a and an angle θ according to some embodiments of the present disclosure;

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FIG. 69 is a schematic diagram illustrating a relative relationship between an acoustic dipole and an ear according to some embodiments of the present disclosure;

FIG. 70 is a schematic diagram illustrating a structure of an earphone facing a side of an ear according to some embodiments of the present disclosure;

FIG. 71 is a schematic diagram illustrating a structure of an earphone according to some embodiments of the present disclosure;

FIG. 72 is a schematic diagram illustrating a frequency response curve of an earphone according to some embodiments of the present disclosure;

FIG. 73 is a schematic diagram illustrating a structure of a rear cavity of an earphone according to some embodiments of the present disclosure;

FIG. 74 is a schematic diagram illustrating a frequency response curve of an earphone according to some embodiments of the present disclosure; and

FIG. 75 is a schematic diagram illustrating structures of earphones in a wearing state respectively according to some embodiments of the present disclosure.

The meanings of the mark numbers in the figures are as followed:

110, open housing; 120-1, panel; 120-2, transducer; 120-3, linking component; 210, first frame; 220, second frame; 230, moving coil; 240, inner magnetic component; 250, outer magnetic component; 260, panel; 270, vibration unit; 10-0, housing; 10-1, sidewall; 10-2, bottom; 21, panel; 22, transducer; 23, linking component; 24, elastic component; 30, sound guiding hole.

DETAILED DESCRIPTION

Followings are some further detailed illustrations about this disclosure. The following examples are for illustrative purposes only and should not be interpreted as limitations of the claimed invention. There are a variety of alternative techniques and procedures available to those of ordinary skill in the art, which would similarly permit one to successfully perform the intended invention. In addition, the figures just show the structures relative to this disclosure, not the whole structure.

To explain the scheme of the embodiments of this disclosure, the design principles of this disclosure will be introduced here. FIG. 3 illustrates the principles of sound interference according to some embodiments of the present disclosure. Two or more sound waves may interfere in the space based on, for example, the frequency and/or amplitude of the waves. Specifically, the amplitudes of the sound waves with the same frequency may be overlaid to generate a strengthened wave or a weakened wave. As shown in FIG. 3, sound source 1 and sound source 2 have the same frequency and locate in different locations in the space. The sound waves generated from these two sound sources may encounter in an arbitrary point A. If the phases of the sound wave 1 and sound wave 2 are the same at point A, the amplitudes of the two sound waves may be added, generating a strengthened sound wave signal at point A; on the other hand, if the phases of the two sound waves are opposite at point A, their amplitudes may be offset, generating a weakened sound wave signal at point A.

This disclosure applies above-noted principles of sound wave interference to a bone conduction speaker and discloses a bone conduction speaker that can reduce sound leakage. This disclosure also applies above-noted principles of sound wave interference to an air conduction speaker and

discloses an air conduction speaker that can reduce sound leakage and/or an earphone including the air conduction speaker.

Embodiment One

FIGS. 4A and 4B are schematic structures of an exemplary bone conduction speaker. The bone conduction speaker may include a housing 10-0, a panel 21, and a transducer 22. The transducer 22 may be inside the housing 10-0 and configured to generate vibrations. The housing 10-0 may have one or more sound guiding holes 30. The sound guiding hole(s) 30 may be configured to guide sound waves inside the housing 10-0 to the outside of the housing 10-0. In some embodiments, the guided sound waves may form interference with leaked sound waves generated by the vibrations of the housing 10-0, so as to reducing the amplitude of the leaked sound. The transducer 22 may be configured to convert an electrical signal to mechanical vibrations. For example, an audio electrical signal may be transmitted into a voice coil that is placed in a magnet, and the electromagnetic interaction may cause the voice coil to vibrate based on the audio electrical signal. As another example, the transducer 22 may include piezoelectric ceramics, shape changes of which may cause vibrations in accordance with electrical signals received.

Furthermore, the panel 21 may be connected to the transducer 22 and configured to vibrate along with the transducer 22. The panel 21 may stretch out from the opening of the housing 10-0, and touch the skin of the user and pass vibrations to auditory nerves through human tissues and bones, which in turn enables the user to hear sound. In some embodiments, the panel 21 may be in contact with human skin directly, or through a vibration transfer layer made of specific materials (e.g., low-density materials). The linking component 23 may reside between the transducer 22 and the housing 10-0, configured to fix the vibrating transducer 120-2 inside the housing. The linking component 23 may include one or more separate components, or may be integrated with the transducer 22 or the housing 10-0. In some embodiments, the linking component 23 is made of an elastic material.

The transducer 22 may drive the panel 21 to vibrate. The transducer 22, which resides inside the housing 10-0, may vibrate. The vibrations of the transducer 22 may drives the air inside the housing 10-0 to vibrate, producing a sound wave inside the housing 10-0, which can be referred to as "sound wave inside the housing." Since the panel 21 and the transducer 22 are fixed to the housing 10-0 via the linking component 23, the vibrations may pass to the housing 10-0, causing the housing 10-0 to vibrate synchronously. The vibrations of the housing 10-0 may generate a leaked sound wave, which spreads outwards as sound leakage.

The sound wave inside the housing and the leaked sound wave are like the two sound sources in FIG. 3. In some embodiments, the sidewall 10-1 of the housing 10-0 may have one or more sound guiding holes 30 configured to guide the sound wave inside the housing 10-0 to the outside. The guided sound wave through the sound guiding hole(s) 30 may interfere with the leaked sound wave generated by the vibrations of the housing 10-0, and the amplitude of the leaked sound wave may be reduced due to the interference, which may result in a reduced sound leakage. Therefore, the design of this embodiment can solve the sound leakage problem to some extent by making an improvement of

setting a sound guiding hole on the housing, and not increasing the volume and weight of the bone conduction speaker.

In some embodiments, one sound guiding hole 30 is set on the upper portion of the sidewall 10-1. As used herein, the upper portion of the sidewall 10-1 refers to the portion of the sidewall 10-1 starting from the top of the sidewall (contacting with the panel 21) to about the 1/2 height of the sidewall.

FIG. 4C is a schematic structure of the bone conduction speaker illustrated in FIGS. 4A-4B. The structure of the bone conduction speaker is further illustrated with mechanics elements illustrated in FIG. 4C. As shown in FIG. 4C, the linking component 23 between the sidewall 10-1 of the housing 10-0 and the panel 21 may be represented by an elastic element 23 and a damping element in the parallel connection. The linking relationship between the panel 21 and the transducer 22 may be represented by an elastic element 24.

Outside the housing 10-0, the sound leakage reduction is proportional to

$$\left(\iint_{S_{hole}} P_{ds} - \iint_{S_{housing}} P_e ds \right), \quad (1)$$

wherein S_{hole} is the area of the opening of the sound guiding hole 30, $S_{housing}$ is the area of the housing 10-0 (e.g., the sidewall 10-1 and the bottom 10-2) that is not in contact with human face.

The pressure inside the housing may be expressed as $P = P_a + P_b + P_c + P_e$, (2) wherein P_a , P_b , P_c and P_e are the sound pressures of an arbitrary point inside the housing 10-0 generated by side a, side b, side c and side e (as illustrated in FIG. 4C), respectively. As used herein, side a refers to the upper surface of the transducer 22 that is close to the panel 21, side b refers to the lower surface of the panel 21 that is close to the transducer 22, side c refers to the inner upper surface of the bottom 10-2 that is close to the transducer 22, and side e refers to the lower surface of the transducer 22 that is close to the bottom 10-2.

The center of the side b, O point, is set as the origin of the space coordinates, and the side b can be set as the $z=0$ plane, so P_a , P_b , P_c and P_e may be expressed as follows:

$$P_a(x, y, z) = -j\omega\rho_0 \iint_{S_a} W_a(x'_a, y'_a) \cdot \frac{e^{jkR(x'_a, y'_a)}}{4\pi R(x'_a, y'_a)} dx'_a, dy'_a - P_{aR}, \quad (3)$$

$$P_b(x, y, z) = -j\omega\rho_0 \iint_{S_b} W_b(x', y') \cdot \frac{e^{jkR(x', y')}}{4\pi R(x', y')} dx', dy' - P_{bR}, \quad (4)$$

$$P_c(x, y, z) = -j\omega\rho_0 \iint_{S_c} W_c(x'_c, y'_c) \cdot \frac{e^{jkR(x'_c, y'_c)}}{4\pi R(x'_c, y'_c)} dx'_c, dy'_c - P_{cR}, \quad (5)$$

$$P_e(x, y, z) = -j\omega\rho_0 \iint_{S_e} W_e(x'_e, y'_e) \cdot \frac{e^{jkR(x'_e, y'_e)}}{4\pi R(x'_e, y'_e)} dx'_e, dy'_e - P_{eR}, \quad (6)$$

wherein $R(x', y') = \sqrt{(x-x')^2 + (y-y')^2 + z^2}$ is the distance between an observation point (x, y, z) and a point on side b $(x', y', 0)$; S_a , S_b , S_c , and S_e are the areas of side a, side b, side c and side e, respectively;

$R(x'_a, y'_a) = \sqrt{(x-x'_a)^2 + (y-y'_a)^2 + (z-z_a)^2}$ is the distance between the observation point (x, y, z) and a point on side a (x'_a, y'_a, z_a) ;

$R(x'_c, y'_c) = \sqrt{(x-x'_c)^2 + (y-y'_c)^2 + (z-z_c)^2}$ is the distance between the observation point (x, y, z) and a point on side c (x'_c, y'_c, z_c) ;

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$R(x'_e, y'_e) = \sqrt{(x-x'_e)^2 + (y-y'_e)^2 + (z-z_e)^2}$ is the distance between the observation point (x, y, z) and a point on side e (x'_e, y'_e, z_e);
 $k = \omega/u$ (u is the velocity of sound) is wave number, ρ_0 is an air density, ω is an angular frequency of vibration;
 P_{aR}, P_{bR}, P_{cR} and P_{eR} are acoustic resistances of air, which respectively are:

$$P_{aR} = A \cdot \frac{z_a r + j\omega \cdot z_a \cdot r'}{\varphi} + \delta, \quad (7)$$

$$P_{bR} = A \cdot \frac{z_b \cdot r + j\omega \cdot z_b \cdot r'}{\varphi} + \delta, \quad (8)$$

$$P_{cR} = A \cdot \frac{z_c \cdot r + j\omega \cdot z_c \cdot r'}{\varphi} + \delta, \quad (9)$$

$$P_{eR} = A \cdot \frac{z_e \cdot r + j\omega \cdot z_e \cdot r'}{\varphi} + \delta, \quad (10)$$

wherein r is the acoustic resistance per unit length, r' is the sound quality per unit length, z_a is the distance between the observation point and side a, z_b is the distance between the observation point and side b, z_c is the distance between the observation point and side c, z_e is the distance between the observation point and side e.

$W_a(x, y)$, $W_b(x, y)$, $W_c(x, y)$, $W_e(x, y)$ and $W_d(x, y)$ are the sound source power per unit area of side a, side b, side c, side e and side d, respectively, which can be derived from following formulas (11):

$$F_e = F_a = F - k_1 \cos \omega t - \iint_{S_d} W_a(x, y) dx dy - \iint_{S_e} W_e(x, y) dx dy - f$$

$$F_b = -F + k_1 \cos \omega t + \iint_{S_b} W_b(x, y) dx dy - \iint_{S_e} W_e(x, y) dx dy - L$$

$$F_c = F_d = k_2 \cos \omega t - \iint_{S_c} W_c(x, y) dx dy - f - \gamma$$

$$F_d = F_b - k_2 \cos \omega t - \iint_{S_d} W_d(x, y) dx dy \quad (11)$$

wherein F is the driving force generated by the transducer 22, F_a, F_b, F_c, F_d and F_e are the driving forces of side a, side b, side c, side d and side e, respectively. As used herein, side d is the outside surface of the bottom 10-2. S_d is the region of side d, f is the viscous resistance formed in the small gap of the sidewalls, and $f = \eta \Delta s (dv/dy)$.

L is the equivalent load on human face when the panel acts on the human face, γ is the energy dissipated on elastic element 24, k_1 and k_2 are the elastic coefficients of elastic element 23 and elastic element 24 respectively, η is the fluid viscosity coefficient, dv/dy is the velocity gradient of fluid, Δs is the cross-section area of a subject (board), A is the amplitude, φ is the region of the sound field, and δ is a high order minimum (which is generated by the incompletely symmetrical shape of the housing);

The sound pressure of an arbitrary point outside the housing, generated by the vibration of the housing 10-0 is expressed as:

$$P_d = -j\omega\rho_0 \iint W_d(x'_d, y'_d) \cdot \frac{e^{jkR(x'_d, y'_d)}}{4\pi R(x'_d, y'_d)} dx'_d dy'_d, \quad (12)$$

wherein $R(x'_d, y'_d) = \sqrt{(x-x'_d)^2 + (y-y'_d)^2 + (z-z_d)^2}$ is the distance between the observation point (x, y, z) and a point on side d (x'_d, y'_d, z_d).

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P_a, P_b, P_c and P_e are functions of the position, when we set a hole on an arbitrary position in the housing, if the area of the hole is S_{hole} , the sound pressure of the hole is $\iint_{S_{hole}} Pds$.

In the meanwhile, because the panel 21 fits human tissues tightly, the power it gives out is absorbed all by human tissues, so the only side that can push air outside the housing to vibrate is side d, thus forming sound leakage. As described elsewhere, the sound leakage is resulted from the vibrations of the housing 10-0. For illustrative purposes, the sound pressure generated by the housing 10-0 may be expressed as $\iint_{S_{housing}} P_d ds$.

The leaked sound wave and the guided sound wave interference may result in a weakened sound wave, i.e., to make $\iint_{S_{hole}} Pds$ and $\iint_{S_{housing}} P_d ds$ have the same value but opposite directions, and the sound leakage may be reduced. In some embodiments, $\iint_{S_{hole}} Pds$ may be adjusted to reduce the sound leakage. Since $\iint_{S_{hole}} Pds$ corresponds to information of phases and amplitudes of one or more holes, which further relates to dimensions of the housing of the bone conduction speaker, the vibration frequency of the transducer, the position, shape, quantity and/or size of the sound guiding holes and whether there is damping inside the holes. Thus, the position, shape, and quantity of sound guiding holes, and/or damping materials may be adjusted to reduce sound leakage.

Additionally, because of the basic structure and function differences of a bone conduction speaker and a traditional air conduction speaker, the formulas above are only suitable for bone conduction speakers. Whereas in traditional air conduction speakers, the air in the air housing can be treated as a whole, which is not sensitive to positions, and this is different intrinsically with a bone conduction speaker, therefore the above formulas are not suitable to an air conduction speaker.

According to the formulas above, a person having ordinary skill in the art would understand that the effectiveness of reducing sound leakage is related to the dimensions of the housing of the bone conduction speaker, the vibration frequency of the transducer, the position, shape, quantity and size of the sound guiding hole(s) and whether there is damping inside the sound guiding hole(s). Accordingly, various configurations, depending on specific needs, may be obtained by choosing specific position where the sound guiding hole(s) is located, the shape and/or quantity of the sound guiding hole(s) as well as the damping material.

FIG. 5 is a diagram illustrating the equal-loudness contour curves according to some embodiments of the present disclosure. The horizontal coordinate is frequency, while the vertical coordinate is sound pressure level (SPL). As used herein, the SPL refers to the change of atmospheric pressure after being disturbed, i.e., a surplus pressure of the atmospheric pressure, which is equivalent to an atmospheric pressure added to a pressure change caused by the disturbance. As a result, the sound pressure may reflect the amplitude of a sound wave. In FIG. 5, on each curve, sound pressure levels corresponding to different frequencies are different, while the loudness levels felt by human ears are the same. For example, each curve is labeled with a number representing the loudness level of said curve. According to the loudness level curves, when volume (sound pressure amplitude) is lower, human ears are not sensitive to sounds of high or low frequencies; when volume is higher, human ears are more sensitive to sounds of high or low frequencies. Bone conduction speakers may generate sound relating to different frequency ranges, such as 1000 Hz~4000 Hz, or 1000 Hz~4000 Hz, or 1000 Hz~3500 Hz, or 1000 Hz~3000 Hz, or 1500 Hz~3000 Hz. The sound leakage within the

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above-mentioned frequency ranges may be the sound leakage aimed to be reduced with a priority.

FIG. 4D is a diagram illustrating the effect of reduced sound leakage according to some embodiments of the present disclosure, wherein the test results and calculation results are close in the above range. The bone conduction speaker being tested includes a cylindrical housing, which includes a sidewall and a bottom, as described in FIGS. 4A and 4B. The cylindrical housing is in a cylinder shape having a radius of 22 mm, the sidewall height of 14 mm, and a plurality of sound guiding holes being set on the upper portion of the sidewall of the housing. The openings of the sound guiding holes are rectangle. The sound guiding holes are arranged evenly on the sidewall. The target region where the sound leakage is to be reduced is 50 cm away from the outside of the bottom of the housing. The distance of the leaked sound wave spreading to the target region and the distance of the sound wave spreading from the surface of the transducer 20 through the sound guiding holes 30 to the target region have a difference of about 180 degrees in phase. As shown, the leaked sound wave is reduced in the target region dramatically or even be eliminated.

According to the embodiments in this disclosure, the effectiveness of reducing sound leakage after setting sound guiding holes is very obvious. As shown in FIG. 4D, the bone conduction speaker having sound guiding holes greatly reduce the sound leakage compared to the bone conduction speaker without sound guiding holes.

In the tested frequency range, after setting sound guiding holes, the sound leakage is reduced by about 10 dB on average. Specifically, in the frequency range of 1500 Hz~3000 Hz, the sound leakage is reduced by over 10 dB. In the frequency range of 2000 Hz~2500 Hz, the sound leakage is reduced by over 20 dB compared to the scheme without sound guiding holes.

A person having ordinary skill in the art can understand from the above-mentioned formulas that when the dimensions of the bone conduction speaker, target regions to reduce sound leakage and frequencies of sound waves differ, the position, shape and quantity of sound guiding holes also need to adjust accordingly.

For example, in a cylinder housing, according to different needs, a plurality of sound guiding holes may be on the sidewall and/or the bottom of the housing. Preferably, the sound guiding hole may be set on the upper portion and/or lower portion of the sidewall of the housing. The quantity of the sound guiding holes set on the sidewall of the housing is no less than two. Preferably, the sound guiding holes may be arranged evenly or unevenly in one or more circles with respect to the center of the bottom. In some embodiments, the sound guiding holes may be arranged in at least one circle. In some embodiments, one sound guiding hole may be set on the bottom of the housing. In some embodiments, the sound guiding hole may be set at the center of the bottom of the housing.

The quantity of the sound guiding holes can be one or more. Preferably, multiple sound guiding holes may be set symmetrically on the housing. In some embodiments, there are 6-8 circularly arranged sound guiding holes.

The openings (and cross sections) of sound guiding holes may be circle, ellipse, rectangle, or slit. Slit generally means slit along with straight lines, curve lines, or arc lines. Different sound guiding holes in one bone conduction speaker may have same or different shapes.

A person having ordinary skill in the art can understand that, the sidewall of the housing may not be cylindrical, the sound guiding holes can be arranged asymmetrically as

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needed. Various configurations may be obtained by setting different combinations of the shape, quantity, and position of the sound guiding. Some other embodiments along with the figures are described as follows.

In some embodiments, the leaked sound wave may be generated by a portion of the housing 10-0. The portion of the housing may be the sidewall 10-1 of the housing 10-0 and/or the bottom 10-2 of the housing 10-0. Merely by way of example, the leaked sound wave may be generated by the bottom 10-2 of the housing 10-0. The guided sound wave output through the sound guiding hole(s) 30 may interfere with the leaked sound wave generated by the portion of the housing 10-0. The interference may enhance or reduce a sound pressure level of the guided sound wave and/or leaked sound wave in the target region.

In some embodiments, the portion of the housing 10-0 that generates the leaked sound wave may be regarded as a first sound source (e.g., the sound source 1 illustrated in FIG. 3), and the sound guiding hole(s) 30 or a part thereof may be regarded as a second sound source (e.g., the sound source 2 illustrated in FIG. 3). Merely for illustration purposes, if the size of the sound guiding hole on the housing 10-0 is small, the sound guiding hole may be approximately regarded as a point sound source. In some embodiments, any number or count of sound guiding holes provided on the housing 10-0 for outputting sound may be approximated as a single point sound source. Similarly, for simplicity, the portion of the housing 10-0 that generates the leaked sound wave may also be approximately regarded as a point sound source. In some embodiments, both the first sound source and the second sound source may approximately be regarded as point sound sources (also referred to as two-point sound sources).

FIG. 4E is a schematic diagram illustrating exemplary two-point sound sources according to some embodiments of the present disclosure. The sound field pressure p generated by a single point sound source may satisfy Equation (13):

$$p = \frac{j\omega\rho_0}{4\pi r} Q_0 \exp j(\omega t - kr), \quad (13)$$

where ω denotes an angular frequency, ρ_0 denotes an air density, r denotes a distance between a target point and the sound source, Q_0 denotes a volume velocity of the sound source, and k denotes a wave number. It may be concluded that the magnitude of the sound field pressure of the sound field of the point sound source is inversely proportional to the distance to the point sound source.

It should be noted that, the sound guiding hole(s) for outputting sound as a point sound source may only serve as an explanation of the principle and effect of the present disclosure, and the shape and/or size of the sound guiding hole(s) may not be limited in practical applications. In some embodiments, if the area of the sound guiding hole is large, the sound guiding hole may also be equivalent to a planar sound source. Similarly, if an area of the portion of the housing 10-0 that generates the leaked sound wave is large (e.g., the portion of the housing 10-0 is a vibration surface or a sound radiation surface), the portion of the housing 10-0 may also be equivalent to a planar sound source. For those skilled in the art, without creative activities, it may be known that sounds generated by structures such as sound guiding holes, vibration surfaces, and sound radiation surfaces may be equivalent to point sound sources at the spatial scale discussed in the present disclosure, and may have consistent

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sound propagation characteristics and the same mathematical description method. Further, for those skilled in the art, without creative activities, it may be known that the acoustic effect achieved by the two-point sound sources may also be implemented by alternative acoustic structures. According to actual situations, the alternative acoustic structures may be modified and/or combined discretionarily, and the same acoustic output effect may be achieved.

The two-point sound sources may be formed such that the guided sound wave output from the sound guiding hole(s) may interfere with the leaked sound wave generated by the portion of the housing 10-0. The interference may reduce a sound pressure level of the leaked sound wave in the surrounding environment (e.g., the target region). For convenience, the sound waves output from an acoustic output device (e.g., the bone conduction speaker) to the surrounding environment may be referred to as far-field leakage since it may be heard by others in the environment. The sound waves output from the acoustic output device to the ears of the user may also be referred to as near-field sound since a distance between the bone conduction speaker and the user may be relatively short. In some embodiments, the sound waves output from the two-point sound sources may have a same frequency or frequency range (e.g., 800 Hz, 1000 Hz, 1500 Hz, 3000 Hz, etc.). In some embodiments, the sound waves output from the two-point sound sources may have a certain phase difference. In some embodiments, the sound guiding hole includes a damping layer. The damping layer may be, for example, a tuning paper, a tuning cotton, a nonwoven fabric, a silk, a cotton, a sponge, or a rubber. The damping layer may be configured to adjust the phase of the guided sound wave in the target region. The acoustic output device described herein may include a bone conduction speaker or an air conduction speaker. For example, a portion of the housing (e.g., the bottom of the housing) of the bone conduction speaker may be treated as one of the two-point sound sources, and at least one sound guiding holes of the bone conduction speaker may be treated as the other one of the two-point sound sources. As another example, one sound guiding hole of an air conduction speaker may be treated as one of the two-point sound sources, and another sound guiding hole of the air conduction speaker may be treated as the other one of the two-point sound sources. It should be noted that, although the construction of two-point sound sources may be different in bone conduction speaker and air conduction speaker, the principles of the interference between the various constructed two-point sound sources are the same. Thus, the equivalence of the two-point sound sources in a bone conduction speaker disclosed elsewhere in the present disclosure is also applicable for an air conduction speaker.

In some embodiments, when the position and phase difference of the two-point sound sources meet certain conditions, the acoustic output device may output different sound effects in the near field (for example, the position of the user's ear) and the far field. For example, if the phases of the point sound sources corresponding to the portion of the housing 10-0 and the sound guiding hole(s) are opposite, that is, an absolute value of the phase difference between the two-point sound sources is 180 degrees, the far-field leakage may be reduced according to the principle of reversed phase cancellation.

Merely by way of example, the sound transmitted to the outside of the housing through a first sound guiding hole may be simply regarded as a first sound formed by a monopole sound source (also referred to as a "single point sound source"). The sound transmitted to the outside of the

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housing through a second sound guiding hole may be simply regarded as a second sound formed by a monopole sound source. The second sound may be opposite to the first sound in phase, which may be reversed and canceled in the far-field. That is, an "acoustic dipole" may be formed to reduce sound leakage. In some embodiments, the speaker may be worn by the user through a fixing assembly (e.g., a fixing assembly 20 illustrated in FIG. 75). For example, the speaker may be implemented as an earphone including the fixing assembly. In some embodiments, the earphone may be configured such that when the earphone is in the wearing state, the sound guiding holes (or the two monopole sound sources) meet certain conditions, and the acoustic output device may output different sound effects in the near field and the far field. For example, a connection line of the two monopole sound sources may be directed to an ear hole (or a hearing position) of the user such that the user can hear a sufficiently loud sound. More description regarding the fixing assembly and/or the earphone may be found elsewhere in the present disclosure. See, e.g., FIGS. 17-75 and relevant descriptions thereof.

In some embodiments, the interference between the guided sound wave and the leaked sound wave at a specific frequency may relate to a distance between the sound guiding hole(s) and the portion of the housing 10-0. For example, if the sound guiding hole(s) are set at the upper portion of the sidewall of the housing 10-0 (as illustrated in FIG. 4A), the distance between the sound guiding hole(s) and the portion of the housing 10-0 may be large. Correspondingly, the frequencies of sound waves generated by such two-point sound sources may be in a mid-low frequency range (e.g., 1500-2000 Hz, 1500-2500 Hz, etc.). Referring to FIG. 4D, the interference may reduce the sound pressure level of the leaked sound wave in the mid-low frequency range (i.e., the sound leakage is low).

Merely by way of example, the low frequency range may refer to frequencies in a range below a first frequency threshold. The high frequency range may refer to frequencies in a range exceed a second frequency threshold. The first frequency threshold may be lower than the second frequency threshold. The mid-low frequency range may refer to frequencies in a range between the first frequency threshold and the second frequency threshold. For example, the first frequency threshold may be 1000 Hz, and the second frequency threshold may be 3000 Hz. The low frequency range may refer to frequencies in a range below 1000 Hz, the high frequency range may refer to frequencies in a range above 3000 Hz, and the mid-low frequency range may refer to frequencies in a range of 1000-2000 Hz, 1500-2500 Hz, etc. In some embodiments, a middle frequency range, a mid-high frequency range may also be determined between the first frequency threshold and the second frequency threshold. In some embodiments, the mid-low frequency range and the low frequency range may partially overlap. The mid-high frequency range and the high frequency range may partially overlap. For example, the mid-high frequency range may refer to frequencies in a range above 3000 Hz, and the mid-low frequency range may refer to frequencies in a range of 2800-3500 Hz. It should be noted that the low frequency range, the mid-low frequency range, the middle frequency range, the mid-high frequency range, and/or the high frequency range may be set flexibly according to different situations, and are not limited herein.

In some embodiments, the frequencies of the guided sound wave and the leaked sound wave may be set in a low frequency range (e.g., below 800 Hz, below 1200 Hz, etc.). In some embodiments, the amplitudes of the sound waves

generated by the two-point sound sources may be set to be different in the low frequency range. For example, the amplitude of the guided sound wave may be smaller than the amplitude of the leaked sound wave. In this case, the interference may not reduce sound pressure of the near-field sound in the low-frequency range. The sound pressure of the near-field sound may be improved in the low-frequency range. The volume of the sound heard by the user may be improved.

In some embodiments, the amplitude of the guided sound wave may be adjusted by setting an acoustic resistance structure in the sound guiding hole(s) 30. The material of the acoustic resistance structure disposed in the sound guiding hole 30 may include, but not limited to, plastics (e.g., high-molecular polyethylene, blown nylon, engineering plastics, etc.), cotton, nylon, fiber (e.g., glass fiber, carbon fiber, boron fiber, graphite fiber, graphene fiber, silicon carbide fiber, or aramid fiber), other single or composite materials, other organic and/or inorganic materials, etc. The thickness of the acoustic resistance structure may be 0.005 mm, 0.01 mm, 0.02 mm, 0.5 mm, 1 mm, 2 mm, etc. The structure of the acoustic resistance structure may be in a shape adapted to the shape of the sound guiding hole. For example, the acoustic resistance structure may have a shape of a cylinder, a sphere, a cubic, etc. In some embodiments, the materials, thickness, and structures of the acoustic resistance structure may be modified and/or combined to obtain a desirable acoustic resistance structure. In some embodiments, the acoustic resistance structure may be implemented by the damping layer.

In some embodiments, the amplitude of the guided sound wave output from the sound guiding hole may be relatively low (e.g., zero or almost zero). The difference between the guided sound wave and the leaked sound wave may be maximized, thus achieving a relatively large sound pressure in the near field. In this case, the sound leakage of the acoustic output device having sound guiding holes may be almost the same as the sound leakage of the acoustic output device without sound guiding holes in the low frequency range (e.g., as shown in FIG. 4D).

Embodiment Two

FIG. 6 is a flowchart of an exemplary method of reducing sound leakage of a bone conduction speaker according to some embodiments of the present disclosure. At 601, a bone conduction speaker including a panel 21 touching human skin and passing vibrations, a transducer 22, and a housing 10-0 is provided. At least one sound guiding hole 30 is arranged on the housing 10-0. At 602, the panel 21 is driven by the transducer 22, causing the panel 21 to vibrate. At 603, a leaked sound wave due to the vibrations of the housing is formed, wherein the leaked sound wave transmits in the air. At 604, a guided sound wave passing through the at least one sound guiding hole 30 from the inside to the outside of the housing 10-0. The guided sound wave interferes with the leaked sound wave, reducing the sound leakage of the bone conduction speaker.

The sound guiding holes 30 are preferably set at different positions of the housing 10-0.

The effectiveness of reducing sound leakage may be determined by the formulas and method as described above, based on which the positions of sound guiding holes may be determined.

A damping layer is preferably set in a sound guiding hole 30 to adjust the phase and amplitude of the sound wave transmitted through the sound guiding hole 30.

In some embodiments, different sound guiding holes may generate different sound waves having a same phase to reduce the leaked sound wave having the same wavelength. In some embodiments, different sound guiding holes may generate different sound waves having different phases to reduce the leaked sound waves having different wavelengths.

In some embodiments, different portions of a sound guiding hole 30 may be configured to generate sound waves having a same phase to reduce the leaked sound waves with the same wavelength. In some embodiments, different portions of a sound guiding hole 30 may be configured to generate sound waves having different phases to reduce the leaked sound waves with different wavelengths.

Additionally, the sound wave inside the housing may be processed to basically have the same value but opposite phases with the leaked sound wave, so that the sound leakage may be further reduced.

Embodiment Three

FIGS. 7A and 7B are schematic structures illustrating an exemplary bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10-0, a panel 21, and a transducer 22. The housing 10-0 may cylindrical and have a sidewall and a bottom. A plurality of sound guiding holes 30 may be arranged on the lower portion of the sidewall (i.e., from about the $\frac{2}{3}$ height of the sidewall to the bottom). The quantity of the sound guiding holes 30 may be 8, the openings of the sound guiding holes 30 may be rectangle. The sound guiding holes 30 may be arranged evenly or evenly in one or more circles on the sidewall of the housing 10-0.

In the embodiment, the transducer 22 is preferably implemented based on the principle of electromagnetic transduction. The transducer 22 may include components such as magnetizer, voice coil, and etc., and the components may be located inside the housing and may generate synchronous vibrations with a same frequency.

FIG. 7C is a diagram illustrating reduced sound leakage according to some embodiments of the present disclosure. In the frequency range of 1400 Hz–4000 Hz, the sound leakage is reduced by more than 5 dB, and in the frequency range of 2250 Hz–2500 Hz, the sound leakage is reduced by more than 20 dB.

In some embodiments, the sound guiding hole(s) at the lower portion of the sidewall of the housing 10-0 may also be approximately regarded as a point sound source. In some embodiments, the sound guiding hole(s) at the lower portion of the sidewall of the housing 10-0 and the portion of the housing 10-0 that generates the leaked sound wave may constitute two-point sound sources. The two-point sound sources may be formed such that the guided sound wave output from the sound guiding hole(s) at the lower portion of the sidewall of the housing 10-0 may interfere with the leaked sound wave generated by the portion of the housing 10-0. The interference may reduce a sound pressure level of the leaked sound wave in the surrounding environment (e.g., the target region) at a specific frequency or frequency range.

In some embodiments, the sound waves output from the two-point sound sources may have a same frequency or frequency range (e.g., 1000 Hz, 2500 Hz, 3000 Hz, etc.). In some embodiments, the sound waves output from the first two-point sound sources may have a certain phase difference. In this case, the interference between the sound waves generated by the first two-point sound sources may reduce a

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sound pressure level of the leaked sound wave in the target region. When the position and phase difference of the first two-point sound sources meet certain conditions, the acoustic output device may output different sound effects in the near field (for example, the position of the user's ear) and the far field. For example, if the phases of the first two-point sound sources are opposite, that is, an absolute value of the phase difference between the first two-point sound sources is 180 degrees, the far-field leakage may be reduced.

In some embodiments, the interference between the guided sound wave and the leaked sound wave may relate to frequencies of the guided sound wave and the leaked sound wave and/or a distance between the sound guiding hole(s) and the portion of the housing 10-0. For example, if the sound guiding hole(s) are set at the lower portion of the sidewall of the housing 10-0 (as illustrated in FIG. 7A), the distance between the sound guiding hole(s) and the portion of the housing 10-0 may be small. Correspondingly, the frequencies of sound waves generated by such two-point sound sources may be in a high frequency range (e.g., above 3000 Hz, above 3500 Hz, etc.). Referring to FIG. 7C, the interference may reduce the sound pressure level of the leaked sound wave in the high frequency range.

Embodiment Four

FIGS. 8A and 8B are schematic structures illustrating an exemplary bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10-0, a panel 21, and a transducer 22. The housing 10-0 is cylindrical and have a sidewall and a bottom. The sound guiding holes 30 may be arranged on the central portion of the sidewall of the housing (i.e., from about the $\frac{1}{3}$ height of the sidewall to the $\frac{2}{3}$ height of the sidewall). The quantity of the sound guiding holes 30 may be 8, and the openings (and cross sections) of the sound guiding hole 30 may be rectangle. The sound guiding holes 30 may be arranged evenly or unevenly in one or more circles on the sidewall of the housing 10-0.

In the embodiment, the transducer 21 may be implemented preferably based on the principle of electromagnetic transduction. The transducer 21 may include components such as magnetizer, voice coil, etc., which may be placed inside the housing and may generate synchronous vibrations with the same frequency.

FIG. 8C is a diagram illustrating reduced sound leakage. In the frequency range of 1000 Hz~4000 Hz, the effectiveness of reducing sound leakage is great. For example, in the frequency range of 1400 Hz~2900 Hz, the sound leakage is reduced by more than 10 dB; in the frequency range of 2200 Hz~2500 Hz, the sound leakage is reduced by more than 20 dB.

It's illustrated that the effectiveness of reduced sound leakage can be adjusted by changing the positions of the sound guiding holes, while keeping other parameters relating to the sound guiding holes unchanged.

Embodiment Five

FIGS. 9A and 9B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10-0, a panel 21 and a transducer 22. The housing 10-0 is cylindrical, with a sidewall and a bottom. One or more perforative sound guiding holes 30 may be along the circumference of the bottom. In some embodiments, there may be 8 sound

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guiding holes 30 arranged evenly or unevenly in one or more circles on the bottom of the housing 10-0. In some embodiments, the shape of one or more of the sound guiding holes 30 may be rectangle.

In the embodiment, the transducer 21 may be implemented preferably based on the principle of electromagnetic transduction. The transducer 21 may include components such as magnetizer, voice coil, etc., which may be placed inside the housing and may generate synchronous vibration with the same frequency.

FIG. 9C is a diagram illustrating the effect of reduced sound leakage. In the frequency range of 1000 Hz~3000 Hz, the effectiveness of reducing sound leakage is outstanding. For example, in the frequency range of 1700 Hz~2700 Hz, the sound leakage is reduced by more than 10 dB; in the frequency range of 2200 Hz~2400 Hz, the sound leakage is reduced by more than 20 dB.

Embodiment Six

FIGS. 10A and 10B are schematic structures of an exemplary bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10-0, a panel 21 and a transducer 22. One or more perforative sound guiding holes 30 may be arranged on both upper and lower portions of the sidewall of the housing 10-0. The sound guiding holes 30 may be arranged evenly or unevenly in one or more circles on the upper and lower portions of the sidewall of the housing 10-0. In some embodiments, the quantity of sound guiding holes 30 in every circle may be 8, and the upper portion sound guiding holes and the lower portion sound guiding holes may be symmetrical about the central cross section of the housing 10-0. In some embodiments, the shape of the sound guiding hole 30 may be circle.

The shape of the sound guiding holes on the upper portion and the shape of the sound guiding holes on the lower portion may be different; One or more damping layers may be arranged in the sound guiding holes to reduce leaked sound waves of the same wave length (or frequency), or to reduce leaked sound waves of different wave lengths.

FIG. 10C is a diagram illustrating the effect of reducing sound leakage according to some embodiments of the present disclosure. In the frequency range of 1000 Hz~4000 Hz, the effectiveness of reducing sound leakage is outstanding. For example, in the frequency range of 1600 Hz~2700 Hz, the sound leakage is reduced by more than 15 dB; in the frequency range of 2000 Hz~2500 Hz, where the effectiveness of reducing sound leakage is most outstanding, the sound leakage is reduced by more than 20 dB. Compared to embodiment three, this scheme has a relatively balanced effect of reduced sound leakage on various frequency range, and this effect is better than the effect of schemes where the height of the holes are fixed, such as schemes of embodiment three, embodiment four, embodiment five, and so on.

In some embodiments, the sound guiding hole(s) at the upper portion of the sidewall of the housing 10-0 (also referred to as first hole(s)) may be approximately regarded as a point sound source. In some embodiments, the first hole(s) and the portion of the housing 10-0 that generates the leaked sound wave may constitute two-point sound sources (also referred to as first two-point sound sources). As for the first two-point sound sources, the guided sound wave generated by the first hole(s) (also referred to as first guided sound wave) may interfere with the leaked sound wave or a portion thereof generated by the portion of the housing 10-0 in a first region. In some embodiments, the sound waves

output from the first two-point sound sources may have a same frequency (e.g., a first frequency). In some embodiments, the sound waves output from the first two-point sound sources may have a certain phase difference. In this case, the interference between the sound waves generated by the first two-point sound sources may reduce a sound pressure level of the leaked sound wave in the target region. When the position and phase difference of the first two-point sound sources meet certain conditions, the acoustic output device may output different sound effects in the near field (for example, the position of the user's ear) and the far field. For example, if the phases of the first two-point sound sources are opposite, that is, an absolute value of the phase difference between the first two-point sound sources is 180 degrees, the far-field leakage may be reduced according to the principle of reversed phase cancellation.

In some embodiments, the sound guiding hole(s) at the lower portion of the sidewall of the housing 10-0 (also referred to as second hole(s)) may also be approximately regarded as another point sound source. Similarly, the second hole(s) and the portion of the housing 10-0 that generates the leaked sound wave may also constitute two-point sound sources (also referred to as second two-point sound sources). As for the second two-point sound sources, the guided sound wave generated by the second hole(s) (also referred to as second guided sound wave) may interfere with the leaked sound wave or a portion thereof generated by the portion of the housing 10-0 in a second region. The second region may be the same as or different from the first region. In some embodiments, the sound waves output from the second two-point sound sources may have a same frequency (e.g., a second frequency).

In some embodiments, the first frequency and the second frequency may be in certain frequency ranges. In some embodiments, the frequency of the guided sound wave output from the sound guiding hole(s) may be adjustable. In some embodiments, the frequency of the first guided sound wave and/or the second guided sound wave may be adjusted by one or more acoustic routes. The acoustic routes may be coupled to the first hole(s) and/or the second hole(s). The first guided sound wave and/or the second guided sound wave may be propagated along the acoustic route having a specific frequency selection characteristic. That is, the first guided sound wave and the second guided sound wave may be transmitted to their corresponding sound guiding holes via different acoustic routes. For example, the first guided sound wave and/or the second guided sound wave may be propagated along an acoustic route with a low-pass characteristic to a corresponding sound guiding hole to output guided sound wave of a low frequency. In this process, the high frequency component of the sound wave may be absorbed or attenuated by the acoustic route with the low-pass characteristic. Similarly, the first guided sound wave and/or the second guided sound wave may be propagated along an acoustic route with a high-pass characteristic to the corresponding sound guiding hole to output guided sound wave of a high frequency. In this process, the low frequency component of the sound wave may be absorbed or attenuated by the acoustic route with the high-pass characteristic.

FIG. 10D is a schematic diagram illustrating an acoustic route according to some embodiments of the present disclosure. FIG. 10E is a schematic diagram illustrating another acoustic route according to some embodiments of the present disclosure. FIG. 10F is a schematic diagram illustrating a further acoustic route according to some embodiments of the present disclosure. In some embodiments, structures such as a sound tube, a sound cavity, a sound resistance, etc.,

may be set in the acoustic route for adjusting frequencies for the sound waves (e.g., by filtering certain frequencies). It should be noted that FIGS. 10D-10F may be provided as examples of the acoustic routes, and not intended be limiting.

As shown in FIG. 10D, the acoustic route may include one or more lumen structures. The one or more lumen structures may be connected in series. An acoustic resistance material may be provided in each of at least one of the one or more lumen structures to adjust acoustic impedance of the entire structure to achieve a desirable sound filtering effect. For example, the acoustic impedance may be in a range of 5 MKS Rayleigh to 500 MKS Rayleigh. In some embodiments, a high-pass sound filtering, a low-pass sound filtering, and/or a band-pass filtering effect of the acoustic route may be achieved by adjusting a size of each of at least one of the one or more lumen structures and/or a type of acoustic resistance material in each of at least one of the one or more lumen structures. The acoustic resistance materials may include, but not limited to, plastic, textile, metal, permeable material, woven material, screen material or mesh material, porous material, particulate material, polymer material, or the like, or any combination thereof. By setting the acoustic routes of different acoustic impedances, the acoustic output from the sound guiding holes may be acoustically filtered. In this case, the guided sound waves may have different frequency components.

As shown in FIG. 10E, the acoustic route may include one or more resonance cavities. The one or more resonance cavities may be, for example, Helmholtz cavity. In some embodiments, a high-pass sound filtering, a low-pass sound filtering, and/or a band-pass filtering effect of the acoustic route may be achieved by adjusting a size of each of at least one of the one or more resonance cavities and/or a type of acoustic resistance material in each of at least one of the one or more resonance cavities.

As shown in FIG. 10F, the acoustic route may include a combination of one or more lumen structures and one or more resonance cavities. In some embodiments, a high-pass sound filtering, a low-pass sound filtering, and/or a band-pass filtering effect of the acoustic route may be achieved by adjusting a size of each of at least one of the one or more lumen structures and one or more resonance cavities and/or a type of acoustic resistance material in each of at least one of the one or more lumen structures and one or more resonance cavities. It should be noted that the structures exemplified above may be for illustration purposes, various acoustic structures may also be provided, such as a tuning net, tuning cotton, etc.

In some embodiments, the interference between the leaked sound wave and the guided sound wave may relate to frequencies of the guided sound wave and the leaked sound wave and/or a distance between the sound guiding hole(s) and the portion of the housing 10-0. In some embodiments, the portion of the housing that generates the leaked sound wave may be the bottom of the housing 10-0. The first hole(s) may have a larger distance to the portion of the housing 10-0 than the second hole(s). In some embodiments, the frequency of the first guided sound wave output from the first hole(s) (e.g., the first frequency) and the frequency of second guided sound wave output from second hole(s) (e.g., the second frequency) may be different.

In some embodiments, the first frequency and second frequency may associate with the distance between the at least one sound guiding hole and the portion of the housing 10-0 that generates the leaked sound wave. In some embodiments, the first frequency may be set in a low frequency

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range. The second frequency may be set in a high frequency range. The low frequency range and the high frequency range may or may not overlap.

In some embodiments, the frequency of the leaked sound wave generated by the portion of the housing 10-0 may be in a wide frequency range. The wide frequency range may include, for example, the low frequency range and the high frequency range or a portion of the low frequency range and the high frequency range. For example, the leaked sound wave may include a first frequency in the low frequency range and a second frequency in the high frequency range. In some embodiments, the leaked sound wave of the first frequency and the leaked sound wave of the second frequency may be generated by different portions of the housing 10-0. For example, the leaked sound wave of the first frequency may be generated by the sidewall of the housing 10-0, the leaked sound wave of the second frequency may be generated by the bottom of the housing 10-0. As another example, the leaked sound wave of the first frequency may be generated by the bottom of the housing 10-0, the leaked sound wave of the second frequency may be generated by the sidewall of the housing 10-0. In some embodiments, the frequency of the leaked sound wave generated by the portion of the housing 10-0 may relate to parameters including the mass, the damping, the stiffness, etc., of the different portion of the housing 10-0, the frequency of the transducer 22, etc.

In some embodiments, the characteristics (amplitude, frequency, and phase) of the first two-point sound sources and the second two-point sound sources may be adjusted via various parameters of the acoustic output device (e.g., electrical parameters of the transducer 22, the mass, stiffness, size, structure, material, etc., of the portion of the housing 10-0, the position, shape, structure, and/or number (or count) of the sound guiding hole(s) so as to form a sound field with a particular spatial distribution. In some embodiments, a frequency of the first guided sound wave is smaller than a frequency of the second guided sound wave.

A combination of the first two-point sound sources and the second two-point sound sources may improve sound effects both in the near field and the far field.

Referring to FIGS. 4D, 7C, and 10C, by designing different two-point sound sources with different distances, the sound leakage in both the low frequency range and the high frequency range may be properly suppressed. In some embodiments, the closer distance between the second two-point sound sources may be more suitable for suppressing the sound leakage in the far field, and the relative longer distance between the first two-point sound sources may be more suitable for reducing the sound leakage in the near field. In some embodiments, the amplitudes of the sound waves generated by the first two-point sound sources may be set to be different in the low frequency range. For example, the amplitude of the guided sound wave may be smaller than the amplitude of the leaked sound wave. In this case, the sound pressure level of the near-field sound may be improved. The volume of the sound heard by the user may be increased.

Embodiment Seven

FIGS. 11A and 11B are schematic structures illustrating a bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10-0, a panel 21 and a transducer 22. One or more perforative sound guiding holes 30 may be set on upper and lower portions of the sidewall of the housing 10-0 and on the bottom of the housing 10-0. The

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sound guiding holes 30 on the sidewall are arranged evenly or unevenly in one or more circles on the upper and lower portions of the sidewall of the housing 10-0. In some embodiments, the quantity of sound guiding holes 30 in every circle may be 8, and the upper portion sound guiding holes and the lower portion sound guiding holes may be symmetrical about the central cross section of the housing 10-0. In some embodiments, the shape of the sound guiding hole 30 may be rectangular. There may be four sound guiding holes 30 on the bottom of the housing 10-0. The four sound guiding holes 30 may be linear-shaped along arcs, and may be arranged evenly or unevenly in one or more circles with respect to the center of the bottom. Furthermore, the sound guiding holes 30 may include a circular perforative hole on the center of the bottom.

FIG. 11C is a diagram illustrating the effect of reducing sound leakage of the embodiment. In the frequency range of 1000 Hz~4000 Hz, the effectiveness of reducing sound leakage is outstanding. For example, in the frequency range of 1300 Hz~3000 Hz, the sound leakage is reduced by more than 10 dB; in the frequency range of 2000 Hz~2700 Hz, the sound leakage is reduced by more than 20 dB. Compared to embodiment three, this scheme has a relatively balanced effect of reduced sound leakage within various frequency range, and this effect is better than the effect of schemes where the height of the holes are fixed, such as schemes of embodiment three, embodiment four, embodiment five, and etc. Compared to embodiment six, in the frequency range of 1000 Hz~1700 Hz and 2500 Hz~4000 Hz, this scheme has a better effect of reduced sound leakage than embodiment six.

Embodiment Eight

FIGS. 12A and 12B are schematic structures illustrating a bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10-0, a panel 21 and a transducer 22. A perforative sound guiding hole 30 may be set on the upper portion of the sidewall of the housing 10-0. One or more sound guiding holes may be arranged evenly or unevenly in one or more circles on the upper portion of the sidewall of the housing 10-0. There may be 8 sound guiding holes 30, and the shape of the sound guiding holes 30 may be circle.

After comparison of calculation results and test results, the effectiveness of this embodiment is basically the same with that of embodiment one, and this embodiment can effectively reduce sound leakage.

Embodiment Nine

FIGS. 13A and 13B are schematic structures illustrating a bone conduction speaker according to some embodiments of the present disclosure. The bone conduction speaker may include an open housing 10-0, a panel 21 and a transducer 22.

The difference between this embodiment and the above-described embodiment three is that to reduce sound leakage to greater extent, the sound guiding holes 30 may be arranged on the upper, central and lower portions of the sidewall 10-1. The sound guiding holes 30 are arranged evenly or unevenly in one or more circles. Different circles are formed by the sound guiding holes 30, one of which is set along the circumference of the bottom 10-2 of the housing 10-0. The size of the sound guiding holes 30 are the same.

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The effect of this scheme may cause a relatively balanced effect of reducing sound leakage in various frequency ranges compared to the schemes where the position of the holes are fixed. The effect of this design on reducing sound leakage is relatively better than that of other designs where the heights of the holes are fixed, such as embodiment three, embodiment four, embodiment five, etc.

Embodiment Ten

The sound guiding holes **30** in the above embodiments may be perforative holes without shields.

In order to adjust the effect of the sound waves guided from the sound guiding holes, a damping layer (not shown in the figures) may locate at the opening of a sound guiding hole **30** to adjust the phase and/or the amplitude of the sound wave.

There are multiple variations of materials and positions of the damping layer. For example, the damping layer may be made of materials which can damp sound waves, such as tuning paper, tuning cotton, nonwoven fabric, silk, cotton, sponge or rubber. The damping layer may be attached on the inner wall of the sound guiding hole **30**, or may shield the sound guiding hole **30** from outside.

More preferably, the damping layers corresponding to different sound guiding holes **30** may be arranged to adjust the sound waves from different sound guiding holes to generate a same phase. The adjusted sound waves may be used to reduce leaked sound wave having the same wavelength. Alternatively, different sound guiding holes **30** may be arranged to generate different phases to reduce leaked sound wave having different wavelengths (i.e., leaked sound waves with specific wavelengths).

In some embodiments, different portions of a same sound guiding hole can be configured to generate a same phase to reduce leaked sound waves on the same wavelength (e.g., using a pre-set damping layer with the shape of stairs or steps). In some embodiments, different portions of a same sound guiding hole can be configured to generate different phases to reduce leaked sound waves on different wavelengths.

The above-described embodiments are preferable embodiments with various configurations of the sound guiding hole(s) on the housing of a bone conduction speaker, but a person having ordinary skills in the art can understand that the embodiments don't limit the configurations of the sound guiding hole(s) to those described in this application.

In the past bone conduction speakers, the housing of the bone conduction speakers is closed, so the sound source inside the housing is sealed inside the housing. In the embodiments of the present disclosure, there can be holes in proper positions of the housing, making the sound waves inside the housing and the leaked sound waves having substantially same amplitude and substantially opposite phases in the space, so that the sound waves can interfere with each other and the sound leakage of the bone conduction speaker is reduced. Meanwhile, the volume and weight of the speaker do not increase, the reliability of the product is not comprised, and the cost is barely increased. The designs disclosed herein are easy to implement, reliable, and effective in reducing sound leakage.

In general, a sound quality of a bone conduction speaker may be affected by various factors, such as, a physical property of components of the bone conduction speaker, a vibration transfer relationship between the components, a vibration transfer relationship between the bone conduction speaker and external environment, a vibration transfer effi-

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ciency of the vibration transfer system, or the like. The components of the bone conduction speaker may include a vibration generation element (such as the transducer **22**), a component for fixing the speaker (such as headset bracket/headset lanyard), a vibration transfer component (such as the panel **21** and a vibration transfer layer covering an outer side of the panel **21**). The vibration transfer relationships between the components and between the bone conduction speaker and external environment may be determined by the manner that the bone conduction speaker is in contact with a user (such as clamping force, contacting area, contacting shape). FIG. **14** is an equivalent diagram illustrating the vibration generation and vibration transfer system of the bone conduction speaker. The equivalent system of a bone conduction speaker may include a fixed end **1401**, a sensor terminal **1402**, a vibration unit **1403**, and a transducer **1404**. The fixed end **1401** may be connected to the vibration unit **1403** through a transfer relationship K1 (i.e., k_4 in FIG. **14**); the sensor terminal **1402** may be connected to the vibration unit **1403** through the transfer relationship K2 (i.e., R_3 and k_3 in FIG. **14**); the vibration unit **1403** may be connected to the transducer **1404** through the transfer relationship K3 (R_4 , k_5 in FIG. **14**).

The vibration unit **1403** may include a panel (e.g., the panel **21**) and a transducer (e.g., the transducer **22**). The transfer relationships K1, K2 and K3 may be used to describe the relationships between the corresponding components in the equivalent system of the bone conduction speaker (described in detail below). Vibration equations of the equivalent system may be expressed as:

$$m_3 x_3'' + R_3 x_3' - R_4 x_4' + (k_3 + k_4) x_3 + k_5 (x_3 - x_4) = f_0, \quad (14),$$

$$m_4 x_4'' + R_4 x_4' - k_5 (x_3 - x_4) = f_4, \quad (15),$$

where, m_3 is an equivalent mass of the vibration unit **1403**; m_4 is an equivalent mass of the transducer **1404**; x_3 is an equivalent displacement of the vibration unit **1403**; x_4 is an equivalent displacement of the transducer **1404**; k_3 is an equivalent elastic coefficient formed between the sensor terminal **1402** and the vibration unit **1403**; k_4 is an equivalent elastic coefficient formed between the fixed ends **1401** and the vibration unit **1403**; k_5 is an equivalent elastic coefficient formed between the transducer **1404** and the vibration unit **1403**; R_3 is an equivalent damping formed between the sensor terminal **1402** and the vibration unit **1403**; R_4 is an equivalent damping formed between the transducer **1404** and the vibration unit **1403**; h and f_4 are interaction forces between the vibration unit **1403** and the transducer **1404**. The equivalent amplitude of the vibration unit A_3 is:

$$A_3 = - \frac{m_4 \omega^2}{(m_3 \omega^2 + j \omega R_3 - (k_3 + k_4 + k_5)) \cdot f_0, \quad (16)} \\ (m_4 \omega^2 + j \omega R_4 - k_5) - k_5 (k_5 - j \omega R_4)$$

where f_0 is a unit driving force, and ω is a vibration frequency. The factors affecting the frequency response of the bone conduction speaker may include the vibration generation (including but not limited to, the vibration unit, the transducer, the housing, and the connection means between each other, such as m_3 , m_4 , k_5 , R_4 in equation (16)), and the vibration transfer (including but not limited to, the way being in contact with skin, the property of headset bracket/headset lanyard, such as k_3 , k_4 , R_3 in equation (16)). The frequency response and the sound quality of the bone

conduction speaker may also be affected by changes of the structure of each component and the parameter of the connection between each component of the bone conduction speaker; for example, changing the size of the clamping force may be equivalent to changing k_4 , changing the bond with glue may be equivalent to changing R_4 and k_5 , and changing hardness, elasticity, damping of relevant materials may be equivalent to changing k_3 and R_3 .

In an embodiment, the location of the fixed end **1401** may refer to a point or an area relatively fixed at a location in the vibration process, and the point or area may be deemed as the fixed end. The fixed end may be consisted of certain components, or may also be determined by the structure of the bone conduction speaker. For example, the bone conduction speaker may be suspended, adhered, or absorbed around a user's ear, or may attach to a man's skin through special design for the structure or the appearance of the bone conduction speaker.

The sensor terminal **1402** may be an auditory system of a person for receiving a sound signal. The vibration unit **1403** may be used to protect, support, and connect the transducer. The vibration unit **1403** may include a vibration transfer layer for transmitting vibrations to a user, a panel being in contact with a user directly or indirectly, and a housing for protecting and supporting other vibration generation components. The transducer **1404** may generate sound vibrations.

The transfer relationship K1 may connect the fixed end **1401** and the vibration unit **1403**, which refers to the vibration transfer relationship between the fixed end and the vibration generation portion. K1 may be determined based on the shape and the structure of the bone conduction speaker. For example, the bone conduction speaker may be fixed on a user's head by a U-shaped headset bracket/the headset lanyard. The bone conduction speaker may also be set on a helmet, a fire mask or a specific mask, a glass, or the like. Different structures and shapes of the bone conduction speaker may affect the transfer relationship K1. Further, the structure of the bone conduction speaker may include the material, mass, etc., of different parts of the bone conduction speaker. The transfer relationship K2 may connect the sensor terminal **1402** and the vibration unit **1403**.

K2 may depend on the component of the transfer system. The transfer may include but not limited to transferring sound through a user's tissue to the user's auditory system. For example, when the sound is transferred to the auditory system through the skin, subcutaneous tissue, bones, etc., the physical properties of various parts and mutual connection relationships between the various parts may have impacts on K2. Further, the vibration unit **1403** may be in contact with tissue. In various embodiments, the contact surface may be the vibration transfer layer or the side surface of the panel. The shape and the size of the contact surface, and the force between the vibration unit **1403** and tissue may influence the transfer coefficient K2.

The transfer coefficient K3 between the vibration unit **1403** and the transducer **1404** may be dependent on the connection property inside the vibration generation unit of the bone conduction speaker. The transducer and the vibration unit may be connected rigidly or flexibly, or changing the relative position of the connector between the vibration unit, and the transducer may affect the transducer for transferring vibrations to the vibration unit, especially the transfer efficiency of the panel, thereby affecting the transfer relationship K3.

When the bone conduction speaker is used, the sound generation and transferring process may affect the sound

quality that a user feels. For example, the fixed end, the sense terminal, the vibration unit, the transducer and transfer relationship K1, K2 and K3, etc., mentioned above, may have impacts on the sound quality. It should be noted that K1, K2, and K3 are merely descriptions for the connection manners involved in different parts of the apparatus or the system may include but not limited to physical connection manner, force conduction manner, sound transfer efficiency, etc.

The descriptions of the equivalent system of bone conduction speaker are merely a specific embodiment, and it should not be considered as the only feasible embodiment. Apparently, those skilled in the art, after understanding the basic principles of bone conduction speaker, may make various modifications and changes on the type and detail of the vibrations of the bone conduction speaker, but these changes and modifications are still in the scope described above. For example, K1, K2, and K3 described above may refer to a simple vibration or mechanical transfer mode, or they may also include a complex non-linear transfer system. The transfer relationship may be formed by a direct connection between each portion or may be transferred via a non-contact manner.

The transfer relationship K2 between the sensor terminal **1402** and the vibration unit **1403** may also affect the frequency response of the bone conduction system. The volume of a sound heard by a user's ear depends on the energy received by a user's cochlea. The energy may be affected by various parameters during its transmission, which may be expressed by the following equation:

$$P = \iint_S \alpha \cdot f(a, R) \cdot L \cdot ds, \quad (17),$$

where P is linear to the energy received by the cochlea, S is the area of a contact surface between the bone conduction speaker and a user's face, α is a coefficient for dimension change, $f(a, R)$ denotes an effect of an acceleration a of a point on the contact surface and tightness R of contact between contact surface and a user's skin on energy transmission, L refers to the damping of any contacting points on the transmission of mechanical wave, i.e., a transmission impedance of a unit area.

In terms of (17), the transmission impedance L may have an impact on the sound transmission, and the vibration transmission efficiency of the bone conduction system may relate to the transmission impedance L . The frequency response curve of the bone conduction system may be a superposition of frequency response curves of multiple points on the contact surface. Factors that change the impedance may include the size of the energy transmission area, the shape of the energy transmission area, the roughness of the energy transmission area, the force on the energy transmission area, or a distribution of the force on the energy transmission area, etc. For example, the transmission effect of sound may change when changing the structure and shape of the vibration unit **1403**, thus changing the sound quality of the bone conduction speaker. Merely by way of example, the transmission effect of sound may be changed by changing the corresponding physical characteristic of the contact surface of the vibration unit **1403**.

A well-designed contact surface may have a gradient structure, and the gradient structure may refer to an area with various heights on the contact surface. The gradient structure may be a convex/concave portion or a sidestep that exists on an outer side (towards a user) or inner side (backward a user) of the contact surface. An embodiment of a vibration unit of the bone conduction speaker may be illustrated in FIG. 15A. A convex/concave portion (not shown in FIG. 15A) may

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exist on a contact surface **1501** (an outer side of the contact surface). During the operation of the bone conduction speaker, the convex/concave portion may be in contact with a user's face, changing the forces between different positions on the contact surface **1501** and a user's face. A convex portion may be in contact with a user's face in a tighter manner; thus the force on the skin and tissue of a user that contact with the convex portion may be larger, and the force on the skin and tissue that contact with a concave portion may be smaller accordingly. For example, three points A, B, and C on the contact surface **1501** in FIG. **15A** may be located on a non-convex portion, an edge of a convex portion, and a convex portion, respectively. When being in contact with a user's skin, clamping forces F_A , F_B , and F_C on the three points may be $F_C > F_A > F_B$. In some embodiments, a clamping force on the point B may be 0; i.e., the point B may not be in contact with the skin of a user. The skin and tissue of a user's face may have different impedances and responses under different forces. The part of a user's face under a larger force may correspond to a smaller impedance rate and have a high-pass filtering characteristic for an acoustic wave. The part under a smaller force may correspond to a larger impedance rate, and have a low-pass filtering characteristic for an acoustic wave. Different parts of the contact surface **1501** may correspond to different impedance characteristics L. Different parts may correspond to different frequency responses for sound transmission. The transmission effect of the sound via the entire contact surface may be equivalent to a sum of transmission effect of the sound via each part of the contact surface. A smooth curve may be formed when the sound transmits into a user's brain, which may avoid exorbitant harmonic peak under a low frequency or a high frequency, thus obtaining an ideal frequency response across the whole bandwidth. Similarly, the material and thickness of the contact surface **1501** may have an effect on the transmission effect of the sound, thus affecting the sound quality. For example, when the contact surface is soft, the transmission effect of the sound in the low frequency range may be better than that in the high frequency range, and when the contact surface is hard, the transmission effect of the sound in the high frequency range may be better than that in the low frequency range.

FIG. **15B** shows response curves of the bone conduction speaker with different contact areas. The dotted line corresponds to the frequency response of the bone conduction speaker having a convex portion on the contact surface. The solid line corresponds to the frequency response of the bone conduction speaker having a non-convex portion of the contact surface. In a low-intermediate frequency range, the vibration of the non-convex portion may be weakened relative to that of the convex portion, which may form one "pit" on the frequency response curve, indicating that the frequency response is not ideal and may influence the sound quality.

The above descriptions of the FIG. **15B** are merely the explanation for a specific embodiment, and those skilled in the art, after understanding the basic principles of bone conduction speaker, may make various modifications and changes on the structure and the components to achieve different frequency response effects.

It should be noted that for those skilled in the art, the shape and the structure of the contact surface may not be limited to the descriptions above. In some embodiments, the convex portion or the concave portion may be located at an edge of the contact surface or may be located at the center of the contact surface. The contact surface may include one or more convex portions or concave portions. The convex

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portion and/or concave portion may be located on the contact surface. The material of the convex portion or the concave portion may be different from the material of the contact surface, such as flexible material, rigid material, or a material easy to produce a specific force gradient. The material may be memory material or non-memory material; the material may be a single material or composite material. The structure pattern of the convex portion or concave portion of the contact surface may include but not limited to axial symmetrical pattern, central symmetrical pattern, symmetrical rotational pattern, asymmetrical pattern, etc. The structure pattern of the convex portion or the concave portion on the contact surface may include one pattern, two patterns, or a combination of two or patterns. The contact surface may include but not limited to a certain degree of smoothness, roughness, waviness, or the like. The distribution of the convex portions or the concave portions on the contact surface may include but not limited to axial symmetry, the center of symmetry, rotational symmetry, asymmetry, etc. The convex portion or the concave portion may be set at an edge of the contact surface or may be distributed inside the contact surface.

It should be noted that, the gradient structure on the contact surface in a bone conduction speaker disclosed in the present disclosure is also applicable for an air conduction speaker. For example, the air conduction speaker may include a gradient structure that exists on an outer side (towards a user) or inner side (backward a user) of a contact surface between the air conduction speaker and the user's face. In some embodiments, the gradient structure on the outer side of the contact surface may match the shape of the user's auricle (e.g., the shape of *fossa triangularis*, the shape of anthelix, etc.) such that the user such can wear the air conduction speaker more comfortably. Optionally or additionally, the air conduction speaker or the bone conduction speaker may include one or more sound guiding holes. The one or more sound guiding holes may be configured to guide sound waves inside a housing of the air conduction speaker or the bone conduction speaker through the one or more sound guiding holes to an outside of the housing. The one or more sound guiding holes may be located on a same wall or different walls of the housing. Merely by way of example, the one or more sound guiding holes may include two sound guiding holes. One sound guiding hole may be located on the contact surface of the air conduction speaker. The other sound guiding hole may be located on a wall (e.g., a sidewall) of the housing different from the contact surface.

1604-1611 in FIG. **16** are embodiments of the structure of the contact surface.

1604 in FIG. **16** shows multiple convex portions with similar shapes and structures on the contact surface. The convex portions may be made of a same material or similar materials as other parts of the panel, or different materials. In particular, the convex portions may be made of a memory material and the material of the vibration transfer layer, wherein the proportion of the memory material may be not less than 10%. Preferably, the proportion may be not less than 50%. The area of a single convex portion may be 1%-80% of the total area, preferably 5%-70%, and more preferably 8%-40%. The sum of the area of the convex portions may be 5%-80% of the total area, preferably 10%-60%. There may be at least one convex portion, preferably one convex portion, more preferably two convex portions, and further preferably at least five convex portions. The shapes of the convex portions may be circular, oval, triangular, rectangular, trapezoidal, irregular polygons or other similar patterns, wherein the structures of the convex

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portions may be symmetrical, or asymmetrical, the distribution of the convex portions may be symmetrically distributed or asymmetrically distributed, the number of the convex portions may be one or more, the heights of the convex portions may be the same or different, and the height distribution of the convex portions may form a certain gradient.

1605 in FIG. 16 shows an embodiment of convex portions on the contact surface with two or more structure patterns. There may be one or more convex portions of different patterns. Shapes of the two or more convex portions may be circular, oval, triangular, rectangular, trapezoidal, irregular polygons, other shapes, or a combination of any two or more shapes. The material, quantity, size, symmetry of the convex portions may be similar to that as illustrated in **1604**.

1606 in FIG. 16 shows an embodiment that the convex portions may be distributed at edges of the contact surface or in the contact surface. The number of the convex portions located at edges of the contact surface may be 1% to 80% of the total number of the convex portions, preferably 5%-70%, more preferably 10%-50%, and more preferably 30%-40%. The material, quantity, size, shape, or symmetry of the convex portions may be similar to **1604**.

1607 in FIG. 16 shows a structure pattern of concave portions on the contact surface. The structures of the concave portions may be symmetrical or asymmetrical, the distribution of the concave portions may be symmetrical or asymmetrical, the number of the concave portions may be one or more than one, the shapes of the concave portions may be same or different, and the concave portions may be hollow. The area of a single concave portion may be not less than 1%-80% of the total area of the contact surface, preferably 5%-70%, and more preferably 8%-40%. The sum of the area of all concave portions may be 5%-80% of the total area, preferably 10%-60%. There may be at least one concave, preferably one, more preferably two, and more preferably at least five. The shapes of the concave portions may be circular, oval, triangular, rectangular, trapezoidal, irregular polygons or other similar patterns.

1608 in FIG. 16 shows a contact surface including convex portions and concave portions. There may be one or more convex portions and one or more concave portions. The ratio of the number of the concave portions to the convex portions may be 0.1%-100%, preferably 1%-80%, more preferably 5%-60%, further preferably 10%-20%. The material, quantity, size, shape, or symmetry of each convex portion or each concave portion may be similar to **1604**.

1609 in FIG. 16 shows an embodiment of the contact surface having a certain waviness. The waviness may be formed by two or more convex/concave portions. Preferably, the distances between adjacent convex/concave portions may be equal. More preferably, the distances between convex/concave portions may be presented in an arithmetic progression.

1610 in FIG. 16 shows an embodiment of a convex portion having a large area on the contact surface. The area of the convex portion may be 30%-80% of the total area of the contact surface. Preferably, a part of an edge of the convex portion may substantially contact with a part of an edge of the contact surface.

1611 in FIG. 16 shows a first convex portion having a large area on the contact surface, and a second convex portion on the first convex portion may have a smaller area. The area of the convex portion having a larger area may be 30%-80% of the total area, and the area of the convex portion having a smaller area may be 1%-30% of the total

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area, preferably 5%-20%. The area of the smaller area may be 5%-80% that of the larger area, preferably 10%-30%.

The above descriptions of the contact surface structure of the bone conduction speaker are merely a specific embodiment, and it may not be considered the only feasible implementation. Apparently, those skilled in the art, after understanding the basic principles of bone conduction speaker, may make various modifications and changes in the type and detail of the contact surface of the bone conduction speaker, but these changes and modifications are still within the scope described above. For example, the count of the convex portions and the concave portions may not be limited to that of the FIG. 16, and modifications made on the convex portions, the concave portions, or the patterns of the contact surface may remain in the descriptions above. Moreover, the contact surface of at least one vibration unit of the bone conduction speaker may have the same or different shapes and materials. The effect of vibrations transferred via different contact surfaces may have differences due to the properties of the contact surfaces, which may result in different sound effects.

Referring to FIG. 17, FIG. 17 is a schematic diagram illustrating a front view of a structure of a contour of a user's ear according to some embodiments of the present disclosure.

As shown in FIG. 17, in addition to an external ear canal **101** and a nearby concha cavity **102** of an ear **100** of a user, parts of the ear **100** such as a concha boat **103**, a triangular fossa **104**, etc., may also be used to meet the wearing requirements of earphones because they have a certain depth and volume in a three-dimensional space. In other words, by rationally designing the structure of the earphone, the wearing of the earphone may also be achieved with the help of the ear **100** of the user except for the external ear canal **101**, and the external ear canal **101** of the user may be "liberated", thereby increasing the physical health of the user, and reducing the probability of traffic accidents. Accordingly, the present disclosure provides an earphone that mainly uses an upper part of the ear **100** of the user (including a region where the concha boat **103**, the triangular fossa **104**, an antihelix **105**, a scapha **106**, a helix **107**, etc., are located) to realize the wearing of the earphone. In some embodiments, in order to improve the wearing comfort and reliability of the earphone, an earlobe **108** of the user and other parts may also be further used. Further, for ease of description, some relatively special physiological positions on the ear **100** may be further identified. The special physiological positions may include an upper ear root LA connecting a front edge of the helix **107** and the head, a Darwin's nodule LB on the helix **107**, a helix notch LC of an end of the antihelix **105** close to the earlobe **108** and facing the concha cavity **102**, an intertragic notch LD of an end of the concha cavity **102** close to the earlobe **108**, etc. Due to individual differences among users, physiological positions such as Darwin's nodules may not be obvious or even non-existent on some users' ears, but this does not indicate that other users' ears do not have the physiological position.

It should be noted that although the external ear canal has a certain depth to extend to a tympanic membrane, for ease of description and in combination with FIG. 17, unless otherwise specified in the present disclosure, the external ear canal specifically may refer to an entrance away from the tympanic membrane, that is, an ear hole. Further, the "front side of the ear" mentioned in the present disclosure may be a concept relative to the "rear side of the ear". The front side of the ear may refer to a side of the ear facing away from the head, as shown in FIG. 17, and the rear side of the ear may

refer to a side of the ear facing the head. Both the front side and the rear side of the ear are relative to the ear of the user.

Referring to FIG. 18 to FIG. 21, FIG. 18 is a schematic diagram illustrating a front view of a structure of an exemplary earphone according to some embodiments of the present disclosure. FIG. 19 is a schematic diagram illustrating a left side view of the structure of the earphone in FIG. 18. FIG. 20 is a schematic diagram illustrating a front side view of the earphone in FIG. 18 in a wearing state. FIG. 21 is a schematic diagram illustrating a rear side view of the earphone in FIG. 18 in a wearing state. In some embodiments, the speaker of the present disclosure may be implemented as the earphone illustrated in FIGS. 18-21. In some embodiments, the “speaker” and the “earphone” can be used interchangeably. It should be noted that three directions of X, Y, and Z of the earphone are shown in FIG. 18 mainly to show three planes of XY, XZ, and YZ, so as to facilitate the corresponding illustration in the following description. Therefore, all directional indications (such as up, down, left, right, front, back . . .) in the present disclosure are mainly used to explain a relative position relationship between components, movement states of the components, or the like, in a specific posture (as shown in FIG. 18). If the specific posture changes, the directional indications may change accordingly.

As shown in FIG. 18 and FIG. 19, the earphone 10 may include a hook-shaped component 11, a connecting component 12, and a holding component 13. The connecting component 12 may connect the hook-shaped component 11 and the holding component 13, so that the earphone 10 may be curved in a three-dimensional space when the earphone 10 is in a non-wearing state (that is, in a natural state). In other words, in the three-dimensional space, the hook-shaped component 11, the connecting component 12, and the holding component 13 may not be coplanar. In such cases, as shown in FIG. 20 and FIG. 21, when the earphone 10 is in the wearing state, the hook-shaped component 11 may be mainly configured to hang between a rear side of an ear and a head of a user, and the holding component 13 may be mainly configured to contact a front side of the ear, thereby allowing the holding component 13 and the hook-shaped component 11 to cooperate to clamp the ear. For example, the connecting component 12 may extend from the head along a coronal axis of the user towards an outside of the head to cooperate with the hook-shaped component 11 to provide the holding component 13 with a pressing force on the front side of the ear. The holding component 13 may specifically press against a region where the concha boat, the triangular fossa, the antihelix, and other parts are located under the action of the pressing force so that the earphone 10 may not cover the external ear canal of the ear when the earphone 10 is in the wearing state. As another example, when the earphone 10 is in the wearing state, a projection of the holding component 13 on the ear of the user may mainly fall within a range of the helix of the ear. Further, the holding component 13 may be arranged at a side of the external ear canal of the ear close to the top of the head of the user and in contact with the helix and/or the antihelix. In this way, the holding component 13 may be prevented from covering the external ear canal, thereby liberating the two ears of the user. A contact area between the holding component 13 and the ear may also be increased, thereby improving the wearing comfort of the earphone 10.

It should be noted that based on standards of ANSI: 53.36, 53.25 and IEC: 60318-7, a simulator (e.g., GRAS 45BC KEMAR) with head and (left and right) ears may be made. Therefore, the description of “a user wears an earphone” or

“an earphone is in a wearing state” may refer to that the earphone is worn on the ear of the simulator mentioned above. Accordingly, the “wearing state” mentioned in the present disclosure may refer to a normal wearing state of the earphone after being worn on the ear of the simulator mentioned above. For ease of description, the normal wearing state may further be illustrated from a perspective of the front side and the rear side of the ear, such as the normal wearing state shown in FIG. 20 and FIG. 21, and another example of the normal wearing state shown in FIG. 25 and FIG. 26. Of course, due to individual differences among users, an actual wearing state of the earphone 10 may be different from the normal wearing state mentioned above.

For adult male users, the thickness of the ears may be relatively thick (commonly known as “thick ears”). By rationally designing (exemplary illustrations may be described below) structural parameters, such as a shape, a size, or the like, of the connecting component 12, and the connection relationship with the hook-shaped component 11 and the holding component 13, it may ensure that the earphone 10 fits the ear as much as possible to improve the wearing stability of the earphone 10, and the earphone 10 can be prevented from over-clamping the helix near the upper ear root, that is, the upper ear root may be naturally bypassed to improve the wearing comfort of the earphone 10. Further, for users such as children, minors, or adult women, the thickness of the ears may be often relatively thin (commonly known as “thin ears”). In particular, compared to the thickness of the ears of adult men, in order to increase the fit of the earphone 10 with the ears of the user when the earphone 10 is in the wearing state, the size of the connecting component 12 may be small. For example, the connecting component 12 may be an arc transition between the holding component 13 and the hook-shaped component 11.

Further, the earphone 10 may also include a core 14, a mainboard 15, and a battery 16. The core 14 may be mainly used to convert an electrical signal into the corresponding mechanical vibration (that is, “sound generation”), and may be electrically connected to the mainboard 15 and the battery 16 through corresponding conductors. The mainboard 15 may be mainly used to control the sound generation of the core 14, and the battery 16 may be mainly used to provide power for the sound generation of the core 14. The earphone 10 described in the present disclosure may also include a sound transmitter such as a microphone, or a pickup device, and may also include a communication device such as a Bluetooth device, or an NFC (Near Field Communication) device, which may be electrically connected to the mainboard 15 and the battery 16 through the corresponding conductors to realize corresponding functions. In some embodiments, the core 14 may include a transducer (e.g., the transducer 120-1, the transducer 22, etc.). In some embodiments, the “core” and the “transducer” can be used interchangeably.

For example, the core 14 may be fixed to the holding component 13. When the earphone 10 is in the wearing state, the core 14 may be pressed against the ears of the user tightly under the action of the pressing force. Further, when the earphone 10 is in the wearing state, as shown in FIG. 20, since the holding component 13 is mainly arranged at the front side of the ear of the user, in addition to fixing the core 14, the holding component 13 may also be configured with some function buttons (not shown in FIG. 18) that facilitate the interaction between the user and the earphone 10. The mainboard 15 may also be arranged in the holding component 13 to shorten a wiring distance between the core 14 and the mainboard 15, and the wiring distance between function

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keys, or the like, and the mainboard 15. It should be noted that since the holding component 13 may be configured with the core 14, the mainboard 15, the function buttons, or the like, and arranged in front of the ear of the user when the earphone 10 is in the wearing state, so that the battery 16 may be arranged in the hook-shaped component 11 and mainly be arranged between the rear side of the ear and the head of the user when the earphone 10 is in the wearing state (as shown in FIG. 21). In this way, the capacity of the battery 16 may be increased to improve the endurance of the earphone 10, and the weight of the earphone 10 may also be balanced to improve the stability and comfort of the earphone 10 in terms of wearing.

Further, the inventor(s) of the present disclosure has discovered in a long-term study that a weight ratio of a total weight of the holding component 13 to a total weight of a part of the hook-shaped component 11 corresponding to the battery 16 (hereinafter referred to as a battery part) may be within 4:1, preferably within 3:1, and more preferably within 2.5:1. Combined with FIG. 18 and FIG. 19, in some embodiments, the total weight of the holding component 13 may be the weight of the holding component 13 and the weight of structural components such as the core 14, the mainboard 15, etc., therein. The total weight of the battery part may be the weight of the battery part and the weight of structural components such as the battery 16 therein. It may be easy for those skilled in the art to know that the structural components in the holding component 13 and the structural components in the battery part may be changed according to design needs. The adjustments to the structural components in different parts may be included in the technical solution of the present disclosure, and the weight ratio may not be affected, which is not repeated herein. At this time, the weight of the earphone 10 may be more evenly distributed at two ends of the earphone 10, and the ears of the user may also serve as a fulcrum to support the earphone 10 when the earphone 10 is in the wearing state so that the earphone 10 may at least not slip off when the earphone 10 is in the wearing state. Certainly, the ears of the user may bear most of the weight of the earphone 10, which may easily cause discomfort when worn for a long time. Therefore, the hook-shaped component 11, the connecting component 12, the holding component 13, and other structures may be made of soft materials (such as polycarbonate, polyamide, acrylonitrile-butadiene-styrene copolymer, silica gel, etc.) to improve the wearing comfort of the earphone 10. Further, in order to improve the structural strength of the earphone 10, elastic metal wires such as spring steel, titanium alloy, titanium-nickel alloy, chromium-molybdenum steel, aluminum alloy, copper alloy, etc., may also be arranged in the structure of the hook-shaped component 11, the connecting component 12, the holding component 13, or other structures.

It should be noted that to take into account the comfort and stability of the earphone 10 in terms of wearing, the following improvements may also be made:

1) The connecting component 12 and the battery part may be made of relatively hard material, and a middle part between the connecting component 12 and the battery part may be made of relatively soft materials mentioned above. In some embodiments, the middle part may also adopt a "soft-wrapped-hard" structure. For example, when the user wears the earphone 10, a region of the hook-shaped component 11 that is in contact with the user may be made of the relatively soft materials mentioned above, and the rest of the regions may be made of the relatively hard materials mentioned above. Different materials may be formed by tech-

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nologies such as a two-color injection molding technology, a spray paint technology, etc. The relatively hard materials mentioned above may include, but are not limited to, polycarbonate (PC), polyamides (PA), acrylonitrile-butadiene-styrene copolymer (ABS), polystyrene (PS), High Impact Polystyrene (HIPS), Polypropylene (PP), Polyethylene Terephthalate (PET), Polyvinyl Chloride (PVC), Polyurethanes (PU), Polyethylene (PE), Phenol-Formaldehyde (PF), Poly (ester sulfones), PES, Polyvinylidene chloride (PVDC), Polymethyl Methacrylate (PMMA), Poly-ether-ether-ketone (PEEK), or the like, or a mixture of at least two thereof, or a mixture formed with reinforcing agents such as glass fibers, carbon fibers, etc. Further, the spray paint may specifically be rubber hand-feel paint, elastic hand-feel paint, plastic elastic paint, or the like.

2) Since the earphone 10 is worn by the user, a part of the earphone 10 may be in contact with the skin of the user (hereinafter referred to as a skin contact region). Moreover, the material of the skin contact region may generally affect the comfort of the user when wearing the earphone 10 for a long time. Thus, the skin contact region may be made of the relatively soft materials mentioned above, and the other regions may be made of the relatively hard materials mentioned above. Different materials may be formed by technologies such as a two-color injection molding, a spray paint technology, etc.

In some embodiments, the Shore hardness of the relatively softer materials may be in a range of 45-85A, 30-60D, preferably may be in a range of 50-60A, 40-50D. Both the relatively softer materials and the relatively hard materials may cover the elastic metal wires.

In some embodiments, in order to take into account the comfort, stability, and the appearance of the earphone 10, the hook-shaped component 11 may also adopt a "soft-wrapped-hard" structure. Specifically, a cavity for accommodating components such as a part of an elastic metal wire 115 of the hook-shaped component 11, the battery 16, etc., may first be formed by using the relatively hard material as a cavity wall (also be referred to as an inner layer) of the cavity. Then the cavity wall may be wrapped by the relatively soft material, so as to form an outer layer of the hook-shaped component 11, thereby improving the user's comfort when wearing the earphone. In some embodiments, the Rockwell hardness of the material of the inner layer (also be referred to as inner layer material) of the hook-shaped component 11 may be in a range of 20-50 HRC, preferably may be in a range of 30-40 HRC, and more preferably may be 36 HRC. In some embodiments, the inner layer may be made of titanium alloy. The elastic modulus of the inner layer may be in a range of 28-42 GPa, and preferably may be in a range of 30-35 GPa. In some embodiments, a Poisson's ratio of the inner layer material may be in a range of 0.1-0.5, preferably may be in a range of 0.2-0.4, and more preferably may be 0.33. In some embodiments, a density of the inner layer material may be in a range of 6-7 g/cm³, and preferably may be in a range of 6.45-6.48 g/cm³. In some embodiments, during a process that the user is wearing the earphone 10, since the user may stretch and/or twist the hook-shaped component 11, the inner layer may be made of memory alloy. An Austenite finish (AF) temperature of the memory alloy may be in a range of -25-0° C., preferably may be -20° C. Further, the fatigue life of the memory alloy may exceed 10 thousands times measured based on a back and forth measurement process. The outer layer of the hook-shaped component 11 may be much softer than the inner layer to improve the user's comfort when wearing the earphone. In some embodiments, in order to improve the wearing stability of the

earphone, that is, to prevent the earphone from sliding, the surface of the outer layer may be rough to increase the frictional assistance of sliding. In some embodiments, a roughness of the surface of the outer layer may be in a range of 0.1-3 μm , and preferably may be in a range of 1-2 μm . In some embodiments, a coefficient of friction of the surface of the output layer may be in a range of 0.1-1.0.

Further, different users may have large differences in age, gender, and gene-controlled trait expression. As a result, the ears and heads of different users may be of different sizes and shapes. In such cases, the hook-shaped component 11 may be rotatable with respect to the connecting component 12, or the holding component 13 may be rotatable with respect to the connecting component 12, or a part of the connecting component 12 may be rotatable with respect to the other part of the connecting component 12, so that a relative positional relationship of the hook-shaped component 11, the connecting component 12, and the holding component 13 in the three-dimensional space can be adjusted, thus the earphone 10 may adapt to different users, that is, to increase the applicability of the earphone 10 to users in terms of wearing. For example, the connecting component 12 may be made of deformable materials such as a soft steel wire. The user may bend the connecting component 12 to rotate one part relative to the other part to adjust the relative positions of the hook-shaped component 11, the connecting component 12, and the holding component 13 in the three-dimensional space, thereby satisfying the wearing needs. As another example, the connecting component 12 may be configured with a rotating shaft mechanism 121, through which the user may also adjust the relative positions of the hook-shaped component 11, the connecting component 12, and the holding component 13 in the three-dimensional space to satisfy the wearing needs. The detailed structure of the rotating shaft mechanism 121 may be within the understanding of those skilled in the art, which may not be described in detail herein. Further, if the hook-shaped component 11 and the connecting component 12 are movably connected by the rotating shaft mechanism 121, the hook-shaped component 11 may rotate relative to the connecting component 12. If the holding component 13 and the connecting component 12 are movably connected by the rotating shaft mechanism 121, the holding component 13 may rotate relative to the connecting component 12. If a part of the connecting component 12 is movably connected with another part of the connecting component 12 by the rotating shaft mechanism 121, the part of the connecting component 12 may be rotated relative to another part of the connecting component 12.

Referring to FIG. 22, FIG. 22 is a schematic diagram illustrating a mechanical model of the earphone in FIG. 18 in a wearing state. It should be noted that the YZ plane in FIG. 22 may be regarded as a plane where the head of the user is located. The ABC section in FIG. 22 may be regarded as the hook-shaped component. The CD section in FIG. 22 may be regarded as the connecting component. The DEF section in FIG. 22 may be regarded as the holding component. Further, point C in FIG. 22 may correspond to a region where an upper proximal end of the ear in FIG. 17 is located (e.g., a region indicated by the dashed box C in FIG. 17).

As shown in FIG. 20 to FIG. 22, when the earphone 10 is in the wearing state, the ABC section may be mainly arranged at the rear side of the ear of the user, the DEF section may be mainly arranged at the front side of the ear of the user, and the CD section may be mainly adapted to the thickness of the ear of the user. In such cases, the BC section, the CD section, and the DEF section may form a structure

similar to a "clip", so that the earphone 10 may be clamped on the ear of the user, thereby forming a basic wearing state. The following description may be an exemplary description of the force and stability of the earphone 10 in terms of wearing:

As shown in FIG. 22, in a direction from a first connection point C between the hook-shaped component 11 and the connecting component 12 to a free end of the hook-shaped component 11 (for example, the end where the point A in FIG. 22 is located), the hook-shaped component 11 may be bent toward the head of the user, and form a first contact point B and a second contact point A with the head. The first contact point B may be arranged between the second contact point A and the first connection point C. It should be noted that the first contact point B and the second contact point A may be both defined points in the mechanical model. In actual wearing, due to differences in the physiological structures of the heads and ears of different users, a certain impact on the actual wearing of the earphone 10 may exist. The position of the earphone 10 that is in contact with the head when the earphone 10 is actually worn may correspond to the free end of the hook-shaped component 11, or any point between the free end and the first contact point B. In some embodiments, the AB section may also partially or entirely abut against the head of the user. The mechanical model and the actual wearing stability principle may be the same as the technical solutions mentioned above. Those skilled in the art may easily learn, adjust, and combine the content based on the technical solutions of the present disclosure without creative work, which may not be repeated herein. In this way, the hook-shaped component 11 may form a lever structure with the first contact point B as a fulcrum. The free end of the hook-shaped component 11 may be pressed against the head of the user, the head of the user may provide a force directed to the outside of the head at the second contact point A, which may be transformed into a force directed to the head at the first connection point C through the lever structure, and the holding component 13 may be provided with a pressing force on the front side of the ear through the connecting component 12.

It should be noted that in order to enable the free end of the hook-shaped component 11 to press against the head of the user when the earphone 10 is in the wearing state, and to enable the head of the user to provide a force directed to the outside of the head at the second contact point A, at least the following conditions may be satisfied: an angle formed between the free end of the hook-shaped component 11 and the YZ plane when the earphone 10 is in the non-wearing state may be greater than an angle formed between the free end of the hook-shaped component 11 and the YZ plane when the earphone 10 is in the wearing state. The larger the angle formed between the free end of the hook-shaped component 11 and the YZ plane when the earphone 10 is in the non-wearing state, the tighter the free end of the hook-shaped component 11 may press against the head of the user when the earphone 10 is in the wearing state, and the larger the force directed to the outside of the head at the second contact point A provided by the head of the user correspondingly.

It should be noted that when the free end of the hook-shaped component 11 is pressed against the head of the user, in addition to making the head of the user provide a force directed to the outside of the head at the second contact point A, it may also cause at least the BC section of the hook-shaped component 11 to form another pressing force on the rear side of the ear, which may cooperate with the pressing force formed by the holding component 13 on the front side

of the ear, so as to form a “front and rear pinching” pressing effect on the ear of the user, thereby improving the stability of the earphone 10 in terms of wearing.

Further, the battery 16 may be mainly arranged at the AB section of the hook-shaped component 11 so as to overcome the weight of the holding component 13, and structures therein such as the core 14, and the mainboard 15, thereby improving the stability of the earphone 10 in terms of wearing. In some embodiments, the surface of the hook-shaped component 11 in contact with the ear and/or the head of the user may be set as a frosted surface, a textured surface, or the like, to increase the friction between the hook-shaped component 11 and the ear and/or the head of the user, and overcome the self-weight of the holding component 13 and structures therein such as the core 14, the mainboard 15, or the like, thereby improving the stability of the earphone 10 in terms of wearing. Further, the free end of the hook-shaped component 11 (especially a region where the point A is located) may be deformed, so that when the earphone 10 is in the wearing state, the free end of the hook-shaped component 11 may be pressed against the head of the user and deformed. In such cases, the contact area between the free end of the hook-shaped component 11 and the head of the user may be enlarged, thereby improving the comfort and stability of the earphone 10 in terms of wearing. For example, the hook-shaped component 11 may be formed by two-color injection molding, and the elastic modulus of the free end (especially the region where the point A is located) may be smaller than that of other regions, so as to increase the deformability of the free end. As another example, the free end of the hook-shaped component 11 may be configured with one or more holes 111 in a hollow structure to increase the deformability of the free end. The hole(s) 111 may be through-hole(s) and/or blind hole(s). A count of the hole(s) 111 may be one or more, and an axial direction of the hole(s) 111 may be perpendicular to the contact area between the free end of the hook-shaped component 11 and the head of the user.

It should be noted that to take into account the comfort and stability of the earphone 10 in terms of wearing, the following improvements may also be made. FIG. 32 is a schematic diagram illustrating surface structures of a skin contact area of a battery part according to some embodiments of the present disclosure.

1) The skin contact region of the battery part may be formed with a texture structure. As shown in diagram (a) in FIG. 32, the texture structure may include a plurality of strip-shaped protrusions 112a spaced apart along a length direction of the hook-shaped component 11. As shown in diagram (b) in FIG. 32, the texture structure may also include a plurality of dot-shaped protrusions 112b spaced apart along the length direction of the hook-shaped component 11. In some embodiments, the texture structure may also be a grid-like shape.

2) As shown in diagram (c) in FIG. 32, the skin contact region of the battery part may also be configured with a semi-spindle protrusion 112c extending along the length direction of the hook-shaped component 11. Taking the free end of the hook-shaped component 11 as a reference, in a direction close to the free end of the hook-shaped component 11 (the direction shown by the arrow in FIG. 32), a protrusion height of each part of the semi-spindle protrusion 112c relative to the hook-shaped component 11 may gradually increase and then gradually decrease. In this way, during a process that the user is wearing the earphone 10, the semi-spindle protrusion 112c and the skin of the user may generate as little resistance as possible. After the user

finishes wearing the earphone 10, the semi-spindle protrusion 112c and the skin of the user may generate as much resistance as possible to prevent the earphone 10 from falling off.

3) When the skin contact region of the battery part is arranged as a frosted surface, a material with relatively good skin affinity may be preferred.

All kinds of protrusions mentioned above may be selected from a material with a relatively soft texture, a relatively large damping coefficient, and a certain degree of skin-friendliness. Further, through the various embodiments described above, a coefficient of friction of the skin contact region of the battery part may be in a range of 0.1-1.0.

Merely by way of example, a linear distance between the projection of the point C on the YZ plane and the projection of the EF segment on the YZ plane may be in a range of 10-17 mm, preferably may be in a range of 12-16 mm, and more preferably may be in a range of 13-15 mm. The angle between the projection of the BC segment on the XY plane and the projection of the DE segment on the XY plane may be in a range of 0-25°, preferably may be in a range of 0-20°, and more preferably may be in a range of 2-20°. Further, the angle between the AB segment and a normal line passing through the point B of the XY plane may be in a range of 0-25°, preferably may be in a range of 0-20°, and more preferably may be in a range of 2-20°. In some embodiments, a linear distance between the projection of point C on the XY plane and the projection of the EF segment on the XY plane may be in a range of 2-4 mm, and preferably may be 2.8 mm. In other embodiments, a linear distance between the projection of point C on the XY plane and the projection of the EF segment on the XY plane may be in a range of 1-4 mm and preferably may be 2.5 mm. Therefore, the connecting component 12 may bypass the upper ear root of the ear when the earphone 10 is in the wearing state, thereby improving the wearing comfort of the earphone 10.

Based on the above detailed description, according to an aspect of the present disclosure, the weight of the earphone 10 may be distributed reasonably and evenly, so that the ear of the user may serve as a fulcrum to support the earphone 10 when the earphone 10 is in the wearing state. According to another aspect of the present disclosure, the connecting component 12 may be arranged between the hook-shaped component 11 and the holding component 13 of the earphone 10, so that when the earphone 10 is in the wearing state, the connecting component 12 may cooperate with the hook-shaped component 11 to provide the holding component 13 with a pressing force on the front side of the ear, thus the earphone 10 may be firmly attached to the ear of the user when in the wearing state. Such a setting may improve the stability of the earphone 10 in terms of wearing, and the reliability of the earphone 10 in terms of sound generation.

Referring to FIG. 23 to FIG. 27, FIG. 23 is a schematic diagram illustrating a front view of a structure of an earphone according to some embodiments of the present disclosure. FIG. 24 is a schematic diagram illustrating a left side view of the earphone in FIG. 23. FIG. 25 is a schematic diagram illustrating a front side view of the earphone in FIG. 23 in a wearing state. FIG. 26 is a schematic diagram illustrating a rear side view of the earphone in FIG. 23 in a wearing state. FIG. 27 is a schematic diagram illustrating a mechanical model of the earphone in FIG. 23 in a wearing state. It should be noted that the YZ plane in FIG. 27 may be regarded as the plane where the head of the user is located. The ABC section in FIG. 27 may be regarded as the hook-shaped component, the CD section in FIG. 27 may be regarded as the connecting component, and the DEF section

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in FIG. 27 may be regarded as the holding component. Further, the Point C in FIG. 27 may correspond to a region where the upper proximal end of the ear in FIG. 17 is located (a region indicated by the dashed box C in FIG. 17).

As shown in FIG. 20 to FIG. 22, when the earphone 10 is in the wearing state, the ABC section may be mainly located at the rear side of the ear of the user, the DEF section may be mainly located at the front side of the ear of the user, and the CD section may be mainly configured to adapt to the thickness of the ear of the user. In such cases, the BC section, the CD section, and the DEF section may form a structure similar to a "clip", so that the earphone 10 may be clamped on the ear of the user, thereby forming a basic state of wearing. The following description may be an exemplary description of the force and stability of the earphone 10 in terms of wearing:

The main difference from the embodiments mentioned above may be that, in the embodiment, as shown in FIG. 23 and FIG. 24, the hook-shaped component 11 may be closer to the holding component 13 as a whole, so that when the earphone 10 is in the wearing state, as shown in FIG. 25 and FIG. 26, the free end of the hook-shaped component 11 away from the connecting component 12 may act on the rear side of the ear of the user instead of pressing against the head of the user.

As shown in FIG. 27, in a direction from the first connection point C between the hook-shaped component 11 and the connecting component 12 to the free end of the hook-shaped component 11 (for example, the end where point A is located in FIG. 27), the hook-shaped component 11 may be bent toward the rear side of the ear to form a first contact point B with the rear side of the ear. The holding component 13 (e.g., a core housing 131 of the holding component 13) may form a second contact point F with the front side of the ear. For the earphone 10, in the natural state (that is, the non-wearing state), a distance between the first contact point B and the second contact point F along the extending direction (i.e., the coronal axis) of the connecting component 12 may be less than a distance between the first contact point B and the second contact point F along the extending direction of the connecting component 12 in the wearing state, thereby providing the holding component 13 with the pressing force against the front side of the ear. In other words, when the earphone 10 is in the natural state, the distance between the first contact point B and the second contact point F along the extending direction of the connecting component 12 may be less than the thickness of the ear of the user, so that the earphone 10 may be clipped to the ear of the user like a "clip" in the wearing state.

Further, a first line BC may be provided between the first contact point B and the first connection point C, and a second line EF may be provided between the second contact point F and the second connection point E of the holding component 13 and the connecting component 12.

Further, the hook-shaped component 11 may also extend in a direction away from the connecting component 12, that is, an overall length of the hook-shaped component 11 may be extended, so that when the earphone 10 is in the wearing state, the hook-shaped component 11 may also form a third contact point A with the rear side of the ear. The first contact point B may be located between the first connection point C and the third contact point A, and close to the first connection point C. For the earphone 10, in the natural state, the distance between the projections of the first contact point B and the third contact point A on a reference plane perpendicular to the extending direction of the connecting component 12 (e.g., the YZ plane in FIG. 27) may be less than the

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distance between the projections of the first contact point B and the third contact point A on the reference plane perpendicular to the extending direction of the connecting component 12 (e.g., the YZ plane in FIG. 27) in the wearing state. With the arrangement mentioned above, not only can the free end of the hook-shaped component 11 press against the rear side of the ear of the user, but also the ABC section can be in a C shape, wherein the third contact point A may also be arranged in a region of the ear near the earlobe, thus the hook-shaped component 11 may clamp the ear of the user in a vertical direction (as indicated by arrow Z in FIG. 27) to overcome the self-weight of the holding component 13. In addition, after the overall length of the hook-shaped component 11 is extended, the hook-shaped component may not only clamp the ear of the user in the vertical direction but also increase the contact area between the hook-shaped component 11 and the ear of the user, that is, the friction between the hook-shaped component 11 and the ear of the user may be increased, thereby improving the stability of the earphone 10 in terms of wearing.

It should be noted that to take into account the comfort and stability of the earphone 10 in terms of wearing, the following improvements may also be made:

1) Since the hook-shaped component 11 needs to match different ears of users, and the different ears of users may have different sizes and shapes, the free end of the hook-shaped component 11 (for example, the battery part) may be prone to hang in the air when a user with small ears wears the earphone 10. That is, the hook-shaped component 11 and the ear of the user may form the first contact point B only. Accordingly, in combination with FIG. 23 and FIG. 24, for the hook-shaped component 11, an outer diameter of the battery part may be larger than that of the other middle parts, that is, a step difference may exist, thereby forming a structure of progressive necking. With the arrangement mentioned above, in combination with FIG. 25 and FIG. 26, when the user wears the earphone 10, the hook-shaped component 11 may not only form the first contact point B with the ear of the user but also the free end of the hook-shaped component 11 may form the third contact point A with the ear of the user. That is, the battery part may form the third contact point A with the ear of the user under any circumstances. Obviously, to adapt to a wide user group, a plurality of progressive necking structures may be distributed at intervals along the length direction of the hook-shaped component 11.

2) In the same situation, a ratio of the length of the battery part to a long diameter of the outer diameter of the battery part may also affect the attachment of the hook-shaped component 11 to the ear of the user. The inventor(s) of the application has discovered in long-term research that, in combination with FIG. 23 and FIG. 24, the ratio of the length to the long diameter of the outer diameter of the battery part may be within 6:1, preferably may be within 4:1. At this time, the hook-shaped component 11 may not only form the first contact point B with the ear of the user, but the free end may also form the third contact point A with the ear of the user. That is, the battery part may fit the user's ears.

Referring to FIG. 28, FIG. 28 is a schematic diagram of a top view of a structure of an earphone according to some embodiments of the present disclosure.

Based on the description mentioned above, in combination with FIG. 17, the ear 100 of the user may generally have recessed regions such as the concha cavity 102, the concha boat 103, the triangular fossa 104, the scapha 106, or the like. Correspondingly, the ear 100 of the user may also generally have protruding regions such as the antihelix 105,

the helix 107, a helix feet 109, or the like. Based on the concave and convex structures of the ear 100, the earphone 10 may also be tightly attached with the corresponding positions of the ear 100 by ways of elastic clamping, elastic abutting, hooking and covering, or the like, thereby improving the comfort and reliability of the earphone 10 in terms of wearing.

Further, in combination with FIG. 18 to FIG. 21, outer surfaces of the holding component 13 may be defined as follows: 1) a side of the holding component 13 in contact with the skin of the user may be defined as an inner surface; 2) a side of the holding component 13 opposite to the inner surface in the X-direction may be defined as an outer surface; 3) a side of the holding component 13 facing the positive direction of the Z-direction may be defined as an upper surface; 4) a side of the holding component 13 facing the negative direction of the Z-direction may be defined as a lower surface; 5) a side of the holding component 13 facing the negative direction of the Y direction may be defined as a rear surface. If the holding component 13 does not have a cubic structure such as shown in FIG. 18 and FIG. 19, but has a structure such as a cylinder, an elliptic cylinder, or the like, the upper surface, the lower surface, and the rear surface may be uniformly defined as a peripheral surface.

The main difference from any embodiments mentioned above may be that in the present embodiment, the holding component 13 may not only press against the front side of the ear of the user, but may also be further extended and held in the concha boat and/or the triangular fossa of the ear. With the arrangement mentioned above, the holding component 13 may be stopped and blocked by the helix of the ear at least in the extending direction of the connecting component 12, so as to prevent the holding component 13 from turning out when the earphone 10 is in the wearing state, thereby improving the stability of the earphone 10 in terms of wearing.

Merely by way of example, as shown in FIG. 28, the earphone 10 may further include an extending component 17 connected to the holding component 13. In the extending direction of the connecting component 12 (as indicated by arrow X in FIG. 28), the extending component 17 and the holding component 13 may have a gap, and the gap may be smaller than or equal to the thickness of the helix of the ear. With the arrangement mentioned above, when the earphone 10 is in the wearing state, the extending component 17 may extend into the concha boat and/or the triangular fossa of the ear. At this time, since the concha boat and/or the triangular fossa have a certain depth and volume in the three-dimensional space, the holding component 13 may be hooked by the helix of the ear when the extending component 17 extends into the concha boat and/or the triangular fossa to prevent the holding component 13 from turning out when the earphone 10 is in the wearing state, thereby improving the stability of the earphone 10 in terms of wearing. At the same time, the holding component 13 may be pressed against the front side of the ear under the action of the pressing force mentioned above. The holding component 13 and the front side of the ear may cooperate with each other, which is beneficial to increase the stability of the earphone 10 in terms of wearing.

FIG. 33 is a schematic diagram illustrating different structures of the extending component in FIG. 28. In some embodiments, in combination with diagram (a) in FIG. 33, the extending component 17 may be mainly arranged at the inner surface and/or the lower surface of the holding component 13 and arranged to be able to extend into the concha cavity 102 after the user wears the earphone 10. At this time,

the extending component 17 may be tightly attached with the concha cavity 102 and the surrounding body tissues in an elastically abutting manner.

In other embodiments, in combination with diagram (b) in FIG. 33, the extending component 17 may be mainly arranged at the inner surface of the holding component 13 and arranged to be able to extend into the concha boat 103 after the user wears the earphone 10. At this time, the extending component 17 may be tightly attached with the concha boat 103 and the surrounding body tissues in a manner of elastic clamping and/or elastic abutting.

In other embodiments, in combination with diagram (c) in FIG. 33, the extending component 17 may be mainly arranged at the upper surface of the holding component 13 and arranged to be able to extend into the triangular fossa 104 after the user wears the earphone 10. At this time, the extending component 17 may be tightly attached with the triangular fossa 104 and the surrounding body tissues in a manner of elastic clamping and/or elastic abutting.

In other embodiments, in combination with diagrams (d) or (e) in FIG. 33, the extending component 17 may be mainly arranged at the upper surface and/or the rear surface of the holding component 13 and arranged so as to be able to extend into the scapha 106 after the user wears the earphone 10. At this time, the extending component 17 may be tightly attached with the scapha 106 and the surrounding body tissues in a manner of elastic clamping and/or elastic abutting.

In other embodiments, in combination with diagram (f) in FIG. 33, the extending component 17 may be mainly arranged at the rear surface of the holding component 13 and arranged to be able to bend and extend from the front side of the ear 100 to the rear side of the ear 100 after the user wears the earphone 10 to hook the helix 107. At this time, the extending component 17 may be tightly attached with the helix 107 and the surrounding body tissues in a manner of hooking and covering.

In other embodiments, in combination with diagram (g) in FIG. 33, the extending component 17 may be mainly arranged on the hook-shaped component 11, for example, a position of the hook-shaped component 11 close to the battery part. The extending component 17 may be configured to be able to bend and extend from the rear side of the ear 100 to the front side of the ear 100 after the user wears the earphone 10 to hook the antihelix 105. At this time, the extending component 17 may be tightly attached with the antihelix 105 and the surrounding body tissues in a manner of hooking and covering.

In other embodiments, in combination with diagram (h) in FIG. 33, the extending component 17 may be mainly arranged on the hook-shaped component 11, such as the battery part, and configured to be able to bend and extend from the rear side of the ear 100 to the front side of the ear 100 after the user wears the earphone 10, thereby hooking the helix 107. At this time, the extending component 17 may be tightly attached with the helix 107 and the surrounding body tissues in a manner of hooking and covering.

It should be noted that structural parameters such as a size and a shape of the extending component 17 may be profiled and designed according to the matching requirements between the extending component 17 and the ear 100, which may not be limited herein. Further, the extending component 17 and the corresponding structural component on the earphone 10 may be integrally formed, that is, the extending component 17 and the corresponding structural component on the earphone 10 may not be detached. In some embodiments, the extending component 17 and the corresponding

structural component on the earphone 10 may also be connected in a detachable manner. For example, the holding component 13 or the corresponding position of the battery part may be configured with a mounting hole, and the extending component 17 may be embedded in the mounting hole. As another example, the extending component 17 may be integrally formed with another elastic sleeve, so that the extending component 17 may be sleeved at a corresponding position on the holding component 13 or the hook-shaped component 11 through the elastic sleeve.

Further, in combination with FIG. 28, the size of the holding component 13 in the Y direction may be in a range of 22-34 mm, preferably may be in a range of 24-28 mm, and more preferably may be 26 mm, so that the holding component 13 may be pressed on the front side of the ear 100. At this time, in combination with FIG. 33, a height size of the extending component 17 in the Z direction may be in a range of 4-8 mm, and the length of the projection of the extending component 17 on the XY plane may be in a range of 8-15 mm, and the width of the projection may be in a range of 2-5 mm.

Referring to FIG. 29, FIG. 29 is a schematic diagram illustrating a front view of a structure of an earphone according to some embodiments of the present disclosure.

The main difference from any of the embodiments mentioned above may be that in the present embodiment, the holding component 13 may be a multi-section structure to facilitate adjustment of the relative position of the core 14 on the overall structure of the earphone 10. With the arrangement mentioned above, when the earphone 10 is in the wearing state, an external ear canal of the ear may not be covered, and the core 14 may be as close as possible to the external ear canal.

Merely by way of example, as shown in diagram (a) in FIG. 29, the holding component 13 may include a first holding section 131a, a second holding section 132a, and a third holding section 133a connected end to end in sequence. An end of the first holding section 131a away from the second holding section 132a may be connected to the connecting component 12. The third holding section 133a may be mainly configured to set up structural assemblies such as the core 14, the mainboard 15, or the like. Further, the second holding section 132a may be folded back relative to the first holding section 131a and maintains a distance therebetween to make the first holding section 131a and the second holding section 132a be in a U-shaped structure.

Merely by way of example, as shown in diagram (b) in FIG. 29, the holding component 13 may include a first holding section 131b, a second holding section 132b, and a third holding section 133b connected end to end in sequence. An end of the first holding section 131b away from the second holding section 132b may be connected to the connecting component 12. The third holding section 133b may be mainly configured to set up structural assemblies such as the core 14, the mainboard 15, or the like. Further, the second holding section 132b may be bent relative to the first holding section 131b, so that the third holding section 133b and the first holding section 131b may be provided with a distance.

Referring to FIG. 14 and FIG. 15, FIG. 30 is a schematic diagram illustrating a structure of an earphone according to some embodiments of the present disclosure. FIG. 31 is a schematic diagram illustrating a mechanical model of the earphone in FIG. 30 in a wearing state. It should be noted that the YZ plane in FIG. 31 may be regarded as the plane where the head of the user is located. The BC section in FIG. 31 may be regarded as the hook-shaped component, the CD

section in FIG. 31 may be regarded as the connecting component, the DEF section in FIG. 31 may be regarded as the holding component, and the GH section in FIG. 31 may be regarded as the extending component. Further, the point C in FIG. 31 may correspond to the region where the upper proximal end of the ear in FIG. 17 is located (e.g., the region indicated by the dashed box C in FIG. 17).

The main difference from any of the embodiments mentioned above may be that in the present embodiment, as shown in FIG. 30, the length of the hook-shaped component 11 may be relatively short, and the angle between the hook-shaped component 11 and the connecting component 12 may be relatively small. The extending component 17 may be connected to the holding component 13 and have a gap with the holding component 13. The gap may be less than or equal to the thickness of the helix of the ear. With the arrangement mentioned above, when the earphone 10 is in the wearing state, the hook-shaped component 11 may cooperate with the connecting component 12 so that the holding component 13 can be hung on the front side of the ear of the user, and the extending component 17 may extend into the concha boat and/or the triangular fossa of the ear to prevent the holding component 13 from turning out, thereby improving the stability of the earphone 10 in terms of wearing. In the embodiment, the extending component 17 that can be extended into the concha boat of the ear may be taken as an example for illustration.

As shown in FIG. 31, the point B may hook the depression on the rear side of the ear, and the point C may be regarded as the fulcrum, so that the hook-shaped component 11 may overcome the weight of the holding component 13, thereby preventing the holding component 13 from falling from the ear of the user. At this time, the friction between the hook-shaped component 11 and the ear may be increased to improve the stability of the earphone 10 in terms of wearing. Further, the point H may hook the helix of the ear, and the point G may be regarded as another fulcrum, so that the extending component 17 may overcome the weight of the holding component 13, thereby preventing the holding component 13 from turning out of the ear of the user. At this time, the friction between the extending component 17 and the ear may be increased to improve the stability of the earphone 10 in terms of wearing.

Based on the related description mentioned above, different users may have large differences in age, gender, and gene-controlled trait expression. As a result, the ears and heads of different users may be of different sizes and shapes. On the basis of any of the embodiments mentioned above, the following improvements may also be made to related structures of the earphone 10 so that the earphone 10 may meet the wearing needs of a wider user group and enable different users to have good comfort and stability when wearing the earphone 10.

Referring to FIG. 34, FIG. 34 is a schematic diagram illustrating a structure of an earphone according to some embodiments of the present disclosure.

The main difference from any of the embodiments mentioned above may be that in the present embodiment, in combination with FIG. 34, the free end of the hook-shaped component 11 may also be configured with an elastic structure 18. The elastic structure 18 may be made of soft materials, have a certain structural strength, and may take into account the comfort of the user wearing the earphone 10. Further, the elastic structure 18 may be tubular, and may be detachably sleeved on the free end of the hook-shaped component 11. In such cases, the elastic structure 18 may be used as an accessory of the earphone 10 to facilitate the user

to install or disassemble according to actual usage requirements. In some embodiments, a part of the elastic structure **18** contacting the user may be configured with a texture structure and/or a matte surface.

Merely by way of example, the elastic structure **18** may include a first tubular part **181** and a second tubular part **182** that are integrally connected with each other. The first tubular part **181** and the second tubular part **182** may be in a bent shape, and a bending angle may be reasonably designed according to actual usage requirements. In some embodiments, the elastic structure **18** may have a certain memory performance at least at the bending position thereof, so that the user may flexibly adjust the bending angle through bending, turning, or the like. With the arrangement mentioned above, during the process that the user wears the earphone **10**, the elastic structure **18** may hook the ear socket of the ear from the rear side of the ear of the user to prevent the earphone **10** from falling off.

Further, both the first tubular part **181** and the second tubular part **182** may have a hollow tubular shape, and the first tubular part **181** and the second tubular part **182** may be in communication with each other or not in communication with each other. The first tubular part **181** and the second tubular part **182** may both be sleeved on the free end of the hook-shaped component **11**. In the embodiment, the first tubular part **181** and the second tubular part **182** not connecting with each other may be taken as an example for illustrative description, the structural strength of the elastic structure **18** at a bending position may be improved. The length (L1) of the first tubular part **181** and the length (L2) of the second tubular part **182** may not be equal, so that the user may select one of the first tubular part **181** and the second tubular part **182** to be sleeved on the free end of the hook-shaped component **11** according to actual usage requirements, thereby adjusting the actual total length of the hook-shaped component **11** and the elastic structure **18**. In such cases, the elastic structure **18** may partially or completely cover the battery part. In combination with FIG. **34**, in the embodiment, the elastic structure **18** partially covering the battery part may be taken as an example for illustrative description, for example, the elastic structure **18** may cover half of the battery part.

In the long-term study, the inventors of the present disclosure discovered that, in combination with FIG. **34**, when a length difference between the length (L1) of the first tubular part **181** and the length (L2) of the second tubular part **182** is within the range of 2.0-8.0 mm, the elastic structure **18** may hook the ear socket on the rear side of the ear when different users wear the earphone **10**. In some embodiments, the length difference may be within the range of 3.5 to 7.0 mm.

Based on the detailed description mentioned above, after the free end of the hook-shaped component **11** is sheathed with the elastic structure **18**, the outer diameter of the battery part may also be increased. That is, the actual outer diameter of the free end of the hook-shaped component **11** may be changed, so that an opening angle of the outer auricle of different user groups may be adapted, especially the "wind ears", thereby solving the problems of rotation and eversion of the earphone **10**. By designing the wall thickness of the first tubular part **181** and/or the second tubular part **182**, a difference may be formed between the elastic structure **18** and the battery part, so as to achieve a technical effect similar to the progressive necking mentioned above.

Referring to FIG. **35** and FIG. **36**, FIG. **35** is a schematic diagram illustrating a perspective view of a structure of the hook-shaped component in FIG. **24**. FIG. **36** is a schematic

diagram illustrating a cross-sectional structure of the elastic metal wire in FIG. **35** on a reference plane perpendicular to an extending direction of the hook-shaped component. It should be noted that the elastic metal wire shown in FIG. **35** may be generally embedded in the hook-shaped component, or the like, which is not visible. In order to facilitate the description, the elastic metal wire may be shown as externally visible, for example, part of the material covering the elastic metal wire may be removed.

Based on the related description above, the hook-shaped component **11**, the connecting component **12**, the holding component **13**, or other structures may also be configured with an elastic metal wire **115** such as a spring steel wire, a titanium alloy wire, a titanium nickel alloy wire, a chromium-molybdenum steel wire, or the like, to improve the structural strength of the earphone **10**. Generally, the cross-section of the elastic metal wire **115** may be circular.

In combination with FIG. **35** and FIG. **36**, the elastic metal wire **115** may have a flat sheet structure, so that the elastic metal wire **115** may have different deformability in various directions. The cross-section of the elastic metal wire **115** may be a rounded rectangle as shown in diagram (a) in FIG. **36**. The cross-section of the elastic metal wire **115** may also be an ellipse as shown in diagram (b) in FIG. **36**. Merely by way of example, a ratio of a long side (or a major axis, L3) of the elastic metal wire **115** to a short side (or a minor axis, L4) may be within the range of 4:1-6:1, preferably may be 5:1. Further, in combination with diagram (c) in FIG. **36**, for the cross-section of the elastic metal wire **115** with the rounded rectangle shown in diagram (a) in FIG. **36**, the elastic metal wire **115** may also be made into an arc shape in a minor axis direction through a process such as stamping, pre-bending, or the like, so that the elastic metal wire **115** may store a certain amount of elastic potential energy. For example, an original state of the elastic metal wire **115** may be in a curled state, and after straightening, the state of the elastic metal wire **115** may be made into the arc shape in the minor axis direction through the process of stamping, so that the elastic metal wire **115** may store certain internal stress and maintain a straight shape to become a "memory metal wire". When receiving a relatively small external force, the elastic metal wire **115** may return to the curled state, so that the hook-shaped component **11** may attach to and cover the ear of the user. Merely by way of example, a ratio of an arc height (L5) of the elastic metal wire **115** to the long side (L3) may be within the range of 0.1-0.4.

With the arrangement mentioned above, under the action of the elastic metal wire **115** with the flat sheet structure, the hook-shaped component **11** have a strong rigidity in the X direction, thereby making the hook-shaped component **11** and the holding component **13** cooperate to form an elastic clamp for the ear **100** of the user. In addition, the hook-shaped component **11** may have strong elasticity due to the bending along the length direction, so that the hook-shaped component **11** may be elastically pressed against the ear or the head of the user.

Referring to FIG. **37**, FIG. **37** is a schematic diagram illustrating a front view of a structure of an earphone according to some embodiments of the present disclosure.

The main difference from any of the embodiments mentioned above may be that in the present embodiment, in combination with FIG. **37**, in order to improve the comfort and stability of the earphone **10** in terms of wearing, a connection position between the connecting component **12** and the holding component **13** may also be adjusted. For example, the connecting component **12** may be mainly

connected with the lower edge of the holding component 13, so that the upper half of the holding component 13 (as indicated by the dashed frame in FIG. 37) may not be restricted by the connecting component 12, thereby compensating the turning moment of the holding component 13 facing away from the outside of the ear 100.

Referring to FIG. 38 and FIG. 39, FIG. 38 is a schematic diagram illustrating a structure of a rotating shaft assembly according to an embodiment of the present disclosure. FIG. 39 is a schematic diagram illustrating a structure of the rotating shaft assembly in FIG. 38 before and after assembly. It should be noted that the rotating shaft assembly shown in FIG. 38 may be generally embedded in the connecting component, or the like, which is not visible. In order to facilitate the description, the rotating shaft assembly may be shown as externally visible, for example, a part of the material covering the rotating shaft assembly may be removed.

As shown in FIG. 38, a rotating shaft mechanism 121 may be arranged to be a bending metal elastic sheet. One end of the rotating shaft mechanism 121 may be connected to the hook-shaped component 11, and the other end may be used as part of the connecting component 12. For example, the metal elastic sheet may be integrated with the connecting component 12 through a process of metal insert injection molding, and connected with the hook-shaped component 11. With the arrangement mentioned above, the metal elastic sheet may be deformed under the action of the external force F, so that the hook-shaped component 11 may be switched between a first usage state (for example, as indicated by the solid line in FIG. 38) and a second usage state (for example, as indicated by the dashed line in FIG. 38) relative to the holding component 13. That is, the hook-shaped component 11 may rotate relative to the holding component 13.

Merely by way of example, in combination with FIG. 39, the metal elastic sheet may include a first deformed part 1211, a second deformed part 1212, and an intermediate connecting component 1213. Before the installation of the metal elastic sheet, in combination with diagram (a) in FIG. 39, the first deformed part 1211 and the second deformed part 1212 may be bent and connected to two ends of the intermediate connecting component 1213, respectively. Further, after the metal elastic sheet is installed, in combination with diagram (b) in FIG. 39, the free end of the first deformed part 1211 away from the intermediate connecting component 1213 and the free end of the second deformed part 1212 away from the intermediate connecting component 1213 may be directly hinged to form a triangular structure, and curved along the length direction of the hook-shaped component 11, or further connected to the elastic metal wire in the hook-shaped component 11. With the arrangement mentioned above, the metal elastic sheet may store a certain amount of elastic potential energy after installation, so that the metal elastic sheet may deform under the action of the external force F.

Further, before the installation of the metal elastic sheet, in combination with diagram (a) in FIG. 39, the length of the first deformed part 1211 and the length of the second deformed part 1212 may be equal (denoted as L6), and may be greater than the length of the intermediate connecting component 1213 (L7). L3 and L4 may satisfy the following relationship: $0.1 \leq L7/L6 \leq 0.6$. In some embodiments, the thickness of the metal elastic sheet may be in a range of 0.1-0.8 mm.

Referring to FIG. 40 to FIG. 44, FIG. 40 is a schematic diagram illustrating a structure of a rotating shaft assembly according to another embodiment of the present disclosure.

FIG. 41 is a schematic diagram illustrating a disassembled structure of the rotating shaft assembly in FIG. 40 according to an embodiment of the present disclosure. FIG. 42 is a schematic diagram illustrating a cross-sectional structure of the rotating shaft assembly in FIG. 41. FIG. 43 is a schematic diagram illustrating a disassembled structure of the rotating shaft assembly in FIG. 40 according to another embodiment of the present disclosure. FIG. 44 is a schematic diagram illustrating a cross-sectional structure of the rotating shaft assembly in FIG. 43.

Merely by way of example, in combination with FIG. 40, the rotating shaft mechanism 121 may include a first connecting seat 1214, a second connecting seat 1215, a rotating shaft 1216, and an elastic assembly 1217. The first connecting seat 1214 may be part of the connecting component 12. The second connecting seat 1215 may be connected to the hook-shaped component 11 (or the metal elastic wire 115 therein). The second connecting seat 1215 may also be used as a part of the hook-shaped component 11. Further, the first connecting seat 1214 and the second connecting seat 1215 may be connected by the rotating shaft 1216, so that the first connecting seat 1214 and the second connecting seat 1215 may rotate relatively, and the hook-shaped component 11 may rotate relative to the connecting component 12 and the holding component 13 through the rotating shaft mechanism 121. In combination with FIG. 41 to FIG. 44, the elastic assembly 1217 may be arranged to be elastically supported between the first connecting seat 1214 and the second connecting seat 1215 so as to maintain the state of the hook-shaped component 11 after rotating relative to the holding component 13. With the arrangement mentioned above, when the user wears the earphone 10, the hook-shaped component 11 may be adjusted to be more attachable to the ear 100, thereby improving the comfort and stability of the earphone 10 in terms of wearing.

In some embodiments, in combination with FIG. 41 and FIG. 42, the second connecting seat 1215 may be partially inserted into the first connecting seat 1214, so that the rotating shaft 1216 may pass through the first connecting seat 1214 and the second connecting seat 1215 at the same time to realize the rotational cooperation. Further, the first connecting seat 1214 may be configured with an accommodating cavity 12141 with an open end, and the elastic assembly 1217 may include an elastic member 12171 and a supporting and holding member 12172. The elastic member 12171 may be arranged in the accommodating cavity 12141. One end of the supporting and holding member 12172 may partially extend into the accommodating cavity 12141 to support and hold the elastic member 12171. The other end of the supporting and holding member 12172 may support and hold the second connecting seat 1215.

It should be noted that in order to facilitate the elastic assembly 1217 to be elastically supported between the first connecting seat 1214 and the second connecting seat 1215, the elastic member 12171 may be in a compressed state after the rotating shaft mechanism 121 is assembled. In such cases, when the user wears the earphone 10, especially when the ear 100 of the user is large, the hook-shaped component 11 and the elastic metal wire 115 therein may be forced to rotate relative to the holding component 13, or have a tendency to rotate, thereby causing the second connecting seat 1215 to rotate relative to the first connecting seat 1214, and causing the supporting and holding member 12172 to compress the elastic member 12171. Based on Newton's third law, the elastic member 12171 may react to the supporting and holding member 12172 to support and hold

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the second connecting seat **1215**, thereby at least making the hook-shaped component **11** be attached to the ear **100** of the user more closely.

In other embodiments, in combination with FIG. **43** and FIG. **44**, the end of the supporting and holding member **12172** facing away from the elastic member **12171** may be set in a spherical structure, a columnar structure, or the like. The end of the second connecting seat **1215** away from the elastic metal wire **115** may be configured with a plurality of grooves distributed along a circumferential direction of the rotating shaft **1216**. The supporting and holding member **12172** may be partially clamped into the grooves under the action of the elastic force of the elastic member **12171**. In other words, after the hook-shaped component **11** rotates to different angles relative to the holding component **13**, the supporting and holding member **12172** may be locked into different grooves, respectively, thereby achieving the purpose of multi-level adjustment.

Referring to FIG. **45**, FIG. **45** is a schematic diagram illustrating a cross-sectional structure of an earphone in an XY plane according to some embodiments of the present disclosure.

In some embodiments, the earphone **10** may be an air conduction earphone. Taking the earphone **10** being the air conduction earphone as an example, the holding component, the core, the mainboard, or other structural parts may be exemplarily described:

In combination with FIG. **45**, the holding component **13** may include an inner housing **131c** and an outer housing **132c**, which may be connected to form a cavity structure for accommodating structural components such as the core **14**, the mainboard **15**, or the like. In some embodiments, the speaker of the present disclosure may be implemented as the earphone **10** or a portion (e.g., the holding component **13**) thereof. In such cases, the inner housing **131c** and an outer housing **132c** may be regarded as an implementation of the housing (e.g., the housing **10-0**) of the speaker. It should be noted that when the user wears the earphone **10**, the inner housing **131c** may be mainly in contact with the ear **100** of the user. Since a large number of electronic components with different sizes and shapes are often integrated on the mainboard **15**, the inside of the cavity of the holding component **13** may become extremely complicated, which may easily affect the acoustic performance of the earphone **10**. Accordingly, in the embodiment, a partition **133c** may be arranged in the holding component **13** to separate the core **14** and the mainboard **15** and form a cavity **200c** independent of the mainboard **15**. The cavity **200c** may have a relatively smooth inner wall. With the arrangement mentioned above, since the cavity **200c** can be protected from the influence of the mainboard **15** and the electronic components thereon, the acoustic performance of the earphone **10** may be effectively improved.

Merely by way of example, the partition **133c** may be directly connected to the core **14**, for example, the partition **133c** and the core **14** may be glued together to directly form the cavity **200c**. The inner wall of the cavity **200c** formed by the partition **133c** and the core **14** may avoid sharp structures such as right angles, sharp corners, or the like, as much as possible. Further, edges of the partition **133c** and the core **14** may also be wrapped with an elastic member (not shown in the figure), thereby forming an interference fit with the inner wall of the holding component **13** to achieve acoustic sealing.

Based on the description mentioned above, in the wearing state, the earphone **10** may be clamped on the ear. In order

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to increase the stability and comfort in terms of wearing, the earphone **10** may elastically clamp the ear.

Merely by way of example, in combination with FIG. **46**, FIG. **46** is a schematic diagram illustrating a structure of an earphone on a side facing away from an ear according to some embodiments of the present disclosure. The hook-shaped component **11** may include an elastic component **112** connected to the connecting component **12** and a battery part **113** arranged at the free end of the hook-shaped component **11**. The battery part **113** may be at least used to set the battery **16** of the earphone **10**. The battery **16** may be arranged in a columnar shape. In order to facilitate the setting of the battery **16**, the battery part **113** may be made of hard materials, such as a hard plastic material. Considering the wearing comfort, at least the region of the battery part **113** that is in contact with the skin of the user may be configured with an elastic covering layer, sprayed with elastic paint, or the like. Further, compared with the battery part **113**, the elastic component **112** may have a certain elastic deformation ability, so that the hook-shaped component **11** may be deformed under the action of an external force, thereby generating a displacement relative to the holding component **13** to allow the hook-shaped component **11** to cooperate with the holding component **13** to elastically clamp the ear. During a process that the user is wearing the earphone **10**, the user may first apply a little force to make the hook-shaped component **11** deviate from the holding component **13**, so that the ear may extend between the holding component **13** and the hook-shaped component **11**. After a wearing position is suitable, the user may let go to allow the earphone **10** to clamp the ear elastically. In some embodiments, the wearing position of the earphone **10** on the ear may also be further adjusted according to the actual wearing situations. In some embodiments, a distance that the hook-shaped component **11** deviates from the holding component **13** in the X direction may be in a range of 10-50 mm, and preferably may be in a range of 20-30 mm. A force between the hook-shaped component **11** and the holding component **13** in the X direction may be in a range of 0.18-0.6 N. In some embodiments, a distance that the hook-shaped component **11** deviates from the holding component **13** in the Y direction may be in a range of 3-10 mm, preferably may be in a range of 4-8 mm, and more preferably may be 5 mm. A force between the hook-shaped component **11** and the holding component **13** in the Y direction may be in a range of 0.18-0.37 N.

In some embodiments, a ratio of the length of the elastic component **112** to the length of the hook-shaped component **11** may be greater than or equal to 48%, and preferably may be greater than or equal to 60%. A radial size in any direction on the cross-section of the elastic component **112** may be smaller than or equal to 5 mm, and preferably may be smaller than or equal to 4 mm. In such cases, the elastic component **112** may be arranged in a slender structure, so that the elastic component **112** may have an excellent elastic deformation ability, thereby causing the earphone **10** elastically clamp the ear relatively well. In addition, an area of the cross-section of the elastic component **112** may be as small as possible, which can leave a corresponding wearing space for myopia glasses, hyperopia glasses, or smart glasses such as AR, VR, MR, or the like, thereby taking into account of the other wearing needs of the user. Further, since the hook-shaped component **11** is mainly hung between the head and the ear of the user, the cross-section of the elastic component **112** may be circular or elliptical, so that at least the elastic component **112** may make good contact with the

ear and/or the head, and may be as close as possible to a boundary line between the ear and the head, thereby increasing the stability of wearing.

In some embodiments, a cross-sectional area of at least a part of the battery part 113 may be greater than the maximum cross-sectional area of the elastic component 112, so that the battery part 113 may be configured with the battery 16 with a relatively large capacity to increase the endurance of the earphone 10. In some embodiments, the battery part 113 may be arranged in a columnar shape, and the ratio of the length to the outer diameter may be less than or equal to 6.

Based on the related description above, for the hook-shaped component 11, since the elastic component 112 and the battery part 113 have different uses, the cross-sectional areas of the elastic component 112 and the battery part 113 may be quite different. Accordingly, the hook-shaped component 11 may further include a transition part 114 between the elastic component 112 and the battery part 113. A cross-sectional area of the transition part 114 may be between the cross-sectional area of the elastic component 112 and the cross-sectional area of the battery part 113, and gradually increase in a direction from the elastic component 112 to the battery part 113. In such cases, not only can the uniformity of the hook-shaped component 11 be increased in appearance, but also can make the hook-shaped component 11 good contact with the ear and/or the head. Further, since there are generally multiple bulges on the rear side of the ear, for example, a concha boat bulge corresponding to the concha boat and a concha cavity bulge corresponding to the concha cavity, and the concha cavity bulge is generally closer to the earlobe than the concha boat bulge, so that the transition part 114 may be configured with a profile depression corresponding to a rear contour of the ear on a side facing the ear, thereby helping the hook-shaped component 11 to form an effective contact with the rear side of the ear. For example, the profile depression may be in contact with the concha cavity bulge of the ear. In short, the bulges on the rear side of the ear may be avoided through the profile depression, so as to prevent the bulges on the rear side of the ear from pushing up the hook-shaped component 11, and make the hook-shaped component 11 good contact with the ear. In some embodiments, for the transition part 114, on a reference cross-section set along a central axis of the battery part 113, a radius of curvature of the profile depression may be smaller than a radius of curvature of the other side of the transition part 114 facing away from the ear. That is, a degree of curvature of the profile depression may be greater, so that the hook-shaped component 11 may adapt to various bulges and depressions on the rear side of the ear. The other regions of the transition part 114 may be mainly configured to smooth the gap between the elastic component 112 and the battery part 113 as quickly as possible, thereby increasing the uniformity of the hook-shaped component 11 in appearance.

It may be well known that in the fields of medicine, anatomy, or the like, three basic sections including a sagittal plane, a coronal plane, and a horizontal plane of the human body may be defined, respectively, and three basic axes including a sagittal axis, a coronal axis, and a vertical axis may also be defined. As used herein, the sagittal plane may refer to a section perpendicular to the ground along a front and rear direction of the body, which divides the human body into left and right parts. The coronal plane may refer to a section perpendicular to the ground along a left and right direction of the body, which divides the human body into front and rear parts. The horizontal plane may refer to a

section parallel to the ground along an up and down direction of the body, which divides the human body into upper and lower parts. Correspondingly, the sagittal axis may refer to an axis that passes through the coronal plane perpendicular to the front and rear direction of the body. The coronal axis may refer to an axis that passes through the sagittal plane perpendicular to the left and right direction of the body. The vertical axis may refer to an axis that passes vertically through the horizontal plane along the up and down direction of the body.

Based on the related description above, the weight and the distribution of the weight of the earphone 10 may affect the stability of wearing to a certain extent. The weight of the hook-shaped component 11 may be mainly concentrated at the battery part 113. In some embodiments, a weight ratio of the total weight of the holding component 13 to the total weight of the battery part 113 may be smaller than or equal to 4. FIG. 47 is a schematic diagram illustrating a structure of an earphone on a side facing an ear according to some embodiments of the present disclosure. In combination with FIG. 47, in the wearing state and viewed from the side of the holding component 13 away from the ear, the battery part 113 may be at least partially arranged at a side of a first reference plane (denoted as RP1) facing directly in front of the user. The first reference plane may pass through the contact point (denoted as CP0) between the holding component 13 and the ear and may be parallel to the coronal plane. In such cases, it may be beneficial to reduce a moment of the center of gravity of the battery part 113 relative to, for example, the upper ear root, to prevent the battery part 113 from turning over due to excessive weight and/or excessive moment in the wearing state, thereby increasing the stability of wearing. Further, the battery part 113 may also intersect with a second reference plane (denoted as RP2). The second reference plane may pass through a first position point (denoted as CP1) of the elastic component 112 closest to the top of the head of the user along the vertical axis and may be parallel to the coronal plane. Further, an inner edge of the hook-shaped component 11 or the connecting component 12 facing the ear may have a second position point (denoted as CP2) farthest from the contact point between the holding component 13 and the ear. The battery part 113 may further intersect with a third reference plane (denoted as RP3). The third reference plane may pass through the second position point and be parallel to the coronal plane. The second position point may be on the connecting component 12, or on the boundary between the hook-shaped component 11 and the connecting component 12, which may be exemplarily illustrated in the following description. In such cases, it may be beneficial for the center of gravity of the battery part 113 and the center of gravity of the holding component 13 to be arranged at the same side of the first reference plane, thereby increasing the stability of wearing.

For ease of description and in combination with FIG. 46, the holding component 13 may have a thickness direction, a length direction, and a height direction orthogonal to each other which may be labeled as "X", "Y" and "Z" in sequence, respectively. The thickness direction may be defined as a direction in which the holding component 13 is close to or away from the ear in the wearing state. The length direction may be defined as a direction in which the holding component 13 is close to or away from the front of the user in the wearing state. The height direction may be defined as a direction in which the holding component 13 is close to or away from the top of the head of the user in the wearing state. In the wearing state, the height direction may be

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parallel to the vertical axis. The thickness direction and the length direction may be parallel to the horizontal plane.

FIG. 48 is a schematic diagram illustrating a structure of an earphone viewed from a side on a top of a user's head according to some embodiments of the present disclosure. In some embodiments, as shown in FIG. 46 to FIG. 48, an orthographic projection of a section of the hook-shaped component 11 close to the connecting component 12 on the reference plane perpendicular to the thickness direction (for example, the plane where YZ is located) and an orthographic projection of the holding component 13 on the reference plane mentioned above may partially overlap. The section of the hook-shaped component 11 close to the connecting component 12 may be an elastic component 112 with a much greater elastic deformation capacity than that of the battery part 113, or may be a rigid structure that is arranged between the battery part 113 and the connecting component 12 and has the same elastic deformation ability as that of the battery part 113. In such cases, not only can the holding component 13 and the hook-shaped component 11 elastically clamp the ear from the front side of the ear and the rear side of the ear, but the clamping force can be mainly manifested as compressive stress, thereby increasing the stability and comfort of wearing. In addition, it may also be beneficial for the center of gravity of the battery part 113 to be close to the face of the user, thereby increasing the stability of wearing. In other embodiments, such as the earphone shown in FIG. 4 and FIG. 21, or the earphone shown in FIG. 25 and FIG. 26, the orthographic projection of the hook-shaped component 11 on the reference plane perpendicular to the thickness direction and the orthographic projection of the holding component 13 on the reference plane mentioned above may also be spaced apart from each other.

Merely by way of example, in combination with FIG. 46 and FIG. 47, the orthographic projection of the elastic component 112 on the reference plane and the orthographic projection of the holding component 13 on the reference plane may partially overlap, and the orthographic projection of the battery part 113 on the reference plane and the orthographic projection of the holding component 13 on the reference plane may be spaced apart from each other. In such cases, it may be advantageous for the holding component 13 and the hook-shaped component 11 to elastically clamp the ear from the front and rear direction.

Further, a radius of curvature of an edge of the orthographic projection of the elastic component 112 and the transition part 114 on the reference plane facing the ear in a direction away from the battery part 113 from the connecting component 12 to the hook-shaped component 11 may be gradually increased first and then gradually decreased. The gradual increase in the radius of curvature of the edge may make the hook-shaped component 11 fit the contour shape of the rear side of the ear. Further, the gradual decrease in the radius of curvature of the edge may increase a bending degree of the hook-shaped component 11 close to the end of the battery part 113, so that the battery part 113 may be brought closer to the holding component 13, which facilitates the hook-shaped component 11 to hook the rear side of the ear to increase the stability of wearing. Further, the radius of curvature of the edge may be gradually increased and then gradually decreased in a continuous manner, or gradually increased first and then gradually decreased in a stepped changing manner, or combined in two manners mentioned above. For example, the edge may include a plurality of sections. Each section may have a radius of curvature, and in a direction from the connecting component 12 to the battery part 113, the radii of curvature of the

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plurality of sections may be gradually increased first and then gradually decreased, which may also be referred as a stepped change. To increase the stability of wearing, the section with the largest radius of curvature among the plurality of sections may partially overlap with the orthographic projection of the holding component 13 on the reference plane.

Merely by way of example, the edge of the orthographic projection of the elastic component 112 or the transition part 114 on the reference plane toward the ear may have a first section (denoted as 11A). The starting point of the first section (denoted as CP3) may be a connection point between the elastic component 112 and the connecting component 12, and the end point (for example, CP1) may be a highest point of the elastic component along the height direction in the wearing state. A radius of curvature of the first section may be in a range of 8 mm to 10 mm. The starting point of the first section may coincide with the second position point, or farther away from the connecting component 12 than the second position point, which may be illustrated in the following description. Further, the edge of the elastic component 112 or the transition part 114 may also have a second section (denoted as 11B). The starting point of the second section may be the end point of the first section. A distance between the end point of the second section (denoted as CP4) and the highest point in the length direction may be in a range of 8 mm to 11 mm, and a distance between the end point of the second section and the highest point in the height direction may be in a range of 7 mm to 10 mm. A radius of curvature of the second section may be in a range of 9 mm to 12 mm. Further, the edge of the elastic component 112 or the transition part 114 may also have a third section (denoted as 11C). The starting point of the third section may be the end of the second section. A distance between the end point of the third section (denoted as CP5) in the length direction and the highest point may be in a range of 9 mm to 12 mm, and a distance between the end point of the third section and the highest point in the height direction may be in a range of 19 mm to 21 mm. A radius of curvature of the third section may be in a range of 29 mm to 36 mm. Further, the edge of the elastic component 112 or the transition part 114 may also have a fourth section (denoted as 11D). The starting point of the fourth section may be the end of the third section. A distance between the end point of the fourth section (denoted as CP6) in the length direction and the highest point may be in a range of 7 mm to 10 mm, and a distance between the end point of the fourth section and the highest point in the height direction may be in a range of 25 mm to 32 mm. A radius of curvature of the fourth section may be in a range of 19 mm to 25 mm. Further, the edge of the elastic component 112 or the transition part 114 may also have a fifth section (denoted as 11E). The starting point of the fifth section may be the end of the fourth section. A distance between the end point of the fifth section (denoted as CP7) and the highest point in the length direction may be smaller or equal to 2 mm, and a distance between the end point of the fifth section and the highest point in the height direction may be in a range of 30 mm to 38 mm. A radius of curvature of the fifth section may be in a range of 9 mm to 13 mm. The fifth section may be configured with the profile depression, and a radius of curvature of the profile depression may also be smaller than the radius of curvature of the fourth section.

It should be noted that the end point of the second section, that is, the starting point of the third section, may be an intersection point between the orthographic projection of the elastic component 112 on the reference plane and the upper

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edge of the holding component 13. Similarly, the end point of the third section, that is, the starting point of the fourth section, may be another intersection point between the orthographic projection of the elastic component 112 on the reference plane and the lower edge of the holding component 13. In such cases, the orthographic projection of the third section on the reference plane may all fall on the holding component 13. FIG. 58 is a schematic diagram illustrating a structure of an earphone on a side facing away from an ear according to some embodiments of the present disclosure. Further, in combination with FIG. 58, the boundary between the elastic component 112 and the transition part 114 may be located in the fourth section. Correspondingly, the starting point of the section of the hook-shaped component 11 close to the connecting component 12 may be the boundary between the hook-shaped component 11 and the connecting component 12. The end point of the section of the hook-shaped component 11 close to the connecting component 12 may be another intersection point between the orthographic projection of the elastic component 112 on the reference plane and the lower edge of the holding component 13.

FIG. 49 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure. In combination with FIG. 49, the hook-shaped component 11 may include the elastic metal wire 115, a battery compartment 1161, and a wire 117. One end of the elastic metal wire 115 may be connected to the connecting component 12, and the other end may be connected to the battery compartment 1161. The wire 117 may extend from the battery compartment 1161 to the connecting component 12 and the holding component 13 along with the elastic metal wire 115. The elastic metal wire 115 may make the hook-shaped component 11 have a certain elastic deformation ability. The battery compartment 1161 may be at least used to accommodate the battery 16. The wire 117 may be at least used to realize the electrical connection between the battery compartment 1161 and the electronic components in the holding component 13. Further, the hook-shaped component 11 may also include an elastic covering body 118, such as silica gel. The elastic covering body 118 may at least cover the elastic metal wire 115 and the wire 117 to increase the appearance quality and the wearing comfort. A cross-sectional area of the battery compartment 1161 may be greater than the cross-sectional area of the elastic component 112 formed by the elastic metal wire 115 and the elastic covering body 118, preferably greater than a sum of the cross-sectional areas of the elastic metal wire 115, the wire 117, and the elastic covering body 118.

Further, the hook-shaped component 11 may further include a transition member 1162 connected to the elastic metal wire 115, so that the elastic metal wire 115 may be connected to the battery compartment 1161 through the transition member 1162. For example, the transition member 1162 and the elastic metal wire 115 may be formed by a metal insert injection process. The battery compartment 1161 may be arranged in a cylindrical structure with an open end to facilitate the placement of structural members such as the battery 16. The transition member 1162 may be buckled with the open end of the battery compartment 1161. In other embodiments, the transition member 1162 and the battery compartment 1161 may be integrally formed. An end of the battery compartment 1161 away from the transition member 1162 may be arranged in an open shape and may be sealed by a cover plate. A cross-sectional area of the transition member 1162 may gradually increase in a direction along the length of the hook-shaped component 11 and away from

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the connecting component 12. Correspondingly, the elastic covering body 118 may also cover the transition member 1162. The profile depression may be formed in the transition member 1162 and appear through the elastic covering body 118. In other words, the transition member 1162 may be configured with the profile depression corresponding to the rear contour of the ear on the side facing the ear. On a reference plane set along a central axis of the battery compartment 1161, the radius of curvature of the profile depression may be smaller than the radius of curvature of the other side of the transition member 1162 facing away from the ear. That is, the bending degree of the profile depression may be greater, so that the transition part 114 may avoid the bulge on the rear side of the ear.

Based on the related description mentioned above, in combination with FIG. 58, for the hook-shaped component 11, the elastic component 112 may correspond to a part of the elastic metal wire 115 exposed to the connecting component 12 and the transition member 1162, and mainly include the elastic covering body 118, the elastic metal wire 115, and the wire 117 covered by the elastic covering body 118. The battery part 113 may correspond to the battery compartment 1161, and mainly include the battery compartment 1161 and the battery 16 therein. The transition part 114 may correspond to the transition member 1162, and mainly include the elastic covering body 118 and the transition member 1162 covered by the elastic covering body 118. In other words, the length of the elastic component 112 may be a length of the part of the elastic metal wire 115 exposed from the connecting component 12 and the transition member 1162 and covered by the elastic covering body 118.

Further, the earphone 10 may further include a processing circuit and a detecting member 1163 coupled with the processing circuit. The detecting member 1163 may be used to detect whether the hook-shaped component 11 is hung between the rear side of the ear and the head. The processing circuit may be used to determine whether the earphone 10 is in the wearing state according to the detection result of the detecting member 1163. The processing circuit may be integrated on the mainboard 15. The detecting member 1163 may be a sensing element arranged on the hook-shaped component 11 (for example, the transition member 1162 or the battery compartment 1161) facing the ear. The sensing element may include a capacitive sensing element, an inductive sensing element, a resistance sensing element, or the like, or any combination thereof. Merely by way of example, the detecting member 1163 may be a capacitive sensing element, and may be arranged in the profile depression of the transition member 1162.

In some application scenarios, when the detecting member 1163 detects that the earphone 10 is in the wearing state, the processing circuit may generate a first control signal for controlling the earphone 10 to switch to a playing state. When the detecting member 1163 does not detect that the earphone 10 is in the wearing state, the processing circuit may generate a second control signal for controlling the earphone 10 to switch to a pause state. In such cases, not only may the power of the earphone 10 be saved, but also the interactivity of the earphone 10 may be increased.

In other application scenarios, the earphone 10 may include a first earphone and a second earphone that are arranged in a pair and are communicatively connected. For example, the first earphone and the second earphone may be worn on the left and right ears of the user, respectively, and each of the first earphone and the second earphone may be configured with the detecting member 1163. The processing circuit may determine and select one of the first earphone

and the second earphone as the main earphone to be communicatively connected with an audio source device (such as a mobile phone, a tablet, a smartwatch, etc.) according to detection results of the detecting member 1163 in the first earphone and the second earphone. When the user uses two earphones at the same time, one of the earphones may be selected as the main earphone to be communicatively connected with the audio source device according to a predetermined rule, and the other one may be selected as an auxiliary earphone to be communicatively connected with the main earphone. When the user only uses one of the two earphones, the earphone in usage may be regarded as the main earphone.

In combination with FIG. 46 and FIG. 48, the side of the holding component 13 facing the ear may include a first region 13A and a second region 13B. The second region 13B may be farther away from the connecting component 12 than the first region 13A. That is, the second region 13B may be arranged at the free end of the holding component 13 away from the connecting component 12. Based on the related description mentioned above, the orthographic projection of the section (e.g., the elastic component 112) of the hook-shaped component 11 close to the connecting component 12 along the thickness direction described above may partially overlap the second region 13B. Further, the first region 13A may be configured with a sound hole 1311. The second region 13B may be convex toward the ear compared to the first region 13A and used to contact with the ear, thereby allowing the sound hole 1311 to be spaced from the ear in the wearing state. In short, the free end of the holding component 13 may be configured in a convex hull structure. Since the core 14 may generate a sound transmitted to the ear through the sound hole 1311, the convex hull structure may prevent the ear from blocking the sound hole 1311 and cause the sound generated by the core 14 to be weakened or even fail to be output. Merely by way of example, a maximum protrusion height of the second region 13B relative to the first region 13A in the thickness direction may be greater than or equal to 1 mm, and a smooth transition may be made between the two regions. It should be noted that if it is only for the sound hole 1311 to be spaced from the ear in the wearing state, the second region 13B protruding toward the ear compared to the first region 13A may also be another region of the holding component 13, for example, a region between the sound hole 1311 and the connecting component 12. Further, since the concha cavity and the concha boat have a certain depth and are connected with the ear hole, the orthographic projection of the sound hole 1311 on the ear along the thickness direction may at least partially fall in the concha cavity and/or the concha boat. Merely by way of example, the holding component 13 may be arranged at a side of the ear hole close to the top of the head of the user and be contact the antihelix. The orthographic projection of the sound hole 1311 on the ear along the thickness direction may at least partially fall in the concha boat.

FIG. 63 is a schematic diagram illustrating a cross-sectional structure of an earphone according to some embodiments of the present disclosure. Further, in combination with FIG. 46 and FIG. 63, the holding component 13 may form a front cavity 200 and a rear cavity 300 of the earphone 10 on opposite sides of the core 14, respectively. The sound hole 1311 may communicate with the front cavity 200 and output a sound to the ear. The holding component 13 may also be configured with a pressure relief hole 1312 communicating with the rear cavity 300. The pressure relief hole 1312 may be farther away from the ear hole than the sound hole 1311. The pressure relief hole 1312 may allow air

to enter and exit the rear cavity 300 freely to make the change of air pressure in the front cavity 200 not be blocked by the rear cavity 300 as much as possible, thereby improving the sound quality of the sound output to the ear through the sound hole 1311. Moreover, since phases of sounds output to the outside of the earphone 10 through the sound hole 1311 and the pressure relief hole 1312 are opposite, the phases may be reversed and canceled in the far-field away from the ear, that is, an "acoustic dipole" may be formed to reduce sound leakage. An angle between a line between the center of the pressure relief hole 1312 and the center of the sound hole 1313 and the thickness direction may be between 0° and 50°. Preferably, the angle may be between 0° and 40°. Further, the holding component 13 may also be configured with a sound adjusting hole 1313 communicating with the rear cavity 300. The sound adjusting hole 1313 may be used to destroy a high-pressure region of a sound field in the rear cavity 300, so that a wavelength of a standing wave in the rear cavity 300 may be shortened, thereby increasing the resonance frequency of the sound output to the outside of the earphone 10 through the pressure relief hole 1312, for example, greater than 4 kHz, to reduce the sound leakage. Preferably, the sound adjustment hole 1313 and the pressure relief hole 1312 may be arranged at opposite sides of the core 14, respectively. For example, the sound adjusting hole 1313 and the pressure relief hole 1312 may be arranged opposite to each other in the height direction to destroy the high-pressure region of the sound field in the rear cavity 300 to the greatest extent. An opening direction of the pressure relief hole 1312 may face the top of the head of the user. For example, an angle between the opening direction and the vertical axis may be between 0° and 10°, to allow the pressure relief hole 1312 to be farther away from the ear hole than the sound adjusting hole 1313, thereby making it difficult for the user to hear the sound output to the outside of the earphone 10 through the pressure relief hole 1312 to reduce the sound leakage. The pressure relief hole 1312 may have a first center in the length direction. The sound adjusting hole 1313 may have a second center in the length direction, and the second center may be farther away from the center of the sound hole 1311 in the length direction than the first center, so as to increase the distance between the sound adjusting hole 1313 and the sound hole 1311 as much as possible, thereby weakening the anti-phase cancellation between the sound output to the outside of the earphone 10 through the sound adjusting hole 1313 and the sound transmitted to the ear through the sound hole 1311. In other words, the orthographic projection of the sound adjusting hole 1313 in the height direction and the orthographic projection of the second region 13B in the thickness direction may at least partially intersect with each other, so as to be as far away from the sound hole 1311 as possible. In some embodiments, the sound hole 1311, the pressure relief hole 1312, and the sound adjusting hole 1313 may also referred to as sound guiding holes (e.g., the sound guiding holes 30 on the housing 10-0 of the speaker).

Briefly, when the user wears the earphone 10, the user mainly listens to the sound transmitted to the ear hole through the sound hole 1311. Other acoustic holes, such as the pressure relief hole 1312 and the sound adjusting hole 1313, may be mainly used to make the sound as possible as to have the sound quality of bass diving and treble penetration. Therefore, a ratio of the size of an outlet end of the pressure relief hole 1312 in the length direction (for example, as indicated by L1 in FIG. 48) to the size of an end of the rear cavity 300 near the pressure relief hole 1312 in the length direction (for example, as indicated by L2 in FIG.

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61) may be greater than or equal to 0.9. A size relationship between the sizes in the thickness direction may also be the same or similar. Therefore, the rear cavity 300 may be connected to the outside of the earphone 10 as large as possible to minimize the blocking of the rear cavity 300 to the front cavity 200. In addition, the resonance frequency of the sound output to the outside of the earphone 10 through the pressure relief hole 1312 may be shifted to a high frequency as much as possible.

It should be noted that since the structural parts such as a core housing 131 have a certain thickness, holes including the sound hole 1311, the pressure relief hole 1312, or the sound adjusting hole 1312 arranged on the core housing 131 may have a certain depth. Thus, with respect to the accommodating cavity formed by the core housing 131, the hole described in the present disclosure may have an inlet end close to the accommodating cavity and an outlet end far away from the accommodating cavity. A partition 137 and the connecting holes arranged on the partition 137 described in the following may be similar to the illustration mentioned above, which may not be repeated herein.

In combination with FIG. 46 to FIG. 48, in the natural state, and viewed from a side of the earphone 10 facing the top of the head of the user in the wearing state, for example, viewed along the height direction, the holding component 13 may be spaced apart from at least the section of the hook-shaped component 11 close to the connecting component 12 in the thickness direction, and the connecting component 12 may be arranged in an arc shape and connected between the holding component 13 and the hook-shaped component 11. In such cases, the connecting component 12 may cause the holding component 13 arranged at the front side of the ear and the hook-shaped component 11 arranged at the rear side of the ear always be spaced apart from each other at least in a section close to the connecting component 12 in the thickness direction, so that the earphone 10 can bypass the upper ear root and the nearby tissues in the wearing state, thereby preventing the earphone 10 from over-clamping the helix near the upper ear root and causing discomfort.

Merely by way of example, the connecting component 12 and the holding component 13 may be connected along the length direction. At least part of the connecting component 12 may extend away from the free end of the holding component 13 along the length direction and the height direction at the same time in a direction from one end connecting the holding component 13 to the other end connecting the hook-shaped component 11 to convex toward the face of the user as a whole, so that a height difference between the hook-shaped component 11 and the holding component 13 in the height direction may be eliminated in a smooth transition manner. In some embodiments, at least part of the connecting component 12 may also extend away from the free end of the holding component 13 along the length direction in the direction from one end connecting the holding component 13 to the other end connecting the hook-shaped component 11. In addition, the connecting component 12 and/or the section of the hook-shaped component 11 close to the connecting component 12 may also extend away from the free end of the holding component 13 in the thickness direction, so that the holding component 13 and the section of the hook-shaped component 11 close to the connecting component 12 can be arranged at intervals in the thickness direction. FIG. 53 is a schematic diagram illustrating a structure of an earphone at a side facing away from an ear according to some embodiments of the present disclosure. FIG. 54 is a schematic diagram illustrating a structure of an earphone viewed from a side on a top of a

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user's head according to some embodiments of the present disclosure. In some embodiments, in combination with FIG. 53 and FIG. 54, the connecting component 12 may further extend close to the free end of the holding component 13 along the length direction and extend away from the free end of the holding component 13 along the height direction at the same time in the direction from one end connecting the holding component 13 to the other end connecting the hook-shaped component 11. That is, the connecting component 12 may form a circuitously extending structure in the three-dimensional space. FIG. 59 is a schematic diagram illustrating a structure of an earphone viewed from a side on a top of a user's head according to some embodiments of the present disclosure. In other embodiments, in combination with FIG. 58 and FIG. 59, the connecting component 12 may only extend away from the free end of the holding component 13 along the length direction and the height direction at the same time in the direction from one end connecting the holding component 13 to the other end connecting the hook-shaped component 11. That is, a first half of the circuitously extending structure may be formed. The section of the hook-shaped component 11 close to the connecting component 12 (for example, the elastic component 112) may continue to extend close to the free end of the holding component 13 along the length direction in a direction away from the connecting component 12 and extend away from the free end of the holding component 13 along the height direction at the same time. That is, a second half of the circuitously extending structure may be formed, thereby cooperating with the first half of the circuitously extending structure to form the circuitously extending structure in the three-dimensional space. In other embodiments, the circuitously extending structure may have only the first half part or the second half part.

In some embodiments, the section of the hook-shaped component 11 close to the connecting component 12 (for example, the elastic component 112), the edge of the connecting component 12 or the holding component 13 toward the ear may be arranged in a shape of a circuitous arc. In a reference direction that passes through a roundabout inflection point of the circuitous arc (for example, CP2) and is parallel to the length direction, the minimum width W1 of the circuitous arc along the thickness direction at a position 3 mm away from the roundabout inflection point may be in a range of 1 mm to 5 mm.

In other embodiments, in the thickness direction, the minimum distance between the section (e.g., the elastic component 112) of the hook-shaped component 11 close to the connecting component 12 and the holding component 13 may be greater than 0, and smaller than or equal to 5 mm.

In other embodiments, in the thickness direction, a distance W2 between the center of the sound hole 1311 (denoted as O0) and the section of the hook-shaped component 11 close to the connecting component 12 (for example, the elastic component 112) may be between 3 mm and 6 mm.

In other embodiments, in the thickness direction, a distance W3 between the second region 13B and the section of the hook-shaped component 11 close to the connecting component 12 (for example, the elastic component 112) may be between 1 mm and 5 mm.

FIG. 50 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure. In combination with FIG. 50 and FIG. 48, the holding component 13 may include the core housing 131 connected to the connecting component 12. Structural components such as the core 14 and the mainboard 15 may

be fixed in the accommodating space of the core housing 131. In some embodiments, the speaker of the present disclosure may be implemented as the earphone illustrated in FIG. 50. In such cases, the core housing 131 may be regarded as an implementation of the housing 10-0 of the speaker. Merely by way of example, the core housing 131 may include a first housing 1314 and a second housing 1315 that are arranged opposite to each other in the thickness direction. The first housing 1314 may be closer to the ear than the second housing 1315. In some embodiments, the first housing 1314 and the second housing 1315 may also be arranged opposite to each other in a vibration direction of the core 14. The vibration direction may be parallel to the thickness direction. Specifically, the core 14 may be fixed on a side of the first housing 1314 facing the second housing 1315 to form the front cavity 200. The second housing 1315 may be buckled with the first housing 1314 and surround the core 14 to form the rear cavity 300. Correspondingly, the sound hole 1311 may be arranged on the first housing 1314, for example, on a side of the first housing 1314 facing the ear. The pressure relief hole 1312 and the sound adjusting hole 1313 may be arranged on opposite sides of the second housing 1315, respectively. For example, the pressure relief hole 1312 and the sound adjusting hole 1313 may be arranged opposite to each other in the height direction. Based on the related description mentioned above, a ratio of the size of the outlet end of the pressure relief hole 1312 in the length direction to the size of the second housing 1315 in the length direction may be greater than or equal to 0.55. Preferably, the ratio may be between 0.8 and 1, so that the rear cavity 300 communicates with the outside of the earphone 10 as much as possible while taking into account the structural strength of the second housing 1315.

In some embodiments, in combination with FIG. 50, the connecting component 12 may include a third housing 122 connected to an end of the elastic metal wire 115 away from the battery compartment 1161. For example, the third housing 122 and the end of the elastic metal wire 115 may be formed by a metal insert injection molding process. The size of the second housing 1315 or the third housing 122 in the length direction may be less than that of the first housing 1314. The size of the second housing 1315 may be much larger than the size of the third housing 122. The second housing 1315 may be buckled with the first housing 1314, and the orthographic projection of the second housing 1315 in the thickness direction may be partially overlapped with that of the first housing 1314. The third housing 122 may be buckled with the part of the first housing 1314 located at the periphery of the orthographic projection of the second housing 1315. In short, the third housing 122 may be buckled on the same side of the second housing 1315 and the first housing 1314. In addition, most of the first housing 1314 may be used as the housing of the holding component 13, and a small part may be used as the housing of the connecting component 12. In a specific embodiment, a ratio of the maximum size of the third housing 122 in the length direction to the size of the second housing 1315 in the length direction may be less than or equal to 0.4.

Based on the related description mentioned above, in combination with FIG. 53 and FIG. 54, in the natural state, and viewed from the side of the earphone 10 facing the top of the head of the user in the wearing state, for example, viewed along the height direction, the first housing 1314 and the elastic metal wire 115 may be spaced apart in the thickness direction. The third housing 122 may be arranged in the arc shape and connect the first housing 1314 and the elastic metal wire 115, thereby allowing the holding com-

ponent 13 arranged at the front side of the ear and the hook-shaped component 11 arranged at the rear side of the ear to be spaced apart from each other in the thickness direction at least in the section close to the connecting component 12. Further, the third housing 122 may first extend away from the second housing 1315 along the length direction and the height direction at the same time in a direction from one end connecting the first housing 1314 to the other end connecting the elastic metal wire 115, and then extend close to the second housing 1315 along the length direction and extend away from the second housing 1315 along the height direction, thereby allowing the height difference between the hook-shaped component 11 and the holding component 13 in the height direction to be eliminated in a smooth transition manner. In such cases, the second position point may fall on the connecting component 12, and the starting point of the first section may be farther away from the connecting component 12 than the second position point. The part of the first housing 1314 that is used as the housing of the connecting component 12 may have the same or similar changing trend with the third housing 122. Thus, the connecting component 12 may form a circuitously extending structure in the three-dimensional space. In combination with FIG. 54, a parting line (denoted as PL1) may be provided between the third housing 122 and the first housing 1314. The third housing 122 and the first housing 1314 may be separately molded and then buckled together, so as to solve the problem that the housing of the connecting component 12 is difficult to mold due to its circuitously extending structure in the three-dimensional space, thereby increasing production efficiency and reducing production costs.

FIG. 57 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure. In some embodiments, in combination with FIG. 57, the third housing 122 and the first housing 1314 may be integrally formed, and a connecting plug hole may also be formed. Further, the connecting component 12 may also include a connecting plug member 123. One end of the connecting plug member 123 may be connected to the hook-shaped component 11, and the other end may be plugged and fixed in the connecting plug hole, thereby realizing the connection between the hook-shaped component 11 and the connecting component 12. Specifically, an end of the connecting plug hole 123 away from the third housing 122 may be connected (e.g., be molded by a metal insert injection process) to the other end of the elastic metal wire 115 away from the battery compartment 1161. Further, the connecting component 12 may also include a locking member 124. A part of the connecting plug member 123 inserted into the third housing 122 may be locked with the third housing 122 by the locking member 124, which is convenient for assembly and increases the reliability of assembly. In some embodiments, the locking member 124 may be a wedge arranged in a column shape or a sheet shape.

Based on the related description mentioned above, in combination with FIG. 58 and FIG. 59, the third housing 122 may extend away from the second housing 1315 along the length direction and the height direction at the same time in a direction from one end connecting the first housing 1314 to the other end connecting the connecting plug member 123. The section of the elastic metal wire 115 exposed to the connecting plug member 123 and close to the connecting plug member 123 may further extend close to the second housing 1315 along the length direction and extend away from the second housing 1315 along the height direction at the same time in a direction away from the connecting plug

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member 123. Correspondingly, the third housing 122 may also extend away from the second housing 1315 in the thickness direction at the same time. The section of the elastic metal wire 115 exposed to the connecting plug member 123 and close to the connecting plug member 123 may continue to extend away from the second housing 1315 along the thickness direction. In such cases, the second position point may fall on the boundary between the hook-shaped component 11 and the connecting component 12, and the starting point of the first section may coincide with the second position point. The part of the first housing 1314 that is used as the housing of the connecting component 12 and the part of the connecting plug member 123 exposed to the third housing 122 may have the same or similar changing trend with the third housing 122. Thus, the connecting component 12 may be allowed to form only the first half of the circuitously extending structure, and the hook-shaped component 11 may continue to form the second half of the circuitously extending structure, thereby cooperating to form the circuitously extending structure in the three-dimensional space. Therefore, in combination with FIG. 58, a parting line (denoted as PL2) may be provided between the connecting plug member 123 and the third housing 122 (and/or the first housing 1314). The connecting plug member 123 and the third housing 122 (and/or the first housing 1314) may be formed separately and then plugged in to solve the problem that the housing of the connecting component 12 is difficult to mold due to the circuitously extending structure in the three-dimensional space, thereby increasing the production efficiency and reducing the production cost.

It should be noted that the housings of the connecting component 12 and the holding component 13 may also be divided according to other dividing manners. For example, the housing of the holding component 13 may be divided into two housings with substantially equal orthographic projection areas along the thickness direction. The housing of the connecting component 12 may be divided into two housing along the roundabout inflection point or may include only one housing, and the other housing may be composed of the elastic metal wire 115, and the housings may be assembled accordingly.

Based on the related description mentioned above, in combination with FIG. 50 and FIG. 48, the holding component 13 may need to be in contact with the front side of the ear, in particular, the free end of the holding component 13 may further need to form a contact point (for example, CP0) with the antihelix of the ear. Thus, a side of the core housing 131 facing the ear may be configured with a flexible covering structure 132 that does not cover at least the sound hole 1311. For example, the flexible covering structure 132 may be configured with a through-hole corresponding to the sound hole 1311. The Shore hardness of the flexible covering structure 132 may be less than the Shore hardness of the core housing 131, so that the holding component 13 can be in contact with the ear through the flexible covering structure 132. That is, the flexible covering structure 132 may be elastically supported between the core housing 131 and the ear, thereby improving the wearing comfort. Further, based on the dividing and splicing manner of the housings of the connecting component 12 and the holding component 13, to increase the appearance quality of the earphone 10, the flexible covering structure 132 may be directly attached to the first housing 1314, the third housing 122, or the like, through an injection molding process. In some embodiments, the flexible covering structure 132 may cover the first housing 1314, the third housing 122, or the like, through a gluing connection manner. Since the hook-shaped compo-

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nent 11 may also be configured with the elastic covering body 118, the elastic covering body 118 and the flexible covering structure 132 may be formed by the one injection molding process, or be separately formed by two injection molding processes. The materials of the two processes may be the same or different. It should be noted that without special descriptions, the present application mainly describes the part where the flexible covering structure 132 and the elastic covering 118 are in contact with the user's skin.

In some embodiments, the flexible covering structure 132 may be at least partially arranged at the side of the holding component 13 away from the free end of the connecting component 12 and facing the ear, that is, the second region 13B. Correspondingly, the orthographic projection of the elastic component 112 on the reference plane (for example, the plane where YZ is located) and the orthographic projection of the flexible covering structure 132 on the reference plane may partially overlap with each other. Further, the thickness of the flexible covering structure 132 may be designed differently. For example, the flexible covering structure 132 corresponding to the second region 13B may be relatively thick, so that the free end of the holding component 13 may protrude toward the ear, and have good flexibility. In some embodiments, if only for the second region 13B to protrude toward the ear compared to the first region 13A, a side of the first housing 1314 toward the ear may also be designed with a thickness difference. Thus, the first housing 1314 may also include a first region and a second region, so as to correspond to the first region 13A and the second region 13B on the side of the holding component 13 facing the ear, respectively.

In some embodiments, a side of the flexible covering structure 132 facing the core housing 131 may be recessed with blind hole(s) 1321 spaced from each other. The blind hole 1321 may be mainly used to provide a deformation space for the flexible covering structure 132 to allow the flexible covering structure 132 to undergo more deformation under pressure in the wearing state, thereby further improving the wearing comfort. In some embodiments, a count of the blind holes 1321 may be multiple, for example at least two, which may be spaced apart from each other to form a bone position to support the own structure, thereby having both elastic deformation and structural strength. In other embodiments, the count of the blind holes 1321 may also be only one. In such cases, by controlling the elastic modulus, thickness, size of the blind hole 1321, and other parameters of the flexible covering structure 132, the blind hole 1321 may also have elastic deformation and structural strength at the same time. To make the flexible covering structure 132 have the blind hole(s) 1321, the core housing 131 (e.g., the part of the first housing 1314 corresponding to the second region 13B) may be configured with through-hole(s) 13141 corresponding to and communicating with the blind hole(s) 1321, respectively. The through-hole(s) 13141 may be used for inserting molding cores of the flexible covering structure 132. In such cases, the plurality of through-holes 13141 may cause the part of the first housing 1314 corresponding to the second region 13B to be arranged in a honeycomb or grid shape so as to balance the structural strength of the first housing 1314 in the region and the support for the flexible covering structure 132. Further, the outer side of the first housing 1314 may also be configured with protrusions surrounding the through-holes 13141 along the honeycomb or grid structure. In some embodiments, the protrusions may be embedded in the flexible covering structure 132. In some embodiments, the flexible covering structure 132 may be

partially embedded in the through-holes 13141 to increase a bonding area of the flexible covering structure 132 between the second region 13B and the first housing 1314, thereby increasing the bonding strength. Thus, the first housing 1314 may have the corresponding through-holes 13141 during the molding process, and the molding cores of the flexible covering structure 132 may be inserted into the through-holes 13141 after the molding is completed. The molding cores may protrude from the first housing 1314, and the maximum protrusion height may depend on the actual requirements of the convex hull structure. The flexible covering structure 132 may be directly molded on the first housing 1314 through the injection molding process, and then the molding cores may be drawn out. Correspondingly, the holding component 13 may further include a cover plate 1316 arranged in the core housing 131. For example, the cover plate 1316 may be fixedly arranged at an inner side of the first housing 1314 away from the flexible covering structure 132 to seal the through-holes 13141, thereby allowing the first housing 1314 and the cover plate 1316 to surround the core 14 to form the front cavity 200. The cover plate 1316 may be supported on the honeycomb or grid structure of the first housing 1314.

Merely by way of example, a first flange 13142 may be arranged on an inner wall surface of the first housing 1314 away from the flexible covering structure 132. A second flange 13161 may be arranged on an inner wall surface of the cover plate 1316 away from the flexible covering structure 132. Two ends of the second flange 13161 and two ends of the first flange 13142 may extend oppositely and respectively to form an annular flange by splicing. In such cases, the core 14 may be held on the annular flange to form the front cavity 200. The first housing 1314 may be configured with a sink groove in the second region 13B. The cover plate 1316 may be embedded into the sink groove to allow the inner wall surface of the cover plate 1316 to be flush with the inner wall surface of the first housing 1314 away from the flexible covering structure 132, thereby causing an inner cavity surface of the front cavity 200 to be as flat as possible. Further, a glue groove may be arranged on the inner wall surface of the first housing 1314 away from the flexible covering structure 132. The glue groove may be arranged at the edge of the sink groove and surrounded by a plurality of through-holes 13141. The cover plate 1316 may be glued with the first housing 1314 through the glue in the glue groove. In short, the first flange 13142 and the glue groove may be both arranged on the inner side of the first housing 1314 away from the flexible covering structure 132, however, the former may mainly correspond to the first region 13A, and the latter may mainly correspond to the second region 13B.

It should be noted that in other embodiments such as the flexible covering structure 132 does not have the blind holes 1321, or the flexible covering structure 132 is formed separately and then connected to (e.g., through glue) the core housing 131, the first housing 1314 may not need to be configured with the through-holes 13141, and the corresponding cover plate 1316 may not be provided. In such cases, the first flange 13142 may be a complete annular flange, and the front cavity 200 may be formed by supporting and holding by the core 14 on the annular flange.

In other embodiments, in combination with FIG. 57, the flexible covering structure 132 may include an inner flexible body 1322 arranged on the core housing 131 and an outer flexible body 1323 at least covering the inner flexible body 1322. The inner flexible body 1322 may be arranged in the second region 13B. The outer flexible body 1323 may cover

the inner flexible body 1322, the first housing 1314, the third housing 122, or the like. In such cases, the flexible covering structure 132 may be in contact with the ear through the outer flexible body 1323. In short, the flexible covering structure 132 may also be configured as a double-layer structure, so as to adjust the thickness and softness of a part of the flexible covering structure 132 corresponding to the second region 13B. Correspondingly, the orthographic projection of the elastic component 112 on the reference plane (for example, the plane where YZ is located) and the orthographic projection of the inner flexible body 1322 on the reference plane may partially overlap with each other. Similarly, the sound hole 1311 may be arranged between the inner flexible body 1322 and the connecting component 12. Further, the inner flexible body 1322 may also protrude toward the ears. That is, the inner flexible body 1322 may protrude from the core housing 131 (specifically, the first housing 1314) to facilitate the flexible covering structure 132 to form the convex hull structure.

Merely by way of example, the blind hole(s) 1321 may be arranged in the inner flexible body 1322, and the function and forming manner may be the same as or similar to those described above, which may not be repeated herein. The number of the blind hole(s) 1321 may be multiple, so that the inner flexible body 1322 may have the bone positions arranged in the honeycomb shape or the grid shape, or may have a plurality of bone positions arranged at intervals. In other embodiments, the blind hole(s) 1321 may further penetrate the inner flexible body 1322 to be a through-hole. Similarly, gaps between the bone positions, that is, the blind holes 1321, may be used to provide a deformation space for the flexible covering structure 132. In some embodiments, the materials of the inner flexible body 1322 and the outer flexible body 1323 may be silica gel with zero degrees.

Merely by way of example, the Shore hardness of the inner flexible body 1322 may be less than the Shore hardness of the outer flexible body 1323 to allow the part of the flexible covering structure 132 corresponding to the second region 13B to be softer. A side of the outer flexible body 1323 facing the core housing 131 may be recessed with the blind hole(s) 1321. The inner flexible body 1322 may be arranged in the blind hole(s) 1321 and in contact with the outer flexible body 1323. In other words, the blind hole(s) 1321 may be arranged in the outer flexible body 1323 so as to accommodate the more flexible inner flexible body 1322. Specifically, the part of the first housing 1314 corresponding to the second region 13B may be configured with the through-holes 13141 for inserting the molding cores of the outer flexible body 1323. In such cases, the outer flexible body 1323 may be formed on the first housing 1314 through the injection molding process, and the molding cores may be drawn out after the outer flexible body 1323 is molded, so that the outer flexible body 1323 may form the corresponding blind holes 1321, thereby forming an accommodating region. The inner flexible body 1322 may be arranged in the blind hole(s) 1321 through the through-hole(s) 13141. That is, the inner flexible body 1322 may be arranged in the accommodating region, and the through-hole(s) 13141 may be sealed by the cover plate 1316. A side of the cover plate 1316 facing the inner flexible body 1322 may be partially embedded in the through-hole(s) 13141 to increase the sealing performance of the accommodating region. Further, the number of the blind hole(s) 1321 may be one, and the number of the through-hole(s) 13141 may also be one. In such cases, when an opening area of the through-hole 13141 is relatively large, the cover plate 1316 may be extended to partially overlap with the first housing 1314 in the first

region 13A, so as to increase a supporting area of the cover plate 1316 by the first housing 1314. The cover plate 1316 may be configured with a communicating hole 13162 connecting the sound hole 1311 and the front cavity 200 to avoid blocking the sound hole 1311. In a specific embodiment, the material of the outer flexible body 1323 may be silica gel with 30-50 degrees, and the material of the inner flexible body 1322 may be silica gel with zero degrees, and the outer flexible body 1323 and the inner flexible body 1322 may be formed in the accommodating region through a glue dropping process. In another specific embodiment, the material of the outer flexible body 1323 may be silica gel with 30-50 degrees, and the material of the inner flexible body 1322 may be silica gel with 0-10 degrees, and the outer flexible body 1323 and the inner flexible body 1322 may be pre-formed into a block to be filled in the accommodating region. In some embodiments, when the inner flexible body 1322 can withstand the impact force of the outer flexible body 1323 during the molding process, the first housing 1314 may not be configured with the through-hole 13141, and the corresponding cover plate 1316 may not be provided.

Based on the detailed description mentioned above, structural components such as the first housing 1314, the outer flexible body 1323, the inner flexible body 1322, the cover plate 1316, etc., may form a housing assembly, that is, the structural components can be modularized to facilitate assembly.

In combination with FIG. 46, the earphone 10 may further include a microphone 125 and a microphone 133 arranged on the holding component 13 and/or the connecting component 12. The two microphones 125 and 133 may be electrically connected to the mainboard 15. A distance between the microphone 125 and the microphone 133 in the length direction may be greater than a distance between the microphone 125 and the microphone 133 in the height direction. The distance between the two microphones 125 and 133 can be set to be as large as possible when the size of the earphone 10 is relatively determined, thus interference between the two microphones 125 and 133 may be avoided, and the sound pickup effect and/or the noise reduction effect of the earphone 10 may be increased. Further, a line between the orthographic projection of the microphone 125 on the reference plane (for example, the plane where YZ is located) and the orthographic projection of the microphone 133 on the reference plane may pass through the orthographic projection of the core 14 on the reference plane. In other words, if the orthographic projection of the core 14 on the reference plane is arranged in a rectangle shape, the two microphones 125 and 133 may be arranged substantially along the diagonal of the core 14.

In some embodiments, the microphone 125 may be arranged at the connecting component 12, and the microphone 133 may be arranged at the free end of the holding component 13 away from the connecting component 12. The microphone 125 may be closer to the mouth of the user than the microphone 133, which is mainly used to pick up the voice of the user. In some embodiments, the earphone 10 may also include the processing circuit, which may be integrated on the mainboard 15, and may designate the microphone 125 as the main microphone and the microphone 133 as the auxiliary microphone. The sound signal collected by the auxiliary microphone may be used to reduce the noise of the sound signal collected by the main microphone, thereby increasing the sound pickup effect. At least one of the two microphones 125 and 133 may also be used

from the earphone 10 to the ear, or only one microphone for sound pickup or noise reduction may be provided.

Merely by way of example, the microphone 125 may be arranged between the third housing 122 and the first housing 1314, and the microphone 133 may be arranged between the second housing 1315 and the first housing 1314. The sides of the third housing 122 and the second housing 1315 away from the first housing 1314 may be respectively configured with through-holes for microphones to collect sounds.

In other embodiments, the earphone 10 may also include a stick microphone 134 that is detachably connected to the free end (i.e., the battery part 113) of the holding component 13 or the hook-shaped component 11 away from the connecting component 12. The free end of the stick microphone 134 may be configured with a microphone 1341 electrically connected to the mainboard 15. Compared with the microphone 125 and the microphone 133, the stick microphone 134 may cause the microphone 1341 closer to the mouth of the user, which is beneficial to increase the sound pickup effect. In the present disclosure, the detachable connection of the stick microphone 134 and the holding component 13 may be taken as an example for illustration. For example, a main rod 1342 of the stick microphone 134 and the second housing 1315 may be detachably connected by ways of buckle, magnetism, or the like. As another example, the main rod 1342 and the second housing 1315 may be detachably connected by a type-C plug-in manner, so as to shorten a wiring distance between the microphone 1341 and the mainboard 15.

Further, in addition to the microphone 1341 on the stick microphone 134, the earphone 10 may also be configured with other microphones, such as the microphone 125 and/or the microphone 133. The processing circuit may use the microphone 1341 as the main microphone when the stick microphone 134 is connected to the holding component 13, and use at least one of the microphone 133 and the microphone 125 as the auxiliary microphone. The sound signal collected by the auxiliary microphone may be used to reduce the noise of the sound signal collected by the main microphone, thereby increasing the sound pickup effect. Correspondingly, the processing circuit may switch the microphone 133 and the microphone 125 to an enabled state when the stick microphone 134 is separated from the holding component 13. One of the microphone 133 and the microphone 125 may be used as the main microphone, and the other may be used as the auxiliary microphone. In some embodiments, the processing circuit may also switch at least one of the microphone 133 and the microphone 125 to a disabled state when the stick microphone 134 is connected to the holding component 13, so as to save power while taking into account sound pickup and/or noise reduction.

In combination with FIG. 46 and FIG. 47, the earphone 10 may further include a first charging electrode 126 arranged at the holding component 13 or the connecting component 12 and a second charging electrode 1164 arranged at the hook-shaped component 11. One of the first charging electrode 126 and the second charging electrode 1164 may be used as a positive charging electrode, and the other may be used as a negative charging electrode. In the present disclosure, for illustration purposes, the first charging electrode 126 may be used as the positive charging electrode and the second charging electrode 1164 may be used as the negative charging electrode. In such cases, the earphone 10 may be charged by the two charging electrodes. In addition, the shortest distance between the two charging electrodes may be greatly increased, which helps prevent short circuits between the charging electrodes caused by sweat, water

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droplets, dust, or the like. In some embodiments, in the case of satisfying the short-circuit prevention, the two charging electrodes may also be arranged in one of the hook-shaped component 11, the connecting component 12, and the holding component 13. Further, the two charging electrodes may be set to be invisible in the wearing state. For example, both charging electrodes may face the skin of the user, so as to take into account the appearance quality of the earphone 10.

Merely by way of example, the first charging electrode 126 may be arranged at the connecting component 12, and the second charging electrode 1164 may be arranged at the battery part 116. Specifically, the first charging electrode 126 may be at least partially arranged at the periphery of the second housing 1315, for example, arranged between the third housing 122 and the first housing 1314. Correspondingly, the second charging electrode 1164 may be arranged in the battery compartment 1161, for example, at the bottom of the battery compartment 1161 away from the open end. The first charging electrode 126 may be arranged in a column shape, and the second charging electrode 1164 may be arranged in a strip shape. The length direction of the second charging electrode 1164 may extend along the circumferential direction of the battery compartment 1161. Further, the first housing 1314 and the battery compartment 1161 may be respectively configured with through-holes that allow the charging electrodes to be exposed, so that the charging electrodes can be in contact with output electrodes on a charging box. Compared with the charging electrode with the column shape, the charging electrode with the strip shape may have a larger contact area with the output electrode, which may increase the reliability of the charging electrode.

It should be noted that the connecting component 12 may be provided with a plurality of first charging electrodes 126 arranged at intervals. For example, the connecting component 12 may be provided with two first charging electrodes 126 so that after one of the first charging electrodes 126 fails, the other can still be available. Further, a magnetic adsorption member, such as a magnet, may also be arranged near each of the two charging electrodes to allow the earphone 10 to make good contact with the output electrode(s) on the charging box by ways of magnetic adsorption. For the charging box, the position(s) of the output electrode(s) may be adjusted with the change of the charging electrode(s) on the earphone 10.

FIG. 51 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure. In combination with FIG. 51, since the second housing 1315 is farther away from the ears than the first housing 1314, the second housing 1315 may be configured with interactive components such as a physical button, a display, a touch circuit board, or the like, to facilitate the user to interact with the earphone 10.

Merely by way of example, the second housing 1315 may include a bottom wall 13151 arranged opposite to the first housing 1314 and a side wall 13152 connected to the bottom wall 13151. The side wall 13152 may extend toward the first housing 1314. A side of the bottom wall 13151 facing the first housing 1314 may be configured with a flexible touch circuit board 135 electrically connected to the mainboard 15. The flexible touch circuit board 135 may include a capacitive flexible touch circuit board, a resistive flexible touch circuit board, a pressure-sensitive flexible touch circuit board, or the like, which is not limited herein. In such cases, the interaction with the earphone 10 can be realized, and there may be no need to arrange an additional through-hole on the core housing 131, thereby increasing the waterproof

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and dustproof performance. Specifically, the flexible touch circuit board 135 may include a touch part 1351 for receiving touch operations and an electrical connection part 1352 for connecting with the mainboard 15. For example, the flexible touch circuit board 135 may be buckled with the mainboard 15 via a BTB connector. A ratio of an area of the touch part 1351 to an area of the bottom wall 13151 may be greater than or equal to 70%. Based on the related description mentioned above, a side of the side wall 13152 close to the third housing 122 may be opened to facilitate the splicing of the second housing 1315 and the third housing 122. The pressure relief hole 1312 and the sound adjusting hole 1313 may be arranged on the side wall 13152, and specifically arranged on the opposite sides of the open end, respectively.

Further, the bottom wall 13151 may be configured with a sink groove 13153, and the touch part 1351 may be attached to the bottom of the sink groove 13153. In such cases, the second housing 1315 may be equivalent to being partially thinned to increase the sensitivity of the flexible touch circuit board 135. In some embodiments, the mainboard 15 may also be connected to the second housing 1315. The flexible touch circuit board 135 may be pressed on the bottom wall 13151 through an elastic pad 1353. Thus, the touch part 1351 may be in close contact with the bottom wall 13151, and the touch part 1351 may be prevented from being crushed. The depth of the sink groove 13153 may be greater than or equal to the thickness of the touch part 1351, and smaller than a sum of the thicknesses of the touch part 1351 and the elastic pad 1353, so as to increase the pressing and holding effect.

In some embodiments, the bottom wall 13151 may be configured with a plurality of hot melt columns 13154 arranged at the periphery of the sink groove 13153 and extend toward the mainboard 15. For example, the number of the hot melt columns may be three. A connection line between the orthographic projections of at least two of the plurality of hot melt columns 13154 on the bottom wall 13151 may pass through the orthographic projection of the touch part 1351 on the bottom wall 13151. Correspondingly, the mainboard 15 may be configured with a connecting hole corresponding to each hot melt column 13154 to allow the mainboard 15 to be sleeved and fixed on the hot melt column 13154 through the connecting hole. In short, if the touch part 1351 is arranged in a rectangular shape, at least two hot melt columns 13154 may be arranged substantially along the diagonal of the touch part, so as to increase the uniformity of the force distribution of the mainboard 15. In other embodiments, the hot melt column 13154 may also be replaced with a screw, a buckle, or the like, which is not limited herein.

Based on the related description mentioned above, the microphone 133 may be directly arranged at a side of the mainboard 15 away from the bottom wall 13151 through the SMT process. Correspondingly, the bottom wall 13151 may be configured with a flange 13155 arranged at the periphery of the sink groove 13153. The flange 13155 may extend toward the mainboard 15 and have a sound pickup hole communicating with the outside of the earphone 10. The mainboard 15 may be pressed on the flange 13155 to allow the microphone 133 to collect sound signals through the sound pickup hole. In some embodiments, a silicone sleeve 13156 may be sleeved on the flange 13155 to allow the mainboard 15 to be elastically supported on the flange 13155 through the silicone sleeve 13156. As a result, not only the sealing of the sound path of the microphone 133 can be increased, but also the uniformity of the force distribution on the mainboard 15 can be increased.

In some embodiments, a metal antenna pattern may be arranged on the second housing **1315** to serve as a communication antenna of the earphone **10**. Correspondingly, the bottom wall **13151** may be configured with an antenna contact point **13157** arranged at the periphery of the sink groove **13153** and electrically connected to the metal antenna pattern. The mainboard **15** may be configured with a metal elastic sheet for elastic contact with the antenna contact point **13157**. In short, the mainboard **15** may be connected to the antenna contact point **13157** through the metal elastic sheet to avoid unnecessary welding, thereby reducing the difficulty of assembly and saving the internal space of the core housing **131**.

As stated above, the connection between the mainboard **15** and the second housing **1315** may not only realize the fixation of the mainboard **15**, but also realize the pressing and holding of the flexible touch circuit board **135**, the sealing of the sound path of the microphone **133**, and the electrical connection between the mainboard **15** and the metal antenna pattern. That is, multiple purposes may be achieved at one stroke.

Based on the related description mentioned above, in combination with FIG. **51** and FIG. **57**, electronic components arranged in the hook-shaped component **11** may be electrically connected to the mainboard **15** through the wire **117**. Since the electronic components arranged in the connecting component **12** are relatively close to the mainboard **15**, the electronic components may be directly electrically connected to the mainboard **15** through leads of the electronic components. The wire **117** may be arranged in a plurality of strands, and may include a positive lead and a negative lead of the battery **16**, a signal line and a shielding line of the detecting member **1163**, and a negative lead of the second charging electrode **1164**. In some embodiments, the shielding wire of the detecting member **1163** and the wire of the second charging electrode **1164** may be a same wire to simplify the wiring. Further, since a size of the mainboard **15** is limited and there are many electronic components integrated on the mainboard **15**, the wire **117** or other leads may be welded to the flexible circuit board **136** first, and then buckled and connected to the mainboard **15** through the flexible circuit board **136**, which is beneficial to enlarge the size of each pad and increase the spacing between each two pads, thereby reducing the difficulty of welding and increasing the reliability of welding.

Merely by way of example, the flexible circuit board **136** may include at least a first connection region **1361** for electrical connection with the battery **16** and a second connection region **1362** for electrical connection with the mainboard **15**. The second connection region **1362** may be arranged along the main surface of the mainboard **15** to facilitate the buckling connection of the flexible circuit board **136** and the mainboard **15**. Further, the first connection region **1361** may be bent toward the side of the mainboard **15** relative to the second connection region **1362**, and may be configured with a plurality of pads. That is, the welding may occur on the side of the mainboard **15**. As a result, since there is no interference from the electronic components on the main surface of the mainboard **15**, the difficulty of welding may be reduced. Moreover, due to the thin thickness, the flexible circuit board **136** may be partially bent toward the side of the mainboard **15**, which may also save the internal space of the core housing **131**. Based on the related description mentioned above, the plurality of pads arranged in the first connection region **1361** may include a first pad and a second pad respectively used to weld the positive electrode lead and the negative electrode lead of the

battery **16**. The plurality of pads arranged in the first connection region **1361** may further include a third pad and a fourth pad respectively used to weld the positive electrode lead and the negative electrode lead of the charging electrode. The plurality of pads arranged in the first connection region **1361** may further include a fifth pad and a sixth pad respectively used to weld the signal line and the shielding line of the detecting member **1163**. Since the shielding wire of the detecting member **1163** and the lead of the second charging electrode **1164** can be a same lead, one of the fourth pad and the sixth pad may be omitted, which is beneficial to enlarge the sizes of other pads and the spacing between every two pads.

Based on the related description mentioned above, since the microphone **125** can be arranged at the connecting component **12** so as to be closer to the mainboard **15**, the flexible circuit board **136** may be further extended to the connecting component **12**. The flexible circuit board **136** may further include a third connection region **1363** connected to the first connection region **1361**. The third connection region **1363** may be bent in a direction away from the mainboard **15** compared to the first connection region **1361**, so that the third connection region **1363** can be attached to the first housing **1314** and/or the third housing **122**. The microphone **125** may be arranged in the third connection region **1363** through the SMT process. The first connection region **1361** and the third connection region **1363** may be perpendicular to the main surface of the mainboard **15**, respectively. The second connection region **1362** may be parallel to the main surface of the mainboard **15**.

Different from the first connection region **1361**, the second connection region **1362** may be buckled with the mainboard **15** by ways of the BTB connector. The flexible circuit board **136** may further include a transition region **1364** connecting the first connection region **1361** and the second connection region **1362**. The transition region **1364** and the second connection region **1362** may be arranged at the same side of the mainboard **15**. A length of the transition region **1364** may be greater than the minimum distance between the first connection region **1361** and the second connection region **1362**, so that the first connection region **1361** can be buckled with the mainboard **15**. Merely by way of example, the transition region **1364** may be arranged in a multi-segment bending structure, and arranged along the main surface of the mainboard **15**.

In combination with FIG. **51**, the core **14** may include a magnetic circuit system **141** and a coil **142**. The coil **142** may extend into a magnetic gap of the magnetic circuit system **141** and move in a magnetic field formed by the magnetic circuit system **141** in an energized state. The magnetic circuit system **141** may include structural components such as a permanent magnet, a yoke, a bracket, or the like. The specific structure and connection relationship of the magnetic circuit system **141** may be well known to those skilled in the art, which are not repeated herein. Further, if the core **14** is applied to a bone conduction earphone, the coil **142** may be arranged to drive a vibration plate to move. If the core **14** is applied to an air conduction earphone, the coil **142** may be arranged to drive a diaphragm to move. In some embodiments, the coil **142** may also be configured to simultaneously drive the vibration plate and the diaphragm to move. In the present disclosure, the coil **142** driving the diaphragm to move may be taken as an example for illustration. The core **14** may further include a diaphragm **143** connected between the coil **142** and the magnetic circuit system **141**. The diaphragm **143** may generate a sound

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during a vibration process which can be transmitted to the ear through the sound hole 1311.

Further, the core 14 may further include a metal elastic sheet 144 fixed on the periphery of the magnetic circuit system 141. The metal elastic sheet 144 may be electrically connected to the coil 142. The core 14 may be elastically pressed on the mainboard 15 by the metal elastic sheet 144, so that the coil 142 can be electrically connected to a contact point on the mainboard 15. Thus, by replacing welding wires in the related technology with the metal elastic sheet 144, unnecessary welding can be avoided, thereby reducing the difficulty of assembly. In addition, there is no need to reserve a welding space, thereby saving the internal space of the core housing 131. A count of the metal elastic sheet(s) 144 may be two, which can be used as the positive lead and the negative lead of the coil 142, respectively.

FIG. 56 is a schematic diagram illustrating a structure of a core facing a side of a mainboard according to some embodiments of the present disclosure. Merely by way of example, in combination with FIG. 56, the metal elastic sheet 144 may include a fixing part 1441 and an elastic contact part 1442 connected to one end of the fixing part 1441. The fixing part 1441 may be connected to the magnetic circuit system 141. The elastic contact part 1442 may extend toward the fixing part 1441 away from the magnetic circuit system 141. In short, the part of the metal elastic sheet 144 for electrically connecting with the contact point on the mainboard 15 may protrude from the magnetic circuit system 141. Further, the metal elastic sheet 144 may further include a limiting part 1443 connected to the other end of the fixing part 1441. The limiting part 1443 and the elastic contact part 1442 may extend toward a same direction. The elastic contact part 1442 may be further bent and extended toward the limiting part 1443, and the free end of the elastic contact part 1442 may be inserted into a limiting groove of the limiting part 1443, so that the elastic contact part 1442 may store an elastic potential energy in advance, thereby increasing the goodness of the contact between the metal elastic sheet 144 and the contact point on the mainboard 15. In such cases, the height of the middle part of the elastic contact part 1442 relative to the fixing part 1441 may be greater than the height of the free end of the elastic contact part 1442 relative to the fixing part 1441 so as to facilitate the contact with the contact point on the mainboard 15.

Based on the related description mentioned above, the magnetic circuit system 141 may be connected to the side of the first housing 1314 facing the second housing 1315. The mainboard 15 may be connected to the side of the second housing 1315 facing the first housing 1314. The second housing 1315 may be buckled with the first housing 1314, so that the core 14 may elastically press the metal elastic sheet 144 on the mainboard 15, which is simple and reliable, and has high assembly efficiency. Each side of the opposite sides of the magnetic circuit system 141 may be configured with a metal elastic sheet 144 to increase the stability of the core 14 clamped by the second housing 1315 and the mainboard 15 together with the first housing 1314. Correspondingly, the diaphragm 143 may be enclosed with the first housing 1314 to form the front cavity 200. For example, the magnetic circuit system 141 may be supported and held on the annular flange formed by splicing the second flange 13161 and the first flange 13142 mentioned above. The magnetic circuit system 141 may be configured with a through-hole connecting the rear cavity 300 and a side of the diaphragm 143 away from the front cavity 200. In other words, the core 14 (specifically, the diaphragm 143) may divide the accommodating cavity formed by the core housing 131 into the front

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cavity 200 and the rear cavity 300 opposite to each other. The orthographic projection of the sound hole 1311 along the vibration direction of the core 14 may at least partially fall on the diaphragm 143. Further, the mainboard 15 and the core 14 may be stacked in the thickness direction, and the core 14 may be closer to the ear than the mainboard 15 to avoid arranging the through-hole connecting the side of the diaphragm 143 away from the rear cavity 300 and the front cavity 200 on the mainboard 15, thereby simplifying the structure. A ratio of an overlap area between the orthographic projection of the core 14 on the reference plane (for example, the plane where YZ is located) and the orthographic projection of the mainboard 15 on the reference plane to the larger one of an area of the orthographic projection of the mainboard 15 on the reference plane and the area of the orthographic projection of the core 14 on the reference plane may be in a range of 0.8 to 1. For example, the area of the orthographic projection of the core 14 on the reference plane may be substantially equal to the area of the orthographic projection of the mainboard 15 on the reference plane. Specifically, a ratio of an absolute value of a difference between a size of core 14 in the length direction and a size of the mainboard 15 in the length direction to the larger one of the size of the mainboard 15 in the length direction and the size of the core 14 in the length direction may be in a range of 0 to 0.2. A dimensional relationship between the core 14 and the mainboard 15 in the height direction may be the same as or similar to their dimensional relationship in the length direction. Thus, under a condition that a volume of the accommodating cavity formed by the core housing 131 is constant, the core 14 can be as large as possible, which is beneficial to increase the loudness of the earphone 10 and widen the frequency response range of the earphone 10.

It should be noted that, in combination with FIG. 56, although the core 14 has a major axis direction (indicated by Y1) and a minor axis direction (indicated by Z1) orthogonal to each other and perpendicular to the vibration direction of the core 14 (indicated by X1), for ease of description, the vibration direction, the major axis direction, and the minor axis direction in the embodiment provided in the present disclosure may be respectively parallel to the thickness direction, the major axis direction, and the height direction mentioned above. In other embodiments, an angle may be allowed between the vibration direction and the thickness direction, or between the major axis direction of the core 14 and the major axis direction mentioned above, or between the minor axis direction of the core 14 and the height direction. Further, the size of the core 14 in the major axis direction may be greater than or equal to the size of the core 14 in the minor axis direction. Merely by way of example, the orthographic projection of the core 14 on the reference plane perpendicular to the vibration direction may be in a rectangular shape. The major axis direction may be a direction of a long side of the rectangle, and the minor axis direction may be a direction of a short side of the rectangle.

The inventor(s) of the present disclosure has discovered in long-term research that when the mainboard 15 is arranged at the side of the core 14 away from the front cavity 200, a large number of electronic components with different sizes and shapes arranged on the mainboard 15 may affect the sound quality of the earphone 10. FIG. 52 is a schematic diagram illustrating a cross-sectional structure of an earphone according to some embodiments of the present disclosure. FIG. 62 is a schematic diagram illustrating a cross-sectional structure of an earphone according to some embodiments of the present disclosure. Thus, in combina-

tion with FIG. 52 and FIG. 62, the holding component 13 may further include the partition 137 arranged in the core housing 131. The partition 137 may be mainly used to separate the core 14 from the mainboard 15, and may be enclosed with the core 14 to form the rear cavity 300, that is, an independent sound cavity. Specifically, the partition 137 may be arranged between the magnetic circuit system 141 and the mainboard 15, and may be enclosed with the magnetic circuit system 141 to form the rear cavity 300. In other embodiments, the mainboard 15 may be covered by a layer to make the side of the mainboard 15 facing the core 14 as flat as possible.

Merely by way of example, the partition 137 may be connected to the core 14, that is, the partition 137 and the core 14 can be modularized to facilitate assembly. FIG. 55 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure. FIG. 60 is a schematic diagram illustrating a disassembled structure of an earphone according to some embodiments of the present disclosure. Specifically, in combination with FIG. 55 and FIG. 60, the partition 137 may include a bottom wall 1371 and a side wall 1372 connected to the bottom wall 1371. The bottom wall 1371 may be separated from the magnetic circuit system 141. The side wall 1372 may extend toward the core 14 and be connected to the core 14 (specifically, the magnetic circuit system 141), so as to allow the partition 137 to be enclosed with the core 14 to form the rear cavity 300. A side of the partition 137 facing the magnetic circuit system 141 may further be configured with a glue groove 1373 and a positioning column 1374 matched with the magnetic circuit system 141 to facilitate the accurate assembly of the partition 137 with the core 14. Correspondingly, the metal elastic sheet 144 may be arranged at the periphery of the partition 137.

Based on the related description mentioned above, the side wall 1372 may also be configured with a communicating hole that allows the rear cavity 300 to communicate with the outside of the earphone 10, for example, a first communicating hole 1375 connecting the pressure relief hole 1312 and the rear cavity 300, a second communicating hole 1376 connecting the sound adjusting hole 1313 and the rear cavity 300, etc. The partition 137 and the core housing 131 may also elastically support a sealing member that surrounds the communicating hole, so as to seal the sound path communicating between the rear cavity 300 and the outside of the earphone 10.

In the present disclosure, the structural components such as the core housing 131, the core 14, etc., may be generally arranged in a cubic structure or a cylindrical structure, which is not limited herein. In the present disclosure, the core 14 being arranged in a cubic structure may be taken as an example for illustration. A size of the partition 137 in the length direction may be greater than or equal to a size of the partition 137 in the height direction. In combination with FIG. 55, the side wall 1372 may include a first side wall 13721 and a third side wall 13723 spaced apart from each other in the length direction, and a second side wall 13722 and a fourth side wall 13724 spaced apart from each other in the height direction. Further, one of the second side wall 13722 and the fourth side wall 13724 may be configured with the first communicating hole 1375, and the other may be configured with the second communicating hole 1376. Based on the related description mentioned above, the first communicating hole 1375 may be arranged in the second side wall 13722, and the second communicating hole 1376 may be arranged in the fourth side wall 13724. FIG. 61 is a

schematic diagram illustrating a structure of a baffle facing a side of a core according to some embodiments of the present disclosure. It should be noted that, in combination with FIG. 60 and FIG. 61, the second side wall 13722 may also be omitted, and the first communicating hole 1375 may be directly enclosed by the bottom wall 1371, the first side wall 13721, and the third side wall 13723, which may be exemplarily described in the following descriptions.

Further, the third side wall 13723 may be farther away from the sound hole 1311 than the first side wall 13721, that is, farther away from the connecting component 12 and closer to the free end of the holding component 13. A size of the first communicating hole 1375 in the length direction may be greater than a size of the second communicating hole 1376 in the length direction, and sizes of the first communicating hole 1375 and the second communicating hole 1376 in the thickness direction may be equal, so as to adjust an actual area of an effective communication region between the rear cavity 300 and the outside of the earphone 10 through the first communicating hole 1375 and the second communicating hole 1376. The first side wall 13721 and the fourth side wall 13724 may be connected by a first arc-shaped transition wall 13725 to avoid sharp structures such as a right angle, a sharp corner, etc., on the inner wall of the enclosed rear cavity 300, thereby helping to eliminate standing waves. The first arc-shaped transition wall 13725 may be arranged in a shape of a circular arc (referred to as a circular arc shape for brevity). A radius of the circular arc may be greater than or equal to 2 mm. Similarly, the third side wall 13723 and the fourth side wall 13724 may be connected by a second arc-shaped transition wall 13726. A radius of curvature of at least part of the inner wall surface of the first arc-shaped transition wall 13725 may be greater than a radius of curvature of the corresponding part of the inner wall surface of the second arc-shaped transition wall 13726, which may also be possible to avoid sharp structures such as a right angle, a sharp corner, etc., on the inner wall of the enclosed rear cavity 300. In other embodiments, the second arc-shaped transition wall 13726 may be omitted. For example, a part of the fourth side wall 13724 close to the third side wall 13723 may be used to arrange the second communicating hole 1376 so that the second communicating hole 1376 can extend along the length direction to be flush with the inner wall surface of the third side wall 13723.

It should be noted that in the thickness direction, an inner wall surface of the first communicating hole 1375 away from the core 14 may be flush with an inner wall surface of the bottom wall 1371 facing the core 14. The inner wall surface of the second communicating hole 1376 far away from the core 14 may be flush with the inner wall surface of the bottom wall 1371 facing the core 14. That is, the first communicating hole 1375 and the second communicating hole 1376 may extend along the thickness direction to be flush with the inner wall surface of the bottom wall 1371, so as to avoid sharp structures such as a right angle, sharp corner, etc., on the inner wall surface of the enclosed rear cavity 300, thereby helping to eliminate standing waves. Further, the inner wall surface of at least one of the first side wall 13721 and the third side wall 13723 may be arc-shaped when viewed from the height direction, so as to avoid sharp structures such as a right angle, a sharp corner, etc., on the inner wall surface of the enclosed rear cavity 300. In some embodiments, the inner wall surfaces of the side wall 1372 and the bottom wall 1371 may be arc connected.

In some embodiments, in combination with FIG. 55, heights of the second side wall 13722 and the fourth side wall 13724 relative to the bottom wall 1371 may both be

greater than heights of the first side wall **13721** and the third side wall **13723** relative to the bottom wall **1371**, so that the core **14** can be embedded between the second side wall **13722** and the fourth side wall **13724**. The first side wall **13721** and the third side wall **13723** may respectively abut against a side of the core **14** facing the bottom wall **1371**. In the thickness direction, a size of the first communicating hole **1375** may be greater than or equal to a distance between the bottom wall **1371** and the core **14**. The size of the second communicating hole **1376** may be greater than or equal to a distance between the bottom wall **1371** and the core **14** to prevent the inner wall surface of the enclosed rear cavity **300** from appearing with sharp structures such as a right angle, a sharp corner, etc., thereby helping to eliminate standing waves. Further, the holding component **13** may further include a first sealing member **1381** and a second sealing member **1382** elastically supported between the partition **137** and the core housing **131**. For example, the first sealing member **1381** may be elastically supported between the second side wall **13722** and the second housing **1315** and surround the first communicating hole **1375**. As another example, the second sealing member **1382** may be elastically supported between the fourth side wall **13724** and the second housing **1315** and surround the second communicating hole **1376**. Further, an outlet end of the first communicating hole **1375** may be covered with a first acoustic resistance net **1383**, and a side of the first acoustic resistance net **1383** away from the side wall **1372** may also be covered with a protective cover. Similarly, an outlet end of the second communicating hole **1376** may be covered with a second acoustic resistance net **1384**, and a side of the second acoustic resistance net **1384** away from the side wall **1372** may also be covered with a protective cover. The acoustic resistance net may not only increase the waterproof and dustproof performance, but also reduce the sound leakage. The structural strength of the protective cover may be greater than the structural strength of the acoustic resistance net so as to prevent the acoustic resistance net from being punctured by foreign objects. Further, a porosity of the second acoustic resistance net **1384** may be smaller than or equal to a porosity of the first acoustic resistance net **1383**.

Merely by way of example, the first sealing member **1381** may include a first extending part **13811** and a second extending part **13812** connected to the first extending part **13811**. The first extending part **13811** and the second extending part **13812** may be attached and fixed on the side wall **1372** and the bottom wall **1371** away from the rear cavity **300**, respectively, to increase a combined area between the first sealing member **1381** and the partition **137**. Correspondingly, the first extending part **13811** may allow a region of the first acoustic resistance net **1383** corresponding to the first communicating hole **1375** to be exposed. For example, the first extending part **13811** may surround the first communicating hole **1375** and the first acoustic resistance net **1383** thereon, so as to facilitate the communication between the rear cavity **300** and the outside of the earphone **10**. Further, the first extending part **13811** may press and fix the first acoustic resistance net **1383** on the side of the side wall **1372** away from the rear cavity **300** to prevent the first acoustic resistance net **1383** from being separated from the side wall **1372**.

In the embodiment, the structure of the second sealing member **1382** and the connection relationship between the second sealing member **1382** and the partition **137** may be the same as or similar to that of the first sealing member **1381**, which may not be repeated herein. Further, the first

sealing member **1381** and the second sealing member **1382** may be formed on the partition **137** through the injection molding process.

It should be noted that in the embodiment, structural components such as the core **14**, the partition **137** or the acoustic resistance net, the sealing member thereon, etc., may form a loudspeaker assembly, that is, the structural components can be modularized to facilitate assembly.

In other embodiments, in combination with FIG. **60**, the second side wall **13722** may be omitted. The fourth side wall **13724** may be partially used for arranging the second communicating hole **1376**, and the height of the fourth side wall **13724** relative to the bottom wall **1371** may be equal to each height of the first side wall **13721** and the third side wall **13723** relative to the bottom wall **1371** to abut against the magnetic circuit system **141** together with the first side wall **13721** and the third side wall **13723**. The first sealing member **1381** may be embedded in the preset sink groove of the first sealing member **1381** or the second housing **1315** first. Then the first sealing member **1381** may be attached and fixed to the second housing **1315**. Thus, the second housing **1315** and the first sealing member **1381** may clamp the first acoustic resistance net **1383** together, and the subsequent assembly may be performed. The side of the first sealing member **1381** facing the second housing **1315** may be configured with a sink groove for accommodating the first acoustic resistance net **1383**. Similarly, the second sealing member **1382** and the second acoustic resistance net **1384** may also be attached and fixed on the second housing **1315** to form a housing assembly, that is, the second sealing member **1382**, the second acoustic resistance net **1384**, and the second housing **1315** can be modularized to facilitate assembly.

Based on the detailed description mentioned above, to facilitate the description, the following definitions may be made in combination with FIG. **63**: the front cavity **200** may have a first opening **201** that allows the front cavity **200** to communicate with the outside of the earphone **10**, and the rear cavity **300** may have a second opening **301** and a third opening **302** that allow the rear cavity **300** to communicate with the outside of the earphone **10**. Correspondingly, the second opening **301** may be farther away from the ear hole than the first opening **201** and the third opening **302**. Each of the first opening, the second opening, and the third opening may refer to an effective communication region between the front cavity **200** or the rear cavity **300** and the outside of the earphone **10**. That is, a region with the smallest cross section through which the sound is transmitted from the front cavity **200** or the rear cavity **300** to the outside of the earphone **10**. For example, the core **14** may cooperate with the first housing **1314** (and the cover plate **1316**) to form the front cavity **300**, and the first opening **201** may correspond to the sound hole **1311**. In the embodiment in which the earphone **10** is configured with the partition **137**, that is, the partition **137** cooperates with the core **14** to form the rear cavity **300**, if an actual area of the pressure relief hole **1312** is greater than an actual area of the second communicating hole **1376**, the second opening **301** may correspond to the second communicating hole **1376**. If the actual area of the pressure relief hole **1312** is smaller than the actual area of the second communicating hole **1376**, the second opening **301** may correspond to the pressure relief hole **1312**. If the pressure relief hole **1312** and the second communicating hole **1376** are staggered with each other, the second opening **301** may correspond to a portion where the pressure relief hole **1312** and the second communicating hole **1376** are not shielded from each other. The third

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opening 302 may be similar to the second opening 301, and details may not be repeated herein. In an embodiment in which the earphone 10 is not configured with the partition 137, that is, the second housing 1315 cooperates with the core 14 to form the rear cavity 300, the second opening 301 and the third opening 302 may directly correspond to the pressure relief hole 1312 and the sound adjusting hole 1313, respectively. In some embodiments, if the earphone 10 is not configured with at least one of the front cavity 200 or the rear cavity 300, the corresponding opening may naturally no longer exist.

Further, in order to facilitate the description, an effective area described in the present disclosure may be defined as a product of an actual area of an effective communication region and a porosity of the corresponding acoustic resistance net. For example, when the first opening 201 is covered with an acoustic resistance net, the effective area of the first opening 201 may be the product of an actual area of the first opening 201 and a porosity of the acoustic resistance net. When the first opening 201 is not covered with an acoustic barrier, the effective area of the first opening 201 may be the actual area of the first opening 201. The second opening 301 and the third opening 302 may be similar to the first opening 201, and details may not be repeated herein. In the present disclosure, an effective area of the third opening 302 may be smaller than an effective area of the second opening 301.

In some embodiments, in combination with FIG. 55 and FIG. 60, an actual area of the outlet end of the second communicating hole 1376 may be smaller than or equal to an actual area of the outlet end of the first communicating hole 1375, so that an actual area of an effective communication region between the sound adjusting hole 1313 and the rear cavity 300 may be smaller than or equal to an actual area of an effective communication region between the pressure relief hole 1312 and the rear cavity 300. An actual area of the outlet end of the pressure relief hole 1312 may be greater than or equal to the actual area of the outlet end of the first communicating hole 1375. In such cases, a size of the outlet end of the sound adjusting hole 1313 in the length direction may be equal to a size of the outlet end of the pressure relief hole 1312 in the length direction. In some embodiments, the size of the outlet end of the sound adjusting hole 1313 in the thickness direction may be equal to the size of the outlet end of the pressure relief hole 1312 in the thickness direction. Thus, an actual area of an effective communication region between the rear cavity 300 and the outside of the earphone 10 at the sound adjusting hole 1313 or the pressure relief hole 1312 may be adjusted by adjusting the size of the corresponding communicating hole to meet the corresponding acoustic design requirements. In addition, the sound adjusting hole 1313 and the pressure relief hole 1312 may be designed to have little difference in appearance to increase the consistency of the appearance, thereby allowing them to use the acoustic resistance net with the same specification to reduce the count of types of materials or avoid material mixing. In other embodiments, the size of the sound adjusting hole 1313 may be changed with the change of the second communicating hole 1376 to make the sound adjusting hole 1313 look different from the pressure relief hole 1312 in appearance, so as to increase the appearance recognition degree. Further, a porosity of the second acoustic resistance net 1384 may be smaller than or equal to a porosity of the first acoustic resistance net 1383, so that the effective area of the effective communication region between the sound adjusting hole 1313 and the rear cavity 300 can be smaller

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than or equal to the effective area of the effective communication region between the pressure relief hole 1312 and the rear cavity 300.

Further, the effective communication region (for example, the first communicating hole 1375) between the pressure relief hole 1312 and the rear cavity 300 may have a first center (denoted as O1) in the length direction. The effective communication region (for example, the second communicating hole 1376) between the sound adjusting hole 1313 and the rear cavity 300 may have a second center (denoted as O2) in the length direction, and the second center may be farther away from the center of the sound hole 1311 (denoted as O0) than the first center in the length direction. That is, the second center may be closer to the third side wall 13723, so as to increase the distance between the sound adjusting hole 1313 and the sound hole 1311 as much as possible, thereby weakening the anti-phase cancellation between the sound output to the outside of the earphone 10 through the sound hole 1313 and the sound transmitted to the ear through the sound hole 1311.

It should be noted that a center of a hole or an opening in the present disclosure may refer to a position where distances to the circumference of the closed curve surrounding the hole or opening are equal. For a regular shape such as a circle, a rectangle, or the like, the center of the hole or opening described in the present disclosure may be the geometric center. For other irregular shapes, the center of the hole or opening described in the present disclosure may be the centroid.

FIG. 64 is a schematic diagram illustrating a sound field distribution of an acoustic dipole according to some embodiments of the present disclosure. In combination with FIG. 64, the sound transmitted to the outside of the earphone 10 through the first opening 201 may be simply regarded as a first sound formed by a monopole sound source A1. The sound transmitted to the outside of the earphone 10 through the second opening 301 may be simply regarded as a second sound formed by a monopole sound source A2. The second sound may be opposite to the first sound in phase, which may be reversed and canceled in the far-field. That is, an "acoustic dipole" may be formed to reduce sound leakage. Preferably, in the wearing state, a connection line of the two monopole sound sources may be directed to the ear hole (denoted as "hearing position"), so that the user can hear a sufficiently loud sound. A sound pressure at the hearing position (denoted as P_{ear}) may be used to indicate the strength of the sound heard by the user. Further, the sound pressures (denoted as P_{far}) on a spherical surface centered on the hearing position of the user may be used to indicate the strength of the sound leakage of the earphone 10 radiated to the far-field. A variety of statistical manners may be used to obtain P_{far} , such as taking an average value of the sound pressure at each point on the spherical surface, taking the sound pressure distribution at each point on the spherical surface for area classification, etc. Generally, the sound pressure P_{ear} transmitted by the earphone 10 to the ear of the user should be large enough to increase the listening effect, and the sound pressure P_{far} in the far-field should be small enough to increase the sound leakage reduction effect. Therefore, parameter α may be used as an indicator for evaluating the sound leakage reduction or listening effect of the earphone 10, which may be determined according to Equation as follows:

$$\alpha = \frac{|P_{far}|^2}{|P_{ear}|^2}$$

Further, when the earphone **10** is in the wearing state, the orthographic projection of the holding component **13** (for example, a side of the holding component **13** arranged at the ear hole close to the top of the head of the user, which is in contact with the antihelix at the front side of the ear) on the ear may mainly fall within the range of the helix. The first opening **201** may be arranged between the antihelix and the upper ear root, and transmit the sound to the ear hole. Further, since the concha cavity and the concha boat have a certain depth and are connected with the ear hole, the orthographic projection of the first opening **201** on the ear may at least partially fall within the concha cavity and/or the concha boat, so that the sound transmitted to the outside of the earphone **10** through the first opening **201** can be transmitted to the ear hole. FIG. **65** is a schematic diagram illustrating a sound field distribution of an acoustic dipole with a baffle according to some embodiments of the present disclosure. FIG. **66** is a schematic diagram illustrating a sound pressure in far-field of when an acoustic dipole with and without a baffle according to some embodiments of the present disclosure. In addition, in combination with FIG. **65** and FIG. **66**, the ear may also be equivalent to a baffle set near the hearing position, which has an effect of converging and reflecting the sound transmitted to the outside of the earphone **10**, thereby changing the sound field distribution. As a result, it not only helps to increase the sound pressure of the hearing position, but also reduces the sound pressure in the far-field. Specifically, the hearing position may be set between the baffle and the monopole sound source A_1 . The baffle may distort the sound field distribution, thereby increasing the sound pressure at the hearing position. Meanwhile, an entire sound field may still retain a large region for anti-phase cancellation may still, thereby reducing the sound pressure in the far-field. It should be noted that the head of the user may also be used as a part of the baffle. Further, since a distance between each of the two monopole sound sources and the ear may be much smaller than the size of the ear, the ear may achieve an effect similar to an acoustic reflector.

FIG. **67** is a schematic diagram illustrating a theoretical model of an acoustic dipole with a baffle according to some embodiments of the present disclosure. The inventors of the present disclosure have discovered in long-term research that, in a theoretical model of the coordination of the acoustic dipole and the baffle, in combination with FIG. **67**, the parameter α may be mainly affected by the factors including an angle θ between a connection line of the two monopole sound sources (denoted as A_1 - A_2) and a normal line of the baffle, a distance d between the two monopole sound sources, a distance D between the monopole sound source A_1 and the hearing position, a length L of the baffle, a distance B between the baffle and the hearing position, or the like. When the angle θ and the distance d are constant, the greater the length L of the baffle is and the smaller the distance B is, the smaller the parameter α may be, that is, the better the sound leakage reduction effect may be. Based on the related description mentioned above, the ear of the user may be regarded as the baffle, so that the length L may be relatively determined, for example, about 50-80 mm, and the distance B may be about zero. Further, in order to increase the sound pressure at the hearing position to increase the listening effect, the first opening **201** may be generally set as close to the ear hole as possible, that is, the distance D may be generally as small as possible. For example, a distance between the center of the first opening **201** and the center of the ear hole may be smaller than or equal to 16 mm. As another example, a distance between the lower edge of the

holding component **13** facing the ear hole and the highest point (for example, CP1) of the hook-shaped component **11** away from the holding component **13** in the height direction may be greater than or equal to 19 mm. Further, if the distance d is too small, the sound pressure at the hearing position may decrease, which is not conducive to listening sound. If the distance d is too large, the sound pressure in the far-field may increase, which is not conducive to reducing sound leakage. In addition, an actual size of the holding component **13** may also be considered. Therefore, the distance between the center of the second opening **301** and the center of the first opening **201** may be in a range of 7 mm to 15 mm. In a specific embodiment, the distance between the centers of the second opening **301** and the first opening **201** may be 9 mm.

FIG. **68** is a schematic diagram illustrating a relationship between a parameter α and an angle θ according to some embodiments of the present disclosure. Further, in combination with FIG. **68**, taking "without baffle" as a reference, "with baffle" may be obviously beneficial to reduce the parameter α , that is, to increase the sound leakage reduction effect. When the angle $\theta=0^\circ$, the parameter α may reach the minimum value, which indicates that the best sound leakage reduction effect is obtained. In the present disclosure, the angle θ may be within the range of $\pm 80^\circ$. Preferably, the angle θ may be within the range of $\pm 40^\circ$. More preferably, the angle θ may be within the range of $\pm 20^\circ$. In combination with FIG. **63**, considering that the second opening **301** is generally arranged at the side of the first opening **201** away from the ear hole, the angle θ may only take a positive value.

FIG. **69** is a schematic diagram illustrating a relative relationship between an acoustic dipole and an ear according to some embodiments of the present disclosure. Merely by way of example, in combination with FIG. **69** and FIG. **63**, a three-dimensional reference coordinate system (denoted as $X'Y'Z'$) may be established based on any three of the basic sections and any three of the basic axes of the human body that are perpendicular to each other. The angle θ between the connection line of the two monopole sound sources and the normal line of the baffle may be determined by angles between the line A_1 - A_2 and the X' axis, the Y' axis, and the Z' axis, respectively. Based on the related description mentioned above, the connection line A_1 - A_2 between the two monopole sound sources may also be regarded as the connection line (denoted as O_1 - O_0) between the center (for example, O_1) of the second opening **301** and the center (for example, O_0) of the first opening **201**. An angle θ_1 between the connection line O_1 - O_0 and the sagittal plane may be greater than or equal to 10° . Preferably the angle θ_1 may be greater than or equal to 30° . An angle θ_2 between the connection line O_1 - O_0 and the coronal plane may be greater than 0° , preferably the angle θ_2 may be greater than or equal to 4° . An angle θ_3 between the connection line O_1 - O_0 and the horizontal plane may be smaller than or equal to 80° , preferably the angle θ_3 may be smaller than or equal to 60° . In a specific embodiment, the three angles θ_1 , θ_2 , and θ_3 may be 34° , 5° and 56° , respectively.

Further, when the earphone **10** is in the wearing state, the holding component **13** may be close to the front side of the ear, and the first opening **201** on the holding component **13** may face the ear, so that the holding component **13** can be simply regarded as an average normal line of the baffle perpendicular to the first opening **201**. An angle between the connection line O_1 - O_0 and the reference plane perpendicular to the average normal line of the first opening **201** may be between 25° and 55° . The average normal line may be determined according to Equation as below.

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$$\hat{r}_0 = \frac{\iint_S \hat{r} ds}{\left| \iint_S \hat{r} ds \right|};$$

where $\hat{r}_0$ denotes the average normal line; $\hat{r}$ denotes a normal line of any point on a surface, ds denotes a surface element.

When the first opening **210** is a plane, the reference plane perpendicular to the average normal line may be a tangent plane of the first opening **201**. Correspondingly, the average normal line may be parallel to the vibration direction of the core **14** and the thickness direction. Therefore, an angle between the connection line O1-O0 and the vibration direction may be between 0° and 50°, preferably may be between 0° and 40°.

Further, based on the related description mentioned above, the ear may be simply regarded as the baffle cooperating with the acoustic dipole. A reference plane may be determined through at least three physiological positions on the front side of the ear that are not collinear. For example, connection lines between each two of the upper ear root, the intertragic notch, and the Darwin's nodule may form a reference plane (denoted as LA-LB-LD), which may be used to describe the baffle. The angle between the connection line O1-O0 and the reference plane may be between 23° and 53°. In a specific embodiment, the angle between the connection line O1-O0 and the reference plane may be 38°.

Further, when the earphone **10** is in the wearing state, the earphone **10** may form a plurality of contact points with the ear to ensure the stability of wearing. As a result, there may also be positions on the earphone **10** corresponding to the contact points, respectively. In the embodiment in which the hook-shaped component **11** is configured with the elastic component **112**, the elastic deformation of the elastic component **112** before and after wearing may cause a certain deviation in the correspondence relationship, and the deviation may be controlled by the deformability of the elastic component **112**. Therefore, for ease of description, the deviation may be tolerable. Merely by way of example, in combination with FIG. 47 and FIG. 75, the free end of the holding component **13** away from a fixing assembly **20** may have a first reference point (for example, CP0) for contact with the front side of the ear. The fixing assembly **20** may have a second reference point (for example CP3) for contact with the upper ear root and a third reference point (for example CP6) for contact with the ear on the rear side of the ear. Connection lines between each two of the first reference point, the second reference point, and the third reference point may form a reference plane (denoted as CP0-CP3-CP6), and the reference plane may be used to describe the baffle. The angle between the connection line O1-O0 and the reference plane may be between 15° and 45°. In a specific embodiment, the angle between the connection line O1-O0 and the reference plane may be 30°.

It should be noted that compared with the baffle, the front surface of the ear may not be a flat and regular structure. Therefore, the above-mentioned parameters related to the parameter α may be obtained through theoretical analysis and actual measurement. The actual measurement may refer to a measurement performed after the earphone **10** is worn on the simulator (for example, GRAS 45BC KEMAR).

As is known to all, although a frequency range of sounds that can be felt by normal people's ears is between 20 Hz and 20 kHz, it does not mean that all of these sounds can be heard. In general, normal people's ears may mainly hear

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sounds with frequencies below 4 kHz. Thus, on the one hand, a resonant frequency of the first sound transmitted to the outside of the earphone **10** through the first opening **201** may be shifted to a high frequency as much as possible, so that a frequency response curve of the first sound can be as flat as possible in a medium-high frequency band, thereby increasing the listening effect. On the other hand, a resonant frequency of the second sound transmitted to the outside of the earphone **10** through the second opening **301** may also be shifted to the high frequency as much as possible, which can not only reduce the user's sensitivity to the sound leakage, but also make the anti-phase cancellation can be extended to a high frequency band, so as to reduce the sound leakage without affecting the listening effect. Therefore, the frequency response curve of the first sound may have a first lowest resonance peak of the medium-high frequency. The first lowest resonance peak of the medium-high frequency may be a resonance peak with the lowest frequency among all resonance peak frequencies in the medium-high frequency and above frequency bands of the frequency response curve formed by the first opening **201**. Similarly, the frequency response curve of the second sound may have a second lowest resonance peak of the medium-high frequency. The second lowest resonance peak of the medium-high frequency may be a resonance peak with the lowest frequency among all resonant peak frequencies in the medium-high frequencies and above frequency bands of the frequency response curve formed by the second opening **301**. In short, the frequency response curve of the first sound may have a first resonance peak with the lowest frequency in the medium-high frequency band and above frequency band. Similarly, the frequency response curve of the second sound may have a second resonance peak with the lowest frequency in the medium-high frequency band and above frequency band. A peak resonance frequency of the first lowest resonance peak of the medium-high frequency and a peak resonance frequency of the second lowest resonant peak of the medium-high frequency may be greater than or equal to 5 kHz. Preferably, the peak resonance frequency of the first lowest resonance peak of the medium-high frequency and the peak resonance frequency of the second lowest resonant peak of the medium-high frequency may be greater than or equal to 6 kHz. Further, a difference between the peak resonance frequency of the first lowest resonance peak of the medium-high frequency and the peak resonance frequency of the second lowest resonant peak of the medium-high frequency may be smaller than or equal to 1 kHz, so that the anti-phase cancellation may be better performed on the second sound and the first sound in the far-field.

It should be noted that in the present disclosure, a frequency range corresponding to a low-frequency band may be in a range of 20 Hz-150 Hz. A frequency range corresponding to a middle-frequency band may be a range of 150 Hz-5 kHz. A frequency range corresponding to a high-frequency band may be a range of 5 k-20 kHz. A frequency range corresponding to a medium-low frequency band may be a range of 150 Hz-500 Hz. A frequency range corresponding to the medium-high frequency band may be a range of 500 Hz-5 kHz. For a frequency response curve described in the present disclosure, the horizontal axis may represent frequency, and the unit may be Hz. The vertical axis may represent intensity, and the unit may be dB. Further, the first lowest resonance peak of the medium-high frequency may include a resonant peak generated by cavity resonance, and/or a standing wave peak generated by reflection from a cavity surface of a cavity. The second lowest

resonance peak of the medium-high frequency may be similar to the first lowest resonance peak of the medium-high frequency, and details may not be described herein.

Based on the detailed description mentioned above, the user may mainly hear the first sound when wearing the earphone **10**, thus the peak resonance frequency of the first lowest resonance peak of the medium-high frequency may have a great influence on the listening effect. The first lowest resonance peak of the medium-high frequency is studied to improve the listening effect. The resonant peaks of the frequency response curve of the first sound in the medium-high frequency band and above frequency band may be mainly caused by cavity resonance, which generally satisfies the calculation formula of the resonant frequency of the Helmholtz resonant cavity:

$$f_0 = \frac{c_0}{2\pi} \sqrt{\frac{S}{V(l + 1.7r)}}$$

where, f_0 denotes the resonance frequency of the cavity resonance, c_0 denotes a speed of sound in the air, S denotes the actual area of the first opening **201**, V denotes a volume of the front cavity **200**, l denotes a length of the first opening **201**, and r denotes an equivalent radius of the first opening **201**. l generally depends on a wall thickness of the housing.

Obviously, the greater the actual area of the first opening **201** is and the smaller the volume of the front cavity **200** is, the higher the resonance frequency corresponding to cavity resonance may be, that is, the first lowest resonant peak of the medium-high frequency may be easier to shift to a higher frequency. Further, the first opening **201** may be generally covered with an acoustic resistance net to increase the waterproof and dustproof performance and adjust the frequency response curve. Merely by way of example, an effective area of the first opening **201** may be greater than or equal to 2 mm². In a specific embodiment, the actual area of the first opening **201** may be greater than or equal to 7 mm², and a porosity of the acoustic resistance net covered on the first opening **201** may be greater than or equal to 13%. In some embodiments, a pore size may be greater than or equal to 18 μm. Further, the volume of the front cavity **200** may be smaller than or equal to 90 mm³. The volume of the front cavity **200** may be approximately a product of the area of the diaphragm **143** and the depth of the front cavity **200** in the vibration direction of the core **14**. After the specification and model of the core **14** are selected, and on a premise that the vibration stroke of the diaphragm **143** is satisfied, the depth of the front cavity **200** in the vibration direction may be as small as possible. Therefore, the maximum depth of the front cavity **200** in the vibration direction may be smaller than or equal to 3 mm, preferably may be smaller than or equal to 1 mm.

FIG. **70** is a schematic diagram illustrating a structure of an earphone facing a side of an ear according to some embodiments of the present disclosure. Further, in combination with FIG. **70**, when the front cavity **200** is configured as a cubic structure, the cavity surface of the front cavity **200** may form at least a pair of parallel or approximately parallel reflecting surfaces, thereby forming a standing wave. Specifically, when a sound wave is reflected in the cavity, an incident wave and a reflected wave may be superimposed to form a fixed antinode and a fixed node, thereby triggering a standing wave at a specific frequency. In other words, the resonance peaks of the frequency response curve of the first sound in the medium-high frequency band and above fre-

quency band may also be derived from a standing wave, which generally satisfies the calculation equation as follows:

$$f_0 = \frac{c_0}{4L} (2n - 1),$$

where, f_0 denotes a frequency of a standing wave peak, c_0 denotes the speed of sound in the air, L denotes a distance between the center of the first opening **201** and the cavity surface of the front cavity **200**, and n denotes a positive integer.

Obviously, the smaller the distance L is, the higher the frequency corresponding to the standing wave peak may be. That is, the first lowest resonance peak of the medium-high frequency may be easier to shift to a higher frequency. Merely by way of example, on a reference plane perpendicular to the vibration direction of the core **14** (for example, the plane where Y1Z1 is located), the distance between the center of the first opening **201** and the cavity surface of the front cavity **200** may be smaller than or equal to 17.15 mm.

Based on the related description mentioned above, the front cavity **200** may have a first front cavity surface **202** and a third front cavity surface **204** spaced apart from each other in the major axis direction of the core **14**, and a second front cavity surface **203** and a fourth front cavity surface **205** spaced apart from each other in the minor axis direction of the core **14**. The first front cavity surface **202** may be closer to the connecting component **12** than the third front cavity surface **204**. The fourth front cavity surface **205** may be closer to the ear hole than the second front cavity surface **203**. A distance between the first front cavity surface **202** and the third front cavity surface **204** may be greater than or equal to a distance between the second front cavity surface **203** and the fourth front cavity surface **205**. Further, vertical distances from the center of the first opening **201** to the first front cavity surface **202**, the second front cavity surface **203**, the third front cavity surface **204**, and the fourth front cavity surface **205** may be defined as a first distance L_1 , a second distance L_2 , a third distance L_3 , and a fourth distance L_4 , respectively. Assuming that the four vertical distances have the following basic relationship: $L_1 \geq L_2 \geq L_3 \geq L_4$, then frequencies corresponding to the corresponding standing wave peaks may have the following relationship: $f_1 \leq f_2 \leq f_3 \leq f_4$. A first standing wave peak of the first sound in the medium-high frequency band and above frequency band may be determined by the greatest distance among the four vertical distances, so that $L_1 \leq 17.15$. Merely by way of example, the first distance may be smaller than or equal to the third distance, and the fourth distance may be smaller than or equal to the second distance, so that the first opening **201** may be closer to the ear hole.

It should be noted that the first opening **201** may be opposite to the diaphragm **143** in the vibration direction of the core **14**, and a ratio of the size of the first opening **201** in the major axis direction of the core **14** to the size of the first opening **201** in the minor axis direction of the core **14** may be smaller than or equal to 3. For example, the first opening **201** may be set in a circular shape. As another example, the first opening **201** may be set in a racetrack shape.

FIG. **71** is a schematic diagram illustrating a structure of an earphone according to some embodiments of the present disclosure. In combination with FIG. **71**, the earphone **10** may further include a Helmholtz resonant cavity **400** communicating with the front cavity **200**. The Helmholtz resonant cavity **400** may be configured to weaken a peak

resonance intensity of the first lowest resonance peak of the medium-high frequency. That is, a sound energy in the front cavity 200 near the peak resonance frequency may be absorbed to suppress a sudden increase of the peak resonance intensity, so that the frequency response curve can be flatter, and the sound quality may be more balanced. FIG. 72 is a schematic diagram illustrating a frequency response curve of an earphone according to some embodiments of the present disclosure. Merely by way of example, in combination with FIG. 72, a difference between the peak resonance intensity of the first lowest resonant peak of the medium-high frequency when the opening connecting the Helmholtz resonant cavity 400 and the front cavity 200 is in an open state (denoted as "HR_Y") and the peak resonance intensity of the first lowest resonant peak of the medium-high frequency when the opening connecting the Helmholtz resonant cavity 400 and the front cavity 200 is in a closed state (denoted as "HR_N") may be greater than or equal to 3 dB. Further, the opening connecting the Helmholtz resonant cavity 400 and the front cavity 200 may be configured with an acoustic resistance net to further adjust the frequency response curve. A porosity of the acoustic resistance net may be greater than or equal to 3%.

Further, there may be multiple Helmholtz resonance cavities 400 to better absorb the acoustic energy in the front cavity 200 near the peak resonance frequency. The multiple Helmholtz resonance cavities 400 may be arranged in parallel with the front cavity 200, for example, respectively in communication with the front cavity 200. Alternatively, the multiple Helmholtz resonant cavities 400 may be arranged in series with the front cavity 200, for example, communicating with the front cavity 200 through one of the multiple Helmholtz resonant cavities 400.

In some embodiments, in combination with FIG. 52, the Helmholtz resonance cavity 400 may be arranged in the second region 13B, for example, in the flexible covering structure 132. Specifically, the blind hole 1321 in the flexible covering structure 132 may not only provide a deformation space for the flexible covering structure 132, but also serve as the Helmholtz resonant cavity 400. Correspondingly, a communicating hole connecting the Helmholtz resonant cavity 400 and the front cavity 200 may be arranged on the cover plate 1316.

In other embodiments, in combination with FIG. 57, the Helmholtz resonance cavity 400 may be arranged in the connecting component 12, for example, between the third housing 122 and the first housing 1314. Specifically, the first flange may be arranged on the inner wall surface of the first housing 1314 facing the third housing 122, and the third housing 122 may be pressed and held on the first flange to enclose and form the Helmholtz resonance cavity 400. Alternatively, the inner wall surface of the third housing 122 facing the first housing 1314 may be configured with the second flange. The first housing 1314 may be pressed and held on the second flange to enclose and form the Helmholtz resonance cavity 400. In short, the third housing 122 and the first housing 1314 may be buckled together to form the Helmholtz resonance cavity 400. Further, the Helmholtz resonance cavity 400 may be formed by a blow molding process, and then be arranged and fixed in the connecting component 12.

Based on the detailed description mentioned above, in order to shift the resonant frequency of the second sound to the high frequency as much as possible, the rear cavity 300 may adopt the same or similar technical solution as the front cavity 200, which may not be repeated herein. A main difference from the front cavity 200 may be that for a

standing wave, the rear cavity 300 may destroy a high pressure region of the sound field in the rear cavity 300 to shorten the wavelength of the standing wave in the rear cavity 300, thereby making the peak resonant frequency of the second lowest resonant peak of the medium-high frequency as large as possible. In combination with FIG. 63, the third opening 302 may be arranged in the high pressure region of the sound field in the rear cavity 300. For example, the third opening 302 and the second opening 301 may be arranged on opposite sides of the core 14. FIG. 74 is a schematic diagram illustrating a frequency response curve of an earphone according to some embodiments of the present disclosure. Merely by way of example, in combination with FIG. 74, the peak resonance frequency of the second lowest resonant peak of the medium-high frequency when the opening 302 is in the open state (denoted as "Turn-on") may be shifted to a high frequency compared to the peak resonance frequency of the second lowest resonance peak of the medium-high frequency when the third opening 302 is in the closed state (denoted as "Turn-off"), and a shift amount may be greater than or equal to 1 kHz. Further, an effective area of the third opening 302 may be smaller than an effective area of the second opening 301 so as to adjust the peak resonance frequency of the second lowest resonance peak of the medium-high frequency. In some embodiments, the size of the second opening 301 in the major axis direction of the core 14 may be larger than the size of the first opening 201 in the major axis direction of the core 14.

FIG. 73 is a schematic diagram illustrating a structure of a rear cavity of an earphone according to some embodiments of the present disclosure. Based on the related description mentioned above, in combination with FIG. 73, the rear cavity 300 may have a first rear cavity surface 303 and a second rear cavity surface 304 spaced apart from each other in the major axis direction of the core 14. The second opening 302 and the third opening may be spaced apart from each other in the minor axis direction of the core 14. An actual area of the third opening 302 may be smaller than an actual area of the second opening 301, so that an effective area of the third opening 302 may be smaller than an effective area of the second opening 301. A section of at least one of the first rear cavity surface 303 and the second rear cavity surface 304 close to the third opening 302 may be arranged in an arc shape along the vibration direction of the core 14 to avoid sharp structures such as a right angle, a sharp corner etc., on the inner wall of the enclosed rear cavity 300, which is beneficial to eliminate standing waves. Further, at least one of the first cavity surface 303 and the third cavity surface 305 may be arranged in an arc shape along the minor axis direction, which is also beneficial to eliminate standing waves.

Further, the opening direction of the second opening 301 may face the top of the head of the user. For example, an angle between the opening direction and the vertical axis may be between 0° and 10°, so as to allow the second opening 301 to be farther away from the ear hole than the third opening 302. As a result, it can be difficult for the user and other people in the surrounding environment to hear the sound output to the outside of the earphone 10 through the second opening 301, thereby reducing sound leakage. The opening direction of the second opening 301 may refer to a direction where the average normal line is located. Correspondingly, the second opening 301 may have the first center (for example O1) in the major axis direction of the core 14. The third opening 302 may have the second center (such as O2) in the major axis direction. The second center may be farther from the center of the first opening 201 than the first

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center in the major axis direction, so as to increase the distance between the third opening 302 and the first opening 201 as much as possible, thereby weakening the anti-phase cancellation between the sound output to the outside of the earphone 10 through the third opening 302 and the sound transmitted to the ear through the first opening 201. The first rear cavity surface 303 may be closer to the connecting component 12 than the second rear cavity surface 304. A radius of curvature of at least a part of the first rear cavity surface 303 may be greater than a radius of curvature of the corresponding part of the second rear cavity surface 204.

Merely by way of example, the first rear cavity surface 303 may include a first sub-rear cavity surface 3031, a second sub-rear cavity surface 3032, and a third sub-rear cavity surface 3033 that are sequentially connected. The first sub-rear cavity surface 3031 may be closer to the second opening 301 and farther from the second rear cavity surface 304 than the third sub-rear cavity surface 3033. At least the second sub-rear cavity surface 3032 of the second sub-rear cavity surface 3032 and the third sub-rear cavity surface 3033 may be arranged in an arc shape. For example, the second sub-rear cavity surface 3032 may be arranged in a shape of an arc. A radius of the arc may be greater than or equal to 2 mm. In a direction in which the second opening 301 points to the third opening 302, an angle between a tangent line of the second sub-rear cavity surface 3032 and the minor axis direction of the core 14 may gradually increase, and an angle between a tangent line of the third sub-rear cavity surface 3033 and the minor axis direction may keep unchanged or gradually decrease.

It should be noted that the fixing assembly 20 being connected to the holding component 13 described in the present disclosure may be mainly used to cause the holding component 13 to contact the front side of the ear in the wearing state. In some embodiments, the fixing assembly 20 may include the hook-shaped component 11 and the connecting component 12 connected to the hook-shaped component 11 and the holding component 13. The related structure and the connection relationship may refer to the detailed description for any embodiment of the present disclosure, which may not be repeated herein. FIG. 75 is a schematic diagram illustrating structures of earphones in a wearing state respectively according to some embodiments of the present disclosure. In other embodiments, in combination with FIG. 75, the fixing assembly 20 may be arranged in an annular shape and around the ear, for example, as shown in diagram (a) in FIG. 75. In some embodiments, the fixing assembly 20 may also be arranged as an ear hook and a rear hook structure and around the rear side of the head, as shown in diagram (b) in FIG. 75. In some embodiments, the fixing assembly 20 may also be arranged into a head beam structure and around the top of the head, as shown in diagram (c) in FIG. 75. Further, the technical solutions described in the present disclosure may be applied to an earphone, a hearing aid, audio glasses, or other smart glasses such as AR, VR, MR, or the like.

The descriptions may be only part of the embodiments of the present disclosure and may not limit the scope of the present disclosure. Any equivalent device or equivalent process transformation made by using the illustration for the description and drawings of the present disclosure, or directly or indirectly used in other related technical fields, may be included in the scope of the present disclosure with the same principles.

It's noticeable that above statements are preferable embodiments and technical principles thereof. A person having ordinary skill in the art is easy to understand that this

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disclosure is not limited to the specific embodiments stated, and a person having ordinary skill in the art can make various obvious variations, adjustments, and substitutes within the protected scope of this disclosure. Therefore, although above embodiments state this disclosure in detail, this disclosure is not limited to the embodiments, and there can be many other equivalent embodiments within the scope of the present disclosure, and the protected scope of this disclosure is determined by following claims.

What is claimed is:

1. An earphone, comprising: a holding component including a housing; a transducer residing inside the housing and configured to generate vibrations, the vibrations producing a sound wave inside the housing; at least two sound guiding holes located on the housing and configured to guide the sound wave inside the housing through the at least two sound guiding holes to an outside of the housing, wherein the at least two sound guiding holes include a first sound guiding hole and a second sound guiding hole, the first sound guiding hole and the second sound guiding hole are located on different side walls of the housing, and the guided sound waves of the first sound guiding hole and the second sound guiding hole have different phases; and a hook-shaped component configured to hang between a rear side of an ear of a user and a head of the user; and a connecting component configured to connect the hook-shaped component and the holding component and cooperate with the hook-shaped component to provide the holding component with a pressing force on the front side of the ear, so that the hook-shaped component and the holding component cooperate to form a clamping for the ear; and wherein the housing includes a specific side wall where no sound guiding hole is located, and a distance from the first sound guiding hole to the specific side wall of the housing is different from a distance from the second sound guiding hole to the specific side wall of the housing.

2. The earphone of claim 1, wherein the connecting component extend from the head along a coronal axis of the user towards an outside of the head to cooperate with the hook-shaped component.

3. The earphone of claim 1, wherein in a direction from a first connection point between the hook-shaped component and the connecting component to a free end of the hook-shaped component,

the hook-shaped component is bent toward the rear side of the ear to form a first contact point with the rear side of the ear, and

the holding component forms a second contact point with the front side of the ear, wherein a distance between the first contact point and the second contact point along an extending direction of the connecting component in a natural state is smaller than that in a wearing state to provide the holding component with the pressing force on the front side of the ear.

4. The earphone of claim 3, wherein the hook-shaped component further forms a third contact point with the rear side of the ear, the third contact point being located between the first connection point and the first contact point and being closer to the first connection point, wherein

a distance between projections of the first contact point and the third contact point on a reference plane perpendicular to the extending direction of the connecting component in the natural state is smaller than that in the wearing state.

5. The earphone of claim 1, wherein in a direction from a first connection point between the hook-shaped component and the connecting component to a free end of the hook-

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shaped component, the hook-shaped component is bent toward the head to form a first contact point and a second contact point with the head, wherein

the first contact point is located between the second contact point and the first connection point, so that the hook-shaped component forms a lever structure with the first contact point as a fulcrum; and

a force directed to the outside of the head and provided by the head at the second contact point causes a force directed to the head at the first connection point by the lever structure to provide the holding component with the pressing force on the front side of the ear through the connecting component.

6. The earphone of claim 1, wherein the holding component extends and is held in a concha boat of the ear.

7. The earphone of claim 1, wherein an elastic metal wire is arranged inside the hook-shaped component, the elastic metal wire has a major axis direction and a minor axis direction orthogonal to each other on a cross section of the elastic metal wire, and a size of the elastic metal wire in the major axis direction is greater than a size of the elastic metal wire in the minor axis direction, so that the hook-shaped component and the holding component cooperate to form an elastic clamping for the ear.

8. The earphone of claim 7, wherein a ratio of the size of the elastic metal wire in the major axis direction to the size of the elastic metal wire in the minor axis direction is between 4:1 and 6:1.

9. The earphone of claim 7, wherein the elastic metal wire is in a shape of an arc in the minor axis direction, and a ratio of a height of the arc to the size of the elastic metal wire in the major axis direction is within a range of 0.1-0.4.

10. The earphone of claim 1, further including an extending component, wherein

the extending component is arranged on the holding component, and extends into any one of a concha cavity, a concha boat, a triangular fossa, and a scapha of the ear in the wearing state; and/or

the extending component is arranged on the hook-shaped component, and hooks a helix and/or an antihelix of the ear in the wearing state.

11. The earphone of claim 1, wherein the holding component has a thickness direction, a length direction, and a height direction orthogonal to each other, the thickness direction is configured as a direction in which the holding component is close to or away from the ear in the wearing state, and the height direction is configured as a direction in which the holding component is close to or away from a top of the user's head in the wearing state, wherein

in the natural state, and viewed from a side of the earphone facing the top of the user's head in the wearing state, the holding component is at least spaced

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apart from a section of the hook-shaped component close to the connecting component in the thickness direction; and

the connecting component is arranged in a shape of an arc and connected between the holding component and the hook-shaped component.

12. The earphone of claim 11, wherein in the thickness direction, a minimum distance between the section of the hook-shaped component close to the connecting component and the holding component is greater than 0 and smaller than or equal to 5 mm.

13. The earphone of claim 11, wherein

edges of the section of the hook-shaped component close to the connecting component, the connecting component, and the holding component facing the ear are arranged in a shape of a circuitous arc; and

in a reference direction that passes through a roundabout inflection point of the circuitous arc and is parallel to the length direction, a minimum width of the circuitous arc along the thickness direction at a position 3 mm away from the roundabout inflection point is in a range of 1 mm to 5 mm.

14. The earphone of claim 11, wherein the at least two sound guiding holes include a sound guiding hole disposed on a side of the holding component facing the ear, and a distance between a center of the sound guiding hole and the section of the hook-shaped component close to the connecting component in the thickness direction is in a range of 3 mm to 6 mm.

15. The earphone of claim 11, wherein a side of the holding component facing the ear includes a first region and a second region, wherein

the at least two sound guiding holes include a sound guiding hole disposed in the first region; and

the second region is farther away from the connecting component than the first region and is more protruding toward the ear than the first region, so as to allow the sound guiding hole to be spaced from the ear in the wearing state.

16. The earphone of claim 15, wherein a distance between the second region and the section of the hook-shaped component close to the connecting component in the thickness direction is in a range of 1 mm to 5 mm.

17. The earphone of claim 16, wherein an orthographic projection of the section of the hook-shaped component close to the connecting component in the thickness direction partially overlaps the second region.

18. The earphone of claim 15, wherein a maximum protrusion height of the second region relative to the first region in the thickness direction is greater than or equal to 1 mm.

19. The earphone of claim 1, wherein the holding component is in contact with an antihelix of the ear.

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